CONSOLIDATED RESEARCH MANUSCRIPT

Euler's Constant

Positive Kernels, Primitive Arithmetic, and Exact Output Geometry

From positive realization to columnwise budgets
and a sharp theorem for negative prefixes

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2 November 2026

Scope. This single manuscript consolidates the six recovered gamma papers and the additional junction notes following the corpus "gamma TOP10" discussion. Distinct proofs and finite certificates are retained; repeated foundations are shared. A new sharp positivity theorem and its exact output-polytope consequence are proved during the consolidation. The irrationality of Euler's constant is not established. Conditional completion statements keep their existence hypotheses explicit.

Reproducibility. The single TeX source contains the entire manuscript, bibliography, and a compressed verification collection encoded after `\end{document}`. No external TeX inputs, bibliography database, or image files are required to compile the paper. Extraction instructions appear in Appendix C.

Abstract

We consolidate a sequence of investigations into Euler's constant, organizing them around one arithmetic requirement: the error must tend to zero after the actual output denominator and complete gcd reduction have been accounted for. Positive Binet kernels yield explicit rational brackets, exact primitive-error optimization, sharp degree bounds, and target realization. Laguerre coordinates give optimal integerization and unconditional obstructions for several localization, interpolation, acceleration, and autonomous-dilation families. Proper rational kernels lead to exact prime-support ranks, bandwidth constraints, primitive quotient norms, infinite dual formulations, and certified finite lattice minima. Weighted-prefix geometry converts finite positivity problems into compact rational intervals and gives an intrinsic cost bound with no linear restriction on the support horizon. A new sharp theorem allows nonuniform negative prefixes: if $a_j = 1/(16^j j^4 \binom{3j}{j}^2)$ and $\lambda_k \geq 0$ for $n \leq k < M$, then every vector with final prefix $q > 0$ and earlier prefixes $S_k \geq -q\lambda_k$ has a strictly positive kernel exactly when $\sum_{k<M} \lambda_k(a_k - a_{k+1}) \leq a_M$. The endpoint is included. This yields a rational output polytope, constructive strict enlargement of every nondegenerate old prefix interval, and a single-output conditional completion criterion. Finally, individual column clearing gives a uniform convergent arithmetic budget, while binary congruences distinguish thin-block obstructions from a surviving output class in an actual wider integer kernel. Exact finite success, unconditional general lemmas, and unresolved infinite existence are kept separate. No proof of the irrationality of Euler's constant or historical-priority claim is made.

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1 The common objective and the logical status of the results

Write

$$\gamma = \lim_{m\to\infty}(H_m - \log m), \qquad H_m = \sum_{k=1}^{m} \frac{1}{k}, \quad H_0 = 0.$$

For a reduced rational p/q, $q > 0$, the relevant quantity is

$$\mathcal{E}(p/q) = q\left|\gamma - \frac{p}{q}\right| = |q\gamma - p|.$$

Convergence of p/q to γ does not by itself control this quantity. Nor does rapid decay of a rational-coefficient integral before its denominator is cleared.

Proposition 1.1 (The five requirements, for one family)**.** *Suppose that for an unbounded sequence of indices there are integers q_n, p_n with*

$$0 < |q_n\gamma - p_n| \longrightarrow 0.$$

Then γ is irrational. More generally, if integer polynomials W_n have uniformly bounded degree and $0 < |W_n(\gamma)| \to 0$, the same conclusion follows.

Proof. If $\gamma = a/b$, $a, b \in \mathbb{Z}$, $b > 0$, then $b(q_n\gamma - p_n)$ is a nonzero integer tending to zero. In the polynomial case, if the degrees are at most d, use the fixed integerizing factor b^d. □

Corollary 1.2 (Independent integer outputs)**.** *Suppose that for every index in an unbounded sequence there are two linearly independent integer pairs $(q_{i,n}, p_{i,n})$, $i = 1, 2$, with $\max_i |q_{i,n}\gamma - p_{i,n}| \to 0$. Then γ is irrational.*

Proof. The integer polynomial $\sum_{i=1}^{2}(q_{i,n}X - p_{i,n})^2$ has degree at most two, is strictly positive for every real X by independence, and tends to zero at γ. Apply Proposition 1.1. □

The integer assertion in this argument is conditional on the rationality assumption: $q_n\gamma - p_n$ itself has integer *coefficients*, not a known integer value. The five operational requirements are therefore:

1. exact cancellation of the logarithmic coordinates and integer output coefficients;
2. a proof of nonvanishing for the evaluated form;
3. convergence to zero after all denominator clearing and output gcd reduction;
4. these facts for one specified family at infinitely many indices;
5. proofs without an unproved premise supplying any of the preceding properties.

The quantifiers must belong to the same construction. Existence of short log-null inputs and separate existence of outputs with useful parity does not prove a short representative in that output class. Likewise, a positive family and a different small-error family cannot be combined without an explicit common construction.

The main routes in this manuscript. Part I realizes positive targets and optimizes their primitive errors. Part II proves why several natural families fail after reduction. Part III establishes the common proper-kernel arithmetic and exact cancellation structure. Part IV transfers optimization to output space. Part V connects kernel signs, primitive costs, rational intervals, and exact finite repairs. Part VI secures a columnwise smallness budget and studies which arithmetic output classes survive.

What is added by the present consolidation. Theorem 42.2 gives the exact nonuniform negative-prefix threshold. Its individual-vector consequence is

$$\sum_{k=n}^{M-1}(a_k - a_{k+1})(-S_k)_+ \le qa_M \quad \Longrightarrow \quad K_c(u) > 0 \ (u > 0).$$

This is a sufficient test for an individual vector, and an exact threshold for the universal class with prescribed prefix lower bounds. Those two scopes are distinguished. Theorem 42.5 shows that the complete budget, including variable individual deficits, has an exact compact rational output interval. Corollary 42.6 gives explicit rational extrapolations that enlarge both ends of every nondegenerate old nonnegative-prefix interval, while preserving every prime-log cancellation equation and strict positivity. This is an actual finite-window improvement, though its least denominator need not decrease strictly. The resulting single-output completion criterion is Corollary 42.7. It removes the need for an independent determinant partner inside this positive subclass; it does not construct an infinite cheap arithmetic family.

Conventions and notation. All logarithms are natural unless a base is specified. The symbols v_ℓ, den, and gcd denote rational prime valuation, positive reduced denominator, and positive gcd. Empty products are one. In the acceleration sections U_j denotes a core-cancelled rational approximant and X_j its acceleration. The proper-kernel integral majorant is always $\mathcal{U}_j$. The matrices for square forms, prime-log coefficients, and output lattices have separate locally specified roles. All finite computational certificates use exact integer or rational arithmetic; decimal upper bounds are rational bounds, not floating-point sign decisions.

2 Corpus context and the ten motivating themes

The starting discussion asked which gamma-related results and constraints in the Verifiable Mathematical Reasoning Corpus were useful for a new irrationality attempt. The following is a source-grounded thematic inventory, not a purported verbatim recovery of an earlier ranked chat response and not an external ranking of mathematical importance. Corpus sources are cited as project manuscripts. The subsequent constructions below are proved here rather than inferred from those sources' theorem counts.

	Theme and source	Role or limitation in the present work
1	Independent forms and nonvanishing [021] [31]	Two output directions give a positive sum of squares; input rank alone does not give small independent outputs.
2	Primitive arithmetic cost [019] [12]	Optimize $\lvert q\gamma - p\rvert$ after the complete gcd, not only the uncleared analytic remainder.
3	Positive kernels and degree constraints [019,021] [12, 31]	Positivity can be constructive and exact; degree and coefficient height remain different resources.
4	Integral moment coordinates [033] [34]	Exact integerization and moment matrices motivate the Laguerre arithmetic used in Part I.
5	Prime-log cancellation and formal zeros [050] [14]	A nonzero kernel vector can have both formal output coordinates zero. Counting alone does not prove a nonzero gamma form.
6	Coefficient locking [022,023] [32, 33]	Integral identities preserve the arithmetic relation they encode; an unproved transfer or nonvanishing principle cannot be inserted.
7	Joint value independence [044] [13]	Independence of auxiliary special-function values does not identify γ without a valid specialization or elimination step.

	Theme and source	Role or limitation in the present work
8	Critical moving-order limits [044] [13]	Matching limiting constants can accelerate rational approximants; their actual denominator must still be analyzed.
9	Gamma derivatives and finite versus all-order results [062] [35]	A finite half-count and all-order window recovery have different quantifiers; neither singles out γ.
10	Lemniscate coordinates [022] [32]	A change of coordinate can restate a target as an arithmetic condition on an inverse integral. The required condition at that specific point remains a separate theorem.

The source manuscripts that explicitly record a repository revision use commit

`8219b3eb00b83a9bb81bc7b7afbc3c49b4538c9c`.

The supplementary corpus files used for this contextual inventory are the recovered project snapshots. No assertion is made that every later repository revision has been audited. In particular, the ten-group priority table in the first gamma manuscript was assembled from three internal notes [1, 2, 3]; it is different from the corpus-themed inventory above. Its complete mathematical contents are mapped in Appendix B.

Relation to the published literature. The general mechanisms employed here have classical antecedents: Binet's integral, orthogonal polynomials, rational quadratic forms, Farey geometry, integer lattices, and convex duality. Sondow and Zlobin [27] develop logarithmic linear forms and irrationality criteria for γ. Ferguson [28] studies rational approximations via Hankel determinants. The recent preprint of Kalisch et al. [29] gives Apéry-like recurrences. These references provide context; none supplies the unresolved infinite output-height assertion in this paper. We do not claim to have excluded identical results from all prior literature.

Part I

Positive realization and exact primitive geometry

3 Canonical compressed families with exact positive kernels

This section analyzes a complete low-degree family inside the shared-cutoff construction. No irrationality claim is made. The new construction gives positive, strictly monotone rational approximants in two parameters; its primitive cost can also be settled uniformly.

Write

$$d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \qquad \mu_j = \int_0^\infty u^j \, d\mu(u) = \frac{|B_{2j+2}|}{2j+2}.$$

For an integer $n \geq 1$, Binet's identity [4] is

$$H_n - \frac{1}{2n} = \gamma + \log n - \Phi(n^2), \qquad \Phi(z) = \int_0^\infty \frac{d\mu(u)}{u+z}.$$

An elementary moment estimate used below is

$$\frac{\mu_{j+1}}{\mu_j} = \frac{(2j+3)(2j+2)}{4\pi^2} \frac{\zeta(2j+4)}{\zeta(2j+2)} < \frac{(j+1)^2}{4}. \tag{1}$$

Indeed the zeta ratio is less than 1, $\pi > 3$, and $(2j+3)(2j+2) < 9(j+1)^2$.

Theorem 3.1 (An exact positive and doubly monotone canonical family)**.** *For integers $m, t \geq 0$, put*

$$P_m(x) = \prod_{j=1}^{m} (4^j x - 1), \quad A_m = P_m(1), \quad s_m = \frac{P_m'(1)}{P_m(1)}, \quad a = 2^t, \quad b = t + s_m, \quad R = 4^{m+1}.$$

Empty products have value 1, so $s_0 = 0$. Define

$$C_{m,t}(x) = \frac{P_m(x)}{A_m}[1 - b(x-1)] = \sum_i c_i x^i, \qquad r_{m,t} = \sum_i c_i \left(H_{a2^i} - \frac{1}{2a2^i} \right).$$

Then $r_{m,t}$ is rational and

$$E_{m,t} := \gamma - r_{m,t} = \int_0^\infty \frac{u^m\{u + a^2[R + b(R-1)]\}}{\prod_{i=0}^{m+1}(u + a^2 4^i)} \, d\mu(u) > 0. \tag{2}$$

Both coordinate increases are strict:

$$r_{m+1,t} > r_{m,t}, \qquad r_{m,t+1} > r_{m,t}.$$

Moreover, uniformly in both parameters,

$$\frac{a^{-2m-2} 4^{-m(m+1)/2}}{3^{15}} \leq E_{m,t} \leq \mu_m (t+m+3) a^{-2m-2} 4^{-m(m+1)/2}. \tag{3}$$

In particular $r_{m,t} \to \gamma$ along every sequence $m + t \to \infty$.

Proof. The defining polynomial satisfies

$$C_{m,t}(1) = 1, \quad C'_{m,t}(1) = -t, \quad C_{m,t}(4^{-j}) = 0 \quad (1 \le j \le m).$$

Binet's identity therefore cancels all logarithms exactly. To evaluate the remaining kernel, first observe the rational identity

$$\sum_{i=0}^{m} \frac{[x^i]P_m(x)}{u + a^2 4^i} = \frac{A_m u^m}{\prod_{i=0}^{m}(u + a^2 4^i)}. \tag{4}$$

After taking a common denominator, the numerator has degree at most m. Its first m Taylor coefficients at zero vanish because $P_m(4^{-j}) = 0$ for $1 \le j \le m$; its leading coefficient is $P_m(1) = A_m$. This proves the identity. Multiplication by x shifts every index by one. Applying this observation to $C_{m,t}$ gives (2), whose integrand is positive for $u > 0$.

For monotonicity, introduce the direction

$$J_{m,t} := \int_0^\infty \sum_i \frac{[x^i]\{P_m(x)(x-1)^2/A_m\}}{u + a^2 4^i}\, d\mu(u).$$

Two shifts of (4) give exactly

$$J_{m,t} = a^2(R-1) \int_0^\infty \frac{u^m\{4a^2(R-1) - 3u\}}{\prod_{i=0}^{m+2}(u + a^2 4^i)}\, d\mu(u). \tag{5}$$

This integral is strictly positive. Indeed, for the positive measure $u^m\, d\mu$ and the strictly decreasing positive weight $w(u) = \prod_{i=0}^{m+2}(u + a^2 4^i)^{-1}$, the weighted mean of u is at most μ_{m+1}/μ_m. This follows by integrating $(u - v)(w(u) - w(v)) \le 0$ against the product measure. By (1), that mean is less than $(m+1)^2/4$, whereas $4a^2(R-1)/3$ is larger. For instance $(m+1)^2 \le 4^m$ and $4(R-1)/3 \ge R$ suffice.

Direct polynomial subtraction now gives

$$C_{m+1,t} - C_{m,t} = -\frac{R[R + b(R-1)]}{(R-1)^2} \frac{P_m}{A_m}(x-1)^2,$$

$$xC_{m,t+1} - C_{m,t} = -(b+1)\frac{P_m}{A_m}(x-1)^2.$$

In the second identity the common external base is $a = 2^t$; the leading x accounts for the new base $2a$. Both differences of gamma errors are negative multiples of $J_{m,t} > 0$, proving strict monotonicity.

For the upper bound in (3), bound each denominator factor below by $a^2 4^i$. This gives

$$E_{m,t} \le \frac{\mu_{m+1} + a^2[R + b(R-1)]\mu_m}{a^{2m+4} 4^{(m+1)(m+2)/2}}.$$

Since $s_m = m + \sum_{j=1}^{m}(4^j - 1)^{-1} < m + 1$, and (1) implies $\mu_{m+1}/(a^2 R \mu_m) < 1$, this proves the stated upper bound.

For the lower bound, integrate only over $1 \le u \le 2$. On that interval $d\mu(u) > 3^{-12} du$, because $2\pi\sqrt{u} < 12$ and $e < 3$. Also

$$\prod_{i=0}^{m+1}(u + a^2 4^i) \le a^{2m+4} 4^{(m+1)(m+2)/2} \exp\left(\sum_{i \ge 0} \frac{2}{4^i}\right) < 27 a^{2m+4} 4^{(m+1)(m+2)/2}.$$

The numerator is at least $a^2 R$ there. Combining these inequalities proves the lower bound.

Finally,

$$\frac{\mu_{m+1}4^{-(m+1)(m+2)/2}}{\mu_m 4^{-m(m+1)/2}} < \frac{(m+1)^2}{4^{m+2}} \le \frac{1}{16}.$$

Thus the upper bound is at most

$$\frac{t+m+3}{12}\,4^{-t(m+1)}16^{-m} \le \frac{t+m+3}{12}\,4^{-t-m},$$

which tends to zero uniformly as $t+m \to \infty$. □

Theorem 3.2 (Complete compressed classification and raw-error optimum)**.** *Fix m, t and use the preceding notation. Every rational polynomial C of degree at most $m+2$ satisfying*

$$C(1) = 1, \qquad C'(1) = -t, \qquad C(4^{-j}) = 0 \quad (1 \le j \le m)$$

has the unique form

$$C_\kappa(x) = \frac{P_m(x)}{A_m}\big[1 - b(x-1) + \kappa(x-1)^2\big], \qquad \kappa \in \mathbb{Q}.$$

This includes every remainder of degree less than $m+3$ obtained from the shared-cutoff polynomial $(x-1)^2(2x-1)P_m(x)$, whenever these normalized constraints hold.

Writing $d = R-1$, its exact gamma-error kernel is

$$F_\kappa(u) = \frac{u^m(u^2 + a^2A_\kappa u + a^4B_\kappa)}{\prod_{i=0}^{m+2}(u + a^2 4^i)},$$

where

$$A_\kappa = 5R + d(b - 3\kappa), \qquad B_\kappa = 4\{R^2 + bRd + \kappa d^2\}.$$

Consequently $F_\kappa(u) \ge 0$ for all $u \ge 0$ if and only if

$$B_\kappa \ge 0, \qquad A_\kappa \ge 0 \text{ or } A_\kappa^2 \le 4B_\kappa. \tag{6}$$

Among all these pointwise nonnegative kernels, the smallest integral is attained uniquely at

$$\kappa_{\min} = -\frac{R(R+bd)}{d^2}.$$

At this parameter $C_{\kappa_{\min}} = C_{m+1,t}$, so the minimum error is exactly $E_{m+1,t}$. This is an optimum for the unnormalized integral; it is not an optimum for the final reduced-denominator error.

Proof. Divisibility by P_m and the two conditions at 1 give the unique quadratic factor. Three shifts of (4) yield the displayed numerator. A monic quadratic $v^2 + Av + B$ is nonnegative on $v \ge 0$ exactly when $B \ge 0$ and either $A \ge 0$ or its real minimum $B - A^2/4$ is nonnegative; this proves (6).

The condition $B_\kappa \ge 0$ forces $\kappa \ge \kappa_{\min}$. At $\kappa_{\min}$, $B_\kappa = 0$ and $A_\kappa > 0$, so this endpoint is admissible. Its polynomial additionally vanishes at 4^{-m-1}, hence the already proved polynomial identity identifies it with $C_{m+1,t}$. The error is the affine function

$$E(\kappa) = E_{m,t} + \kappa J_{m,t}.$$

The preceding proof established $J_{m,t} > 0$ for every m, t, so the minimum over all admissible parameters is attained uniquely at the least one, as claimed. □

A rational double-zero example. For $m = 0$, $t = 4$, and $\kappa = 80/9$, one has

$$C_\kappa(x) = \frac{125 - 196x + 80x^2}{9}, \qquad a = 16,$$

and the exact positive kernel is

$$F_\kappa(u) = \frac{(u - 6144)^2}{(u + 256)(u + 1024)(u + 4096)}.$$

Thus an interior double zero, exact logarithmic cancellation, and global positivity coexist in a three-node formula. No quality claim about its final rational approximation is inferred from that zero.

Theorem 3.3 (An explicit positive certificate with a square-root logarithmic degree bound)**.** *For every rational $p < \gamma$, an exact rational search constructs a globally positive rational kernel of mass exactly $\gamma - p$. When $1/2 \le p < \gamma$, it constructs a dyadic polynomial C with*

$$C(1) = 1, \qquad C'(1) = 0,$$

whose dyadic kernel is positive for all $u > 0$ and whose corrected harmonic target is exactly p. If a separation bound $0 < \varepsilon \le \gamma - p$ is available, any integer $M \ge 32$ satisfying

$$2^{-M^2/2} \le \varepsilon$$

gives the explicit degree bound $\deg C \le M + 1$ and support bound $n \le 2^{M+1}$. In particular the degree bound is $O(\sqrt{\log(1/\varepsilon)})$ as $\varepsilon \to 0$.

For $p = P/Q$ in lowest terms, the complete primitive integrated form is exactly $Q\gamma - P$, regardless of the intermediate interpolation coefficient denominators. The construction does not select a sequence for which these primitive forms tend to zero.

Proof. For $p < 1/2$, use

$$\frac{1}{u+1} + 12(1/2 - p)K_1(u), \qquad K_1(u) = \frac{1}{u+1} - \frac{1}{u+4} + \frac{1}{u+9/4} - \frac{1}{u+9}.$$

Each difference in K_1 is positive, and Binet's integral combined with digamma duplication [4, 5] gives $\int K_1 \, d\mu = 1/12$, proving this assertion directly.

Write $r_m = r_{m,0}$. The preceding theorem gives $r_0 = 1/2$ and $r_m \uparrow \gamma$, with every r_m rational and exactly computable. If $p = 1/2$, take $C = 1$. Otherwise enumerate these exact rationals until the first $m \ge 1$ with $r_m \ge p$. Define

$$\lambda = \frac{p - r_{m-1}}{r_m - r_{m-1}} \in (0, 1], \qquad C = (1 - \lambda)C_{m-1,0} + \lambda C_{m,0}.$$

All coefficients are rational. The constraints and target are exact by linearity, and the kernel is a convex combination of two strictly positive kernels. The search terminates precisely for $p < \gamma$ in the stipulated range $p \ge 1/2$.

For the quantitative bound, the moment formula gives $\mu_m < (2m + 1)!$. Hence

$$E_{m,0} \le (m + 3)(2m + 1)! \, 2^{-m(m+1)}.$$

For $m \ge 32$, the elementary inequalities $2m + 1 \le 2^{m/4}$ and $m + 3 \le 2^{m/2}$ imply

$$(m + 3)(2m + 1)! \le 2^{m/2}(2m + 1)^{2m+1} \le 2^{m^2/2 + 3m/4}.$$

Consequently $E_{m,0} < 2^{-m^2/2}$. If M is as stated, then $r_M > \gamma - \varepsilon \geq p$, so the search terminates with $m \leq M$. The degree and support bounds follow.

The interpolation also has an exact nonnegative integer form. If the two bracketing targets have primitive pairs $(q_0, p_0), (q_1, p_1)$, put $\Delta = q_0 p_1 - q_1 p_0 > 0$. For $p = P/Q$ reduced inside their interval,

$$u = Qp_1 - Pq_1 \geq 0, \qquad v = Pq_0 - Qp_0 \geq 0,$$

and

$$u(q_0, p_0) + v(q_1, p_1) = \Delta(Q, P).$$

Thus a nonnegative integer combination of the two positive primitive kernels gives exactly $\Delta(Q\gamma - P)$, whose full coefficient gcd is Δ. This explicitly tracks the cancellation of all factors of the determinant, including primes already present in the base targets.

Finally choose an integer D that clears the integrated coefficient pair and is divisible by Q. The pair is $(D, DP/Q)$ and its gcd is D/Q, so complete reduction leaves (Q, P). This arithmetic identity does not bound $Q(\gamma - p)$ for targets that have not yet been selected. □

4 The sharp degree scale for positive dyadic certificates

The exact-target constructions allow arbitrary rational coefficients. Their height may therefore grow without restriction. Nevertheless, positivity imposes a quantitative lower bound that depends only on the dyadic support. Together with the canonical construction, this determines the leading asymptotic constant in the smallest possible certificate degree. This is an analytic complexity statement; it does not give primitive smallness.

Lemma 4.1 (A monic polynomial bound on the half-line)**.** *Let $P \in \mathbb{R}[u]$ be monic of degree $d \geq 0$ and nonnegative on $[0, \infty)$. Then*

$$\int_0^1 P(u)\, du \;\geq\; \mathfrak{c}_d := \frac{1}{(d+1)\binom{d}{\lfloor d/2 \rfloor}^2}.$$

The bound is attained among such polynomials.

Proof. First, every polynomial nonnegative on the half-line has a representation

$$P(u) = S(u)^2 + uT(u)^2, \qquad S, T \in \mathbb{R}[u].$$

Here is a direct factorization proof in the monic case. Every positive real root has even multiplicity, so its factor is a square. A zero root contributes powers of $u = 0^2 + u \cdot 1^2$. A negative root contributes $u + a = (\sqrt{a})^2 + u \cdot 1^2$, where $a > 0$. For a pair of nonreal conjugate roots, put $r = \sqrt{b^2 + c^2}$, with $c \neq 0$, and use

$$(u - b)^2 + c^2 = (u - r)^2 + u\big(\sqrt{2(r - b)}\big)^2.$$

Products remain of the stated form because

$$(S^2 + uT^2)(V^2 + uW^2) = (SV - uTW)^2 + u(SW + TV)^2.$$

This proves the representation, including possible repeated roots.

Write $d = 2k + \alpha$, where $\alpha \in \{0, 1\}$. If $\alpha = 0$, the degree and leading coefficient force $\deg S = k$, with leading coefficient 1 or -1, and $\deg T \leq k - 1$. If $\alpha = 1$, they force $\deg T = k$, with leading coefficient 1 or -1, and $\deg S \leq k$. Thus it suffices to bound the squared norm of a monic polynomial of degree k for the weight u^α on $[0, 1]$.

For completeness, the relevant shifted Jacobi polynomial is

$$J_{k,\alpha}(u) = \frac{1}{k!u^\alpha}\frac{d^k}{du^k}\left[u^{k+\alpha}(u-1)^k\right].$$

The expression is a polynomial, with leading coefficient $B = \binom{2k+\alpha}{k}$. Integration by parts k times shows that it is orthogonal, for the weight u^α, to every polynomial of degree less than k. All boundary terms vanish. A second such calculation, followed by the elementary beta integral, gives

$$\int_0^1 u^\alpha J_{k,\alpha}(u)^2\,du = B\int_0^1 u^{k+\alpha}(1-u)^k\,du = \frac{1}{2k+\alpha+1}.$$

Consequently, for every monic V of degree k, orthogonality yields

$$\int_0^1 u^\alpha V(u)^2\,du \geq \int_0^1 u^\alpha\left(\frac{J_{k,\alpha}(u)}{B}\right)^2 du = \frac{1}{(2k+\alpha+1)B^2} = \mathfrak{c}_d.$$

Applying this to S or T, after changing its sign if necessary, proves the bound. Equality is attained by $P = (J_{k,0}/B)^2$ when d is even, and by $P = u(J_{k,1}/B)^2$ when d is odd. Both are nonnegative on the entire half-line. □

Theorem 4.2 (A uniform lower bound at a fixed dyadic cutoff)**.** *Let $a \geq 1$, let $d \geq 0$ be an integer, and let $c_0, \ldots, c_d$ be real numbers satisfying $\sum_{i=0}^d c_i = 1$. Suppose that*

$$\mathcal{K}(u) = \sum_{i=0}^d \frac{c_i}{u + a^2 4^i} \geq 0 \qquad (u \geq 0).$$

For the Binet measure $d\mu(u) = du/(e^{2\pi\sqrt{u}} - 1)$, one has

$$\int_0^\infty \mathcal{K}(u)\,d\mu(u) \geq \frac{\mathfrak{c}_d}{3^7\prod_{i=0}^d(1+a^2 4^i)} \tag{7}$$

$$\geq \frac{1}{3^8(d+1)a^{2d+2}2^{d^2+3d}}. \tag{8}$$

In particular, the bounds hold without any restriction on coefficient height, coefficient signs, or rational denominators.

Proof. Put

$$D(u) = \prod_{i=0}^d (u + a^2 4^i), \qquad P(u) = D(u)\mathcal{K}(u).$$

The numerator P is a polynomial of degree exactly d, and its leading coefficient is $\sum_i c_i = 1$. Since $D > 0$ on the half-line, P is nonnegative there. On $0 < u \leq 1$, the elementary inequalities $2\pi < 7$ and $e < 3$ give

$$\frac{1}{e^{2\pi\sqrt{u}} - 1} > 3^{-7}, \qquad D(u) \leq \prod_{i=0}^d (1 + a^2 4^i).$$

The preceding lemma, applied after restricting the integral to $[0, 1]$, proves (7).

Furthermore,

$$\prod_{i=0}^d (1 + a^2 4^i) = a^{2d+2}2^{d(d+1)}\prod_{i=0}^d (1 + a^{-2}4^{-i}) < 3a^{2d+2}2^{d(d+1)}.$$

Indeed the last product is at most $2\exp(\sum_{i\geq 1} 4^{-i}) = 2e^{1/3} < 3$. Finally $\binom{d}{\lfloor d/2\rfloor} \leq 2^d$, so $\mathfrak{c}_d \geq 1/((d+1)2^{2d})$. These estimates give (8). □

Theorem 4.3 (Sharp asymptotic degree for exact positive dyadic targets)**.** *For a rational number $p \in [1/2, \gamma)$, let $d_*(p)$ be the least degree of a polynomial $C(x) = \sum_i c_i x^i \in \mathbb{Q}[x]$ satisfying*

$$C(1) = 1, \qquad C'(1) = 0, \qquad \mathcal{K}_C(u) := \sum_i \frac{c_i}{u + 4^i} \geq 0 \quad (u \geq 0),$$

and having the exact corrected harmonic target

$$p = \sum_i c_i \left(H_{2^i} - \frac{1}{2^{i+1}} \right).$$

This minimum exists. As rational p approaches γ from below,

$$\frac{d_*(p)}{\sqrt{\log_2(1/(\gamma - p))}} \longrightarrow 1. \tag{9}$$

No assumption about the rationality or irrationality of γ is used. If $p = P/Q$ is reduced, complete integer normalization still leaves exactly $Q\gamma - P$; the degree conclusion is independent of whether this primitive error is small.

Proof. Binet's identity gives

$$H_{2^i} - \frac{1}{2^{i+1}} = \gamma + i \log 2 - \int_0^\infty \frac{d\mu(u)}{u + 4^i}.$$

The two normalization conditions therefore imply

$$\gamma - p = \int_0^\infty \mathcal{K}_C(u)\, d\mu(u) > 0.$$

Put $\delta = \gamma - p$ and $L = \log_2(1/\delta)$. The preceding theorem with $a = 1$ yields, for every admissible degree d,

$$L \leq d^2 + 3d + \log_2(d + 1) + 8 \log_2 3.$$

In particular $d_*(p) \to \infty$ as $\delta \to 0$, and this inequality implies

$$\liminf_{p \uparrow \gamma} \frac{d_*(p)}{\sqrt{L}} \geq 1.$$

We recall the explicit canonical family to prove the matching upper bound. Set

$$P_m(x) = \prod_{j=1}^{m} (4^j x - 1), \quad A_m = P_m(1), \quad s_m = \frac{P_m'(1)}{P_m(1)}, \quad C_m(x) = \frac{P_m(x)}{A_m}[1 - s_m(x - 1)].$$

For $m = 0$, take $P_0 = A_0 = C_0 = 1$ and $s_0 = 0$. These polynomials have $C_m(1) = 1$, $C_m'(1) = 0$, and degree at most $m + 1$. The partial-fraction identity

$$\sum_i \frac{[x^i]P_m(x)}{u + 4^i} = \frac{A_m u^m}{\prod_{i=0}^m (u + 4^i)}$$

follows by comparing the numerator's leading coefficient and its first m Taylor coefficients at zero. Multiplication by x shifts the indices by one; it follows that, with $R_m = 4^{m+1}$,

$$\mathcal{K}_{C_m}(u) = \frac{u^m\{u + R_m + s_m(R_m - 1)\}}{\prod_{i=0}^{m+1}(u + 4^i)} > 0 \qquad (u > 0).$$

Let r_m be its rational corrected target and put $E_m = \gamma - r_m$. In particular $r_0 = 1/2$.

The moment formula

$$\mu_j = \int_0^\infty u^j \, d\mu(u) = \frac{2(2j+1)!\,\zeta(2j+2)}{(2\pi)^{2j+2}}$$

gives $\mu_j < (2j+1)!$ and $\mu_{j+1}/\mu_j < (j+1)^2/4$. The latter inequality follows from $\zeta(2j+4)/\zeta(2j+2) < 1$, $\pi > 3$, and $(2j+3)(2j+2) < 9(j+1)^2$. Also $s_m = m + \sum_{j=1}^m (4^j - 1)^{-1} < m+1$. Bounding each factor in the denominator of the positive kernel below by 4^i now gives

$$E_m \le \mu_m (m+3) 2^{-m(m+1)} < (m+3)(2m+1)!\, 2^{-m(m+1)}.$$

Consequently

$$\log_2 E_m \le -m^2 + O(m \log(m+1)), \qquad E_m \longrightarrow 0.$$

The implied constant is absolute.

For $p > 1/2$, enumerate the rational numbers r_m until the first index $j \ge 1$ with $r_j \ge p$. Such an index exists because $E_m \to 0$. Minimality gives $r_{j-1} < p \le r_j$, and hence

$$\lambda = \frac{p - r_{j-1}}{r_j - r_{j-1}} \in (0,1], \qquad C = (1-\lambda) C_{j-1} + \lambda C_j$$

is an admissible rational polynomial of degree at most $j+1$. Its positivity and exact target follow from linearity and convexity. For $p = 1/2$, simply use $C = 1$. This proves existence of $d_*(p)$ by exact rational comparisons, without an equality oracle for γ.

Fix $\eta > 0$ and take $m = \lceil (1+\eta)\sqrt{L} \rceil$. As $L \to \infty$,

$$\log_2 E_m \le -(1+\eta)^2 L + O(\sqrt{L} \log L) < -L.$$

Thus $E_m < \delta$, so $r_m > p$, and the first crossing index above satisfies $j \le m$. Therefore

$$\limsup_{p \uparrow \gamma} \frac{d_*(p)}{\sqrt{L}} \le 1 + \eta.$$

Letting $\eta \downarrow 0$ proves (9). Finally, an integer D clearing the integrated rational pair gives $(D, DP/Q)$ with gcd D/Q, leaving exactly (Q, P). This verifies the last assertion without a bound on intermediate coefficient heights. □

A quantitative price for positive support compression. Suppose that an original normalized certificate has error $E > 0$, and a target-changing operation replaces it by a globally nonnegative kernel supported at $a2^i$, $0 \le i \le d$, with the same gamma coefficient 1. Assume the complete logarithm condition is preserved, so both masses are log-free gamma errors. If their targets are p and $\widetilde{p}$, then

$$p - \widetilde{p} = (\gamma - \widetilde{p}) - (\gamma - p) \ge \frac{1}{3^8 (d+1) a^{2d+2} 2^{d^2+3d}} - E.$$

In particular, if E is at most half of this cutoff bound, any such positive compression must lower the target by at least half of the bound. This applies to shared-cutoff compression whenever its normalized output kernel remains globally nonnegative. A certificate merely proving that its integrated error is positive does not suffice for this conclusion.

The sharp degree law concerns a fixed base and a specified pole grid. It does not bound arbitrary rational-node representations, support with a varying external base, or signed kernels with only a positive integral. Nor does it control the denominator Q of a target or prove $Q(\gamma - P/Q) \to 0$.

5 Global positive certificates with prescribed local data

The first three results join three requirements that were previously handled separately: control on the whole half-line, exact local vanishing, and an exact rational integral target. They do not provide a bound for the denominator of a well-chosen target.

For $a > 0$ and $u \geq 0$, put

$$h_a(u) = \frac{1}{u + a^2}, \qquad K_a(u) = h_a(u) + h_{a+1/2}(u) - h_{2a}(u) - h_{2a+1}(u), \tag{10}$$

$$k_a(u) = (1+u)K_a(u), \qquad d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \qquad \ell(f) = \int_0^\infty \frac{f(u)}{1+u}\, d\mu(u). \tag{11}$$

Binet's formula and digamma duplication [4, 5] give

$$\int h_1\, d\mu = \gamma - \frac{1}{2}, \qquad \ell(k_a) = \int K_a\, d\mu = \frac{1}{4a(2a+1)}, \qquad 0 < \ell(1) < \frac{1}{12}. \tag{12}$$

For completeness, Binet gives $I(a) := \int h_a\, d\mu = \log a - \psi(a) - 1/(2a)$. Substituting the duplication identity into $I(a) + I(a+1/2) - I(2a) - I(2a+1)$ yields the rational mass in (12). The total mass of μ is $1/12$, so $0 < \int h_1\, d\mu < 1/12$ follows directly. In particular the masses of K_a are rational when a is rational.

Theorem 5.1 (Density with global and derivative control)**.** *Let $M \geq 0$ be an integer and $L > 0$. The rational linear span of $\{k_a : a \in \mathbb{Q}_{>0}\}$ is dense in the Banach space*

$$X_{M,L} = \{f \in C_0([0,\infty)) : f|_{[0,L]} \in C^M([0,L])\}, \quad \|f\|_X = \|f\|_\infty + \sum_{j=1}^{M} \|f^{(j)}\|_{[0,L]}.$$

No derivative control at infinity is asserted.

Proof. Completeness follows from uniform convergence of the functions and their local derivatives. Suppose a continuous real functional Λ annihilates all rational-node kernels. Hahn–Banach, applied to the embedding of X in $C_0([0,\infty)) \oplus C([0,L])^M$, and the measure representation of the dual give finite signed measures $\nu_0, \ldots, \nu_M$ such that

$$\Lambda(f) = \int f\, d\nu_0 + \sum_{j=1}^{M} \int_{[0,L]} f^{(j)}\, d\nu_j.$$

Apply this integral expression to $g_a(u) = (1+u)/(u+a^2)$ and denote the result by $F(a)$. The expression is meaningful although $g_a \notin C_0$. For $a \geq 1$, $|g_a| \leq 1$ and $g_a(u) \to 0$ at each finite u as $a \to \infty$. Moreover

$$g_a(u) = 1 + \frac{1 - a^2}{u + a^2}$$

shows that all its positive-order derivatives tend uniformly to zero on $[0, L]$. Thus $F(a) \to 0$ by dominated convergence.

Continuity in a extends the annihilation identity from rational nodes to all $a > 0$:

$$F(a) + F(a+1/2) - F(2a) - F(2a+1) = 0.$$

Set $D(a) = F(a) - F(2a)$. Then $D(a+1/2) = -D(a)$, whereas $D(a + j/2) \to 0$. Hence $D = 0$. Iterating $F(a) = F(2a)$ and using the same limit proves $F = 0$.

Write $x = (1+u)^{-1}$. Differentiation with respect to $z = a^2$ at $z = 1$ now gives $\Lambda(x^n) = 0$ for every $n \geq 1$. Differentiation under each measure is justified by uniform bounds in a fixed neighborhood of $z = 1$, including the required local derivatives.

Polynomials in x with zero constant term are dense in X. Here are details for this last approximation step. The change of variable identifies X with functions continuous on $[0,1]$, zero at 0, and C^M on $[b,1]$, where $b=(1+L)^{-1}$; the local derivative norms are equivalent. Extend the Taylor polynomial at b to its left and blend it, in a small interval strictly to the left of b, into a smooth uniform approximation of the remaining continuous function. This gives a global C^M function that is unchanged on $[b,1]$ and arbitrarily close uniformly. A small smooth bump below b restores its value 0 at 0. Approximate its Mth derivative uniformly by a polynomial and integrate M times with the initial derivatives fixed. The resulting polynomial is close in C^M and is still zero at 0. For $M=0$, ordinary polynomial approximation followed by subtraction of the value at 0 suffices. Therefore $\Lambda=0$. This proves real density. Approximation of the finitely many coefficients in each combination by rational numbers proves rational-span density. □

Theorem 5.2 (Exact rational jets and mass with a bounded correction)**.** *Let $T_1,\dots,T_s$ consist of distinct derivative evaluations $f\mapsto f^{(r)}(x)$, with $x\in[0,L]\cap\mathbb{Q}$ and $0\le r\le M$, together with the single functional ℓ. There are positive rational nodes $a_1,\dots,a_s$ for which $C_{ji}=T_j(k_{a_i})$ is an invertible rational matrix. The nodes can be found by exact rational enumeration. For $b\in\mathbb{Q}^s$, define*

$$\mathcal{P}_b(v)=v+\sum_{i=1}^{s}[C^{-1}(b-T(v))]_i k_{a_i}.$$

This map satisfies $T(\mathcal{P}_b(v))=b$ and preserves the rational kernel span. If $T(F)=b$, then

$$\|\mathcal{P}_b(v)-F\|_X\le C_T\|v-F\|_X,\qquad C_T=1+\sum_{i,j}U_i|(C^{-1})_{ij}|V_j,$$

where U_i is any rational upper bound for $\|k_{a_i}\|_X$, and V_j can be taken as 1 for an evaluation and $1/12$ for the mass functional. The upper bound C_T is therefore computable in rational arithmetic.

Proof. The functionals are independent. A smooth positive bump away from the finitely many evaluation points first eliminates the coefficient of ℓ in any relation. Localized polynomial bumps then separate the derivative evaluations. Theorem 5.1 implies full rank on the kernel span. All matrix entries at rational nodes are rational, so enumerating nodes and retaining independent columns terminates. The exact constraint identity follows by multiplication by C. The error bound follows from $b-T(v)=T(F-v)$ and the triangle inequality. Explicit rational norm bounds follow by summing, over the four terms defining k_a,

$$\|g_a\|_\infty\le\max(1,a^{-2}),\qquad \|g_a^{(r)}\|_{[0,L]}\le\frac{r!|1-a^2|}{a^{2r+2}}\quad(r\ge1).$$

Finally $\|\ell\|\le 1/12$ follows from (12). □

Theorem 5.3 (Prescribed zeros and an exact rational target)**.** *Fix finitely many distinct nonnegative rational points x_j and positive integer multiplicities m_j, with m_j even whenever $x_j>0$. Let $d=\sum_j m_j$ and*

$$W(u)=\frac{\prod_j(u-x_j)^{m_j}}{(1+u)^d};$$

use $W=1$ for an empty set. For rational p, the following are equivalent:

(i) $p<\gamma$.

(ii) There exist rational c_i, positive rational a_i, and rational $\eta>0$ such that

$$R(u)=h_1(u)-\sum_i c_iK_{a_i}(u)\ge\frac{\eta W(u)}{1+u}\qquad(u\ge0),\quad\int R\,d\mu=\gamma-p,$$

and the zeros of R on $[0,\infty)$ are exactly the specified points, with exactly the specified multiplicities.

Moreover $R(u) \sim 1/u$ at infinity. If $p = P/Q$ is reduced, complete primitive normalization of its integrated identity gives exactly $Q\gamma - P$, independently of the intermediate coefficient denominators.

Proof. Suppose $\epsilon = \gamma - p > 0$. There exists a smooth function $G > 0$, equal to 1 sufficiently far out, such that $\ell(WG) = \epsilon$. Indeed, let G_{low} be a small positive constant on a long initial interval and 1 in the far tail. Boundedness of W and finiteness of $h_1 d\mu$ show that these masses can tend to 0. On the other hand, $G_{\text{high}} = 1 + C\varphi$, for a nonnegative smooth bump φ away from the zero set, gives arbitrarily large masses. A suitable convex combination realizes ϵ exactly while remaining positive and equal to 1 in the tail.

Put $F_* = 1 - WG$. It belongs to $X_{M,L}$ for $M = \max_j m_j$ and L beyond all the points. At each x_j its value is 1 and its derivatives of orders $1, \dots, m_j - 1$ are zero. Also

$$\ell(F_*) = (\gamma - 1/2) - (\gamma - p) = p - 1/2 \in \mathbb{Q}.$$

For an empty zero set take $M = 0$. Apply Theorems 5.1 and 5.2 to approximate F_* by a rational combination A of the weighted kernels $k_a = (1 + u)K_a$, with all these jets and the mass exact. Near x_j, Taylor's theorem bounds the error by

$$|A(u) - F_*(u)| \le \frac{\|(A - F_*)^{(m_j)}\|}{m_j!} |u - x_j|^{m_j}.$$

The leading coefficient of WG at that point is strictly positive; the parity restriction ensures positivity on both sides of an interior zero. Sufficiently close approximation makes $1 - A$ positive near each point, with precisely that order of vanishing. Away from small disjoint neighborhoods, WG has a positive lower bound, since its limit is 1. Uniform closeness proves positivity there too.

Thus $R = h_1(1 - A)$ has the required zero set and signs. The ratio $(1 - A)/W$ extends positively and continuously across each zero and infinity. Its positive minimum admits a smaller positive rational η, proving the stated global bound. The mass is exact by construction; the tail follows from $K_a(u) = O(u^{-2})$. Conversely, the global bound has positive integral, so it implies $\gamma - p > 0$.

For the arithmetic claim, take a positive integer D clearing the coefficients and both $D/2$ and $D(p - 1/2)$. Then $Q \mid D$, and the integer coefficient pair of the integrated identity is $(D, DP/Q)$. Its gcd is D/Q, giving exactly $Q\gamma - P$. □

Finite verification and effective search. For fixed rational data, enumerate rational kernel combinations and apply the exact projection. The denominators of $R - \eta W/(1+u)$ are positive on the half-line. Clearing them reduces verification to an exact polynomial sign test. Enumerating positive rational η therefore gives a search that terminates exactly when $p < \gamma$. No bound for its degree, height, or running time is asserted. Most importantly, it does not select a sequence for which $Q\gamma - P \to 0$.

Two small certificates can be checked without a search:

$$R_1 = h_1 - \frac{12}{13} K_1 = \frac{u(52u^2 + 325u + 153)}{13(u+1)(u+4)(u+9)(4u+9)}, \tag{13}$$

$$\int R_1 \, d\mu = \gamma - \frac{15}{26};$$

$$R_2 = h_1 + \frac{17}{5640} K_{1/2} - \frac{351}{376} K_1 = \frac{u^2(752u^2 + 4820u + 2943)}{47(u+1)(u+4)(u+9)(4u+1)(4u+9)}, \tag{14}$$

$$\int R_2 \, d\mu = \gamma - \frac{6509}{11280}.$$

Positive numerator coefficients prove the sign globally and give exact orders 1 and 2 at the origin. These are worked finite examples, not evidence for an unproved infinite pattern.

6 Nested rational brackets from the canonical kernels

Define the positive measure and its moments by

$$d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \qquad \mu_j = \int_0^\infty u^j \, d\mu(u) = \frac{|B_{2j+2}|}{2j+2}.$$

Here B_k denotes a Bernoulli number; the polynomial quantity B_m used later is explicitly defined there and has a different role. Binet's identity [4] gives, for integers $n \geq 1$,

$$H_n - \frac{1}{2n} = \gamma + \log n - \int_0^\infty \frac{d\mu(u)}{u + n^2}. \tag{15}$$

For $m \geq 0$, set

$$P_m(x) = \prod_{j=1}^{m} (4^j x - 1), \quad A_m = P_m(1), \quad s_m = \frac{P_m'(1)}{A_m}, \quad C_m(x) = \frac{P_m(x)}{A_m}[1 - s_m(x-1)].$$

Empty products are 1. Put

$$r_m = \sum_i [x^i] C_m(x) \left(H_{2^i} - \frac{1}{2^{i+1}} \right), \qquad E_m = \gamma - r_m.$$

The first values are $r_0 = 1/2$, $r_1 = 31/54$, and $r_2 = 27269/47250$.

Lemma 6.1 (Positive kernels and a sharpened contraction)**.** *For every $m \geq 0$, with $R = 4^{m+1}$ and $\lambda_m = R + s_m(R-1)$,*

$$E_m = \int_0^\infty \frac{u^m (u + \lambda_m)}{\prod_{i=0}^{m+1} (u + 4^i)} \, d\mu(u) > 0. \tag{16}$$

Define

$$\rho_m = \frac{9(m+1)^2}{16 \cdot 4^{m+1}} \leq \frac{9}{64}.$$

Then

$$0 < \frac{E_{m+1}}{E_m} < \frac{2}{3}\rho_m. \tag{17}$$

In particular r_m increases strictly to γ and $E_{m+1}/E_m \to 0$.

Proof. The polynomial has $C_m(1) = 1$ and $C_m'(1) = 0$, so (15) cancels every logarithm exactly. The elementary rational identity

$$\sum_{i=0}^{m} \frac{[x^i] P_m(x)}{u + 4^i} = \frac{A_m u^m}{\prod_{i=0}^{m} (u + 4^i)}$$

follows by taking a common denominator: its numerator has degree at most m, its first m Taylor coefficients vanish because $P_m(4^{-j}) = 0$ for $1 \leq j \leq m$, and its leading coefficient is $P_m(1) = A_m$. Shifting the indices once evaluates the contribution from xP_m and gives (16).

The exact moment formula also gives the stronger estimate

$$\frac{\mu_{m+1}}{\mu_m} = \frac{(2m+3)(2m+2)}{4\pi^2} \frac{\zeta(2m+4)}{\zeta(2m+2)} < \frac{(m+1)^2}{6}.$$

Indeed $(2m+3)(2m+2) \leq 6(m+1)^2$, $\pi > 3$, and the zeta ratio is strictly less than 1. The kernels F_m in (16) satisfy

$$\frac{F_{m+1}(u)}{F_m(u)} = \frac{u(u + \lambda_{m+1})}{(u + 4R)(u + \lambda_m)} \leq \frac{9u}{4R}.$$

Here $s_{m+1} = s_m + R/(R-1)$ and

$$9\lambda_m - \lambda_{m+1} = \frac{R(R-4)}{R-1} + s_m(5R-8) \geq 0.$$

The function $W_m(u) = (u+\lambda_m)/\prod_{i=0}^{m+1}(u+4^i)$ is strictly decreasing. Its logarithmic derivative is negative because $\lambda_m \geq R$. For the measure $u^m d\mu$, multiplication by a decreasing positive function cannot increase the mean of u. This follows by integrating $(u-v)(W_m(u)-W_m(v)) \leq 0$ against its product measure. Consequently

$$\frac{E_{m+1}}{E_m} \leq \frac{9}{4R}\frac{\mu_{m+1}}{\mu_m} < \frac{3(m+1)^2}{8R} = \frac{2}{3}\rho_m.$$

The ratio is uniformly at most $3/32 < 1$, which proves convergence of E_m to zero. Since $\rho_m \to 0$, the remaining assertions follow. □

Theorem 6.2 (Strictly nested rational upper brackets)**.** *Put $w_m = r_{m+1} - r_m > 0$ and*

$$h_m = r_m + \frac{w_m}{1-\rho_m}, \qquad \delta_m = h_m - \gamma.$$

These quantities satisfy

$$r_m < r_{m+1} < \gamma < h_{m+1} < h_m, \qquad r_m \uparrow \gamma, \quad h_m \downarrow \gamma.$$

More precisely,

$$\frac{\rho_m E_m}{3(1-\rho_m)} < \delta_m < \frac{\rho_m E_m}{1-\rho_m}, \tag{18}$$

and

$$0 < \frac{\delta_{m+1}}{\delta_m} < 2\rho_{m+1}\frac{1-\rho_m}{1-\rho_{m+1}} \leq \frac{9}{32}.$$

Every endpoint r_m, h_m is obtained using rational arithmetic only.

Proof. Since $w_m = E_m - E_{m+1}$,

$$\delta_m = \frac{\rho_m E_m - E_{m+1}}{1-\rho_m}.$$

The sharpened ratio gives the lower bound in (18); positivity of E_{m+1} gives its upper bound. Dividing these bounds at successive indices and using (17) gives the displayed contraction. The sequence ρ_m is nonincreasing, since $\rho_{m+1}/\rho_m = (m+2)^2/(4(m+1)^2) \leq 1$. This proves strict decrease, convergence, and the uniform constant. □

For reference, let $\widehat{p}_m/\widehat{q}_m$ be the first mediant accepted by the Stern–Brocot search in $[r_m, r_{m+1}]$. Unimodularity of the two search parents proves that this is a least-denominator fraction in the interval. The search terminates: any fixed rational in the target interval bounds the increasing parent denominator sum until acceptance. All comparisons use the rational endpoints. Thus

$$0 < \widehat{q}_m\gamma - \widehat{p}_m \leq \frac{\widehat{q}_m w_m}{1-\rho_m}. \tag{19}$$

Also $\widehat{q}_m \to \infty$, because finitely many rational numbers have bounded denominator in $[r_0, 1]$, and each fixed one below γ is eventually below r_m. This proves infinitely many distinct targets, without assuming irrationality. The right side of (19) is an explicit rational upper certificate; its convergence to zero is the unresolved condition.

Theorem 6.3 (Least-denominator selection and an exact rationality dichotomy)**.** *Write $r_m = r_{m,0}$ and define the positive rational numbers*

$$w_m = r_{m+1} - r_m, \qquad \rho_m = \frac{9(m+1)^2}{16 \cdot 4^{m+1}} \le \frac{9}{64}.$$

Choose a reduced fraction $\widehat{p}_m/\widehat{q}_m$ of least positive denominator in the closed rational interval $[r_m, r_{m+1}]$. It can be found by the Stern–Brocot interval algorithm using rational comparisons only; take its first accepted mediant as the definite selection. The denominators satisfy $\widehat{q}_m \to \infty$, without any rationality assumption on γ. Set

$$\mathcal{U}_m = \frac{\widehat{q}_m w_m}{1 - \rho_m}.$$

The associated forms have integer coefficients and satisfy the positive rational certificate

$$0 < \widehat{q}_m\gamma - \widehat{p}_m \le \mathcal{U}_m. \tag{20}$$

For every $j \ge m+1$, the sharper fully rational bound

$$\widehat{q}_m\gamma - \widehat{p}_m \le \widehat{q}_m \left(r_j - \frac{\widehat{p}_m}{\widehat{q}_m} + \frac{w_j}{1-\rho_j} \right) \tag{21}$$

also holds.

The limit of the completely rational sequence $\mathcal{U}_m$ exists, with the exact dichotomy

$$\lim_{m\to\infty} \mathcal{U}_m = \begin{cases} 0, & \gamma \notin \mathbb{Q}, \\ 1/b, & \gamma = a/b \text{ in lowest terms.} \end{cases} \tag{22}$$

In the rational case the primitive errors $\widehat{q}_m\gamma - \widehat{p}_m$ are eventually exactly $1/b$. Consequently proving $\mathcal{U}_m \to 0$ would prove gamma irrationality; the theorem does not establish which branch in (22) holds.

Proof. Let F_m denote the kernel in (2) at $t = 0$. Put $E_m = \gamma - r_m > 0$ and $\lambda_m = 4^{m+1} + s_m(4^{m+1} - 1)$. The exact kernels give

$$\frac{F_{m+1}(u)}{F_m(u)} = \frac{u(u + \lambda_{m+1})}{(u + 4^{m+2})(u + \lambda_m)}.$$

Writing $R = 4^{m+1}$ and $d = R - 1$, the identity $s_{m+1} = s_m + R/d$ shows

$$\lambda_{m+1} = 4R + s_m(4R - 1) + \frac{R(4R-1)}{d} \le 9\lambda_m.$$

Therefore $F_{m+1}/F_m \le 9u/(4R)$. The function

$$\frac{u + \lambda_m}{\prod_{i=0}^{m+1}(u + 4^i)}$$

is strictly decreasing, since $\lambda_m \ge 4^{m+1}$ and its logarithmic derivative is negative. The same weighted-mean argument as before thus gives

$$\frac{E_{m+1}}{E_m} \le \frac{9}{4R}\frac{\mu_{m+1}}{\mu_m} < \rho_m.$$

In particular $\rho_m \to 0$, and

$$w_m < E_m \le \frac{w_m}{1 - \rho_m}.$$

These bounds prove (20) and (21), because $\widehat{p}_m/\widehat{q}_m \le r_{m+1} < \gamma$.

For completeness, start the Stern–Brocot search with Farey neighbours $0/1$ and $1/0$. Replace the lower neighbour by the mediant when the mediant is below r_m, and replace the upper neighbour when it is above r_{m+1}. Stop when the mediant lies in the target interval. The elementary integral comparison gives $\gamma \le 3/2 - \log 2 < 1$, so the first mediant $1/1$ replaces the upper infinite endpoint. Thereafter both neighbour denominators are positive. The neighbours $a/b < c/d$ always satisfy $bc - ad = 1$. Every reduced fraction strictly between them has denominator at least $b + d$, because its coefficient representation in these two unimodular columns consists of positive integers. The first accepted mediant therefore has least denominator. Termination follows by keeping any one rational in the nonempty target interval: until acceptance the sum of the neighbour denominators increases and is bounded by that rational's denominator. This also proves that the selection uses no gamma oracle.

For any fixed denominator bound, there are only finitely many reduced fractions in the bounded interval $[r_0, \gamma)$ satisfying that bound. Each is eventually smaller than r_m, because $r_m \uparrow \gamma$. Consequently none can be selected for all sufficiently large m, which proves $\widehat{q}_m \to \infty$ and supplies infinitely many distinct targets.

Suppose first that γ is irrational. Fix $0 < \eta < 1/2$. An irrational rotation visits every fixed open arc with bounded gaps. Here is the elementary compactness argument: the pigeonhole principle makes the subgroup generated by γ dense modulo 1; finitely many translates of a fixed open arc cover the circle. A common shift makes their finitely many integer indices nonnegative. Their maximum then bounds a return time from every starting point.

Apply this to the arc $(\eta/4, \eta/2)$. For all large m, there is an integer

$$\frac{\eta}{2E_m} \le q \le \frac{\eta}{E_m} \quad \text{with} \quad \frac{\eta}{4} < \{q\gamma\} < \frac{\eta}{2}.$$

For $p = \lfloor q\gamma \rfloor$ this gives

$$E_m/4 < \gamma - p/q < E_m.$$

Since $E_{m+1}/E_m \to 0$, the fraction lies in $[r_m, r_{m+1}]$ for all large m. Reduction can only lower its denominator, so the minimality of $\widehat{q}_m$ gives $\widehat{q}_m E_m \le \eta$. As η is arbitrary, $\widehat{q}_m E_m \to 0$ and hence $\mathcal{U}_m \to 0$.

Now suppose $\gamma = a/b$ in lowest terms. Every fraction $p/q < \gamma$ has gap at least $1/(bq)$, so $\widehat{q}_m E_m \ge 1/b$. Choose the smallest integer $q \ge 1/(bE_m)$ in the residue class $aq \equiv 1 \pmod b$. Then

$$\frac{1}{bE_m} \le q < \frac{1}{bE_m} + b, \qquad p = \frac{aq-1}{b} \in \mathbb{Z}, \qquad \gcd(p,q) = 1.$$

Its gap satisfies

$$\frac{E_m}{1 + b^2 E_m} < \gamma - p/q \le E_m.$$

For large m this gap exceeds E_{m+1}, so the fraction is in the target interval. Therefore

$$\frac{1}{b} \le \widehat{q}_m E_m \le \frac{1}{b} + bE_m,$$

and $\widehat{q}_m E_m \to 1/b$. Since $w_m/E_m = 1 - E_{m+1}/E_m \to 1$ and $\rho_m \to 0$, this proves the rational branch of (22). Finally each $\widehat{q}_m\gamma - \widehat{p}_m$ is a positive integral multiple of $1/b$ and is at most $\widehat{q}_m E_m \to 1/b$. It must therefore equal $1/b$ for all sufficiently large m. □

7 Logarithmic Farey reduction of the primitive optimization

Let $\alpha = a/b < \beta = c/d < \xi$, where the endpoint fractions are reduced and $b, d > 0$. Set $v_0 = (b, a)$, $v_1 = (d, c)$ and $\Delta = \det(v_0, v_1) = bc - ad > 0$. The linear functional

$$\mathcal{L}_\xi(q, p) = q\xi - p$$

is strictly positive on every nonzero integer vector in the cone spanned by v_0, v_1. The following reduction replaces enumeration of a fundamental parallelogram of size Δ by a logarithmic number of candidates.

Theorem 7.1 (A logarithmic candidate set for the exact primitive minimum)**.** *One can construct, using integer arithmetic only, a set G of primitive vectors (q,p) with $q>0$ and $\alpha \le p/q \le \beta$ such that*

$$|G| \le \lfloor \log_2 \Delta \rfloor + 2$$

and, simultaneously for every real $\xi > \beta$,

$$\min_{\substack{\alpha \le p/q \le \beta \\ q>0,\ \gcd(p,q)=1}} (q\xi - p) = \min_{(q,p)\in G} (q\xi - p). \tag{23}$$

More generally the same equality holds for every linear functional strictly positive on the nonzero cone. After one extended Euclidean computation for a, b, the construction uses at most $2\lfloor \log_2 \Delta \rfloor + 1$ ceiling-Euclidean updates when $\Delta > 1$. Each collinear run is handled by a single integer division. For $\Delta = 1$, the two endpoint vectors suffice.

The resulting lower envelope is a finite rational piecewise-linear function of ξ. No test for whether ξ equals one of its rational breakpoints is part of the construction.

Proof. The case $\Delta = 1$ follows from the unimodular endpoint basis. Suppose $\Delta > 1$. Choose integers u, v with $bu + av = 1$ and put $r_{\mathrm{raw}} = ud + vc$. A unimodular shear after

$$U = \begin{pmatrix} u & v \\ -a & b \end{pmatrix}$$

sends the two endpoint vectors to $(1,0)$ and (r,Δ), where $r \equiv r_{\mathrm{raw}} \pmod{\Delta}$ and $1 \le r < \Delta$. The last assertion follows because the upper endpoint is primitive, so $\gcd(r,\Delta) = 1$.

In these transformed coordinates start with

$$z_0 = (1,0), \quad z_1 = (1,1), \qquad D_0 = \Delta, \quad D_1 = \Delta - r.$$

While $D_i > 0$, use

$$h_i = \left\lceil \frac{D_{i-1}}{D_i} \right\rceil \ge 2, \qquad z_{i+1} = h_i z_i - z_{i-1}, \qquad D_{i+1} = h_i D_i - D_{i-1}.$$

Then

$$0 \le D_{i+1} < D_i, \qquad \det(z_i, z_{i+1}) = 1, \qquad D_i = \det(z_i, (r,\Delta)).$$

The coordinates are nondecreasing, with increasing second coordinate, so every ray remains in the cone and the terminal vector is exactly (r,Δ): it is a positive parallel primitive vector. The consecutive rays form a unimodular fan. Every integer vector in the cone is therefore a nonnegative integer combination of two adjacent fan vectors. A positive linear functional on such a nonzero combination is at least its value on one of those generators. Thus all fan vectors form an exact candidate set.

When $h_i = 2$, the vector z_i lies on the line segment between z_{i-1} and z_{i+1}. Throughout a run of such updates the vectors form an arithmetic progression, and a linear functional is minimized at one of the run endpoints. Consequently it is enough to retain z_0, the terminal vector, and precisely the vectors z_i for which $h_i \ge 3$.

Every retained interior vertex satisfies $D_i < D_{i-1}/2$. Since the positive determinant remainders decrease, there are at most $\lfloor \log_2 \Delta \rfloor$ such events. This proves the candidate count after applying the inverse unimodular map. All candidate vectors remain primitive, and their first coordinates are positive because they lie in the original cone.

To skip a run with $h_i = 2$, put $g = D_{i-1} - D_i > 0$ and $k = \lfloor D_i/g \rfloor \geq 1$. The next k updates all have quotient 2, and can be replaced by

$$z_{i+k} = z_i + k(z_i - z_{i-1}), \qquad D_{i+k} = D_i - kg.$$

The predecessor is obtained by replacing k by $k-1$. Either the remainder becomes zero or the next quotient is at least 3. There is at most one such run between retained interior vertices, giving the stated update bound. Equality (23) follows from generation and removal of collinear candidates, and rationality of the breakpoints follows by equating two integer affine functions. □

Theorem 7.2 (A terminating relative optimizer without an equality oracle)**.** *Let G be the preceding finite set, and suppose rational brackets $L_j \leq \xi \leq U_j$ are available with $L_j > \beta$ and $U_j - L_j \to 0$. Define*

$$A_j = \min_{(q,p)\in G}(qL_j - p), \qquad B_j = \min_{(q,p)\in G}(qU_j - p).$$

Writing M_ξ for the primitive minimum in (23), one has

$$0 < A_j \leq M_\xi \leq B_j, \qquad B_j - A_j \leq Q_{\max}(U_j - L_j), \quad Q_{\max} = \max_{(q,p)\in G} q.$$

For every rational $\tau > 0$, refinement until $B_j \leq (1+\tau)A_j$ terminates. If $\widehat{v}_j$ is any candidate attaining B_j, then

$$M_\xi \leq \mathcal{L}_\xi(\widehat{v}_j) \leq B_j \leq (1+\tau)M_\xi.$$

At any step all candidates with $qL_j - p > B_j$ may be discarded without losing a true optimum. For the canonical gamma intervals $[r_m, r_{m+1}]$, the rational brackets

$$L_j = r_j, \qquad U_j = r_j + \frac{r_{j+1} - r_j}{1 - \rho_j}, \qquad j \geq m+2,$$

supply such an optimizer. This gives an explicit infinite family of positive primitive forms within any prescribed fixed relative factor of the true interval optimum. It does not prove that those optima tend to zero.

Proof. For each candidate, replacing ξ by L_j or U_j gives the stated lower and upper bounds. If a vector minimizes the lower expression, its upper expression is at most $A_j + Q_{\max}(U_j - L_j)$; minimizing the upper expression proves the difference estimate. The finite set has a strictly positive minimum M_ξ, so both A_j and B_j tend to M_ξ. Therefore the relative stopping condition holds after finitely many refinements. This argument does not require identifying a unique optimum or deciding whether ξ equals a rational breakpoint. The bound for an upper-minimizing candidate follows directly. A discarded vector has actual value greater than B_j, whereas an undiscarded upper-minimizing vector has actual value at most B_j, so discarding is safe. The previously proved canonical gamma bounds verify all hypotheses of the application. □

Theorem 7.3 (A three-fraction certificate for an exact primitive optimum)**.** *Suppose $a/b < c/d$ are Farey neighbours, so $bc - ad = 1$, and*

$$M = \frac{a+c}{b+d} < \xi < \frac{c}{d}.$$

Then for every reduced fraction $p/q < M$ with $q > 0$,

$$q\xi - p \geq b\xi - a > 0,$$

with equality only for $p/q = a/b$. Consequently, whenever

$$\alpha \leq \frac{a}{b} \leq \beta < M < \xi < \frac{c}{d},$$

the fraction a/b is the unique primitive-error minimizer on $[\alpha, \beta]$, and

$$0 < b\xi - a = \frac{1}{d} - b\left(\frac{c}{d} - \xi\right) < \frac{1}{d}. \tag{24}$$

For gamma, both strict comparisons must be certified: choose rational brackets with $M < L_k \leq \gamma \leq U_j < c/d$. Thus three fractions and rational comparisons certify an exact interval optimum without an equality oracle. In the canonical application below, the next interval's selected mediant satisfies $M < r_{m+2}$, while a computed upper bracket lies below c/d.

Proof. First suppose $p/q \leq a/b$. If $q \geq b$, then $q(\xi - p/q) \geq b(\xi - a/b)$, with equality only at the same primitive fraction. If $q < b$, the strict upper bound $\xi < c/d$ gives

$$(q\xi - p) - (b\xi - a) > \frac{cq - dp - 1}{d} \geq 0,$$

because $cq - dp$ is a positive integer.

If $a/b < p/q < M$, unimodularity gives the positive integers

$$s = cq - dp, \qquad t = bp - aq, \qquad (q, p) = s(b, a) + t(d, c).$$

The inequality $p/q < M$ is equivalent to $s > t$. Hence

$$(q, p) = (s - t)(b, a) + t(b + d, a + c).$$

Both associated linear forms are strictly positive because $a/b < M < \xi$. Their positive integer combination has value strictly greater than $b\xi - a$. This proves the minimum statement. Finally $bc - ad = 1$ gives $c/d - a/b = 1/(bd)$ and therefore (24). □

Proposition 7.4 (A quantitative parent-denominator target)**.** *For the least-denominator fraction of $[r_m, r_{m+1}]$, let its final Stern–Brocot parents be $a_m/b_m < c_m/d_m$. Then*

$$\widehat{q}_m = b_m + d_m, \qquad \widehat{q}_m w_m < \frac{1}{b_m} + \frac{1}{d_m},$$

and consequently

$$\mathcal{U}_m < \frac{b_m^{-1} + d_m^{-1}}{1 - \rho_m}. \tag{25}$$

The lower-parent denominators satisfy $b_m \to \infty$ without assuming gamma irrationality. The upper-parent denominators satisfy the exact alternative:

- *if gamma is irrational, then $d_m \to \infty$;*
- *if gamma is rational, then c_m/d_m is eventually exactly gamma, so d_m is eventually its reduced denominator.*

In particular, an independent proof that d_m is unbounded would suffice for condition 3. This proposition does not supply that proof.

Proof. The selected fraction is the mediant of its final parents, so its denominator is $b_m + d_m$. The target interval lies strictly inside their Farey interval of length $1/(b_m d_m)$, proving the quantitative inequalities.

The selected denominators tend to infinity: otherwise a fixed rational would lie in infinitely many intervals converging strictly from below to gamma, forcing that rational to equal gamma, a contradiction. If the lower denominators had a bounded subsequence, finitely many possible lower parents would give a constant subsubsequence. The upper denominators would then tend

to infinity, and the Farey gap would tend to zero, forcing that fixed lower parent to equal gamma. This contradicts its being strictly below every target interval. Thus $b_m \to \infty$.

The same compactness argument shows that a bounded subsequence of d_m forces gamma to be rational. Conversely, if gamma is a reduced rational, the finite Stern–Brocot path reaches it. For all sufficiently late intervals lying strictly below and sufficiently close to it, this rational becomes the right parent during that common initial path. Thereafter mediants with that right parent increase towards gamma. If a mediant overshot the target interval but still lay below gamma, it would replace the right parent. For late intervals this cannot occur before a mediant enters the interval: consecutive such mediants have gaps from gamma of the form $1/[d(b+kd)]$, with successive gap ratio tending to 1, whereas $E_{m+1}/E_m \to 0$. More explicitly, choose the first k with gap at most E_m. Its gap is at least $E_m/(1+d^2E_m)$, which exceeds E_{m+1} for all large m. It therefore lies inside the target interval and the search stops with right parent gamma unchanged. This proves eventual constancy. □

Proposition 7.5 (An exact integer test for the final upper Farey parent)**.** *Let $0 < \alpha = p_0/q_0 < \beta = p_1/q_1 < 1$ be reduced rational endpoints. For a proposed reduced upper parent c/d with $\beta < c/d \le 1$, put*

$$k_i = cq_i - dp_i > 0 \quad (i = 0, 1).$$

Choose $b_0 \in \{0, \dots, d-1\}$ with $b_0 c \equiv 1 \pmod{d}$; for $d = 1$ use $b_0 = 0$. Define the integer

$$B_* = b_0 + d\left\lfloor \frac{q_0 - 1 - b_0 k_0}{dk_0} \right\rfloor.$$

Then c/d is exactly the final upper parent in the Stern–Brocot least-denominator search on $[\alpha, \beta]$ if and only if

$$B_* > 0, \qquad (B_* + d)k_1 \le q_1. \tag{26}$$

When these inequalities hold, the lower parent is

$$\frac{A_*}{B_*}, \qquad A_* = \frac{B_* c - 1}{d},$$

and the selected fraction is $(A_* + c)/(B_* + d)$.

For the canonical endpoints, this translates bounded upper-parent denominators into explicit floor and congruence inequalities involving only their harmonic numerator and denominator pairs. In particular,

$$\gcd(q_i, k_i) = \gcd(q_i, d).$$

The existence of many primes dividing q_i and not dividing d merely makes k_i a unit at those primes; it does not by itself contradict (26).

Proof. Set

$$X = \frac{q_0}{k_0} = \frac{1}{d(c/d - \alpha)}, \qquad Y = \frac{q_1}{k_1} = \frac{1}{d(c/d - \beta)}.$$

A Farey lower neighbour A/B of c/d must satisfy $Bc - Ad = 1$, so $Bc \equiv 1 \pmod{d}$. The conditions $A/B < \alpha$ and $(A + c)/(B + d) \ge \alpha$ are respectively

$$B < X, \qquad B + d \ge X.$$

There is exactly one integer in the residue class $b_0 \bmod d$ satisfying these two inequalities: the largest such integer strictly below X. Since $Bk_0 < q_0$ is equivalent to $Bk_0 \le q_0 - 1$, that integer is exactly B_* in the statement. It must be positive. The remaining condition that the mediant be at most β is $B_* + d \le Y$, which is precisely the second inequality in (26).

Conversely these inequalities give a Farey pair whose open interval contains $[\alpha, \beta]$ and whose mediant belongs to that target interval. Every fraction in the open Farey interval has denominator at least the sum of the parent denominators, so the mediant is the unique least-denominator fraction in the target interval. A reduced fraction in $(0, 1)$ has a unique Farey parent pair in $[0, 1]$ with positive parent denominators: the right denominator is determined by the inverse congruence modulo the mediant denominator. Thus this pair is exactly the final pair from the Stern–Brocot search. The gcd assertion follows from $\gcd(p_i, q_i) = 1$. □

Why positivity and prime-rich denominators alone do not settle the target. The obstruction can be exhibited exactly for the rational limit $\xi = 1/2$. Choose integers D_m divisible by 4, with $D_{m+1} \geq D_m^2$ and $D_m \to \infty$, and put

$$\ell_m = \frac{1}{2} - \frac{1}{D_m}.$$

The endpoint ℓ_m has reduced denominator exactly D_m, so the D_m can be chosen to include any prescribed increasing lists of primes. The errors have ratio $D_m/D_{m+1} \to 0$. They even have positive integral representations against the same Binet measure: since $\mu_0 = 1/12$,

$$\xi - \ell_m = \int \frac{12}{D_m}\, d\mu > 0.$$

Yet the least-denominator rational in $[\ell_m, \ell_{m+1}]$ is exactly

$$\frac{D_m/4}{D_m/2 + 1},$$

and its primitive error is exactly $1/2$ for every m. Indeed odd denominators require $q \geq D_m/2$ and the smallest eligible odd denominator is $D_m/2 + 1$; even denominators require $q \geq D_m$. The displayed fraction belongs to the interval because $D_{m+1} \geq D_m + 2$. Thus arbitrarily fast raw convergence, global positivity, and prescribed prime factors of the endpoint denominators do not by themselves imply primitive-error decay. This example does not imitate every gamma-specific harmonic coefficient identity.

7.1 Exact finite optima and a rational numerical certificate

In each of the eleven canonical intervals $m = 0, \ldots, 10$, the relative optimizer's surviving set reduces to one vector. The same vector is independently certified by the three-fraction criterion. The following selected rows show the actual optimum, the number of logarithmic-sail candidates, and an upward rational rounding of its positive primitive error.

m	Exact optimal target	Candidates	Strict upper error bound
0	$\frac{4}{7}$	3	$4.05096544 \times 10^{-2}$
1	$\frac{15}{26}$	6	$7.60728744 \times 10^{-3}$
2	$\frac{157}{272}$	15	$2.66085322 \times 10^{-3}$
4	$\frac{33841}{58628}$	49	$1.84706856 \times 10^{-6}$
6	$\frac{9062634}{15700603}$	172	$1.53583719 \times 10^{-9}$
8	$\frac{8788062966751}{15224921118955}$	723	$9.04971062 \times 10^{-14}$
10	$\frac{2323006970406828281}{4024504377931443899}$	2728	$3.81603071 \times 10^{-19}$

For $m = 10$, the certifying triple is

$$\frac{2323006970406828281}{4024504377931443899}, \qquad \frac{3378631047479839813}{5853325287102525206}, \qquad \frac{1055624077073011532}{1828820909171081307}.$$

The first and third fractions are Farey neighbours; the middle is their mediant. The first lies in $[r_{10}, r_{11}]$, while the middle lies strictly above r_{11} and strictly below $r_{12} < \gamma$. The rational upper bracket h_{12} is below the third fraction. Thus the first fraction is the unique primitive-error optimum over the entire interval $[r_{10}, r_{11}]$.

A sharper, independently constructed rational Binet bracket proves

$$\mathcal{F}_{10} = 4024504377931443899\,\gamma - 2323006970406828281,$$
$$\frac{38160307067862345768}{10^{38}} < \mathcal{F}_{10} < \frac{38160307067862345769}{10^{38}}.$$

Here is the exact bracket construction used by the verifier. Put

$$\ell_- = 2\sum_{j=0}^{109} \frac{1}{(2j+1)3^{2j+1}}, \qquad \ell_+ = \ell_- + \frac{2}{221 \cdot 3^{221}}\frac{9}{8}.$$

The positive series for $2\,\mathrm{artanh}(1/3)$ gives $\ell_- < \log 2 < \ell_+$. Define

$$S = \sum_{j=0}^{19} \frac{(-1)^j \mu_j}{64^{2j+2}}, \qquad R = \frac{\mu_{20}}{64^{42}}.$$

Finite division of $1/(64^2 + u)$ with an even number of terms shows

$$S < \int_0^\infty \frac{d\mu(u)}{64^2 + u} < S + R.$$

Consequently the entirely rational endpoints

$$g_- = H_{64} - \frac{1}{128} - 6\ell_+ + S, \qquad g_+ = H_{64} - \frac{1}{128} - 6\ell_- + S + R$$

satisfy $g_- < \gamma < g_+$. Comparing the desired integer form at these two endpoints proves the displayed strict inequalities by integer cross multiplication.

These are finite optimum certificates. They neither prove that the interval optima tend to zero nor assert a computational record for excluding rational denominators of γ.

8 Selection and primitive optimization on shrinking intervals

The next two results clarify the relationship between least-denominator selection and optimization of the actual primitive error. The first is a synthesis of the preceding constructions with standard irrational-rotation and Farey arguments. Its interval hypothesis is weaker than the contraction available for the canonical gamma kernels. The second shows that the two selection rules can nevertheless have arbitrarily different finite errors. Neither result determines whether Euler's constant is rational.

For a real number ξ, define

$$\kappa(\xi) = \begin{cases} 0, & \xi \notin \mathbb{Q}, \\ 1/b, & \xi = a/b \text{ in lowest terms, } b > 0. \end{cases}$$

All rational targets below are reduced, with positive denominators. Their primitive errors are $q\xi - p$; positivity follows from $p/q < \xi$.

Theorem 8.1 (A common limiting dichotomy for three selection rules)**.** *Let $\xi \in \mathbb{R}$ and let*

$$I_m = [\alpha_m, \beta_m], \qquad \alpha_m, \beta_m \in \mathbb{Q}, \qquad \alpha_m < \beta_m < \xi.$$

Put

$$E_m = \xi - \alpha_m, \qquad E_m^+ = \xi - \beta_m.$$

Assume

$$E_m \longrightarrow 0, \qquad \limsup_{m\to\infty} \frac{E_m^+}{E_m} < 1. \tag{27}$$

No monotonicity or nesting of the intervals is required.

Choose $p_m/q_m \in I_m$ of least positive denominator and set

$$S_m = q_m\xi - p_m, \qquad M_m = \min_{\substack{p/q\in I_m \\ q>0,\ \gcd(p,q)=1}} (q\xi - p).$$

The minimum defining M_m is attained and is strictly positive. Then

$$q_m \longrightarrow \infty, \qquad \lim q_m E_m = \lim S_m = \lim M_m = \kappa(\xi). \tag{28}$$

If $\xi = a/b$ is rational in lowest terms, then

$$S_m = M_m = 1/b \qquad \text{for all sufficiently large } m. \tag{29}$$

More generally, suppose a reduced target $P_m/Q_m \in I_m$ satisfies

$$M_m \le T_m := Q_m\xi - P_m \le (1+\tau_m)M_m, \qquad 0 < \tau_m < 1. \tag{30}$$

Then $Q_m \to \infty$ and $T_m \to \kappa(\xi)$. In the rational case, $T_m = 1/b$ eventually. If additionally $\tau_m \to 0$, then

$$\frac{T_m}{M_m} \longrightarrow 1. \tag{31}$$

When the endpoints and narrowing rational brackets for ξ are effectively available, the relative rule can be implemented by Theorem 7.2, for any effectively given rational sequence $0 < \tau_m < 1$. It does not require an equality oracle.

Finally, any positive numbers C_m with $C_m/E_m \to 1$ satisfy

$$q_m C_m \longrightarrow \kappa(\xi). \tag{32}$$

In particular this applies to any asymptotically exact rational upper certificate for E_m.

Proof. For a fixed interval, take any admissible primitive pair (q_*, p_*). Since $E_m^+ > 0$, every admissible pair satisfies

$$q\xi - p = q(\xi - p/q) \ge qE_m^+.$$

Pairs with $q > (q_*\xi - p_*)/E_m^+$ cannot improve the chosen pair. Only finitely many fractions in a bounded interval have a bounded denominator, proving existence and positivity of M_m. The same finiteness observation shows that every sequence of rational targets chosen in I_m has denominator tending to infinity: otherwise a fixed rational would occur along a subsequence converging to ξ strictly from below, forcing that rational to equal ξ.

Choose a fixed number $\theta \in (0,1)$ such that

$$E_m^+/E_m \le \theta$$

for all sufficiently large m. We first suppose that ξ is irrational. We use only the bounded-return property of an irrational rotation, not equidistribution. The elementary density of the forward

orbit of an irrational rotation implies that, for every nonempty open arc J of $\mathbb{R}/\mathbb{Z}$, the sets $J - j\xi$, $j \geq 0$, cover the circle. Compactness gives a finite subcover. Consequently there is an integer K_J such that, from every starting point, some iterate with index between 0 and K_J lies in J. The density assertion itself follows from the usual pigeonhole approximation of an irrational rotation by arbitrarily small nonzero rotations.

Fix $h > 0$ with $\theta(1 + h) < 1$, and fix $0 < \eta < 1$. Apply the bounded return property to

$$J = (\theta(1 + h)\eta, \eta).$$

For all sufficiently large m, the interval $[\eta/E_m, (1 + h)\eta/E_m]$ is long enough to contain such a return. Thus there is an integer q with

$$\frac{\eta}{E_m} \leq q \leq \frac{(1 + h)\eta}{E_m}, \qquad \theta(1 + h)\eta < \{q\xi\} < \eta.$$

Put $p = \lfloor q\xi \rfloor$. Then

$$E_m^+ \leq \theta E_m < \xi - p/q < E_m.$$

After reduction the fraction still belongs to I_m and its denominator does not increase. Minimality of q_m therefore gives

$$q_m E_m \leq (1 + h)\eta.$$

Since $\eta > 0$ is arbitrary, $q_m E_m \to 0$. The inequalities

$$0 < M_m \leq S_m \leq q_m E_m$$

prove the other two limits in the irrational case.

Now let $\xi = a/b$ be rational in lowest terms. Every positive primitive error is a positive integer multiple of $1/b$, so

$$q_m E_m \geq S_m \geq M_m \geq 1/b.$$

Choose the smallest integer $q \geq 1/(bE_m)$ in the residue class $aq \equiv 1 \pmod{b}$, and put $p = (aq - 1)/b$. For large m, this is a positive denominator and

$$\frac{1}{bE_m} \leq q < \frac{1}{bE_m} + b, \qquad \gcd(p, q) = 1,$$

because $aq - bp = 1$. Its gap satisfies

$$\frac{E_m}{1 + b^2 E_m} < \xi - p/q \leq E_m.$$

For sufficiently large m, the left bound exceeds θE_m and therefore exceeds E_m^+. This fraction belongs to I_m. Hence

$$\frac{1}{b} \leq q_m E_m \leq \frac{1}{b} + bE_m,$$

which proves $q_m E_m \to 1/b$. Since S_m is a positive integer multiple of $1/b$ and $S_m \leq q_m E_m$, eventually $S_m = 1/b$. The inequalities $1/b \leq M_m \leq S_m$ then give $M_m = 1/b$ as well.

For a relative optimizer, the irrational case follows from $0 < T_m < 2M_m \to 0$. In the rational case, eventually

$$1/b \leq T_m \leq (1 + \tau_m)/b < 2/b.$$

Its membership in $(1/b)\mathbb{Z}$ forces $T_m = 1/b$. If $\tau_m \to 0$, division of (30) by $M_m > 0$ proves (31). The effective assertion uses the finite candidate set of Theorem 7.1 and the terminating refinement rule of Theorem 7.2; for each fixed m, its strict gap $\xi - \beta_m > 0$ lets the available lower rational bracket eventually exceed β_m. Finally (32) follows by multiplying $q_m E_m \to \kappa(\xi)$ by $C_m/E_m \to 1$. □

The exact Farey-parent alternative. Under the hypotheses of Theorem 8.1, the parent criterion also holds for the general intervals, with a short proof that uses the rational stabilization just established. For all sufficiently large m, $q_m > 1$. Let

$$\frac{a_m}{b_m} < \frac{p_m}{q_m} < \frac{c_m}{d_m}, \qquad b_m + d_m = q_m, \quad a_m + c_m = p_m, \quad b_m c_m - a_m d_m = 1,$$

be its canonical Farey parents, whose denominators lie between 1 and $q_m - 1$. They are the final Stern–Brocot parents in the positive-interval application. Since either parent has smaller denominator than q_m, minimality implies

$$a_m/b_m < \alpha_m < \beta_m < c_m/d_m.$$

Their gap is $1/(b_m d_m)$, and consequently

$$q_m(\beta_m - \alpha_m) < \frac{1}{b_m} + \frac{1}{d_m}. \tag{33}$$

One always has $b_m \to \infty$. Indeed, on a subsequence with bounded b_m, one has $d_m = q_m - b_m \to \infty$. The parent gap then tends to zero and forces the lower parents to converge to ξ. A bounded denominator leaves only finitely many fractions near ξ, so some fixed lower parent would equal ξ, contradicting its strict position below $\alpha_m < \xi$. The same argument applied to a bounded subsequence of d_m forces ξ to be rational. Thus irrational ξ implies $d_m \to \infty$.

Conversely, if $\xi = a/b$ is reduced, the theorem gives $aq_m - bp_m = 1$ for all sufficiently large m. Also $q_m > b$ eventually. The right Farey-parent denominator is the unique integer $d \in \{1, \dots, q_m - 1\}$ satisfying $p_m d \equiv -1 \pmod{q_m}$. The displayed determinant identity shows that it is b, with numerator a. Hence

$$\frac{c_m}{d_m} = \xi \quad \text{eventually if } \xi \in \mathbb{Q}. \tag{34}$$

In this class of interval sequences the following statements are therefore equivalent:

$$\begin{aligned} \xi \notin \mathbb{Q} &\iff d_m \to \infty \iff (d_m) \text{ is unbounded} \\ &\iff M_m \to 0 \iff S_m \to 0 \iff q_m E_m \to 0. \end{aligned}$$

These are exact equivalences, not an independent proof of any one of them for $\xi = \gamma$.

Application to the canonical gamma intervals. Take $\alpha_m = r_m$, $\beta_m = r_{m+1}$ and $\xi = \gamma$. Lemma 6.1 supplies the stronger condition $E_m^+/E_m = E_{m+1}/E_m \to 0$, so the theorem applies. The rational upper quantity in Theorem 6.3 is

$$C_m = \frac{r_{m+1} - r_m}{1 - \rho_m}, \qquad \frac{C_m}{E_m} = \frac{1 - E_{m+1}/E_m}{1 - \rho_m} \longrightarrow 1.$$

Thus the original rational upper certificate, the selected primitive errors, and the true interval optima all have the same rationality dichotomy. Choosing, for example, $\tau_m = 1/(m+2)$ in the effective relative optimizer additionally gives an explicit positive primitive sequence whose ratio to the true interval optimum tends to one. Every step terminates using rational comparisons. No uniform running-time bound and no convergence to zero for gamma are inferred.

Theorem 8.2 (Least denominator need not be asymptotically primitive-optimal)**.** *Let* $\varphi = (1 + \sqrt{5})/2$, $\xi = \varphi^{-1}$, *and let* $F_0 = 0$, $F_1 = 1$, $F_{k+2} = F_{k+1} + F_k$ *be the Fibonacci numbers. Put*

$$r_n = \frac{F_{2n}}{F_{2n+1}} \qquad (n \geq 1).$$

For any integers $N > n \geq 1$, the unique least-denominator fraction in $[r_n, r_N]$ is r_n, whereas the unique primitive-error minimizer there is r_N. Their exact errors are

$$F_{2n+1}\xi - F_{2n} = \varphi^{-(2n+1)}, \qquad F_{2N+1}\xi - F_{2N} = \varphi^{-(2N+1)}. \tag{35}$$

Consequently the quotient of the least-denominator error by the true primitive minimum is exactly

$$\varphi^{2(N-n)}. \tag{36}$$

In particular, with $n_m = 2^m$ and $I_m = [r_{n_m}, r_{n_{m+1}}]$, one has

$$\frac{\xi - r_{n_{m+1}}}{\xi - r_{n_m}} \longrightarrow 0, \qquad \frac{\textit{least-denominator primitive error on } I_m}{\textit{least primitive error on } I_m} = \varphi^{2^{m+1}} \longrightarrow \infty.$$

Thus even the stronger contraction used in the canonical gamma construction does not, by itself, make least-denominator selection asymptotically primitive-optimal.

Proof. The recurrence and $1 - \xi = \xi^2$ give

$$F_{k+1}\xi - F_k = (-1)^k \xi^{k+1}. \tag{37}$$

For example the left side at $k+1$ is $-\xi$ times its value at k, and its value at $k = 0$ is ξ. This proves the identity by induction. Consecutive Fibonacci numbers are coprime, so all fractions in the statement are reduced. The even-index identity puts r_n strictly below ξ, while Cassini's identity gives

$$r_{n+1} - r_n = \frac{1}{F_{2n+1}F_{2n+3}} > 0.$$

Use $r_0 = 0$ for the next parent computation. The fractions

$$\ell_n = \frac{F_{2n-2}}{F_{2n-1}}, \qquad u_n = \frac{F_{2n-1}}{F_{2n}}$$

are Farey neighbours, their mediant is r_n, and

$$\ell_n < r_n \leq r_N < \xi < u_n.$$

The last strict inequality follows from (37) at the odd index $2n-1$. Every reduced fraction strictly between two Farey neighbours has denominator at least the sum of their denominators. Equality is possible only for their mediant: the coefficient representation in the two unimodular columns consists of positive integers. Here that sum is F_{2n+1}. Since r_n belongs to the target interval, it is the unique least-denominator fraction there.

For the primitive minimum, the fractions

$$\frac{a}{b} = r_N = \frac{F_{2N}}{F_{2N+1}}, \qquad \frac{c}{d} = \frac{F_{2N+1}}{F_{2N+2}}$$

are Farey neighbours with $bc - ad = 1$. Their mediant is r_{N+1}, and

$$r_N < r_{N+1} < \xi < c/d.$$

The three-fraction principle of Theorem 7.3 therefore gives

$$q\xi - p \geq b\xi - a \quad \text{for every reduced } p/q < r_{N+1},$$

with equality only at $p/q = r_N$. The whole interval $[r_n, r_N]$ is within this range, proving its unique primitive optimum.

Equation (37) now proves (35), and their quotient is (36). For the final interval sequence, write $n = n_m$ and $N = n_{m+1}$. Its raw-gap ratio is exactly

$$\frac{\xi - r_N}{\xi - r_n} = \varphi^{-2(N-n)} \frac{F_{2n+1}}{F_{2N+1}} < \varphi^{-2(N-n)} \longrightarrow 0.$$

Since $N - n = 2^m$, the asserted unbounded error ratio follows. □

What the comparison adds. The Fibonacci example does not assert the same behavior for gamma. It establishes that an asymptotic optimality claim cannot be deduced merely from rational interval endpoints, strict approach from below, and a raw-gap ratio tending to zero. Exact primitive optimization can therefore improve finite errors substantially even when a least-denominator selector already has the correct limiting dichotomy. Conversely, the first theorem shows that switching to a relative optimizer does not eliminate the unresolved rationality branch. An independent all-index arithmetic estimate is still required to prove primitive-error decay for Euler's constant.

9 Primitive obstructions for dyadic targets

9.1 Canonical endpoints

Theorem 9.1 (Uniform primitive-cost divergence of the canonical family)**.** *Let $r_{m,t} = p_{m,t}/q_{m,t}$ be reduced with $q_{m,t} > 0$. Except for $(m,t) = (0,0)$, set*

$$N = 2^{t+m+1}, \quad k = t+m+1, \quad B_{m,t} = tP_m(1) + P_m'(1) > 0, \quad \mathcal{P}_N = \prod_{N/2<\ell\le N,\ \ell \text{ prime}} \ell.$$

Then

$$\frac{\mathcal{P}_N}{\gcd(\mathcal{P}_N, B_{m,t})} \mid q_{m,t} \tag{38}$$

and

$$q_{m,t}(\gamma - r_{m,t}) \ge \frac{\mathcal{P}_N}{3^{15} k\, 2^{2k(k-1)}}. \tag{39}$$

In particular the primitive errors tend to $+\infty$ along every sequence $m+t \to \infty$, although the raw errors tend to zero and the approximants increase strictly in both coordinates.

Proof. The coefficients of $P_m(1)^2 C_{m,t}$ are integers. Its top coefficient is

$$-4^{m(m+1)/2}\{tP_m(1) + P_m'(1)\}.$$

For an odd prime $\ell \in (N/2, N]$ not dividing $B_{m,t}$, only the top harmonic sum H_N contains a denominator divisible by ℓ, and it contains precisely the term $1/\ell$. All correction terms $1/(2n)$ have power-of-two denominators. Thus the integer-coefficient harmonic sum has ℓ-valuation -1. Division by $P_m(1)^2$ can only decrease that valuation. Every such prime therefore divides the actual reduced denominator, proving (38). Here $N \ge 4$, so all primes in the interval are odd.

Moreover,

$$B_{m,t} = (t + s_m)P_m(1) \le k\, 4^{m(m+1)/2}.$$

Use this in (38) and multiply by the lower bound in (3). Since $a = 2^t$,

$$q_{m,t}E_{m,t} \ge \frac{\mathcal{P}_N}{3^{15} k\, 2^{2(m+1)(t+m)}} \ge \frac{\mathcal{P}_N}{3^{15} k\, 2^{2k(k-1)}}.$$

The prime number theorem [7] gives $\log \mathcal{P}_N = N/2 + o(N)$, whereas the logarithm of the remaining denominator in this bound is $O((\log N)^2)$. This proves the uniform divergence. All asymptotic estimates here depend only on N; no parameter-dependent short-interval assertion is used.

The conclusion concerns this explicitly specified canonical family. It does not exclude rational choices of κ in the larger compressed class, different moment prescriptions, or a target-changing operation with a separately proved primitive-error bound. □

9.2 Odd-denominator cost of nonperturbative interpolation

We examine arbitrary rational interpolation between consecutive canonical targets. The interpolation coefficient is allowed to have arbitrarily large numerator and denominator. The result excludes every dyadic rational interpolation, including choices whose heights grow without any restriction. For general coefficients it gives an explicit necessary cost in the odd part of their denominator. It does not exclude arbitrary rational interpolation.

Recall the canonical polynomials and rational targets

$$P_m(x) = \prod_{j=1}^{m} (4^j x - 1), \quad A_m = P_m(1), \quad B_m = P_m'(1), \quad s_m = B_m/A_m,$$

$$C_m(x) = \frac{P_m(x)}{A_m}[1 - s_m(x-1)] = \sum_i c_{m,i} x^i, \qquad r_m = \sum_i c_{m,i} f_i, \quad f_i = H_{2^i} - \frac{1}{2^{i+1}}.$$

Here $P_0 = A_0 = C_0 = 1$, $B_0 = s_0 = 0$, and $r_0 = 1/2$. The exact positive-kernel identity is

$$E_m := \gamma - r_m = \int_0^\infty \frac{u^m\{u + 4^{m+1} + s_m(4^{m+1} - 1)\}}{\prod_{i=0}^{m+1}(u + 4^i)} \frac{du}{e^{2\pi\sqrt{u}} - 1} > 0.$$

It also gives the uniform elementary bound

$$E_m \geq 3^{-15} 2^{-m(m+1)}. \tag{40}$$

Indeed, restrict the integral to $1 \leq u \leq 2$. Its numerator is at least 4^{m+1}, its denominator is less than $27\, 4^{(m+1)(m+2)/2}$, and the measure density is greater than 3^{-12} on that interval.

Theorem 9.2 (Exact two-adic arithmetic of the canonical targets)**.** *For every integer $m \geq 0$,*

$$v_2(r_m) = -1, \qquad v_2(r_{m+1} - r_m) = 2m + 1.$$

Let $0 < \lambda = u/v \leq 1$ be reduced, and write $v = 2^L H$ with H odd. If q is the reduced denominator of $r = r_m + \lambda(r_{m+1} - r_m)$, then

$$v_2(q) = \begin{cases} 1, & 0 \leq L \leq 2m+1, \\ 0, & L = 2m+2, \\ L - 2m - 1, & L \geq 2m+3. \end{cases}$$

In particular $2^L \leq 2^{2m+2} q$ for every such interpolation coefficient.

Proof. Write $P_m(x) = \sum_i a_i x^i$, taking coefficients outside its support to be zero. For $0 \leq i \leq m$,

$$v_2(a_i) = i(i+1).$$

Indeed, in the elementary symmetric sum of $4, 4^2, \ldots, 4^m$, the unique term of least two-adic valuation is the product $4^1 \cdots 4^i$. The number A_m is odd, and for $m \geq 1$ the term a_1 uniquely has least valuation in $B_m = \sum_i i a_i$, so $v_2(B_m) = 2$.

For $m \geq 1$ the coefficients of C_m satisfy

$$c_{m,i} = \frac{(A_m + B_m)a_i - B_m a_{i-1}}{A_m^2}.$$

Consequently $v_2(c_{m,0}) = 0$, $v_2(c_{m,1}) \geq 2$, and

$$v_2(c_{m,i}) = i^2 - i + 2 \quad (2 \leq i \leq m+1).$$

The unique term of least valuation in H_{2^i} is $1/2^i$, hence $v_2(f_i) = -i - 1$. The $i = 0$ term of r_m therefore has valuation -1, the $i = 1$ term has valuation at least 0, and every $i \geq 2$ term has valuation $(i-1)^2 \geq 1$. This proves $v_2(r_m) = -1$; $m = 0$ is direct.

Put $D_m(x) = P_m(x)(x-1)^2/A_m = \sum_i d_i x^i$ and $Z_m = \sum_i d_i f_i$. From $A_m d_i = a_i - 2a_{i-1} + a_{i-2}$ one gets

$$v_2(d_0) = 0, \quad v_2(d_1) = 1, \quad v_2(d_2) = 0, \qquad v_2(d_i) = (i-2)(i-1) \quad (3 \leq i \leq m+2).$$

In Z_m, the $i = 2$ term has valuation -3. The terms $i = 0, 1$ have valuation -1, the term $i = 3$ (when present) has valuation -2, and all terms $i \geq 4$ have valuation at least 1. Thus $v_2(Z_m) = -3$.

Set $R = 4^{m+1}$. Direct polynomial subtraction gives

$$C_{m+1} - C_m = -\kappa_m D_m, \qquad \kappa_m = \frac{R\{R + s_m(R-1)\}}{(R-1)^2}.$$

For $m = 0$ this coefficient has valuation 4. For $m \geq 1$, $v_2(s_m) = 2$ and $v_2(R) \geq 4$, so $v_2(\kappa_m) = 2m + 4$. Therefore $r_{m+1} - r_m = -\kappa_m Z_m$ has valuation $2m + 1$.

If $L > 0$, reducedness makes u odd, so the added term has valuation $2m + 1 - L$. If this is greater or less than -1, comparison with $v_2(r_m) = -1$ gives the displayed denominator exponent. At equality, the sum of two rational numbers of valuation -1 is two-adically integral, so the denominator is odd. If $L = 0$, the added term has valuation at least $2m + 1$, and the first branch again applies. □

Theorem 9.3 (A necessary odd-denominator cost for every rational interpolation)**.** *For $m \geq 0$, put $k = m + 1$, $N = 2^{m+2}$, and*

$$\mathcal{P}_N = \prod_{\substack{N/2 < \ell \leq N \\ \ell \text{ prime}}} \ell, \quad T_m = A_k B_k > 0, \quad G_m = \frac{\mathcal{P}_N}{\gcd(\mathcal{P}_N, T_m)}.$$

Let $0 < \lambda = u/v \leq 1$ be reduced, let H be the odd part of v, and write $r = r_m + \lambda(r_{m+1} - r_m) = p/q$ in lowest terms. For every prime $\ell \mid G_m$, one has the exact formula

$$v_\ell(q) = \max\{0, 1 + v_\ell(v) - v_\ell(u)\}.$$

In particular,

$$G_m \mid qu, \qquad q^2 H \geq \frac{G_m}{2^{2m+2}}. \tag{41}$$

Since

$$T_m < (m+2)2^{2(m+1)(m+2)},$$

the odd denominator must satisfy the completely explicit bound

$$H \geq \frac{\mathcal{P}_N}{(m+2)2^{2(m+1)(m+3)} q^2}. \tag{42}$$

No smallness assumption on λ is used.

Proof. Fix a prime $\ell \mid G_m$. It is odd and does not divide $A_k B_k$. Since $A_m \mid A_k$, the lower target r_m is ℓ-integral: all its harmonic nodes are at most $N/2$, and its coefficient denominators divide A_m^2. The top coefficient of $A_k^2 C_k$ is

$$-4^{k(k+1)/2} B_k.$$

It is an ℓ-unit, and only its harmonic term H_N contains the single nonintegral summand $1/\ell$. The correction $1/(2N)$ is an ℓ-unit. Hence $v_\ell(r_{m+1}) = -1$ and $v_\ell(r_{m+1} - r_m) = -1$.

If $v_\ell(\lambda) < 1$, its product with the increment has negative valuation and dominates the integral base r_m. Otherwise both terms are integral. This proves the exact denominator formula. It implies $\ell \mid qu$ for every $\ell \mid G_m$, hence $G_m \mid qu$. The interval condition gives $u \le v = 2^L H$, and the preceding theorem gives $2^L \le 2^{2m+2}q$. Combining them yields $G_m \le qu \le 2^{2m+2}q^2 H$.

Finally $A_k < 2^{k(k+1)}$ and $B_k = s_k A_k < (k+1)A_k$, because $s_k = k + \sum_{j=1}^{k}(4^j - 1)^{-1} < k+1$. This proves the bound on T_m and then (42). □

Theorem 9.4 (All dyadic interpolations fail, with unrestricted height)**.** *For the preceding rational interpolation with* $0 < \lambda \le 1$*,*

$$q(\gamma - r) \ge \frac{\sqrt{\mathcal{P}_N}}{3^{15}\sqrt{(m+2)H}\,2^{(m+1)(2m+5)}}. \tag{43}$$

Consequently, along every sequence $m \to \infty$ *for which the odd part* $H = H_m$ *of the reduced interpolation denominator satisfies* $\log H_m = o(2^{m+2})$*, the primitive errors tend to* $+\infty$*. More quantitatively, with* $N_m = 2^{m+2}$*,*

$$\liminf_{m\to\infty} \frac{\log(q(\gamma - r))}{N_m} \ge \frac{1}{4} - \frac{1}{2}\limsup_{m\to\infty} \frac{\log H_m}{N_m},$$

whenever the right side is finite. Thus divergence also holds when the last limsup is strictly less than $1/2$*.*

In particular every choice of dyadic rational coefficients $\lambda_m \in [0,1] \cap \mathbb{Z}[1/2]$ *fails the primitive-smallness condition, even if their heights grow arbitrarily rapidly. For any rational-interpolation sequence with primitive error tending to zero, it is necessary that*

$$\liminf_{m\to\infty} \frac{\log H_m}{2^{m+2}} \ge \frac{1}{2}.$$

The conclusion is a necessary condition, not a construction of weights satisfying it with successful primitive errors.

Proof. The error is the convex combination $(1-\lambda)E_m + \lambda E_{m+1}$, so (40) gives

$$\gamma - r \ge 3^{-15}2^{-(m+1)(m+2)}.$$

Combine this with $q \ge \sqrt{G_m/(2^{2m+2}H)}$ and $G_m \ge \mathcal{P}_N/T_m$ to obtain (43). The prime number theorem [7] gives $\log \mathcal{P}_N = N/2 + o(N)$, while $m^2 = o(N)$. Taking logarithms proves the quantitative bound and the stated divergence.

For dyadic coefficients $H = 1$. The case $\lambda = 0$, omitted from the divisibility theorem because its reduced numerator is zero, gives the canonical target r_m itself. For $m \ge 1$ this is the preceding interval's interpolation with coefficient 1, so the same proved bound at index $m-1$ already gives divergence. Therefore allowing zero coefficients at any subsequence does not change the all-dyadic conclusion.

Finally, if the primitive errors tend to zero, then they are eventually at most 1. Squaring (43) gives

$$H_m \ge \frac{\mathcal{P}_{N_m}}{3^{30}(m+2)2^{2(m+1)(2m+5)}}$$

at all sufficiently large nonzero-interpolation indices. Zero interpolation cannot occur infinitely often in a successful sequence. Taking logarithms proves the necessary liminf bound. □

Scope. This argument permits completely nonperturbative weights and arbitrary two-adic denominator growth. It does not restrict the height of the odd denominator to be small by assumption and then declare all interpolation impossible: instead it proves how large that odd part must become for a possible successful sequence. A least-denominator target chosen by a Stern–Brocot algorithm may require precisely such enormous odd interpolation denominators, so that selection procedure is not excluded. The exact arithmetic above does not prove its primitive errors tend to zero.

10 Constructive extensions of the positive square map

The results in this section are auxiliary tools for choosing polynomial square forms. They do not prove the irrationality of Euler's constant, and no claim of historical novelty is made. In particular, a degree bound conditional on separation from a rational number is not a proof of that separation.

Put $H_0 = 0$, $H_j = \sum_{h=1}^{j} 1/h$, and

$$a_j = j!(1 + jH_j), \qquad b_j = jj!.$$

The moment normalization is fixed by the exact identity

$$\int_0^\infty t^j (t-1) \log t \, e^{-t} \, dt = \Gamma'(j+2) - \Gamma'(j+1) = j!(1 + jH_j - j\gamma).$$

For polynomials $f, g \in \mathbb{Q}[t]$, define the rational bilinear forms

$$\mathcal{A}(f,g) = \sum_{i,j} f_i g_j a_{i+j}, \qquad \mathcal{B}(f,g) = \sum_{i,j} f_i g_j b_{i+j},$$

and write $A(q) = \mathcal{A}(q,q)$, $B(q) = \mathcal{B}(q,q)$. Thus

$$I(q) := A(q) - \gamma B(q) = \int_0^\infty q(t)^2 (t-1) \log t \, e^{-t} \, dt > 0 \quad (q \neq 0). \tag{44}$$

For integer $q \neq 0$, set $g(q) = \gcd(A(q), B(q)) > 0$ and $E(q) = I(q)/g(q)$. For rational q, first clear coefficient denominators; the resulting primitive pair and $E(q)$ are independent of that choice. For a real parameter θ, write

$$L_\theta(q) = A(q) - \theta B(q) = \int_0^\infty q(t)^2 (t-1)(\log t + \gamma - \theta) e^{-t} \, dt.$$

We use $1/2 < \gamma < 7/12$, which follows directly from the positive Binet integral $\gamma - 1/2 = \int h_1 \, d\mu < \int d\mu = 1/12$. For nonnegative integer α, our normalization is

$$L_j^{(\alpha)}(t) = \sum_{h=0}^{j} \frac{(-1)^h}{h!} \binom{j+\alpha}{j-h} t^h.$$

Theorem 10.1 (A quantitative polynomial lift of exponential square tests)**.** *Let $m = 2k \geq 2$, $\lambda > 1$ be rational, and $0 \leq \theta \leq 1$. Define*

$$\alpha = m - 1, \quad \sigma = \frac{\lambda - 1}{2}, \quad \varrho = \frac{\lambda - 1}{\lambda + 1}, \quad c = \left(\frac{2}{\lambda + 1}\right)^m, \qquad f(t) = t^k e^{-\sigma t}.$$

For $N \geq 0$, the polynomial

$$q_N(t) = c \, t^k \sum_{j=0}^{N} \varrho^j L_j^{(\alpha)}(t) \in \mathbb{Q}[t]$$

has degree $N+k$. Put

$$z=\frac{1+\varrho^2}{2},\qquad \tau=\frac{\varrho^2}{z},\qquad \mathcal{S}_\alpha(z)=\left[1+64\left(z\frac{d}{dz}+\alpha+4\right)^3\right](1-z)^{-\alpha-1},$$

$$\mathcal{E}_N=6c^2\alpha!\,\tau^{N+1}\mathcal{S}_\alpha(z),\qquad \mathcal{F}=6\left(\frac{(m+2)!}{\lambda^{m+3}}+\frac{(m-1)!}{\lambda^m}\right).$$

These bounds are positive rational numbers and $\mathcal{E}_N\to 0$. The exact exponential test is

$$L_\theta(f)=\frac{m!}{\lambda^{m+2}}\left[(m+1-\lambda)(H_m-\log\lambda-\theta)+1\right], \tag{45}$$

and the entire-ray error satisfies

$$|L_\theta(q_N)-L_\theta(f)|\le\mathcal{E}_N+2\sqrt{\mathcal{E}_N\mathcal{F}}. \tag{46}$$

Consequently a certified negative exponential test can be converted into a negative rational polynomial test by a terminating search in N. Clearing the coefficients then gives an integer polynomial with the same sign.

Proof. The Laguerre generating function and orthogonality [6] give

$$e^{-\sigma t}=c\sum_{j\ge0}\varrho^j L_j^{(\alpha)}(t),\qquad h_j:=\int_0^\infty t^\alpha e^{-t}L_j^{(\alpha)}(t)^2\,dt=\frac{(j+\alpha)!}{j!}.$$

These identities follow, respectively, from $(1-u)^{-\alpha-1}\exp(-tu/(1-u))$ and the Rodrigues formula. In the orthonormal Laguerre basis, multiplication by t has diagonal $2j+\alpha+1$ and off-diagonal $-\sqrt{(j+1)(j+\alpha+1)}$. The sum of absolute values in its jth row is at most $4(j+\alpha+1)$. Since the matrix is tridiagonal, the row sum for the absolute values of its cube is at most $64(j+\alpha+4)^3$. Symmetry and $2|uv|\le u^2+v^2$ therefore imply

$$\int_0^\infty t^\alpha(1+t^3)e^{-t}\left|\sum_j d_jL_j^{(\alpha)}(t)\right|^2dt\le\sum_j h_j[1+64(j+\alpha+4)^3]|d_j|^2.$$

First this holds for finite sums. Applying it to differences of finite tails proves convergence in the displayed weighted norm. The generating function identifies that limit with the exponential, so this also justifies the infinite-tail estimate without a compactness or tail assumption.

For $\theta\in[0,1]$, the signed density factor obeys

$$|(t-1)(\log t+\gamma-\theta)|\le W(t):=6(t^2+t^{-1}).$$

Indeed $|\gamma-\theta|<1$ and $|\log t|\le t+t^{-1}$; bounding separately on $t\le1$ and $t\ge1$ proves the stated inequality. With $\|u\|_W^2=\int W(t)|u(t)|^2e^{-t}dt$, the preceding Laguerre estimate gives

$$\|q_N-f\|_W^2\le6c^2\alpha!\sum_{j>N}\binom{j+\alpha}{\alpha}\varrho^{2j}[1+64(j+\alpha+4)^3]\le\mathcal{E}_N.$$

Here $\varrho^{2j}=\tau^jz^j\le\tau^{N+1}z^j$, and the complete sum in z equals $\mathcal{S}_\alpha(z)$. Also $\|f\|_W^2=\mathcal{F}$. Cauchy–Schwarz applied to $(q_N-f)(q_N+f)$ proves (46).

Finally,

$$\int_0^\infty t^j\log t\,e^{-\lambda t}dt=\frac{j!}{\lambda^{j+1}}(H_j-\gamma-\log\lambda).$$

Substitution and $H_{m+1}=H_m+1/(m+1)$ give (45). This evaluation also shows explicitly why its only unevaluated logarithm is the elementary $\log\lambda$; the eventual polynomial test has exactly the rational coefficients $A(q_N)-\theta B(q_N)$ when θ is rational. □

Theorem 10.2 (An effective degree bound for exclusion and representation)**.** *Let* $0 < \varepsilon \le 1/3$ *and put*

$$m = 2\left\lceil \frac{48}{\varepsilon^2} \right\rceil, \qquad N = 8m^2, \qquad D = N + \frac{m}{2} < 77000\varepsilon^{-4}.$$

For every real $\theta \in [0,1]$ *with* $|\theta - \gamma| \ge \varepsilon$*, there is a rational polynomial* q *of degree at most* D *such that* $L_\theta(q) < 0$*. For rational* θ *this polynomial has an explicit construction using only rational arithmetic and certified bounds for* $\log m$*.*

Let

$$S_d = \{X \in \mathbb{R} : [a_{i+j} - Xb_{i+j}]_{0 \le i,j \le d} \text{ is positive semidefinite}\}.$$

Then

$$S_D \subset (\gamma - \varepsilon, \gamma + \varepsilon). \tag{47}$$

For every rational r *with* $|r - \gamma| \ge \varepsilon$*, there is moreover an integer polynomial* $q \ne 0$ *of degree at most* D *such that* $B(q) \ne 0$ *and* $A(q)/B(q) = r$*. A search restricted to this degree therefore terminates. No coefficient-height bound is asserted.*

Proof. Choose rational $\widetilde{h}$ with

$$|\widetilde{h} - (H_m - \log m - \theta)| \le 1/m, \qquad s = \widetilde{h}/2, \qquad \lambda = m(1+s).$$

For rational θ this uses no digits or assumption about γ. For example, for $u = (m-1)/(m+1)$ the partial sums of $2\sum_{j \ge 0} u^{2j+1}/(2j+1)$ approximate $\log m$ with remainder at most $2u^{2J+3}/((2J+3)(1-u^2))$ after index J. The elementary estimate

$$0 < H_m - \log m - \gamma < 1/m$$

follows by summing $0 < \log(1+1/j) - 1/(j+1) < 1/(j(j+1))$ for $j \ge m$. Consequently $|s| \le 1/3$ and $|s| \ge \varepsilon/4$. In particular $2m/3 \le \lambda \le 4m/3$ and $\lambda > 1$.

Write $h = H_m - \log m - \theta$ and $e = h - 2s$, so $|e| \le 1/m$. On $[-1/3, 1/3]$ the function $g(s) = 2s - \log(1+s)$ satisfies

$$sg(s) \ge \tfrac{1}{2}s^2, \qquad |g(s)| \le \tfrac{5}{4}|s|,$$

as follows from $1/2 \le g'(s) \le 5/4$. The bracket in (45) is thus

$$1 + (1 - ms)(g(s) + e) \le 1 + \tfrac{9}{4}|s| + 1/m - \tfrac{m}{2}s^2 < 2 - \tfrac{m}{2}s^2 \le -1.$$

It follows that $L_\theta(f) \le -\eta$, where $\eta = m!/\lambda^{m+2} > 0$ is rational.

We record estimates establishing a fully explicit truncation order. With the notation of Theorem 10.1, $m \ge 96$ and elementary binomial generating functions give

$$\mathcal{S}_{m-1}(z) \le 257m^3(1-z)^{-m-3}.$$

For clarity, use $(j+m+3)^3 \le 2(j+m)^3$ for $m \ge 32$ and

$$(j+m)^3\binom{j+m-1}{m-1} \le m(m+1)(m+2)\binom{j+m+2}{m+2},$$

then $m(m+1)(m+2) \le 2m^3$. Since

$$1 - z = \frac{2\lambda}{(\lambda+1)^2}, \qquad \tau = 1 - \frac{2\lambda}{\lambda^2+1},$$

one obtains

$$\frac{\mathcal{E}_N}{\eta} < 60000m^7 2^m \exp\left(-\frac{3(N+1)}{4m}\right), \qquad \frac{\mathcal{F}}{\eta} \le 24m.$$

For the first inequality, use

$$6c^2(m-1)!\mathcal{S}_{m-1}(z) < 30000m^6(m-1)!(2/\lambda)^m$$

and $\tau^{N+1} \le \exp(-3(N+1)/(4m))$. For $N = 8m^2$,

$$\mathcal{E}_N < \frac{\eta}{1536m}.$$

Indeed the logarithm of $92160000m^8 2^m e^{-6m}$ is negative for $m \ge 96$; the inequalities

$$\log(92160000) < 27, \qquad 8\log m \le 2m, \qquad m\log 2 < m$$

suffice. Thus

$$\mathcal{E}_N + 2\sqrt{\mathcal{E}_N \mathcal{F}} < \eta/2,$$

and the constructed q_N has $L_\theta(q_N) < -\eta/2$.

Each S_d is a convex interval, contains γ, and the intervals are nested as d grows. The tests at $\theta = 0, 1$ exclude both endpoints, since their distances from γ exceed $1/3$. Therefore $S_D \subset (0,1)$; the other tests give (47). The degree estimate follows from $m \le 98\varepsilon^{-2}$ and $8 \cdot 98^2 + 49 < 77000$.

For rational $r \in [0,1]$ at the stated distance, the rational quadratic form L_r on polynomials of degree at most D has a negative vector and the positive vector 1. For $r < 0$, use the test negative at 0: its B is negative by (44), so it remains negative at every $r < 0$. Likewise the test negative at 1 remains negative for $r > 1$. Thus the form is indefinite in every case. Its dimension is at least five. If singular, a nonzero rational kernel vector is isotropic. Otherwise Meyer's theorem [36] for indefinite rational quadratic forms in at least five variables supplies a rational isotropic vector. Clearing denominators gives $q \in \mathbb{Z}[t] \setminus \{0\}$. Equation (44) forces $B(q) \ne 0$, and hence $A(q)/B(q) = r$. Enumeration of integer coefficient vectors of the bounded degree is a terminating exact algorithm; no bound for its running time or the coefficient height is implied. □

A quantitative enclosure and its scope. For every integer $d \ge 6237000$, the same argument yields the explicit enclosure

$$S_d \subset \left(\gamma - (77000/d)^{1/4}, \gamma + (77000/d)^{1/4}\right).$$

The condition $|r - \gamma| \ge \varepsilon$ remains essential in the representation theorem. Applying it to an arbitrary rational r without first establishing that condition would assume the desired exclusion.

11 Exact arithmetic in Laguerre coordinates

This section makes the coefficient arithmetic of the positive square map explicit in the ordinary Laguerre basis. It provides an optimal universal normalization, a local prime-residue rule, and a full primitive-cost classification for fixed polynomial coefficient schedules. It does not exclude arbitrary growing sequences of coefficient vectors.

Let

$$L_j(t) = \sum_{h=0}^{j} \frac{(-1)^h}{h!}\binom{j}{h} t^h, \qquad H_0 = 0, \quad H_j = \sum_{h=1}^{j} \frac{1}{h}.$$

For a rational polynomial Q, define the rational numbers $A(Q), B(Q)$ by

$$A(Q) - \gamma B(Q) = I(Q) := \int_0^\infty Q(t)^2(t-1)\log t\, e^{-t}\, dt, \qquad B(Q) = \int_0^\infty Q(t)^2(t-1)e^{-t}\, dt.$$

Their rationality also follows from the matrices below. For every nonzero Q, $I(Q) > 0$. Clear denominators in $(A(Q), B(Q))$ by any positive integer D and let $g = \gcd(DA(Q), DB(Q))$. The primitive error is $E(Q) = DI(Q)/g$, independent of the choice of D or nonzero rational scaling of Q. If $B(Q) \neq 0$ and q is the reduced denominator of $A(Q)/B(Q)$, then

$$E(Q) = q\,\frac{I(Q)}{|B(Q)|}. \tag{48}$$

If $B(Q) = 0$, then $E(Q) = 1$.

Theorem 11.1 (Optimal half-lcm normalization in Laguerre coordinates)**.** *Let $Q_c = \sum_{j=0}^{n} c_j L_j$ with $c_j \in \mathbb{Z}$. Then $B(Q_c)$ is an integer. For $n \geq 2$, the smallest positive integer D_n such that $D_n A(Q_c) \in \mathbb{Z}$ for every $c \in \mathbb{Z}^{n+1}$ is*

$$D_n = \frac{\operatorname{lcm}(1, \ldots, n)}{2}.$$

For $n = 0, 1$ it is $D_n = 1$. The exact symmetric Gram matrices are

$$B_{ii} = 2i, \qquad B_{i,i+1} = -(i+1), \qquad B_{ij} = 0 \quad (|i-j| \geq 2), \tag{49}$$

$$A_{ii} = 2iH_i + 2i + 1, \qquad A_{i,i+1} = -(i+1)H_i - \frac{3i}{2} - 1, \tag{50}$$

$$A_{ij} = \frac{i + j + (j-i)^2}{(j-i)((j-i)^2 - 1)} \qquad (j - i \geq 2). \tag{51}$$

Thus $(D_n A(Q_c), D_n B(Q_c))$ is an exact integer coefficient pair, without requiring factorial clearing of the polynomial coefficients. Its gcd must still be removed to obtain its primitive pair.

Proof. The Laguerre recurrence is

$$tL_i = (2i+1)L_i - iL_{i-1} - (i+1)L_{i+1},$$

and the L_i are orthonormal for $e^{-t}dt$. This gives (49). The logarithmic Gram matrix is

$$\int_0^\infty \log t\, L_i(t) L_j(t) e^{-t}\, dt = \begin{cases} H_i - \gamma, & i = j, \\ -1/|i-j|, & i \neq j. \end{cases}$$

One proof differentiates generalized Laguerre orthogonality at parameter zero. The norm derivative is $H_i - \gamma$, and

$$\partial_\alpha L_i^{(\alpha)}\Big|_{\alpha=0} = \sum_{j<i} \frac{L_j}{i-j}.$$

The last identity follows by differentiating the Laguerre generating function; see [6]. All expressions are finite polynomial sums and the differentiated integrals have an integrable majorant. Multiplying this logarithmic matrix by the recurrence for $t - 1$, including the L_{n+1} term when necessary, gives (50)–(51).

Write $d_n = \operatorname{lcm}(1, \ldots, n)$. A quadratic form is integer-valued on all integer vectors exactly when its diagonal coefficients and twice its off-diagonal coefficients are integers. For $k = j - i \geq 2$, the useful partial-fraction identity is

$$2A_{ij} = \frac{2(i+1)}{k-1} - \frac{4i}{k} + \frac{2i}{k+1}.$$

If $i \geq 1$, all displayed denominators are at most n. If $i = 0$, only $2/(k-1)$ remains. Together with (50), this shows that $d_n/2$ clears the quadratic values.

For sharpness, testing the vectors e_i and $e_i + e_j$ forces every denominator in A_{ii} and $2A_{ij}$. Row zero supplies $2/(k-1)$ for $2 \le k \le n$, so for $n \ge 3$ it forces $d_{n-1}/2$. Induct on n, starting at $n = 2$. A new factor is needed only when n is a prime power. For $n \ge 3$,

$$2A_{1n} = \frac{2[n(n-1)+2]}{(n-1)n(n-2)}.$$

If $n = p^a$ for an odd prime p, this has p-adic valuation $-a$. If $n = 2^a$ with $a \ge 2$, its valuation at 2 is $1 - a$. These are exactly the new prime-power factors of $d_n/2$, proving optimality. □

For finite vectors c, d, pad both by zero and write

$$\mathcal{B}(c,d) = \sum_{r\ge 0} 2r\, c_r d_r - \sum_{r\ge 0}(r+1)(c_r d_{r+1} + c_{r+1} d_r).$$

Thus $B(Q_c) = \mathcal{B}(c,c)$.

Theorem 11.2 (Exact residue at a prime in the upper half)**.** *Let p be an odd prime with $n/2 < p \le n$, and let $s = n - p$. For an integer coefficient vector $c = (c_0, \dots, c_n)$ put*

$$d_r = c_{p+r} \ (0 \le r \le s), \qquad h_r = c_r \ (0 \le r \le s+1),$$

and pad these vectors by zero. Then $pA(Q_c)$ is p-integral and

$$pA(Q_c) \equiv \mathcal{B}(d,d) - 2\mathcal{B}(h,d) \pmod{p}. \tag{52}$$

If this residue is nonzero and $B(Q_c) \ne 0$, the prime p divides the actual reduced denominator of $A(Q_c)/B(Q_c)$.

Proof. The normalization theorem shows that $v_p(A(Q_c)) \ge -1$. In the diagonal terms, the residue of pA_{ii} is $2i$ when $i \ge p$, and zero otherwise. In the adjacent cross terms, the residue of $2pA_{i,i+1}$ is $-2(i+1)$ when $i \ge p$, and zero otherwise. These give $\mathcal{B}(d,d)$.

For a gap $k = j - i \ge 2$, the only potentially nonzero residues of $2pA_{ij}$ occur at $k = p-1, p, p+1$. They are, respectively,

$$2i, \qquad -4i, \qquad 2(i+1).$$

The possible exceptional gap $2p - 1$ has $i = 0$ and residue zero. Collecting the three contributing diagonals gives

$$2\sum_{r=0}^{s}(r+1)h_{r+1}d_r - 4\sum_{r=0}^{s} r h_r d_r + 2\sum_{r=1}^{s} r h_{r-1} d_r = -2\mathcal{B}(h,d).$$

At $n = 2p - 1$ the apparent overlapping endpoint has coefficient $s + 1 = p$ and vanishes modulo p, so the same formula applies.

If the residue is nonzero, $v_p(A(Q_c)) = -1$. Since $B(Q_c)$ is an integer, $v_p(A(Q_c)/B(Q_c)) = -1 - v_p(B(Q_c)) < 0$. This proves survival in the reduced denominator, not merely in a chosen clearing factor. □

Theorem 11.3 (Complete primitive-cost classification of fixed polynomial schedules)**.** *Fix a nonzero polynomial $P(X,Y) \in \mathbb{Q}[X,Y]$ and put*

$$Q_N(t) = \sum_{k=0}^{N} P(N,k) L_k(t).$$

These polynomials are nonzero for all sufficiently large N. The quantity $B_N = B(Q_N)$ is a rational polynomial in N. Exactly one of the following holds:

1. *B_N is identically zero. Then $E(Q_N) = 1$ for all sufficiently large N.*
2. *B_N is not identically zero. Then it is eventually nonzero and*

$$\liminf_{N\to\infty} \frac{\log E(Q_N)}{N} \geq \frac{1}{2}.$$

In particular, no fixed polynomial coefficient schedule of this form has primitive error tending to zero.

Proof. Clear the fixed coefficient denominators of P once. This scales each Q_N by the same nonzero rational factor and leaves its primitive error unchanged, so assume $P \in \mathbb{Z}[X, Y]$.

Write $c_k = P(N, k)$ and define the integer-valued rational polynomial

$$F_P(N) = 2(N+1)P(N,N)P(N,N+1) - B_N. \tag{53}$$

Both B_N and $F_P(N)$ are polynomials because sums of polynomial functions over $k = 0, \ldots, N$ are polynomials in N. We first prove that F_P is nonzero; it is essential not to infer this merely from nonconstancy in the second variable.

Let d be the total degree of P. Its highest homogeneous part gives a nonzero real polynomial f such that, uniformly for $x \in [0, 1]$,

$$P(N, Nx) = N^d f(x) + O_P(N^{d-1}).$$

The exact discrete energy identity is

$$B_N = (N+1)c_N^2 - \sum_{k=0}^{N} c_k^2 + \sum_{k=0}^{N-1} (k+1)(c_k - c_{k+1})^2. \tag{54}$$

The last sum is $O_P(N^{2d})$, while the middle sum is a polynomial Riemann sum. Hence

$$B_N = N^{2d+1} \left[f(1)^2 - \int_0^1 f(x)^2 \, dx \right] + O_P(N^{2d}),$$

and therefore

$$F_P(N) = N^{2d+1} \left[f(1)^2 + \int_0^1 f(x)^2 \, dx \right] + O_P(N^{2d}). \tag{55}$$

The coefficient in brackets is strictly positive. Thus F_P has degree exactly $2d + 1$ and is nonzero for all sufficiently large N.

Put $r = \deg_Y P$. Let p be a prime with

$$N/2 < p \leq N, \qquad p > 2r + 2,$$

and let $s = N - p$. In (52), $P(N, p + j) \equiv P(N, j) \pmod{p}$. Thus the residue becomes

$$pA(Q_N) \equiv 2(s+1)P(N,s)P(N,s+1) - B_s(P(N,\cdot)) \pmod{p}.$$

For fixed N the right side is a rational polynomial in s. Its coefficient denominators have no prime factors above $2r + 2$: expand the polynomial summands of degree at most $2r + 1$ in the Newton binomial basis and sum them. Since $s \equiv N \pmod{p}$, we may replace s by N and conclude

$$pA(Q_N) \equiv F_P(N) \pmod{p}. \tag{56}$$

When $B_N \neq 0$, every prime in $(N/2, N]$ not dividing $F_P(N)$ therefore survives in the reduced denominator q_N of $A(Q_N)/B_N$. For all sufficiently large N the fixed lower restriction $p > 2r + 2$

is automatic. Since the product of the distinct excluded primes divides the nonzero integer $F_P(N)$,

$$\log q_N \geq \vartheta(N) - \vartheta(N/2) - \log|F_P(N)| = \left(\frac{1}{2} + o(1)\right) N.$$

Here $\vartheta(x) = \sum_{p \leq x} \log p$ and $\vartheta(x) \sim x$ is the prime number theorem [7].

We must also control the real error; denominator growth alone would not suffice. If Q has degree j and highest Laguerre coefficient c_j, then

$$I(Q) \geq e^{-2} \log 2\, (j+1) c_j^2. \tag{57}$$

Indeed, restrict the positive integral to $t \geq 2$ and use $(t-1)\log t \geq (t-2)\log 2$. After $t = x+2$, the leading monomial coefficient is $(-1)^j c_j / j!$. The monic degree-j orthogonal polynomial for $xe^{-x}dx$ has squared norm $j!(j+1)!$, so the orthogonal decomposition of $Q(x+2)$ gives

$$I(Q) \geq e^{-2} \log 2 \left(\frac{c_j}{j!}\right)^2 j!(j+1)!,$$

which is (57). The norm identity follows directly from the generalized Laguerre Rodrigues formula by j integrations by parts.

For all sufficiently large N, $P(N, Y)$ is a nonzero polynomial of degree at most r. It cannot vanish at the $r+1$ distinct integers $N, N-1, \ldots, N-r$. Consequently $\deg Q_N \geq N - r$, and its highest Laguerre coefficient is a nonzero integer. Thus $I(Q_N)$ has a positive linear lower bound. Since the nonzero polynomial B_N has polynomial size, $I(Q_N)/|B_N|$ is bounded below by an inverse power of N. Equation (48) now proves the exponential primitive-error lower growth.

If B_N is identically zero, positivity gives $A(Q_N) > 0$ whenever $Q_N \neq 0$. Primitive normalization of the rational pair $(A(Q_N), 0)$ then gives error exactly one. □

Scope and an instructive zero-coefficient case. Nonconstant coefficient profiles can satisfy $B_N \equiv 0$. For example, $P(N,k) = 3k - (2N+1)$ has this property; the first branch of the theorem includes it. The proof of (55) remains valid in this case and does not assume that B_N is nonzero.

Let

$$\mathcal{D}_t = t\frac{d^2}{dt^2} + (1-t)\frac{d}{dt}, \qquad \mathcal{D}_t L_k = -kL_k.$$

The theorem covers every fixed polynomial filter $P(N, -\mathcal{D}_t)$ applied to the zero-location kernel $\sum_{k=0}^{N} L_k(t)$. It therefore covers polynomial tapers of any fixed order and every fixed polynomial dependence on N. Coefficients such as $L_k(x_N)$ at nonzero x_N, filters whose degree grows with N, and arbitrary coefficient sequences are outside its scope.

12 Shared-cutoff support compression

Theorem 12.1 (Compression at a shared cutoff and its primitive cost)**.** *For $m \geq 0$ define*

$$T_m(x) = (x-1)^2(2x-1)\prod_{k=1}^{m}(4^k x - 1), \qquad d = m+3, \qquad s = 1 + m(m+1).$$

For every $C \in \mathbb{Z}[x]$, division gives unique $S, R \in \mathbb{Z}[1/2][x]$ with

$$C = T_m S + R, \qquad \deg R < d.$$

If $\deg C = L \geq d$, all coefficient denominators of S and R divide $2^{s(L-d+1)}$. Moreover,

$$\begin{aligned} &R(1) = C(1), && R'(1) = C'(1), && R(1/2) = C(1/2), \\ &R(4^{-k}) = C(4^{-k}) && && (1 \leq k \leq m). \end{aligned} \tag{58}$$

Let $a = 2^t$, where $t \geq 0$ is an integer, and assume

$$Q = C(1) \neq 0, \qquad C'(1) + tQ = 0. \tag{59}$$

For $C(x) = \sum c_i x^i$ and $R(x) = \sum r_i x^i$, define the corrected harmonic targets

$$r_C = \frac{1}{Q} \sum_i c_i \left(H_{a2^i} - \frac{1}{2a2^i} \right), \qquad r_R = \frac{1}{Q} \sum_i r_i \left(H_{a2^i} - \frac{1}{2a2^i} \right). \tag{60}$$

Both give log-free gamma forms. Their target difference is generally nonzero, but it satisfies the exact identity and bound

$$\Delta := r_C - r_R = \frac{1}{Q} \sum_i [x^i](T_m S) H_{a2^i},$$

$$|\Delta| \leq \Delta_{\max} := \frac{\mu_m a^{-2m-2}}{|Q|} T_m^{\mathrm{abs}}(x_0) S^{\mathrm{abs}}(x_0), \qquad x_0 = 4^{-m-1}, \tag{61}$$

where $P^{\mathrm{abs}}(x) = \sum_i |[x^i]P| x^i$.

Let E be the least nonnegative integer with $2^E R \in \mathbb{Z}[x]$, and put

$$N_0 = a2^{d-1}, \qquad D_0 = 2\,\mathrm{lcm}(1, \ldots, N_0), \qquad Z = 2^E D_0 \sum_i r_i \left(H_{a2^i} - \frac{1}{2a2^i} \right) \in \mathbb{Z}.$$

The complete reduced denominator of r_R is exactly

$$q_R = \frac{|Q| 2^E D_0}{\gcd(|Z|, |Q| 2^E D_0)}. \tag{62}$$

In particular,

$$\mathrm{odd}(q_R) \mid \mathrm{odd}(Q)\,\mathrm{odd}\big(\mathrm{lcm}(1, \ldots, N_0)\big).$$

Every odd prime above N_0 not dividing Q is therefore absent after compression, even if it appeared in the original target r_C.

If independently certified rational bounds satisfy

$$0 < \ell \leq \gamma - r_C \leq u, \qquad \Delta_{\max} < \ell,$$

then, with $P_R = q_R r_R \in \mathbb{Z}$,

$$0 < \ell - \Delta_{\max} \leq \gamma - r_R \leq u + \Delta_{\max}, \qquad 0 < q_R \gamma - P_R \leq q_R(u + \Delta_{\max}). \tag{63}$$

This is a checkable transfer of a positive error certificate. It is not an assertion that the last upper bound tends to zero.

Proof. The leading coefficient of T_m is

$$2 \prod_{k=1}^{m} 4^k = 2^{1+m(m+1)} = 2^s.$$

Long division therefore introduces only powers of 2 in denominators. Each of the at most $L-d+1$ steps increases the necessary power by at most s, giving the stated bound. The double root at 1 and the other listed roots give (58). In particular the entire condition (59) is preserved. It is essential to include tQ: for $a > 1$, cancellation of $C'(1)$ alone would leave $Q \log a$.

Subtract the corrected targets. Since $(T_m S)(1/2) = 0$, the inverse first moment vanishes, and the $1/(2n)$ terms cancel in their difference. Also $(T_m S)(1) = (T_m S)'(1) = 0$, so all gamma and

logarithmic terms cancel when (15) is summed over its coefficients. For $z > 0$, the exact finite expansion is

$$\Phi(z) = \sum_{k=0}^{m-1} (-1)^k \mu_k z^{-k-1} + (-1)^m z^{-m} \int \frac{u^m}{u+z}\, d\mu(u).$$

For $m = 0$ the sum is empty. The remainder has absolute value at most $\mu_m z^{-m-1}$. Each of the first m terms vanishes in the coefficient sum because $(T_m S)(4^{-k}) = 0$ for $1 \le k \le m$. Thus

$$|\Delta| \le \frac{\mu_m a^{-2m-2}}{|Q|} (T_m S)^{\mathrm{abs}}(x_0) \le \frac{\mu_m a^{-2m-2}}{|Q|} T_m^{\mathrm{abs}}(x_0) S^{\mathrm{abs}}(x_0),$$

which is (61).

All surviving indices satisfy $a2^i \le N_0$. Each corrected harmonic denominator divides D_0, and 2^E clears all r_i. Hence Z is an integer, and reduction of $Z/(Q2^E D_0)$ gives (62). The odd-part assertion follows immediately. Finally $\gamma - r_R = (\gamma - r_C) + \Delta$, so (63) follows from the triangle inequality and the exact value of q_R. □

An infinite shared-prime cancellation with positive errors. Set the external base $a = 1$, take $m = 0$, and put

$$C_t(x) = 1 + x^t T_0(x), \qquad t \ge 1.$$

The remainder is $R = 1$. Its corrected target is exactly $H_1 - 1/2 = 1/2$, with final denominator 2. The original target is

$$r_{C_t} = \frac{1}{2} + V_t, \qquad V_t = \sum_{i=0}^{3} [x^i] T_0(x) H_{2^{t+i}}.$$

Since $T_0^{\mathrm{abs}}(1/4) = 75/32$, the finite Binet remainder gives

$$|V_t| \le \frac{25}{128\, 4^t} < \frac{3}{40}.$$

On the other hand $h_1(u) > 1 - u$ for $u > 0$, so

$$\gamma - \frac{1}{2} > \mu_0 - \mu_1 = \frac{3}{40}.$$

Every original target and every compressed target therefore has a strictly positive gamma error. For any prime ℓ in $(2^{t+2}, 2^{t+3}]$, only the last harmonic sum in V_t contains a denominator divisible by ℓ, and its coefficient is 2. It contains exactly the single term $1/\ell$ with such a denominator. Thus $v_\ell(r_{C_t}) = -1$. All these primes disappear after compression. For instance the exact reduced denominators for $t = 1, 2$ are respectively 90090 and 9025222055850 before compression, and 2 afterward.

This example explicitly removes a block of primes already present in the base. It lies outside an obstruction whose hypothesis requires that the base avoid that block. At the same time, it changes the target to 1/2, leaving the error $\gamma - 1/2$ fixed. It therefore proves the arithmetic cancellation mechanism without claiming the missing small-error estimate needed for irrationality.

13 Rational localization at the double zero of the positive weight

The positive square weight $(t-1)\log t$ vanishes quadratically at $t = 1$. This section constructs rational polynomials localized at points tending to that zero. The coefficients depend on Laguerre values and lie outside the fixed-polynomial-schedule obstruction. The real integral becomes small

with an explicit order, and its full primitive cost is expressed in terms of the actual reduced denominator. A one-step change of the cutoff also removes a specific adverse prime factor exactly.

Let $L_k^{(1)}$ denote the generalized Laguerre polynomial with parameter 1, and let $L_k = L_k^{(0)}$. For an integer $M \geq 2$ and $\sigma \in \{0, -1\}$, define

$$N = M^2 + \sigma, \qquad x = 1 - \frac{1}{M}, \qquad H = N + 1, \qquad w_k = \left(1 - \frac{k}{H}\right)^2 \quad (0 \leq k \leq N),$$

$$p_{N,x}(t) = \sum_{k=0}^{N} \frac{w_k L_k^{(1)}(x) L_k^{(1)}(t)}{k+1}, \qquad Q_M(t) = t\, p_{N,x}(t). \tag{64}$$

These polynomials have rational coefficients, constructed without using a value or an assumption about the arithmetic nature of γ. Write

$$I_M = A_M - \gamma B_M = \int_0^\infty Q_M(t)^2 (t-1) \log t\, e^{-t} dt > 0.$$

The positivity is strict because Q_M is nonzero. Indeed its ordinary Laguerre coefficient at index zero is $w_0 L_0^{(1)}(x) = 1$.

Theorem 13.1 (Exact rational certificates for the localized family)**.** *Use the inner product $\langle f, g\rangle_1 = \int_0^\infty f(t)g(t)te^{-t}dt$ and put*

$$H_j = \langle p_{N,x}, t^j p_{N,x}\rangle_1, \qquad T_M = H_2 + H_1, \qquad U_M = H_3 - H_2 - H_1 + H_0.$$

Then all these numbers are rational, $B_M = H_2 - H_1$, and

$$\frac{2B_M^2}{T_M} \leq I_M \leq \frac{U_M}{2}. \tag{65}$$

When $B_M \neq 0$, let q_M be the actual reduced denominator of A_M/B_M. The complete primitive error $\mathcal{E}_M$ satisfies the entirely rational certificate

$$\frac{2q_M|B_M|}{T_M} \leq \mathcal{E}_M \leq \frac{q_M U_M}{2|B_M|}. \tag{66}$$

For exact integerization, put

$$F_M = H^2 N! M^N, \qquad D_N = \frac{\operatorname{lcm}(1, \ldots, N+1)}{2}, \qquad S_M = D_N F_M^2.$$

Then $S_M A_M, S_M B_M$ are integers. If $g_M = \gcd(S_M A_M, S_M B_M) > 0$, the identities

$$\mathcal{E}_M = \frac{S_M I_M}{g_M}, \qquad q_M = \frac{S_M |B_M|}{g_M} \quad (B_M \neq 0)$$

use the complete gcd, not merely a chosen clearing factor.

Proof. For every $t > 0$,

$$\frac{2(t-1)^2}{t+1} \leq (t-1)\log t \leq \frac{(t-1)^2(t+1)}{2t}.$$

For the lower inequality use $\log t = 2\operatorname{artanh}((t-1)/(t+1))$. For the upper one, the derivative of $(t - t^{-1})/2 - \log t$ is $(t-1)^2/(2t^2) \geq 0$, and its value at 1 is zero. Multiplication by $Q_M(t)^2 = t^2 p_{N,x}(t)^2$ and integration gives the upper bound $U_M/2$. Cauchy–Schwarz gives

$$B_M^2 \leq \left(\int_0^\infty \frac{t(t-1)^2}{t+1} p_{N,x}(t)^2 te^{-t}dt\right)\left(\int_0^\infty t(t+1)p_{N,x}(t)^2 te^{-t}dt\right) \leq \frac{I_M T_M}{2}.$$

This proves (65). Since $I_M > 0$,

$$\mathcal{E}_M = q_M \left|\gamma - \frac{A_M}{B_M}\right| = q_M \frac{I_M}{|B_M|},$$

which gives (66).

The elementary identity

$$tL_k^{(1)}(t) = (k+1)(L_k(t) - L_{k+1}(t))$$

shows that, with $z_k = w_k L_k^{(1)}(x)$,

$$Q_M = \sum_{k=0}^{N} z_k (L_k - L_{k+1}).$$

Its ordinary Laguerre coefficient vector is $(z_0, z_1 - z_0, \dots, z_N - z_{N-1}, -z_N)$. The finite coefficient formula for $L_k^{(1)}(x)$ shows that F_M clears this vector. Lemma 11.1, applied to an integer Laguerre vector of length $N+2$, proves that S_M clears (A_M, B_M). The remaining identities follow by complete reduction of this pair. □

Theorem 13.2 (Uniform localization and its exact arithmetic threshold)**.** *For either $\sigma = 0$ or $\sigma = -1$ in (64), let*

$$\kappa = \frac{128e}{315\pi} > 0.$$

Then, as $M \to \infty$,

$$B_M = -\kappa + O\left(\frac{\log M}{M}\right), \qquad T_M = 2\kappa M + O(\log M), \qquad U_M = O(M^{-1}).$$

Consequently

$$I_M = \Theta(M^{-1}), \qquad \gamma - \frac{A_M}{B_M} = \Theta(M^{-1})$$

with the second quantity positive for all sufficiently large M. For the actual reduced denominator,

$$\mathcal{E}_M = \Theta\left(\frac{q_M}{M}\right). \tag{67}$$

Thus this particular family would satisfy primitive-error decay exactly if $q_M = o(M)$; equivalently, $Mg_M/S_M \to \infty$. Neither condition is assumed in this analytic estimate. The later two-adic obstruction in Part II excludes it for the repaired family.

Proof. All estimates below are uniform for x in the compact interval $[1/2, 1]$. We use the established Laguerre asymptotic [8]

$$L_k^{(\alpha)}(x) = \frac{e^{x/2} k^{\alpha/2 - 1/4}}{\sqrt{\pi}\, x^{\alpha/2 + 1/4}} \left[\cos\left(2\sqrt{kx} - \left(\frac{\alpha}{2} + \frac{1}{4}\right)\pi\right) + O(k^{-1/2})\right] \tag{68}$$

for $\alpha = 0, 1$. Its compact-x uniformity is essential because $x = 1 - 1/M$ varies with M.

Let $\phi_k = L_k^{(1)}/\sqrt{k+1}$, an orthonormal basis for the inner product $\langle\ ,\ \rangle_1$. Multiplication by t has Jacobi matrix

$$J_{kk} = 2k + 2, \qquad J_{k,k+1} = -\sqrt{(k+1)(k+2)}.$$

The coefficient vector of $p_{N,x}$ is $c_k = w_k L_k^{(1)}(x)/\sqrt{k+1}$ for $k \le N$, and zero otherwise. Write

$$S = \|c\|^2, \qquad g = (J - x)c, \qquad \delta = \langle c, g\rangle.$$

First, squaring (68) for $\alpha = 1$ gives

$$S = \kappa(x)\sqrt{N} + O(\log(N+2)), \qquad \kappa(x) = \frac{128e^x}{315\pi x^{3/2}}. \tag{69}$$

Here the nonoscillating sum is

$$\frac{e^x}{2\pi x^{3/2}} \sum_{k=1}^{N} k^{-1/2} \left(1 - \frac{k}{N+1}\right)^4,$$

whose leading coefficient is

$$\frac{e^x}{2\pi x^{3/2}} \int_0^1 u^{-1/2}(1-u)^4 du = \frac{128e^x}{315\pi x^{3/2}}.$$

The asymptotic remainders sum to $O(\log(N+2))$. For the oscillating sum, comparison with its integral has the same error bound: the phase derivative gives an integrable-on-finite-interval bound $O(1/t)$ for the derivative of the summand. Substitution $t = y^2$ bounds the integral by $O(1)$ using integration by parts. The frequency $4\sqrt{x}$ stays bounded away from zero, so all constants are uniform.

Second, the exact Jacobi localization identity is

$$\delta = \sum_{k=0}^{N} L_k^{(1)}(x) L_{k+1}^{(1)}(x)(w_k - w_{k+1})^2, \qquad w_{N+1} = 0.$$

The point bounds from (68) and $|w_k - w_{k+1}| \le 2/(N+1)$ imply

$$|\delta| = O(N^{-1/2}). \tag{70}$$

No sign of the individual products is used.

We also need a weighted residual bound, not just weak concentration. Put $H = N+1$, $L_{-1}^{(1)} = 0$, and

$$U_k = L_k(x) + L_{k+1}(x), \qquad V_k = L_{k-1}^{(1)}(x) + L_{k+1}^{(1)}(x).$$

Direct substitution of the quadratic taper gives, for $0 \le k \le N$,

$$g_k = \sqrt{k+1}\left[\frac{2(H-k)}{H^2}U_k - \frac{1}{H^2}V_k\right], \qquad g_{N+1} = -\frac{\sqrt{N+2}}{H^2}L_N^{(1)}(x).$$

All later coefficients vanish. The identities

$$L_{k+1}^{(1)} - L_k^{(1)} = L_{k+1}, \qquad L_{k+1} - L_k = -\frac{x}{k+1}L_k^{(1)}$$

and (68) give

$$U_k = O((k+1)^{-1/4}), \quad U_{k+1} - U_k = O((k+1)^{-3/4}),$$
$$V_k = O((k+1)^{1/4}), \quad V_{k+1} - V_k = O((k+1)^{-1/4}).$$

It follows that $g_k = O((k+1)^{1/4}/H)$ in the bulk, while $g_N, g_{N+1} = O(H^{-5/4})$ at the endpoint. Thus $\|g\|^2 = O(N^{-1/2})$.

For the additional factor of J, use the exact nonnegative energy identity

$$\langle g, Jg\rangle = g_0^2 + \sum_{k\ge 0}\left(\sqrt{k+1}\,g_k - \sqrt{k+2}\,g_{k+1}\right)^2.$$

The displayed difference bounds show that the bulk differences are $O((k+1)^{1/4}/H)$; the two endpoint contributions are $O(H^{-3/4})$ before squaring. Therefore

$$\langle g, (J+1)g\rangle = O(N^{-1/2}). \tag{71}$$

This controls the required unbounded observable on the full half-line.

The exact moment identities now give

$$B_M = (x^2-x)S + (2x-1)\delta + \|g\|^2,$$
$$T_M = (x^2+x)S + (2x+1)\delta + \|g\|^2.$$

Since $\sqrt{N} = M + O(M^{-1})$, $x = 1 - M^{-1}$, and $\kappa(1) = \kappa$, these prove the stated asymptotics of B_M and T_M. Finally,

$$U_M = \langle (J-1)c, (J+1)(J-1)c\rangle \le 2\langle g, (J+1)g\rangle + 2(x-1)^2\langle c, (J+1)c\rangle = O(M^{-1}).$$

The rational sandwich proves $I_M = \Theta(M^{-1})$: its lower bound is asymptotic to κ/M, and its upper bound is $O(M^{-1})$. Since $|B_M| \to \kappa$, both remaining assertions follow. □

Theorem 13.3 (An adverse prime factor and its removal by one cutoff step)**.** *For the family above:*

1. *If $\sigma = 0$ and $M = p^e$ is any prime power, then*

$$v_p(A_M/B_M) = -2e.$$

Thus the actual reduced denominator contains M^2. Along every prime-power sequence $M \to \infty$, the primitive error is bounded below by a positive constant times M.

2. *If $\sigma = -1$ and $M = p^e$ with p odd, then*

$$v_p(q_M) \le 2e - 2.$$

For $p \ge 5$ this improves to

$$v_p(q_M) \le \max(0, 2e-3).$$

In particular, at $M = p$ every odd prime p, including $p = 3$, is absent from the actual reduced denominator.

These statements concern the indicated prime factor only. The second case does not bound the cost of all remaining denominator factors.

Proof. Put $u_k = L_k - L_{k+1}$ and $z_k = w_k L_k^{(1)}(x)$, so $Q_M = \sum_{k=0}^{N} z_k u_k$. For $M = p^e$, $v_p(x) = -e$. The finite Laguerre formula gives

$$L_k^{(1)}(x) = \frac{(-x)^k}{k!}(1 + p^e\mathbb{Z}_p).$$

Every nonleading term divided by the leading term is an integer times x^{-j} for some $j \ge 1$. In particular the displayed leading unit is nonzero. Consequently, for $1 \le j \le N$,

$$v_p\left(\frac{z_{N-j}}{z_N}\right) = je + v_p\left(\frac{N!}{(N-j)!}\right) + 2v_p(j+1). \tag{72}$$

The ordinary Laguerre Gram matrices give exactly

$$B(u_N) = 6N + 4, \tag{73}$$
$$A(u_N) = (6N+4)H_N + 7N + 8, \tag{74}$$
$$2\mathcal{A}(u_N, u_{N-1}) = -2(4N+1)H_N - \frac{32N+7}{3}. \tag{75}$$

Here $\mathcal{A}$ denotes the bilinear form associated with A.

First take $N = p^{2e}$. Every ratio in (72) has valuation at least $3e$. Since $H_{p^{2e}}$ has the unique summand $1/p^{2e}$ of lowest p-adic valuation,

$$v_p(H_{p^{2e}}) = -2e.$$

For odd p, (73) is a unit and (74) has valuation $-2e$. For $p = 2$, their valuations are respectively 2 and $2 - 2e$.

Write $Q_M/z_N = u_N + \sum_{j\geq 1}(z_{N-j}/z_N)u_{N-j}$. In integer Laguerre coordinates of degree $N + 1$, the denominators of quadratic coefficients divide $\mathrm{lcm}(1, \ldots, N + 1)/2$. Its valuation is $2e$ for odd p and $2e - 1$ for $p = 2$. Thus the perturbation in A has valuation at least e for odd p and $e + 1$ for $p = 2$. The perturbation in B has valuation at least $3e$. Both are strictly higher than the corresponding leading valuations. Division gives $v_p(A_M/B_M) = -2e$ in every case. The real lower bound then follows from (67).

Now take $N = p^{2e} - 1$ with p odd. Pairing j and $p^a - j$ in the sum over p-units shows

$$H_{p^a-1} = \frac{1}{p}H_{p^{a-1}-1} + V_a, \qquad v_p(V_a) \geq a.$$

Also $v_p(H_{p-1}) \geq 1$. Hence

$$v_p(H_N) \geq 2 - 2e.$$

For $p \geq 5$, pairing modulo p^2 and using $\sum_{j=1}^{p-1} j^{-2} \equiv 0 \pmod{p}$ gives $v_p(H_{p-1}) \geq 2$, improving this to $v_p(H_N) \geq 3 - 2e$.

The leading value $B(u_N) = 6p^{2e} - 2$ is a p-adic unit. Every ratio in (72) has valuation at least je, so all changes to B are divisible by p. Thus the normalized B remains a unit.

For A, all linear perturbation terms with $j \geq 2$ and all quadratic perturbation terms are p-integral, because their numerator valuations are at least $2e$ and the global quadratic coefficient denominators have valuation at most $2e$. Only $j = 1$ needs a local check. For $p \geq 5$, (75) has valuation at least $\min(0, 3 - 2e)$, and multiplication by z_{N-1}/z_N, of valuation e, cannot worsen that bound.

For $p = 3$, $v_3(4N + 1) = 1$. The adjacent coefficient in (75) therefore has valuation at least $\min(3-2e, -1)$. After multiplication by the ratio, its valuation is at least $\min(3-e, e-1) \geq 2-2e$. This includes $p = 3, e = 1$. The leading value (74) satisfies the same required lower bounds. Since the normalized B is a unit, the stated bounds for the reduced denominator follow. □

What the one-step repair accomplishes. The two cutoffs have the same rigorously proved real error order $\Theta(1/M)$. At $M = p$, changing $N = p^2$ to $N = p^2 - 1$ removes the entire selected prime-power denominator p^2 rather than merely changing a clearing factor. The exact checks give the following comparison:

$M = p$	digits of q $(N = p^2)$	digits of q $(N = p^2 - 1)$	$v_p(q)$ $(N = p^2)$	$v_p(q)$ $(N = p^2 - 1)$
3	20	17	2	0
5	86	79	2	0
7	184	177	2	0

The remaining denominators in these examples are large. No estimate $q_M = o(M)$ is inferred from this local improvement.

13.1 The repaired cutoff still fails on two fixed-prime towers

The cutoff change $M^2 \mapsto M^2 - 1$ removes the selected prime entirely when M itself is an odd prime. It does not have that effect along prime-power towers with a fixed prime. The following additional consequence gives exact valuations on two such towers. It is a further obstruction within the present construction, not a claim that every modified localization schedule fails.

Proposition 13.4 (Exact denominator costs after repair on the 3- and 5-power towers)**.** *Use the localized family with $\sigma = -1$, so $N = M^2 - 1$, and let q_M be the actual reduced denominator of A_M/B_M. Then, for every integer $e \geq 1$,*

$$\begin{aligned} M = 3^e : &\qquad v_3(q_M) = 2e - 2, \\ M = 5^e : &\qquad v_5(q_M) = \max(0, 2e - 3). \end{aligned}$$

In particular,

$$\frac{q_M}{M} \geq \frac{M}{9} \quad (M = 3^e), \qquad \frac{q_M}{M} \geq \frac{M}{125} \quad (M = 5^e, e \geq 2).$$

Together with Theorem 13.2, these bounds imply that the primitive errors tend to $+\infty$ on both fixed-prime towers, even with the repaired cutoff.

Proof. First we determine the relevant harmonic valuations exactly. For an odd prime p and an integer $a \geq 2$, write

$$H_{p^a-1} = \frac{1}{p} H_{p^{a-1}-1} + V_{p,a}, \qquad V_{p,a} = \sum_{\substack{1 \leq j < p^a \\ p \nmid j}} \frac{1}{j}.$$

Pairing j with $p^a - j$ gives

$$\frac{1}{j} + \frac{1}{p^a - j} = \frac{p^a}{j(p^a - j)},$$

so $v_p(V_{p,a}) \geq a$. The exact initial values $H_2 = 3/2$ and $H_4 = 25/12$ now imply, by induction,

$$v_3(H_{3^a-1}) = 2 - a, \qquad v_5(H_{5^a-1}) = 3 - a \qquad (a \geq 1). \tag{76}$$

Indeed, at each step $a \geq 2$, the inherited valuations are $2 - a$ and $3 - a$, respectively, both strictly smaller than the unit-block lower bound a. Thus no cancellation can change the leading valuation.

We next check that these harmonic valuations survive the complete quadratic form. Use the notation

$$u_k = L_k - L_{k+1}, \qquad z_k = \left(\frac{N + 1 - k}{N + 1}\right)^2 L_k^{(1)}\left(1 - \frac{1}{M}\right).$$

For $M = p^e$, the leading-term valuation of the Laguerre polynomial shows $z_N \neq 0$. Normalize the polynomial by this coefficient:

$$\frac{Q_M}{z_N} = u_N + \sum_{j=1}^{N} t_j u_{N-j}, \qquad t_j = \frac{z_{N-j}}{z_N}.$$

The exact coefficient estimate proved in Theorem 13.3 is

$$v_p(t_j) = je + v_p\left(\frac{N!}{(N-j)!}\right) + 2v_p(j+1) \geq je. \tag{77}$$

For $j = 1$ and either $p = 3$ or $p = 5$, equality is $v_p(t_1) = e$, because $N = p^{2e} - 1$ and 2 are p-adic units.

The exact leading coefficients are

$$B(u_N) = 6N + 4 = 6p^{2e} - 2, \qquad A(u_N) = (6p^{2e} - 2)H_N + 7p^{2e} + 1.$$

The first is a p-adic unit for both primes. For $e \geq 2$, (76) gives the strictly negative valuations

$$v_3(A(u_N)) = 2 - 2e, \qquad v_5(A(u_N)) = 3 - 2e. \tag{78}$$

There is no cancellation within this leading expression: the harmonic term has negative valuation, while $7p^{2e} + 1$ is a unit.

All coefficients of the quadratic form B in integer Laguerre coordinates are integers. Therefore every change from $B(u_N)$ after substituting (77) is divisible by p, and

$$v_p(B(Q_M/z_N)) = 0.$$

For A, the half-lcm clearing factor in degree $N + 1 = p^{2e}$ shows that every quadratic coefficient has valuation at least $-2e$ for these odd primes. Consequently, every linear perturbation $t_j\, 2\mathcal{A}(u_N, u_{N-j})$ with $j \geq 2$ has valuation at least $je - 2e \geq 0$, and every quadratic perturbation in the t_j has valuation at least $(j + k)e - 2e \geq 0$.

It remains to check the single adjacent linear perturbation. Its coefficient is exactly

$$C_N := 2\mathcal{A}(u_N, u_{N-1}) = -2(4N + 1)H_N - \frac{32N + 7}{3}.$$

For $p = 3$, the factor $4N + 1 = 4 \cdot 3^{2e} - 3$ has valuation one, while $32N + 7 = 32 \cdot 3^{2e} - 25$ is a unit. Hence

$$v_3(t_1 C_N) \geq e + \min(3 - 2e, -1) = \min(3 - e, e - 1) > 2 - 2e \qquad (e \geq 2).$$

For $p = 5$, $4N + 1$ and 3 are units, and (76) gives

$$v_5(t_1 C_N) \geq e + (3 - 2e) = 3 - e > 3 - 2e \qquad (e \geq 2).$$

Thus every perturbation has valuation strictly greater than the corresponding negative leading valuation in (78). The full normalized A retains that valuation. Since the normalized B is a unit and quadratic homogeneity cancels z_N^2 in A_M/B_M, this proves

$$v_3(A_M/B_M) = 2 - 2e, \qquad v_5(A_M/B_M) = 3 - 2e \qquad (e \geq 2).$$

Reduction of a rational number makes its denominator valuation the negative part of its valuation, giving the asserted exponents for $e \geq 2$. At $e = 1$, the odd-prime removal conclusion of Theorem 13.3 gives exponent zero in both cases, which is exactly the formula stated here.

The resulting divisibilities are $M^2/9 \mid q_M$ when $M = 3^e$, and $M^2/125 \mid q_M$ when $M = 5^e$ and $e \geq 2$. They imply the displayed bounds for q_M/M. Finally the established estimate $\mathcal{E}_M = \Theta(q_M/M)$ gives divergence of the primitive error on both towers. □

Small exact checks and their role. The leading-unit checks at $e = 2$ reduce to the finite rational sums H_{80} and H_{624}. Exact arithmetic gives

(p, M, N)	$v_p(H_N)$	$p^{-v_p(A(u_N))}A(u_N) \pmod p$	$B(u_N) \pmod p$
$(3, 9, 80)$	-2	2	1
$(5, 25, 624)$	-1	4	3

The perturbation bounds above show that the same scaled A residues hold for $A(Q_M/z_N)$, and that the B residues also remain unchanged. Thus these checks certify denominator exponents 2 and 1, respectively. The all-index assertions follow from the valuation proof, not extrapolation from these checks.

13.2 A fresh-prime test after the cutoff repair

The removal of a prime dividing the localization parameter does not control primes of the size of the cutoff. The following exact test detects many such primes in the *actual* reduced denominator. Here $A(Q)$ and $B(Q)$ have their positive-square meaning, $A(Q)-\gamma B(Q)=\int_0^\infty Q(t)^2(t-1)\log t\,e^{-t}\,dt$.

Proposition 13.5 (Fresh-prime denominator test)**.** *Let $M\ge 2$, $H=M^2$, $N=H-1$, $x=(M-1)/M$, and*

$$w_k=\left(\frac{H-k}{H}\right)^2,\qquad Q(t)=\sum_{k=0}^{N} w_k L_k^{(1)}(x)\bigl(L_k(t)-L_{k+1}(t)\bigr).$$

Let ℓ be an odd prime with $H/2<\ell\le H$, and set $s=N-\ell$. Define the truncated polynomial

$$W_{M,\ell}(t)=\sum_{r=0}^{s} w_r L_r^{(1)}(x)\bigl(L_r(t)-L_{r+1}(t)\bigr),\qquad R_{M,\ell}=x^2B(W_{M,\ell})\pmod{\ell}.$$

Then $s\ge 0$, $s\le \ell-2$, the residue on the right is defined, and the two numbers on the left of

$$\ell^3A(Q)\equiv \ell^2B(Q)\equiv R_{M,\ell}\pmod{\ell} \tag{79}$$

are ℓ-integral. If $R_{M,\ell}\neq 0$, then

$$v_\ell(A(Q))=-3,\qquad v_\ell(B(Q))=-2,\qquad v_\ell\Bigl(\operatorname{den}\frac{A(Q)}{B(Q)}\Bigr)=1.$$

In particular, if $B(Q)\neq 0$, the completely computable integer

$$G_M=\prod_{\substack{\ell\ \text{odd prime},\ H/2<\ell\le H\\ R_{M,\ell}\neq 0}}\ell$$

divides the actual reduced denominator q_M.

Proof. Since $H=M^2$ is composite, $\ell\le H-1$, and $H<2\ell$ gives $0\le s\le \ell-2$. Such a prime divides neither M nor $M-1$, so x,H are ℓ-adic units. The finite Laguerre formula, Wilson's theorem, Lucas's congruence, and $x^\ell\equiv x$ give

$$\ell L_{\ell+r}^{(1)}(x)\equiv xL_r^{(1)}(x)\pmod{\ell}\qquad(0\le r\le s). \tag{80}$$

For completeness, in $L_n^{(1)}(x)=\sum_{j=0}^{n}(-1)^j\binom{n+1}{j+1}x^j/j!$, the terms $j<\ell$ disappear after multiplication by ℓ. Writing $j=\ell+a$ in the remaining terms gives

$$\frac{\ell}{(\ell+a)!}\equiv-\frac{1}{a!},\qquad\binom{\ell+r+1}{\ell+a+1}\equiv\binom{r+1}{a+1},$$

which proves (80).

Write $z_k=w_kL_k^{(1)}(x)$ and let c be the ordinary Laguerre coefficient vector of Q:

$$c=(z_0,z_1-z_0,\dots,z_N-z_{N-1},-z_N).$$

The vector $\widetilde{c}=\ell c$ is ℓ-integral, vanishes modulo ℓ in indices $0,\dots,\ell-1$, and in indices $\ell,\dots,H$ is congruent to x times the *complete* ordinary Laguerre coefficient vector of $W_{M,\ell}$, including its final coefficient $-z_s$. This uses $w_{\ell+r}\equiv w_r$. The exact Gram form

$$B(c)=\sum_{i=0}^{H}2ic_i^2-2\sum_{i=0}^{H-1}(i+1)c_ic_{i+1}$$

therefore gives $B(\widetilde{c}) \equiv x^2 B(W_{M,\ell})$.

All entries of ℓA are ℓ-integral because $H < 2\ell$. For indices $i, j \geq \ell$, its diagonal residues are $2i$, its adjacent cross-term residues are $-2(i+1)$, and its remaining cross-term residues vanish. Indeed $\ell H_i \equiv 1$ for $\ell \leq i \leq H$ and

$$2A_{ij} = \frac{2(i+1)}{j-i-1} - \frac{4i}{j-i} + \frac{2i}{j-i+1} \qquad (j-i \geq 2).$$

The only possible exceptional gap within the high block is $j - i = \ell - 1$. It requires $H = 2\ell - 1$, $i = \ell$, and $j = 2\ell - 1$; the potential residue contains $i \equiv 0$ and is zero. All terms involving a low coefficient vanish because that coefficient is divisible by ℓ. Consequently $\ell A(\widetilde{c}) \equiv B(\widetilde{c})$. Quadratic homogeneity proves (79), and a nonzero residue gives the three exact valuations. □

Scope and a zero-residue example. The nonzero-residue condition is sufficient, and is not necessary. For $M = 5$, $N = 24$, $\ell = 17$, $s = 7$, one has $x \equiv 11 \pmod{17}$ and $B(W_{5,17}) \equiv 0 \pmod{17}$. Exact rational arithmetic nevertheless gives

$$v_{17}(A(Q)) = -2, \qquad v_{17}(B(Q)) = 0, \qquad v_{17}(q_5) = 2.$$

Thus a zero residue does not certify cancellation of the prime. The accompanying verifier checks the full rational forms and actual reduced denominators for $2 \leq M \leq 11$. Its separate finite-field scan over $2 \leq M \leq 30$ has 786 eligible pairs, of which 781 have nonzero residue. The five zero pairs (M, ℓ) are $(5, 17)$, $(17, 197)$, $(19, 241)$, $(23, 487)$, and $(24, 449)$. These finite data do not establish a lower bound for G_M as $M \to \infty$. Combined with the proved localization estimate, the proposition gives $\mathcal{E}_M \gg G_M/M$ for sufficiently large M; it supplies no unconditional global growth rate for G_M, and no irrationality proof.

Part II
Arithmetic obstructions after localization and acceleration

14 A two-adic obstruction for every even localization parameter

Use the localized positive-square family defined and analyzed in Section 13: $M \geq 2$, $N = M^2 + \sigma$ with $\sigma \in \{0, -1\}$, $x = 1 - 1/M$, $u_k = L_k - L_{k+1}$,

$$z_k = \left(\frac{N+1-k}{N+1}\right)^2 L_k^{(1)}(x), \qquad Q_M = \sum_{k=0}^{N} z_k u_k.$$

Write $A_M = A(Q_M)$ and $B_M = B(Q_M)$ for the rational forms satisfying

$$A_M - \gamma B_M = \int_0^\infty Q_M(t)^2 (t-1) \log t \, e^{-t} dt > 0.$$

The earlier all-prime-power obstruction for $\sigma = 0$ does not address arbitrary even M, and the odd-prime cutoff repair does not address the prime 2. The following proposition covers both issues.

Theorem 14.1 (Exact two-adic denominator cost for even parameters)**.** *Let $M \geq 4$ be even, $\sigma \in \{0, -1\}$, and let $q_M = \operatorname{den}(A_M/B_M)$ be the positive reduced denominator. Then $B_M \neq 0$ and*

$$\boxed{v_2(q_M) = \lfloor \log_2(M^2 + \sigma) \rfloor.}$$

Consequently $q_M > (M^2 + \sigma)/2$. Using the established uniform real estimate $\mathcal{E}_M = \Theta(q_M/M)$, the primitive errors are bounded below by a positive constant times M for all sufficiently large even M, with either cutoff. In particular no subsequence consisting of even parameters can prove irrationality through this localized family.

Proof. Put $e = v_2(M) \geq 1$ and $a = \lfloor \log_2 N \rfloor$. Since $v_2(x) = -e$, the finite Laguerre formula gives

$$L_k^{(1)}(x) = \frac{(-x)^k}{k!}(1 + 2^e \mathbb{Z}_2).$$

Indeed every nonleading term divided by the leading term is an integer multiple of x^{-j} for some $j \geq 1$. Thus $z_N \neq 0$, and for $1 \leq j \leq N$ the ratios $t_j = z_{N-j}/z_N$ satisfy

$$v_2(t_j) = je + v_2\left(\frac{N!}{(N-j)!}\right) + 2v_2(j+1).$$

If $\sigma = -1$, then N is odd and $v_2(N-1) = 1$. The displayed formula gives $v_2(t_1) = e + 2 \geq 3$ and $v_2(t_2) = 2e + 1 \geq 3$; for $j \geq 3$ it gives $v_2(t_j) \geq je \geq 3$. If $\sigma = 0$, then $v_2(N) = 2e$. For $j = 1, 2$ the bounds are respectively $3e + 2 \geq 5$ and $4e \geq 4$; for $j = 3$ the factor N already gives a bound $5e \geq 5$, and for $j \geq 4$ the bound $je \geq 4$ suffices. Hence

$$\min_{j \geq 1} v_2(t_j) \geq \begin{cases} 4, & \sigma = 0, \\ 3, & \sigma = -1. \end{cases}$$

There is exactly one integer in $[1, N]$ divisible by 2^a, namely 2^a itself. Therefore $v_2(H_N) = -a$. The exact top-row formulas are

$$B(u_N) = 6N + 4, \qquad A(u_N) = (6N+4)H_N + 7N + 8.$$

For $\sigma = 0$, these have valuations 2 and $2 - a$: $6M^2 + 4$ has valuation 2, while $7M^2 + 8$ has valuation at least 2, strictly greater than $2 - a$. For $\sigma = -1$, their valuations are 1 and $1 - a$: $6M^2 - 2$ has valuation 1, and $7M^2 + 1$ is odd; the condition $M \geq 4$ ensures $a \geq 3$.

Now normalize

$$Q_M/z_N = u_N + \sum_{j=1}^{N} t_j u_{N-j}.$$

The quadratic form B has integer coefficients in integer Laguerre coordinates. Its perturbation has valuation at least 4 when $\sigma = 0$ and at least 3 when $\sigma = -1$. Hence the valuations of the full normalized B remain 2 and 1 respectively; in particular it is nonzero.

The half-lcm normalization theorem says that all coefficients of the quadratic form A in these coordinates are cleared by $\operatorname{lcm}(1, \ldots, N+1)/2$. For $\sigma = 0$ its two-adic valuation is $a - 1$, because the even integer $N = M^2$ cannot be one less than a power of 2. The perturbation of A then has valuation at least $4 - (a - 1) = 5 - a > 2 - a$. For $\sigma = -1$, the clearing factor has valuation at most a, including the case that M^2 is a power of 2. The perturbation has valuation at least $3 - a > 1 - a$. Quadratic perturbation terms only improve these bounds. Consequently the full normalized A retains valuation $2 - a$ or $1 - a$ respectively.

Quadratic homogeneity cancels z_N^2 in A_M/B_M; in either case

$$v_2(A_M/B_M) = -a.$$

Taking the reduced denominator proves the exact formula. Since $2^a \leq N < 2^{a+1}$, one has $q_M \geq 2^a > N/2$. The primitive-error assertion follows from the previously proved two-sided estimate, whose constants are uniform in M. □

Scope. The proposition excludes all even parameters in this specific family. The next section treats odd M for the repaired cutoff. Other node schedules or taper choices are outside this statement.

15 A complete two-adic obstruction for the repaired localization family

The one-step cutoff repair does not yield a sequence with small primitive error. In fact the prime 2 alone obstructs every parameter of the repaired family, including parameters at which the prime dividing the node denominator has disappeared from the primitive denominator.

Write $\mathcal{A}(c, d)$ for the symmetric bilinear rational form with ordinary Laguerre Gram entries

$$a_{ii} = 2iH_i + 2i + 1, \qquad a_{i,i+1} = -(i+1)H_i - \frac{3i}{2} - 1,$$

$$a_{ij} = \frac{i + j + (j-i)^2}{(j-i)((j-i)^2 - 1)} \quad (j - i \geq 2),$$

and put $A(c) = \mathcal{A}(c, c)$. The second form is

$$B(c) = 2\sum_{i=0}^{D} i c_i^2 - 2\sum_{i=0}^{D-1} (i+1) c_i c_{i+1}.$$

The established half-lcm theorem says that the coefficients of the quadratic polynomial $\operatorname{lcm}(1, \ldots, D)A(c)/2$ are integers. Consequently, for $a = \lfloor \log_2 D \rfloor$, the quadratic polynomial $\widetilde{A}(c) = 2^{a-1}A(c)$ has coefficients in $\mathbb{Z}_2$. Here and below an off-diagonal quadratic coefficient is twice the corresponding bilinear coefficient.

Theorem 15.1 (Exact cost at every odd repaired parameter)**.** *Let $M \geq 3$ be odd, $D = M^2$, $N = D - 1$, $x = 1 - 1/M$, and*

$$z_k = (D-k)^2 L_k^{(1)}(x), \qquad Q = \sum_{k=0}^{D-1} z_k (L_k - L_{k+1}).$$

The omitted common factor D^{-2} has no effect on $A(Q)/B(Q)$. Let $a = \lfloor \log_2 D \rfloor = \lfloor \log_2 N \rfloor$. Then

$$v_2(B(Q)) = 2, \qquad v_2(A(Q)) = \begin{cases} 3-a, & M \equiv 3 \pmod 4, \\ 4-a, & M \equiv 1 \pmod 4. \end{cases}$$

In particular, the actual reduced denominator q_M satisfies

$$\boxed{v_2(q_M) = \begin{cases} a-1, & M \equiv 3 \pmod 4, \\ a-2, & M \equiv 1 \pmod 4. \end{cases}}$$

Combining this with the even-parameter theorem gives

$$q_M > \frac{M^2 - 1}{8} \qquad (M \geq 3)$$

for the entire repaired family. The already established real estimate $\mathcal{E}_M = \Theta(q_M/M)$ therefore gives $\mathcal{E}_M \gg M$. Thus no subsequence of this repaired family satisfies primitive-error decay.

Proof. We first prove three uniform congruences with $D \equiv 1 \pmod 8$, $D \geq 9$, and arbitrary $x \in 2\mathbb{Z}_2$. Set

$$z_k(x) = (D-k)^2 L_k^{(1)}(x), \quad c(x) = (z_0, z_1 - z_0, \ldots, z_{D-1} - z_{D-2}, -z_{D-1}).$$

The finite Laguerre formula and the identity $v_2(j!) = j - s_2(j)$, where s_2 is binary digit sum, show that every $L_k^{(1)}(x)$ lies in $\mathbb{Z}_2$, and $L_k^{(1)}(x) \equiv k+1 \pmod 2$. Therefore z_{2r} is odd and z_{2r+1} is divisible by 8. In particular $c_{2r} \equiv z_{2r} \pmod 4$ and $c_{2r+1} \equiv -z_{2r} \pmod 4$; at endpoints the missing z is zero. Thus all c_i are odd, and

$$\frac{B(c)}{2} \equiv \sum_{i=0}^{D} i + \sum_{i=0}^{D-1} (i+1) = D(D+1) \equiv 2 \pmod 4.$$

For an even i this uses $c_i c_{i+1} \equiv -1 \pmod 4$; for an odd i the coefficient $i+1$ is even, so its product with any odd number is congruent modulo 4 to its product with -1. This proves $v_2(B(c)) = 2$.

The polynomial base case and the branch $x \in 4\mathbb{Z}_2$. Put

$$Z_j(k) = (D-k)^2 \binom{k+1}{j+1}, \qquad c_k^{[j]} = Z_j(k) - Z_j(k-1) \quad (0 \leq k \leq D).$$

In particular $c_k^{[0]} = 3k^2 - (4D+1)k + D^2$. The exact polynomial evaluations recorded below imply

$$\widetilde{A}(c^{[0]}) \equiv D + 7 \pmod{16},$$
$$2^{a-1}\, 2\mathcal{A}(c^{[0]}, c^{[1]}) \equiv 2 \pmod 4, \qquad 2^{a-1}\, 2\mathcal{A}(c^{[0]}, c^{[2]}) \equiv 0 \pmod 2.$$

If $x \in 4\mathbb{Z}_2$, the finite Laguerre formula gives

$$c(x) \equiv c^{[0]} - x c^{[1]} + \frac{x^2}{2} c^{[2]} \pmod{16}.$$

Indeed every term of index $j \geq 3$ has two-adic valuation at least 5. Using the integrality of the coefficients of $\widetilde{A}$ and expanding this quadratic polynomial proves

$$\boxed{\widetilde{A}(c(x)) \equiv D + 7 + 2x \pmod{16} \quad (x \in 4\mathbb{Z}_2).}$$

The quadratic term in $xc^{[1]}$ is divisible by 16; the cross term in $x^2c^{[2]}/2$ is also divisible by 16 by the second cross congruence. The signs $-2x$ and $+2x$ agree modulo 16 here.

The branch $x \equiv 2 \pmod 4$**.** Write

$$L = 2^a, \quad h = 2^{a-1}, \quad T = (D-1)/2 = h + t.$$

The assumptions on D imply $a \geq 3$, $0 \leq t < h$, and $t \equiv 0 \pmod 4$. Let $\rho_r = s_2(\lfloor r/2 \rfloor) \pmod 2$ for $0 \leq r \leq T$. We claim

$$z_{2r}(x) \equiv z_{2r}(0) + 2\rho_r \pmod 4.$$

To see this, in

$$L_k^{(1)}(x) = \sum_{j=0}^{k} \binom{k+1}{j+1} \frac{(-x)^j}{j!}$$

the terms of positive index contributing modulo 4 have $j = 2^b$. Each such factor $(-x)^j/j!$ is 2 modulo 4. For $k = 2r$ the parity of $\binom{2r+1}{2^b+1}$ is the b-th binary digit of $2r+1$ when $b \geq 1$, whereas the exceptional index $j = 1$ has binomial coefficient $\binom{2r+1}{2}$, also the first binary digit. Those two contributions cancel modulo 2. The remaining sum is the sum of the digits in positions $b \geq 2$, namely ρ_r. This binary-binomial rule follows directly by expanding $(1+X)^n = \prod_{b:n_b=1}(1+X^{2^b})$ over $\mathbb{F}_2$. Multiplication by the odd square $(D-2r)^2 \equiv 1 \pmod 4$ proves the claim.

Let $u_i = L_i - L_{i+1}$. For $0 \leq r < s \leq T$, denote the cross coefficient in the even-indexed u basis by

$$C_{rs} = 2^{a-1}\, 2\mathcal{A}(u_{2r}, u_{2s}).$$

Every C_{rs} is even. The following is the complete list of cases in which $C_{rs}/2$ is odd:

$$\begin{aligned} &s - r = h - 1, && r \text{ odd},\\ &s - r = h + 1, && r \text{ even},\\ &s = r + 1, && r \geq h \text{ even}. \end{aligned}$$

Here pairs must of course lie in $0 \leq r < s \leq T$. For completeness these assertions follow from exact Gram formulas. For $k = s - r \geq 2$,

$$C_{rs} = -\frac{2^a(k^2 + 3k + 6r + 2)}{k(k^2-1)(4k^2-1)}.$$

If k is even, $C_{rs}/2$ is even. If k is odd, the denominator has valuation at most a, and the numerator of $C_{rs}/2$ has valuation at least a. Equality is possible precisely when $k = h - 1$ or $h + 1$. Reducing $k^2 + 3k + 6r + 2$ modulo 4 gives, respectively, $2r$ and $2(r+1)$, proving the first two cases. For $s = r + 1$,

$$\frac{C_{r,r+1}}{2} = 2^a(r+1)H_{2r+1} + 2^{a-2}\frac{25r + 26}{3}.$$

The second term is even. The first is odd exactly when $2r + 1 \geq 2^a$ and r is even, proving the third case.

Regard these pairs as the edges of a finite graph on $0, \ldots, T$. Its odd-degree vertices are exactly

$$\{0, 1, \ldots, t-1\} \cup \{t+1, h+t\}.$$

Indeed the long edges are $(r, h+r+1)$ for even $0 \le r \le t-1$ and $(r, h+r-1)$ for odd $1 \le r \le t+1$; the additional edges pair the high vertices $h, h+1;\ h+2, h+3;\ \ldots; h+t-2, h+t-1$. All high vertices below $h+t$ consequently have even degree. The sum of ρ_r over the odd-degree vertices is 1 modulo 2: the first interval cancels in pairs, while

$$\rho_{t+1} = s_2(t/2) \pmod 2, \qquad \rho_{h+t} = 1 + s_2(t/2) \pmod 2.$$

We now compare $\widetilde{A}(c(x))$ and $\widetilde{A}(c(0))$ modulo 8. Odd-indexed z's vanish modulo 8 and can be omitted. All remaining z's are odd, so their squares are 1 modulo 8, and diagonal terms do not change. Since the cross coefficients are even, only their residues modulo 4 and the residues of the z's modulo 4 matter. Thus

$$\widetilde{A}(c(x)) - \widetilde{A}(c(0)) \equiv 4 \sum_{r<s} \frac{C_{rs}}{2}(\rho_r + \rho_s) \equiv 4 \pmod 8.$$

The base congruence gives $\widetilde{A}(c(0)) \equiv 0 \pmod 8$. We have proved

$$\boxed{\widetilde{A}(c(x)) \equiv 4 \pmod 8 \quad (x \equiv 2 \pmod 4).}$$

Finally, for $D = M^2$ and $x = 1 - 1/M$, if $M \equiv 3 \pmod 4$ then $x \equiv 2 \pmod 4$, giving $v_2(\widetilde{A}) = 2$. If $M \equiv 1 \pmod 4$ then $x \in 4\mathbb{Z}_2$ and

$$D = (1-x)^{-2} \equiv 1 + 2x \pmod{16}.$$

Hence $D + 7 + 2x \equiv 8 \pmod{16}$, giving $v_2(\widetilde{A}) = 3$. Together with $v_2(B) = 2$ these are precisely the asserted valuations. In the second case $M \ge 5$ so $a \ge 4$; all displayed denominator exponents are positive. For odd M this gives $q_M \ge 2^{a-2} > N/8$. The even-parameter theorem gives the stronger $q_M > N/2$ for even $M \ge 4$. □

Exact polynomial identities used in the proof

For $j = 0, 1, 2$, the bilinear values in the proof have the form $\mathcal{A}(c^{[0]}, c^{[j]}) = P_j(D)H_D + R_j(D)$, where

$$P_0 = \frac{D(D+1)(D+2)(2D^2 - 11D - 31)}{15},$$

$$R_0 = -\frac{D(334D^4 - 1915D^3 - 9660D^2 - 9425D - 934)}{900},$$

$$P_1 = \frac{D(D+1)(D+2)(D^3 - 13D^2 - 28D - 20)}{60},$$

$$R_1 = -\frac{D(167D^5 - 3380D^4 - 8455D^3 - 2600D^2 - 2692D - 4640)}{3600},$$

$$P_2 = \frac{D(D+1)(D+2)(2D^4 - 55D^3 - 77D^2 - 254D - 36)}{1260},$$

$$R_2 = -\frac{D}{176400}\big(726D^6 - 41167D^5 + 1316D^4 - 12005D^3 - 472766D^2 - 109368D + 280464\big).$$

They follow by substitution into the displayed Gram entries, ordinary finite sums of polynomials, and the elementary summation identity

$$\sum_{i=0}^{D} P(i)H_i = H_D \sum_{i=0}^{D} P(i) - \sum_{r=1}^{D} \frac{1}{r} \sum_{i=0}^{r-1} P(i).$$

For off-diagonal gaps $d \geq 2$ use $a_{i,i+d} = 1/(d-1) + 2i/(d(d^2-1))$ and partial fractions in d. These operations give polynomial identities, not interpolation claims.

To check the congruences, write $D = 1 + 8u$. Then

$$P_0/16 \equiv 1 + u \pmod 2, \qquad v_2(R_0) \geq 2,$$

$$v_2(P_1) = v_2(P_2) = 1, \qquad R_1, R_2 \in \mathbb{Z}_2.$$

There is a unique term of valuation $-a$ in H_D, so $2^a H_D$ is odd. It follows that $2^{a-1}(P_0 H_D + R_0) \equiv 8(1+u) = D + 7 \pmod{16}$, while $2^{a-1}(P_j H_D + R_j)$ is odd for $j = 1, 2$. These are the three base congruences used above.

What this establishes. This theorem closes a proposed route by proving a uniform arithmetic obstruction. It supplies no irrationality proof for Euler's constant, and says nothing about other cutoffs, rational node schedules, tapers, or signed constructions.

16 Opposite Bessel limits: a nonzero accelerated family and its exact denominator obstruction

Write $H_0 = 0$ and $H_m = \sum_{k=1}^{m} 1/k$. For a rational number x, $\operatorname{den}(x)$ denotes its positive reduced denominator, with $\operatorname{den}(0) = 1$. All logarithms below are natural. The following argument concerns the actual rational approximants in manuscript [044] [13], not merely their limiting constants. It proves that both their simple mirror average and a specified acceleration have nonzero errors for all sufficiently large indices. Their reduced integer errors nevertheless diverge. No assertion of historical priority is made for the elementary denominator mechanism.

16.1 Persistence of a harmonic denominator

Lemma 16.1 (A rational perturbation changes denominator size by at most its own size)**.** *For rational x, y,*

$$\left|\log \operatorname{den}(x+y) - \log \operatorname{den}(x)\right| \leq \log \operatorname{den}(y).$$

For a fixed nonzero rational c, multiplying a rational number by c changes its logarithmic denominator by a bounded amount depending only on c.

Proof. If the reduced denominators of $x, y, x+y$ are b, d, f, respectively, then $f \mid bd$ and $b \mid df$. Thus $b/d \leq f \leq bd$. The scaling assertion follows by clearing the fixed numerator and denominator of c and applying the reverse scaling as well. □

Lemma 16.2 (Harmonic cores with nonzero prime residues)**.** *Fix positive integers $b_1, \ldots, b_s$ and rational numbers $\alpha_1, \ldots, \alpha_s$. Suppose*

$$R(a) := \sum_{i=1}^{s} \alpha_i H_{\lfloor a/b_i \rfloor} \neq 0 \qquad \text{for every integer } a \geq 1.$$

Let M tend to infinity through multiples of all the b_i, and put

$$A_M = \sum_{i=1}^{s} \alpha_i H_{M/b_i}.$$

Then

$$\log \operatorname{den}(A_M) \sim M.$$

The same conclusion holds for $A_M + z_M$ whenever $z_M \in \mathbb{Q}$ and $\log \operatorname{den}(z_M) = o(M)$.

Proof. Fix a positive integer K. For sufficiently large M, every prime ℓ with $M/(K+1) < \ell \le M$ exceeds $\sqrt{M}$ and is coprime to the denominators of the fixed α_i. Put $a = \lfloor M/\ell \rfloor$, so $1 \le a \le K$. Reduction in the local ring $\mathbb{Z}_{(\ell)}$ gives

$$\ell A_M \equiv \sum_i \alpha_i H_{\lfloor M/(b_i \ell) \rfloor} = R(a) \pmod{\ell}.$$

Indeed only terms whose indices are multiples of ℓ survive, and there are no multiples of ℓ^2. The equality of the floors uses $\lfloor x/b \rfloor = \lfloor \lfloor x \rfloor / b \rfloor$ for integral b. The finitely many nonzero rationals $R(1), \dots, R(K)$ have fixed nonzero numerators. All sufficiently large ℓ therefore give $\ell A_M \not\equiv 0 \pmod{\ell}$, hence $v_\ell(A_M) = -1$.

If $\vartheta(x) = \sum_{\ell \le x} \log \ell$, it follows that

$$\log \operatorname{den}(A_M) \ge \vartheta(M) - \vartheta(M/(K+1)) = \left(1 - \frac{1}{K+1} + o(1)\right) M$$

by the prime number theorem. On the other hand, the denominator of A_M divides a fixed constant times $\operatorname{lcm}(1, \dots, M)$, whose logarithm is $\psi(M) \sim M$. First let $M \to \infty$ and then $K \to \infty$ to prove the result. The perturbation assertion follows from Lemma 16.1. □

In particular $\log \operatorname{den}(H_M) \sim M$. This lemma does not say that every prime below M survives: it obtains the asymptotic by first considering a fixed range of the quotient $\lfloor M/\ell \rfloor$.

16.2 The two exact rational families

For integers $n, m \ge 1$, retain the factorial approximants

$$r_{n,m} = \frac{\sum_{k=0}^{n} f_{n,m,k}(H_{m+k} - 2H_k + 2H_{n-k})}{\sum_{k=0}^{n} f_{n,m,k}}, \qquad f_{n,m,k} = \frac{(m+k)!}{k!^2 (n-k)!^2}.$$

For $n \ge 1$ and an integer $p \ge 0$, define

$$T_{n,p} = \frac{\sum_{k=0}^{n} b_{n,p,k}((p+3)H_k - 2H_{n-k} - pH_{n+k})}{\sum_{k=0}^{n} b_{n,p,k}}, \qquad b_{n,p,k} = \frac{\binom{n}{k}^2 \binom{n+k}{k}^p}{k!}.$$

Set

$$n_j = 2^j, \qquad p_j = 3j, \qquad V_j = \frac{r_{n_j, n_j^2} + T_{n_j, p_j}}{2}, \qquad W_j = 4V_{j+2} - 4V_{j+1} + V_j.$$

Thus V_j, W_j are explicit rational numbers. Introduce the positive constants

$$D = \sum_{a \ge 0} \frac{1}{(a!)^2} = I_0(2), \qquad c = \frac{\sum_{a \ge 0} a/(a!)^2}{D} = \frac{I_1(2)}{I_0(2)}.$$

One has $0 < c < 1$, as also follows from the moment calculation below.

Theorem 16.3 (Mirror averaging and a three-term acceleration)**.** *As $j \to \infty$, with $n = n_j$ and $p = p_j$,*

$$V_j - \gamma = \frac{p+6}{8nD^2} + \frac{(1-c)p^2}{32n^2D^2} + O\left(\frac{p}{n^2} + \frac{p^3}{n^3}\right), \tag{81}$$

$$W_j - \gamma \sim \frac{(1-c)p^2}{128n^2D^2} > 0. \tag{82}$$

In particular both errors are strictly positive for all sufficiently large j, and both rational families converge to γ. Their actual reduced denominators satisfy

$$\log \operatorname{den}(V_j) \sim n_j^2 = 4^j, \tag{83}$$
$$\log \operatorname{den}(W_j) \sim 16 n_j^2 = 4^{j+2}. \tag{84}$$

Consequently

$$\log\left(\operatorname{den}(V_j)|V_j - \gamma|\right) \sim 4^j, \qquad \log\left(\operatorname{den}(W_j)|W_j - \gamma|\right) \sim 4^{j+2}.$$

Both denominator-normalized errors tend to $+\infty$.

Proof of the analytic assertions. For $s \geq 0$ put

$$D(s) = \sum_{a \geq 0} \frac{s^a}{(a!)^2}, \qquad F(s) = \sum_{a \geq 0} \frac{H_a s^a}{(a!)^2}, \qquad \theta = s\frac{d}{ds}.$$

The identities

$$\theta^2 D = sD, \qquad \theta^2 F = sF + \theta D$$

follow by shifting the coefficients. Moreover

$$\theta\big(D\theta F - F\theta D\big) = D\theta D,$$

so evaluation at $s = 0$ yields

$$D\theta F - F\theta D = \frac{D^2 - 1}{2}.$$

Use the probability distribution $\Pr(J = a) = 1/((a!)^2 D(1))$, and write $h = \mathbb{E}H_J$. At $s = 1$ these identities give

$$\mathbb{E}J = c, \quad \mathbb{E}J^2 = 1, \quad \mathbb{E}J^3 = 1 + c, \quad \mathbb{E}J^4 = 2 + 2c,$$
$$u := \operatorname{Cov}(H_J, J) = \frac{1 - D^{-2}}{2}, \qquad \operatorname{Cov}(H_J, J^2) = c,$$
$$\operatorname{Cov}(H_J, J^3) = 1 + u, \qquad \operatorname{Cov}(H_J, J^4) = 1 + 2c + 2u.$$

Since $J^2 - J \geq 0$ and its expectation is strictly positive, $c < 1$. For $X = J(J-1)$ and $Y = 1 - 2J$, the displayed moments imply

$$\mathbb{E}X = 1 - c, \qquad \mathbb{E}X^2 = 1, \qquad \operatorname{Cov}(J, X) = c^2, \tag{85}$$
$$\operatorname{Cov}(H_J, X) = c - u, \qquad \operatorname{Cov}(H_J, X^2) = 3c - 1.$$

Reindex the factorial approximants by $a = n - k$ and divide all weights by their term at $k = n$. On the square diagonal their weights are

$$w_{n,a} = \frac{1}{(a!)^2} \prod_{r=0}^{a-1} \frac{(1 - r/n)^2}{1 + 1/n - r/n^2}.$$

For $a \leq n^{1/8}$, Taylor expansion gives

$$w_{n,a} = \frac{1}{(a!)^2}\left(1 - \frac{a^2}{n} + O\left(\frac{(a+1)^4}{n^2}\right)\right),$$

while the corresponding harmonic factor is

$$H_{n^2+n-a} - 2H_{n-a} + 2H_a = -\gamma + 2H_a + \frac{2a}{n} + O\left(\frac{(a+1)^2}{n^2}\right).$$

Thus the $1/n$ term in the quotient is $2\mathbb{E}J - 2\operatorname{Cov}(H_J, J^2) = 0$, and

$$r_{n,n^2} = -\gamma + 2h + O(n^{-2}). \tag{86}$$

The normalized weights for $T_{n,p}$ are exactly

$$v_{n,p,a} = \frac{(n)_a^{p+3}}{(2n)_a^p (a!)^2} = \frac{1}{(a!)^2} \prod_{r=0}^{a-1} \frac{(1 - r/n)^{p+3}}{(1 - r/(2n))^p},$$

where $(n)_a$ is the falling factorial and the second equality uses $n^3 = 2^p$. Put $A = (p+6)/4$. Expanding the logarithm and then the exponential gives, for $a \le n^{1/8}$,

$$v_{n,p,a} = \frac{1}{(a!)^2}\left[1 - \frac{Aa(a-1)}{n} + \frac{A^2a^2(a-1)^2}{2n^2} + O\left(\frac{(p+1)(a+1)^3}{n^2} + \frac{(p+1)^3(a+1)^6}{n^3}\right)\right].$$

The harmonic factor has the expansion

$$(p+3)H_{n-a} - pH_{2n-a} - 2H_a = 3\gamma - 2H_a + \frac{A(1-2a)}{n} + O\left(\frac{(p+1)(a+1)^2}{n^2}\right).$$

Here $3\log n - p\log 2 = 0$ was used exactly.

Both exact families of normalized weights are bounded above by $1/(a!)^2$. Their harmonic factors are $O((p+1)\log(n+1))$ uniformly in $0 \le a \le n$. The tails beyond $n^{1/8}$, including any fixed polynomial weight, are smaller than every power of n^{-1}. All expansions may therefore be summed and divided with the errors shown above; $p = O(\log n)$ throughout.

For a weight perturbation $1 - tX + t^2X^2/2$ and an observable $H + tY$, the quotient expansion is

$$\mathbb{E}H + t\{\mathbb{E}Y - \operatorname{Cov}(H, X)\} + t^2\left\{\tfrac{1}{2}\operatorname{Cov}(H, X^2) - \mathbb{E}X\operatorname{Cov}(H, X) - \operatorname{Cov}(Y, X)\right\} + O(t^3),$$

with the preceding summable errors retained in our application. Take $H = 3\gamma - 2H_J$ and $t = A/n$. The first coefficient is D^{-2}. Using (85), the second coefficient is

$$\begin{aligned} &-\operatorname{Cov}(H_J, X^2) + 2\mathbb{E}X\operatorname{Cov}(H_J, X) + 2\operatorname{Cov}(J, X) \\ &\qquad = -(3c-1) + 2(1-c)(c-u) + 2c^2 = (1-c)D^{-2}. \end{aligned}$$

Consequently

$$T_{n,p} = 3\gamma - 2h + \frac{p+6}{4nD^2} + \frac{(1-c)p^2}{16n^2D^2} + O\left(\frac{p}{n^2} + \frac{p^3}{n^3}\right).$$

Together with (86) this proves (81).

The first correction in (81) is a constant multiple of $(3j+6)2^{-j}$, which the operator $V_j \mapsto 4V_{j+2} - 4V_{j+1} + V_j$ annihilates exactly. For the next term, writing $C = (1-c)/(32D^2)$ gives

$$4\frac{C(p+6)^2}{(4n)^2} - 4\frac{C(p+3)^2}{(2n)^2} + \frac{Cp^2}{n^2} = \frac{C}{n^2}\left(\frac{p^2}{4} - 3p\right).$$

The retained errors are $O(p/n^2 + p^3/n^3) = o(p^2/n^2)$. This proves (82), including its eventual strict sign. □

Proof of the denominator assertions. For the factorial family, clear the common weight factor and put

$$a_{n,m,k} = \binom{n}{k}^2 (m+1)(m+2)\cdots(m+k) \in \mathbb{Z}_{>0}, \qquad Q_r = \sum_{k=0}^{n} a_{n,m,k}.$$

Then

$$r_{n,m} = H_m + z_{n,m},$$

where the denominator of $z_{n,m}$ divides

$$Q_r \operatorname{lcm}(1,\dots,n) \prod_{h=1}^{n} (m+h).$$

Indeed $H_{m+k} - H_m = \sum_{h=1}^{k} 1/(m+h)$; all remaining harmonic indices are at most n. Also

$$Q_r \le (n+1)4^n (m+n)^n.$$

On $m = n^2$ this proves $\log \operatorname{den}(z_{n,n^2}) = O(n \log(n+1))$.

For $T_{n,p}$, multiply all weights by $n!$ and write

$$B_k = \binom{n}{k}^2 \binom{n+k}{k}^p \frac{n!}{k!} \in \mathbb{Z}_{>0}, \qquad Q_T = \sum_{k=0}^{n} B_k.$$

Its denominator divides $Q_T \operatorname{lcm}(1,\dots,2n)$, and

$$Q_T \le (n+1)4^{n(p+1)} n!.$$

For $p = 3\log_2 n$ this gives $\log \operatorname{den}(T_{n,p}) = O(n \log(n+1))$. Therefore

$$V_j = \tfrac{1}{2} H_{n_j^2} + \varepsilon_j, \qquad \log \operatorname{den}(\varepsilon_j) = O(n_j \log(n_j+1)).$$

The harmonic-core and perturbation lemmas prove (83).

For the accelerated sequence, put $M = 16 n_j^2$. Its harmonic core is

$$W_j = 2H_M - 2H_{M/4} + \tfrac{1}{2} H_{M/16} + \widetilde{\varepsilon}_j, \qquad \log \operatorname{den}(\widetilde{\varepsilon}_j) = O(n_j \log(n_j+1)).$$

The residue function of this core is

$$R(a) = 2(H_a - H_{\lfloor a/4 \rfloor}) + \tfrac{1}{2} H_{\lfloor a/16 \rfloor} > 0 \qquad (a \ge 1).$$

Lemma 16.2 therefore gives (84). Finally the logarithms of the positive analytic errors are $O(\log n_j)$, so adding them does not change either quadratic denominator asymptotic. □

The five-condition test for these two families. Their reduced pairs give logarithm-free integer linear forms. The analytic expansions prove nonvanishing for all sufficiently large indices, hence for infinitely many indices without an unproved hypothesis. However, multiplying the error by the actual reduced denominator makes it diverge exponentially on the n_j^2 scale. Thus condition 3 fails for both V_j and W_j. The theorem concerns these specified constructions; it is not a universal obstruction to all gamma approximations or to all nonlinear combinations of the corpus.

17 An exact escape from the large harmonic core

Retain $r_{n,m}, T_{n,p}, V_j, D, c$ from the mirror construction, and write $\operatorname{den}(x)$ for the positive denominator of a reduced rational number x. The following construction cancels H_{n^2} exactly in the finite rational formula. It therefore lies outside the preceding harmonic-core denominator obstruction.

Theorem 17.1 (Exact core cancellation and a further acceleration)**.** *For $j \geq 1$, put $n = 2^j$, $p = 3j$, and define*

$$U_j = \frac{r_{n,n^2} + T_{n,p} - H_{n^2} + 2H_n}{3}, \qquad X_j = 4U_{j+2} - 4U_{j+1} + U_j.$$

Both are explicitly defined rational numbers. Their asymptotic expansions are

$$U_j - \gamma = \frac{1}{n}\left(\frac{p+6}{12D^2} + \frac{1}{3}\right) + \frac{(1-c)p^2}{48D^2n^2} + O\left(\frac{p}{n^2} + \frac{p^3}{n^3}\right), \tag{87}$$

$$X_j - \gamma \sim \frac{(1-c)p^2}{192D^2n^2} > 0. \tag{88}$$

In particular, both converge to γ and both exceed γ for all sufficiently large j. Their reduced denominators satisfy

$$\log \operatorname{den}(U_j) \leq 8n \log n + O(n), \tag{89}$$

$$\log \operatorname{den}(X_j) \leq 56n \log n + O(n). \tag{90}$$

Writing $q_{U,j} = \operatorname{den}(U_j)$ and $q_{X,j} = \operatorname{den}(X_j)$, their exact asymptotic arithmetic thresholds are

$$q_{U,j}|U_j - \gamma| \sim \frac{q_{U,j}p}{12D^2n}, \qquad q_{X,j}|X_j - \gamma| \sim \frac{(1-c)q_{X,j}p^2}{192D^2n^2}.$$

Thus primitive-error decay along any infinite subsequence is equivalent, respectively, to $q_{U,j} = o(n/p)$ and $q_{X,j} = o(n^2/p^2)$ along that subsequence. These are conclusions about the exact threshold, not assumed or proved denominator estimates. For U_j there is also the following exact arithmetic obstruction:

$$v_2(U_j) = \begin{cases} -j, & j \geq 3 \text{ is odd}, \\ 1-j, & j \geq 8,\ j \equiv 0 \pmod 4, \\ 2-j, & j \geq 6,\ j \equiv 6 \pmod 8. \end{cases} \tag{$*$}$$

Consequently

$$\operatorname{den}(U_j)(U_j - \gamma) \longrightarrow +\infty$$

along any sequence of indices tending to infinity outside the residue class $j \equiv 2 \pmod 8$. Thus the full sequence U_j does not satisfy primitive-error decay. The theorem does not settle that decay for the remaining residue class or for the accelerated sequence X_j.

Analytic expansion and exact denominator clearing. First observe the exact identity

$$U_j = \frac{2V_j + G_j}{3}, \qquad G_j = 2H_n - H_{n^2}.$$

The elementary harmonic expansion gives

$$G_j = \gamma + \frac{1}{n} - \frac{2}{3n^2} + O(n^{-4}).$$

Combining this with the two-term expansion of V_j proves (87). The displayed first correction is a linear polynomial in j multiplied by 2^{-j}, so the operator $U_j \mapsto 4U_{j+2} - 4U_{j+1} + U_j$ annihilates it exactly. If $C = (1-c)/(48D^2)$, its action on the second term is

$$\frac{C}{n^2}\left(\frac{p^2}{4} - 3p\right).$$

The retained errors are $O(p/n^2 + p^3/n^3) = o(p^2/n^2)$, proving (88). The coefficient is positive because $0 < c < 1$.

For the arithmetic bound set

$$a_{n,m,k} = \binom{n}{k}^2 (m+1)_k, \qquad Q_r = \sum_{k=0}^{n} a_{n,m,k}, \qquad d_s = \operatorname{lcm}(1, \ldots, s).$$

Here $(m+1)_k = (m+1)(m+2)\cdots(m+k)$ is a rising factorial, with $(m+1)_0 = 1$. These weights are polynomials in m with integer coefficients, and

$$\partial_m a_{n,m,k} = a_{n,m,k}(H_{m+k} - H_m).$$

Their derivatives are integers at integer m. Therefore the exact finite cancellation has the stronger denominator statement

$$\operatorname{den}(r_{n,m} - H_m) \mid Q_r d_n.$$

No product of the integers $m+1, \ldots, m+n$ is needed. For the second family put

$$B_k = \binom{n}{k}^2 \binom{n+k}{k}^p \frac{n!}{k!}, \qquad Q_T = \sum_{k=0}^{n} B_k.$$

Then B_k are integers and $\operatorname{den}(T_{n,p}) \mid Q_T d_{2n}$. It follows that

$$\operatorname{den}(U_j) \mid 3 Q_r Q_T d_{2n}. \tag{$\dagger$}$$

This is an upper bound for the actual reduced denominator; it makes no assumption about the gcd of the displayed integerized pair.

The normalized positive weights used in the mirror proof give the exact bounds

$$Q_r \le D(n^2+1)_n, \qquad Q_T \le D\binom{2n}{n}^p.$$

Indeed their ratios to the respective final summands are bounded by $1/(a!)^2$ after the change of index $a = n - k$. Hence

$$\log Q_r \le 2n \log n + O(1), \qquad \log Q_T \le 6n \log n + O(1),$$

where the second bound uses $\binom{2n}{n} \le 4^n$ and $p = 3\log_2 n$. The standard bound $\log d_{2n} = O(n)$ proves (89). An integer linear combination of rationals has denominator dividing the product of their denominators. Applying this to X_j yields

$$8\big(n \log n + 2n \log(2n) + 4n \log(4n)\big) + O(n) = 56n \log n + O(n),$$

as asserted. □

The two-adic obstruction. All congruences below are in $\mathbb{Z}_2$, after the indicated scaling. Put

$$R_j = r_{n,n^2} - H_{n^2} + 2H_n, \qquad U_j = (R_j + T_{n,p})/3.$$

For $1 \le k < n$, $\binom{n}{k}$ is even; for $k = n$ one has $v_2((n^2+1)_n) = v_2(n!) = n - 1$. Thus, for $j \ge 3$, all $a_{n,n^2,k}$ with $k \ge 1$ are divisible by 4, and $Q_r \equiv 1 \pmod 4$. Using the integer-derivative identity above and the fact that $nH_k \in \mathbb{Z}_2$ for $k \le n$ gives

$$nR_j \equiv 2nH_n(1 + Q_r^{-1}) \equiv 4 \pmod 8. \tag{$\ddagger$}$$

Here nH_n is odd, since precisely one summand in H_n has denominator of two-adic valuation j. In particular $v_2(R_j) = 2 - j$.

We next locate the unique minimum among the valuations of the integer weights B_k. For $1 \le k < n$,

$$v_2\binom{n}{k} = j - v_2(k), \qquad v_2\binom{n+k}{k} = 0.$$

If $a = n - k$ and s_2 denotes the number of ones in a binary expansion, Legendre's factorial formula therefore gives

$$v_2(B_{n-a}) = p + g(a), \qquad g(a) = a - 1 - 2v_2(a) - s_2(a-1), \quad 1 \le a < n.$$

For a power $a = 2^t$, $g(a) = 2^t - 1 - 3t$; its only negative values are

$$g(2) = -2, \qquad g(4) = -3, \qquad g(8) = -2.$$

For completeness, if $a = 2^t u$ with odd $u \ge 3$, then

$$g(a) = 2^t u - 1 - 3t - s_2(u-1) \ge (2^t - 1)u - 3t \ge 0.$$

The endpoint weights have valuations $v_2(B_n) = p$ and $v_2(B_0) = n - 1$. For $j \ge 3$ the latter is strictly greater than $p - 3$. Thus $a = 4$ is the unique minimum, and

$$v_2(Q_T) = p - 3. \tag{§}$$

Write $\widetilde{B}_a = B_{n-a}/2^{p-3}$ and $\widetilde{Q} = Q_T/2^{p-3}$, an odd integer. Also put

$$h'_a = n((p+3)H_{n-a} - 2H_a - pH_{2n-a}).$$

For $a \ge 1$, h'_a is two-adically integral. If j is odd, then p is odd, and h'_4 is odd: nH_{n-4} is even, $2nH_4$ is even, and nH_{2n-4} is odd. All other normalized weights are even. The only possible nonintegral h'_a is at $a = 0$, where $v_2(h'_0) \ge -1$ but $v_2(\widetilde{B}_0) = 3$. Consequently $nT_{n,p}$ is odd. Together with (‡), this proves $v_2(U_j) = -j$ for odd $j \ge 3$.

Suppose next that j is even and $j \ge 6$. Then every h'_a is two-adically integral, including $a = 0$ because p is even. Modulo 8 only the normalized weights at $a = 2, 4, 8$ can contribute: their valuations are respectively $1, 0, 1$, and all other weights have valuation at least 3. For each of these three values of a, grouping harmonic summands by their two-adic denominator gives

$$nH_{n-a} \equiv 2 \pmod 8, \qquad nH_{2n-a} \equiv 1 \pmod 8, \qquad 2nH_a \equiv 0 \pmod 8.$$

For the first congruence, the $k = n/2$ summand gives 2, and the two $k = n/4, 3n/4$ summands sum to a multiple of 8; all remaining summands are multiples of 8. For the second, $k = n$ gives 1, the pair $k = n/2, 3n/2$ sums to $8/3$, and the four terms with $v_2(k) = j - 2$ also sum to a multiple of 8. The omission of at most seven terms near either upper endpoint changes neither congruence. It follows that $h'_a \equiv p + 6 \pmod 8$ for $a = 2, 4, 8$, and hence

$$nT_{n,p} \equiv p + 6 \pmod 8, \qquad nU_j \equiv (p+10)/3 \pmod 8.$$

When $j \equiv 0 \pmod 4$, $p + 10$ has valuation 1; when $j \equiv 6 \pmod 8$, it has valuation 2. This proves the remaining parts of (∗).

Outside $j \equiv 2 \pmod 8$, these formulas imply $\operatorname{den}(U_j) \ge n/4$ for all sufficiently large j. Since (87) gives $U_j - \gamma \sim p/(12D^2 n)$, the primitive error tends to infinity along any such sequence of indices. No statement about the remaining residue class is inferred from this calculation. □

What has and has not been gained. The large harmonic core has been removed from an actual finite rational formula, and the proven denominator upper scale improves from n^2 to $n \log n$. A further exact acceleration preserves eventual strict positivity. Nevertheless the first escaped family still fails the primitive-error test on seven residue classes out of eight. For its remaining class, and for the accelerated family, the displayed upper denominator bounds are far too large to establish primitive-error decay. They provide explicit candidates and exact arithmetic constraints, not an irrationality proof or an assumption that the unknown gcd is large.

Notation for the accelerated family. In this part, U_j denotes the core-cancelled rational approximant defined above, and

$$X_j = 4U_{j+2} - 4U_{j+1} + U_j. \tag{91}$$

In the kernel parts below the rational integral majorant is always written $\mathcal{U}_j$.

18 The accelerated sequence has no factor of two in its denominator

The approximation X_j is defined by (91). The new result below determines its two-adic valuation at every sufficiently large index. In particular, the large two-primary denominator of the unaccelerated family does not survive acceleration. All assertions concern reduced fractions.

Write $O_2(2^r)$ for an element of $2^r\mathbb{Z}_2$, allowing negative integers r. Constants introduced by convergent two-adic series below are not assumed to agree with any real constants bearing similar descriptions.

Theorem 18.1 (Uniform removal of the two-primary denominator)**.** *For every integer $j \geq 6$,*

$$\boxed{v_2(X_j) = 2.}$$

Equivalently, if $X_j = P_j/Q_j$ in lowest terms with $Q_j > 0$, then Q_j is odd and P_j is divisible by 4 but not by 8. This result does not bound the odd part Q_j and does not prove primitive-error decay.

18.1 Three convergent factorial series

Put

$$t_0 = 1, \qquad t_a = (-1)^{a-1}\frac{(a-1)!}{a^2} \quad (a \geq 1),$$

$$\mathfrak{F} = 1 + \sum_{a\geq 1} t_a, \qquad \mathfrak{G} = \sum_{a\geq 1} t_a H_{a-1}, \qquad \mathfrak{W} = \sum_{a\geq 1} t_a \left(H_{a-1} - \frac{2}{a}\right).$$

For $a \geq 2$ let $b(a) = \lfloor \log_2(a-1) \rfloor$, and let $b(1) = 0$. Legendre's factorial identity gives

$$g(a) := v_2(t_a) = a - 1 - s_2(a-1) - 2v_2(a).$$

The following elementary bounds will control all remainders:

$$g(a) - b(a) \geq -4, \qquad g(a) - 2b(a) \geq -6. \tag{92}$$

For $1 \leq a < 32$ they follow from the finite table

a	1	2–3	4–7	8–15	16–31
$\min(g-b)$	0	-2	-4	-4	0
$\min(g-2b)$	0	-2	-5	-6	-3

and for $a \geq 32$ one uses, with $d = \lfloor \log_2 a \rfloor \geq 5$,

$$g(a) - 2b(a) \geq a - 1 - 5d \geq 2^d - 1 - 5d \geq 6.$$

The same estimates show convergence of the three factorial series.

Lemma 18.2 (Exact valuations of the factorial series)**.** *One has*

$$v_2(\mathfrak{F}) = -3, \qquad v_2(\mathfrak{G}) = -3, \qquad v_2(\mathfrak{W}) = -1.$$

Proof. The unique minimum of $g(a)$ is $g(4) = -3$, proving the first assertion. For the second, the only terms below valuation -2 occur at $a = 4, 8$. Their sum is

$$t_4 H_3 + t_8 H_7 = -\frac{11}{16} - \frac{3267}{16} = -\frac{1639}{8}.$$

Every remaining term has valuation at least -2; this follows from the displayed bounds for $a \geq 32$ and direct substitution for $a < 32$. Thus the sum has valuation exactly -3.

For the third assertion, the only negative-valuation terms are at $a = 4, 6, 8$. They sum to

$$t_4 \left(H_3 - \frac{2}{4} \right) + t_6 \left(H_5 - \frac{2}{6} \right) + t_8 \left(H_7 - \frac{2}{8} \right) = -\frac{1}{2} - \frac{13}{2} - \frac{369}{2} = -\frac{383}{2}.$$

All other terms are two-adically integral, again by the bounds and the finite range $a < 32$. This proves the last assertion. □

18.2 A uniform two-adic expansion of the rational weights

For $n = 2^j$, $p = 3j$, reindex the $T_{n,p}$ weights by $a = n - k$ and divide by their final weight. Exactly as in the real analysis,

$$v_{n,p,a} = \frac{(n)_a^{p+3}}{(2n)_a^p (a!)^2}, \qquad 0 \leq a \leq n,$$

where the factorials $(n)_a$ in this display are falling. For $a \geq 1$, the identity $n^3 = 2^p$ gives the exact factorization

$$v_{n,p,a} = t_a \prod_{r=1}^{a-1} \frac{(1 - n/r)^{p+3}}{(1 - 2n/r)^p}.$$

For $r < n$, $v_2(n/r) \geq 1$, so these factors are units. Their convergent expansions over $\mathbb{Z}_2$ have first coefficient $(p-3)n/r$. Since all binomial coefficients involved are integers,

$$v_{n,p,a} = t_a (1 + n(p-3) H_{a-1} + O_2(2^{2(j - b(a))})). \tag{93}$$

The weaker first-order difference is $v_{n,p,a} - t_a = t_a O_2(2^{j-b(a)})$.

Put $S = \sum_{a=0}^{n} v_{n,p,a}$. Summing (93) and using (92) yields, for $j \geq 6$,

$$S = \mathfrak{F} + n(p-3)\mathfrak{G} + O_2(2^{2j-6}). \tag{94}$$

The omitted tails beyond n satisfy the same bounds. Indeed, for $a > n = 2^j$ and $d = \lfloor \log_2 a \rfloor \geq j$, the lower bound $a - 1 - 5d$ is at least $2j - 6$ when $j \geq 6$; it also controls the harmonic factors. In particular $v_2(S) = -3$. Expanding its inverse gives

$$\frac{1}{S} = \frac{1}{\mathfrak{F}} - \frac{n(p-3)\mathfrak{G}}{\mathfrak{F}^2} + O_2(2^{2j}). \tag{95}$$

To check the last error, the remainder in (94) divided by $\mathfrak{F}^2$ has valuation at least $2j$, while the quadratic inverse remainder has valuation at least $2j + 3$.

18.3 The harmonic terms and the affine pole

The sequence $K_j = 2^j H_{2^j}$ converges in $\mathbb{Z}_2$ to a unit

$$\mathfrak{K} = \lim_{j\to\infty} 2^j H_{2^j}, \qquad \mathfrak{K} - K_j \in 2^{2j+2}\mathbb{Z}_2 \quad (j \ge 1).$$

In fact, separating the odd and even terms gives

$$K_{j+1} - K_j = 2^{j+1} \sum_{\substack{1\le r<2^{j+1}\\ r \text{ odd}}} \frac{1}{r}.$$

Pair r with $2^{j+1}-r$ to see that the last sum is divisible by 2^{j+1} for $j \ge 1$. This proves convergence and the stated error.

Set

$$A = \frac{p}{2} + 3, \qquad C = \mathfrak{K} - 1 + \frac{1}{\mathfrak{F}}.$$

For $a \ge 1$, the harmonic observable multiplied by n can be written exactly as

$$n\big((p+3)H_{n-a} - 2H_a - pH_{2n-a}\big) = C_j + nL_a,$$

where

$$C_j = (p+3)(K_j - 1) - \frac{p}{2}(K_{j+1} - 1),$$

$$L_a = -(p+3)\sum_{r=1}^{a-1} \frac{1}{n-r} + p\sum_{r=1}^{a-1} \frac{1}{2n-r} - 2H_a.$$

For $a = 0$, the observable is $C_j + A$. Since $v_2(A) \ge -1$,

$$C_j = A(\mathfrak{K} - 1) + O_2(2^{2j+1}).$$

Expanding the two reciprocal sums gives, uniformly in $1 \le a \le n$,

$$L_a = H_{a-1} - \frac{2}{a} + O_2(2^{j-2b(a)}).$$

Together with (92) and the first-order weight bound, this implies

$$\sum_{a=1}^{n} v_{n,p,a} L_a = \mathfrak{W} + O_2(2^{j-6}), \qquad \frac{\sum_{a=1}^{n} v_{n,p,a} L_a}{S} = \frac{\mathfrak{W}}{\mathfrak{F}} + O_2(2^{j-3}).$$

Combining these identities with (95) proves

$$T_{n,p} = \frac{AC}{n} - \frac{A(p-3)\mathfrak{G}}{\mathfrak{F}^2} + \frac{\mathfrak{W}}{\mathfrak{F}} + O_2(2^{j-3}). \tag{96}$$

The other component $R_j = r_{n,n^2} - H_{n^2} + 2H_n$ has the simpler expansion

$$R_j = \frac{4\mathfrak{K}}{n} + O_2(2^{j-1}). \tag{97}$$

To prove it, use the integer weights $a_k = \binom{n}{k}^2 (n^2+1)_k$, now with a rising factorial, and $Q_r = \sum a_k$. For $1 \le k < n$,

$$v_2(a_k) = 2j - 2v_2(k) + k - s_2(k) \ge 2j - 1.$$

The elementary bound used here is $k - s_2(k) - 2v_2(k) \ge -1$: writing $k = 2^t u$ with u odd, the case $u = 1$ reduces to $2^t - 1 - 2t \ge -1$, and $u \ge 3$ follows from $(2^t - 1)u - 2t \ge -1$. At $k = n$ the valuation is $n - 1 \ge 2j - 1$. Thus $Q_r = 1 + O_2(2^{2j-1})$. The derivative identity $a_k' = a_k \sum_{h=1}^{k} 1/(n^2+h)$ gives $v_2(a_k') \ge j - 1$, since $v_2(n^2+h) \le j$. The remaining noninitial harmonic terms have valuation at least j. Hence $R_j = 4H_n + O_2(2^{j-1})$, and the estimate for K_j proves (97).

Proof of Theorem 18.1. Since $U_j = (R_j + T_{n,p})/3$, the two expansions give

$$U_j = \frac{AC + 4\mathfrak{K}}{3n} - \frac{A(p-3)\mathfrak{G}}{3\mathfrak{F}^2} + \frac{\mathfrak{W}}{3\mathfrak{F}} + O_2(2^{j-3}).$$

The first term is an affine polynomial in j divided by 2^j. The operator $U_j \mapsto 4U_{j+2} - 4U_{j+1} + U_j$ annihilates it exactly. Since $2A(p-3) = p^2 + 3p - 18$, its action on the remaining polynomial gives

$$X_j = \frac{\mathfrak{W}}{3\mathfrak{F}} - \frac{\mathfrak{G}}{6\mathfrak{F}^2}(p^2 + 27p + 126) + O_2(2^{j-3}). \tag{98}$$

The first term has valuation 2. The coefficient $\mathfrak{G}/(6\mathfrak{F}^2)$ has valuation 2, and the integer $p^2 + 27p + 126$ is even, so the second term is divisible by 8. For $j \geq 6$ the error is also divisible by 8. Consequently

$$X_j \equiv 4 \pmod 8,$$

proving $v_2(X_j) = 2$ and the assertion about its reduced fraction. □

Arithmetic meaning. This is a proved infinite cancellation in the actual denominator, not a cancellation of a chosen common multiplier. The theorem removes the two-primary obstacle completely from X_j. It leaves the odd denominator unconstrained, and therefore does not by itself establish condition 3.

19 An exact odd-prime certificate for the reduced denominator

The removal of the two-primary denominator does not control the odd denominator. The following result gives an exact, finite certificate for prime divisors of the reduced denominator of X_j. It also isolates one specific distribution estimate that would prove failure of condition 3. No such distribution estimate is assumed.

For integers $N, m \geq 1$ and $q \geq 1$, define the positive integers

$$Q_r(N,m) = \sum_{k=0}^{N} \binom{N}{k}^2 (m+1)_k, \qquad Q_T(N,q) = \sum_{k=0}^{N} \binom{N}{k}^2 \binom{N+k}{k}^q \frac{N!}{k!}.$$

Here $(m+1)_k = (m+1)\cdots(m+k)$ is a rising factorial, and the empty product is 1. Write $\widehat{R}_N = r_{N,N^2} - H_{N^2} + 2H_N$; the additional parameter of $T_{N,q}$ remains explicit.

Lemma 19.1 (Upper-half prime residues)**.** *Let $N \geq 2$, $q \geq 1$, and let ℓ be an odd prime with $N/2 < \ell \leq N$ and $\ell^2 > 2N$. If $\ell \nmid Q_r(N,N^2)Q_T(N,q)$, then*

$$\ell\widehat{R}_N \equiv 4 \pmod \ell, \qquad \ell T_{N,q} \equiv 3 - \frac{q}{2} \pmod \ell.$$

In these formulas all denominators after multiplication by ℓ are coprime to ℓ. If instead $N < \ell$, $\ell^2 > 2N$, and $\ell \nmid Q_r(N,N^2)Q_T(N,q)$, then both $\widehat{R}_N$ and $T_{N,q}$ are ℓ-adically integral.

Proof. First suppose $N = \ell + s$ with $0 \leq s < \ell$. Put $a_k = \binom{N}{k}^2 (N^2+1)_k$. The integer-derivative identity proved in Section 17 gives

$$\widehat{R}_N = \frac{\sum_{k=0}^{N} (a_k' + 2a_k(H_{N-k} - H_k + H_N))}{Q_r(N,N^2)},$$

where a'_k is the derivative with respect to the rising-factorial parameter, evaluated at the integer N^2. Each a'_k is an integer. For $s < k < \ell$, the binomial coefficient $\binom{N}{k}$ is divisible by ℓ. For $k \geq \ell$, the rising factorial $(N^2+1)_k$ is divisible by ℓ. Consequently

$$Q_r(N, N^2) \equiv \sum_{k=0}^{s} a_k \pmod{\ell}.$$

For $k \leq s$, multiplying the harmonic expression $2(H_{N-k} - H_k + H_N)$ by ℓ gives residue 4. All terms with $k > s$ have zero residue after multiplication by ℓ, and the integer derivative terms also have zero residue. Division by the unit $Q_r(N, N^2)$ proves the first congruence.

For the other family put

$$B_k = \binom{N}{k}^2 \binom{N+k}{k}^q \frac{N!}{k!}, \qquad h_k = (q+3)H_k - 2H_{N-k} - qH_{N+k}.$$

If $k < \ell$, the quotient $N!/k!$ is divisible by ℓ, so $\ell B_k h_k$ has zero residue. If $k = \ell + t$, $0 \leq t \leq s$, and $s + t \geq \ell$, the binomial coefficient $\binom{N+k}{k}$ is divisible by ℓ, and the same conclusion holds. In every remaining case $k = \ell + t$ with $s + t < \ell$,

$$\ell H_k \equiv 1, \qquad \ell H_{N-k} \equiv 0, \qquad \ell H_{N+k} \equiv H_2 = \frac{3}{2} \pmod{\ell}.$$

Thus $\ell h_k \equiv 3 - q/2$ on every weight that can survive modulo ℓ. Dividing the resulting numerator congruence by the unit $Q_T(N, q)$ proves the second formula. The assumption $\ell^2 > 2N$ ensures that all harmonic poles used here are at most simple.

Finally suppose $N < \ell$. The derivative formula shows that $\widehat{R}_N$ is ℓ-adically integral whenever $Q_r(N, N^2)$ is a unit. In $T_{N,q}$ the only possible harmonic pole is in H_{N+k} when $N + k \geq \ell$. For exactly those indices $\binom{N+k}{k}$ is divisible by ℓ, and its positive power q clears that simple pole. Division by the unit $Q_T(N, q)$ therefore gives an integral value. □

Theorem 19.2 (Certified prime product in the actual denominator)**.** *Let $j \geq 2$, $n = 2^j$, $p = 3j$, and put*

$$\Pi_j = \prod_{r=0}^{2} Q_r(2^{j+r}, 2^{2(j+r)}) Q_T(2^{j+r}, 3(j+r)),$$

$$P_j^{\mathrm{band}} = \prod_{\substack{2n < \ell \leq 4n \\ \ell \text{ prime}}} \ell, \qquad D_j = \frac{P_j^{\mathrm{band}}}{\gcd(P_j^{\mathrm{band}}, \Pi_j)}.$$

Then D_j divides the actual reduced denominator of X_j. More precisely, every prime $\ell \mid D_j$ satisfies

$$v_\ell(X_j) = -1, \qquad \ell X_j \equiv \frac{2(8-p)}{3} \pmod{\ell}.$$

Consequently

$$\log \operatorname{den}(X_j) \geq \vartheta(4n) - \vartheta(2n) - \log \gcd(P_j^{\mathrm{band}}, \Pi_j),$$

where $\vartheta(x) = \sum_{\ell \leq x,\ \ell \text{ prime}} \log \ell$.

Proof. Fix a prime $\ell \mid D_j$. Since $2n < \ell$, both U_j and U_{j+1} are ℓ-adically integral by the last assertion of Lemma 19.1. For U_{j+2} take $N = 4n$ and $q = p + 6$ in the first part of that lemma. It gives

$$\ell U_{j+2} \equiv \frac{4 + 3 - (p+6)/2}{3} = \frac{8-p}{6} \pmod{\ell}.$$

Therefore $X_j = 4U_{j+2} - 4U_{j+1} + U_j$ has the claimed residue. For $j \geq 2$ one has $0 < |p-8| < 2n < \ell$, so this residue is nonzero. The poles are at most simple, proving $v_\ell(X_j) = -1$. Multiplying the distinct certified primes proves $D_j \mid \operatorname{den}(X_j)$, and taking logarithms gives the displayed inequality. □

Corollary 19.3 (Two certified primes suffice)**.** *Let $\mathcal{J}$ be an infinite set of indices $j \geq 2$. Suppose that for each sufficiently large $j \in \mathcal{J}$ there are at least two distinct primes $\ell \in (2n, 4n]$ with $\ell \nmid \Pi_j$. Then, at every index in $\mathcal{J}$ satisfying this two-prime hypothesis,*

$$\operatorname{den}(X_j) > 4n^2,$$

and, for all sufficiently large $j \in \mathcal{J}$,

$$\operatorname{den}(X_j)|X_j - \gamma| > \frac{1-c}{96\mathcal{D}^2}\, p^2.$$

Consequently the primitive errors tend to infinity along $\mathcal{J}$. An infinite set of such indices implies that the full sequence has infinite limsup; this alone does not exclude a different successful subsequence. If the two-prime hypothesis holds for every sufficiently large index, the primitive errors tend to infinity along the full sequence and along every infinite subsequence.

The existence of two certified primes is established here only for the finite checked indices. No infinite existence assertion is included in this corollary.

Proof. The two distinct certified primes both exceed $2n$ and their product divides the actual reduced denominator by Theorem 19.2. Thus that denominator is strictly greater than $4n^2$. The real asymptotic established for this same sequence is

$$X_j - \gamma \sim C_\infty \frac{p^2}{n^2}, \qquad C_\infty = \frac{1-c}{192\mathcal{D}^2} > 0.$$

For all sufficiently large j its left side is at least $(C_\infty/2)p^2/n^2$. Multiplication by the denominator lower bound gives the assertion. Since $p = 3j$, the lower bound tends to infinity. □

Corollary 19.4 (A sufficient obstruction stated entirely in integers)**.** *Along any sequence of indices tending to infinity on which*

$$\log \gcd(P_j^{\mathrm{band}}, \Pi_j) = o(n),$$

one has

$$\liminf \frac{\log \operatorname{den}(X_j)}{n} \geq 2, \qquad \operatorname{den}(X_j)|X_j - \gamma| \longrightarrow +\infty.$$

More generally, if the gcd logarithm is at most $(2-\varepsilon)n$ eventually for some fixed $\varepsilon > 0$, the primitive errors diverge exponentially. These gcd estimates are sufficient hypotheses; neither is asserted for an infinite sequence here.

Proof. The prime number theorem [7] gives $\vartheta(4n) - \vartheta(2n) = 2n + o(n)$. Apply the theorem, and use the asymptotic $X_j - \gamma \sim C_\infty p^2/n^2$ with $C_\infty = (1-c)/(192\mathcal{D}^2) > 0$. Its logarithm is $O(\log n)$ and hence cannot offset a positive linear lower bound for the logarithm of the reduced denominator. □

Finite exact checks and their limitation. For $j = 2, \ldots, 10$, the numbers of primes in $(2n, 4n]$ certified by the theorem are respectively

$$2,\ 5,\ 7,\ 12,\ 22,\ 43,\ 74,\ 137,\ 254,$$

out of total prime counts

$$2,\ 5,\ 7,\ 13,\ 23,\ 43,\ 75,\ 137,\ 255.$$

The omitted primes in those checks are 67, 131, 653, and 2113, one at each of $j = 5, 6, 8, 10$. Direct reduction shows that these four primes also divide the corresponding reduced denominators, although the unit-denominator hypothesis does not certify them. These are exact finite computations. They establish no asymptotic bound on the omitted prime product and therefore do not close condition 3.

19.1 Rational certificates for the finite primitive errors

The large finite primitive errors can be certified without inserting a decimal approximation to γ. The following classical enclosure is sufficient; its proof records the sign of the remainder explicitly.

Lemma 19.5 (A rationally usable enclosure for Euler's constant)**.** *For every positive integer M,*

$$H_M - \log M - \frac{1}{2M} + \frac{1}{12M^2} - \frac{1}{120M^4} < \gamma < H_M - \log M - \frac{1}{2M} + \frac{1}{12M^2}.$$

Define the rational numbers

$$\lambda = 2\sum_{a=0}^{30} \frac{1}{(2a+1)3^{2a+1}}, \qquad \rho = \frac{9}{4 \cdot 63 \cdot 3^{63}},$$

$$G_- = H_{64} - 6(\lambda+\rho) - \frac{1}{128} + \frac{1}{12 \cdot 64^2} - \frac{1}{120 \cdot 64^4}, \quad G_+ = H_{64} - 6\lambda - \frac{1}{128} + \frac{1}{12 \cdot 64^2}.$$

Then $G_- < \gamma < G_+$, and both endpoints are given by finite rational arithmetic.

Proof. Binet's digamma identity [4], together with $\psi(M+1) = H_M - \gamma$ and $\psi(M+1) = \psi(M) + 1/M$, gives

$$H_M - \log M - \gamma = \frac{1}{2M} - 2\int_0^\infty \frac{t}{(M^2+t^2)(e^{2\pi t}-1)}\,dt.$$

For $t > 0$,

$$\frac{t}{M^2} - \frac{t^3}{M^4} < \frac{t}{M^2+t^2} < \frac{t}{M^2}.$$

The two needed positive integrals are

$$\int_0^\infty \frac{t}{e^{2\pi t}-1}\,dt = \frac{1}{24}, \qquad \int_0^\infty \frac{t^3}{e^{2\pi t}-1}\,dt = \frac{1}{240};$$

they follow by expanding the positive geometric series and using $\zeta(2) = \pi^2/6$ and $\zeta(4) = \pi^4/90$. Integration of the strict kernel inequalities proves the first claim. Finally

$$\log 2 = 2\sum_{a\ge 0} \frac{1}{(2a+1)3^{2a+1}}, \qquad 0 < \log 2 - \lambda < \rho,$$

where the upper bound follows by replacing every tail denominator $2a+1$ by 63 and summing the geometric series. Take $M = 64$ and $\log 64 = 6\log 2$ in the first enclosure. □

How the finite certificates are verified. The exact finite sums define a reduced fraction $X_j = A_j/B_j$ with $B_j > 0$. If $X_j > G_+$, rational comparison proves $X_j - \gamma > X_j - G_+$. If $X_j < G_-$, it proves $\gamma - X_j > G_- - X_j$. If the relevant positive rational gap exceeds 2^{-k} and B_j has bit length b, then

$$B_j|X_j - \gamma| > 2^{b-1-k}.$$

The supplied exact-arithmetic certificate program performs these comparisons with the displayed rational endpoints and gives

j	b	$\operatorname{sign}(X_j-\gamma)$	k	$b-1-k$
8	135056	$-$	25	135030
9	299332	$+$	23	299308
10	656413	$+$	24	656388

All entries in this table concern actual reduced fractions and exact rational inequalities. They are finite verification results and do not assert any infinite denominator growth.

20 Fixed-shift localization and explicit node identities

The following calculation extends the shift $1 - 1/M$ to every fixed $1 - k/M$, $k > 0$. The forms $A(Q)$ and $B(Q)$ retain their Laguerre normalization from Part I.

20.1 The finite polynomial schedule at the node zero

At $x = 0$, define the unnormalized polynomial

$$Q_D^\circ = \sum_{j=0}^{D-1} (D-j)^2(j+1)(L_j - L_{j+1}).$$

Its ordinary Laguerre coefficient at index j, $0 \le j \le D$, is $C_D(j) = 3j^2 - (4D+1)j + D^2$. Direct finite summation yields

$$A(Q_D^\circ) = P(D)H_D + R(D), \qquad B(Q_D^\circ) = V(D), \tag{99}$$

where

$$P(D) = \frac{D(D+1)(D+2)(2D^2 - 11D - 31)}{15},$$
$$V(D) = -\frac{D(D+1)(D+2)(2D^2 - 11D - 1)}{15}.$$
$$R(D) = -\frac{D(334D^4 - 1915D^3 - 9660D^2 - 9425D - 934)}{900}.$$

Here is a finite-sum verification, including enough intermediate identities to check the polynomials without numerical interpolation. Substitution of C_D into the tridiagonal form gives

$$2\sum_{j=0}^{D} jC_D(j)^2 - 2\sum_{j=0}^{D-1}(j+1)C_D(j)C_D(j+1) = V(D).$$

For the diagonal and adjacent terms of A, use

$$\sum_{j=0}^{m} f(j)H_j = H_m \sum_{j=0}^{m} f(j) - \sum_{r=1}^{m} \frac{1}{r} \sum_{j=0}^{r-1} f(j).$$

The inner sum is a polynomial divisible by r, so the final sum is an ordinary polynomial sum. With $m = D, D-1$ and $H_{D-1} = H_D - 1/D$, this gives the near-diagonal contribution

$$V(D)H_D - \frac{D(30D^5 - 334D^4 - 785D^3 + 810D^2 - 55D - 3266)}{900}.$$

For the remaining gaps $2 \le d \le D$, the summand after summing in j is

$$\sum_{j=0}^{D-d} 2C_D(j)C_D(j+d)A_{j,j+d} = G_D(d) + \frac{r_-}{d+1} + \frac{r_0}{d} + \frac{r_+}{d-1},$$

where

$$G_D(d) = -\frac{9d^4 - (20D^2 + 28D - 7)d^2 + 15D^4 + 20D^3 - 55D^2 - 78D - 12}{15},$$

$$r_- = \frac{D(D+1)(D+2)(D+4)(D^2 - D - 7)}{15}, \quad r_0 = -\frac{2D(D+1)(D+2)(D^3 + D^2 + 3)}{15},$$

$$r_+ = \frac{D(D-2)(D+1)(D+2)(D^2+5D-1)}{15}.$$

These are obtained by expanding the degree-five polynomial in j and taking the three residues at $d = -1, 0, 1$. Sum the polynomial part using

$$\sum_{d=1}^{D} d^2 = \frac{D(D+1)(2D+1)}{6}, \qquad \sum_{d=1}^{D} d^4 = \frac{D(D+1)(2D+1)(3D^2+3D-1)}{30}.$$

The rational parts sum respectively to $r_-(H_D + 1/(D+1) - 3/2)$, $r_0(H_D - 1)$, and $r_+(H_D - 1/D)$. Adding the near contribution gives exactly (99). All these steps are finite polynomial identities and harmonic telescoping identities.

20.2 Normalized shifted nodes and exact rational sandwiches

Fix a positive integer k. For integers $M \geq 2k$, put

$$D = M^2, \quad N = D - 1, \quad x = 1 - k/M, \quad w_j = (1 - j/D)^2 \ (0 \leq j \leq N), \quad w_{N+1} = 0,$$

$$p_M(t) = \sum_{j=0}^{N} \frac{w_j L_j^{(1)}(x) L_j^{(1)}(t)}{j+1}, \qquad Q_M(t) = t p_M(t).$$

These are the *normalized* polynomials. The identity $tL_j^{(1)} = (j+1)(L_j - L_{j+1})$ gives

$$Q_M = D^{-2} Q^\circ_{D,x}, \qquad Q^\circ_{D,x} = \sum_{j=0}^{D-1} (D-j)^2 L_j^{(1)}(x)(L_j - L_{j+1}).$$

Consequently

$$(A(Q^\circ_{D,x}), B(Q^\circ_{D,x}), I(Q^\circ_{D,x})) = D^4 (A(Q_M), B(Q_M), I(Q_M)).$$

All asymptotics below refer to Q_M, not to $Q^\circ_{D,x}$; their target ratios, reduced denominators, and primitive errors agree. The ordinary Laguerre coefficient of Q_M at index zero is 1, so these polynomials are nonzero.

Use $\langle f, g \rangle_1 = \int_0^\infty f(t) g(t) t e^{-t} dt$ and put

$$\mathsf{H}_r = \langle p_M, t^r p_M \rangle_1, \qquad T_M = \mathsf{H}_2 + \mathsf{H}_1, \qquad U_M = \mathsf{H}_3 - \mathsf{H}_2 - \mathsf{H}_1 + \mathsf{H}_0.$$

These are rational, $B_M = B(Q_M) = \mathsf{H}_2 - \mathsf{H}_1$, and

$$\frac{2B_M^2}{T_M} \leq I_M \leq \frac{U_M}{2}, \qquad \frac{2q_M |B_M|}{T_M} \leq \mathcal{E}_M \leq \frac{q_M U_M}{2|B_M|} \quad (B_M \neq 0). \tag{100}$$

Here $I_M = I(Q_M) > 0$ and q_M is the actual reduced denominator of $A(Q_M)/B(Q_M)$.

To prove these inequalities, for $t > 0$ use

$$\frac{2(t-1)^2}{t+1} \leq (t-1)\log t \leq \frac{(t-1)^2(t+1)}{2t}.$$

The first follows from $\log t = 2\,\mathrm{artanh}((t-1)/(t+1))$. For the second, differentiate $(t - t^{-1})/2 - \log t$ and multiply its sign by that of $t - 1$. The upper bound integrates to $U_M/2$. Cauchy–Schwarz gives

$$B_M^2 \leq \left\langle p_M, \frac{t(t-1)^2}{t+1} p_M \right\rangle_1 \langle p_M, t(t+1) p_M \rangle_1 \leq I_M T_M / 2.$$

Finally use $\mathcal{E}_M = q_M I_M / |B_M|$.

20.3 Uniform real localization for every fixed shift

Theorem 20.1 (The real scale and the exact denominator threshold)**.** *For fixed $k > 0$ and the normalized polynomials above, as $M \to \infty$,*

$$B_M = -k\kappa + O_k(\log M/M), \quad T_M = 2\kappa M + O_k(\log M), \quad U_M = O_k(M^{-1}), \qquad \kappa = \frac{128e}{315\pi} > 0.$$

In particular $B_M < 0$ eventually and

$$I_M = \Theta_k(M^{-1}), \qquad \gamma - A(Q_M)/B(Q_M) = \Theta_k(M^{-1}), \qquad \mathcal{E}_M = \Theta_k(q_M/M).$$

Thus primitive-error decay in this family is equivalent to $q_M = o(M)$. These assertions make no assumption about the arithmetic nature of γ.

Proof. The compact-x Laguerre expansion, for $\alpha = 0, 1$, is

$$L_j^{(\alpha)}(x) = \frac{e^{x/2} j^{\alpha/2-1/4}}{\sqrt{\pi}\, x^{\alpha/2+1/4}} \left[\cos\Big(2\sqrt{jx} - (\alpha/2 + 1/4)\pi\Big) + O(j^{-1/2})\right]. \tag{101}$$

Its error is uniform for $x \in [1/2, 1]$; see the standard Laguerre asymptotic expansion [8]. This compact uniformity applies to $x = 1 - k/M$.

The basis $\phi_j = L_j^{(1)}/\sqrt{j+1}$ is orthonormal for $\langle\ ,\ \rangle_1$. Multiplication by t has Jacobi matrix

$$J_{jj} = 2j + 2, \qquad J_{j,j+1} = -\sqrt{(j+1)(j+2)}.$$

The coefficient vector of p_M is $b_j = w_j L_j^{(1)}(x)/\sqrt{j+1}$ for $j \le N$, zero afterwards. Set $S = \|b\|^2$, $g = (J - x)b$, and $\delta = \langle b, g\rangle$. Squaring (101) gives

$$S = \kappa(x)\sqrt{N} + O(\log(N+2)), \qquad \kappa(x) = \frac{128e^x}{315\pi x^{3/2}}. \tag{102}$$

Indeed the nonoscillating sum is $e^x/(2\pi x^{3/2}) \sum_{j=1}^N j^{-1/2}(1 - j/D)^4$. Integral comparison uses $\int_0^1 u^{-1/2}(1-u)^4 du = 256/315$. The squared asymptotic remainders sum to $O(\log(N+2))$. For the oscillating sum, comparison with its integral has the same error: the phase derivative contributes $O(1/t)$ to the absolute derivative. Substitution $t = y^2$ makes the integral bounded by integration by parts, since $4\sqrt{x}$ stays away from zero and the amplitude $(1 - y^2/D)^4$ has bounded variation. All constants are uniform on the stated compact interval.

The eigenvector recurrence for $L_j^{(1)}(x)$ gives the exact identity

$$\delta = \sum_{j=0}^{N} L_j^{(1)}(x) L_{j+1}^{(1)}(x)(w_j - w_{j+1})^2.$$

It follows either by expanding both sides or by summing the Jacobi recurrence with the finitely supported weights w_j. Since $|w_j - w_{j+1}| \le 2/D$ and $L_j^{(1)}(x)L_{j+1}^{(1)}(x) = O((j+1)^{1/2})$,

$$\delta = O(N^{-1/2}). \tag{103}$$

No sign of the individual Laguerre products is assumed.

A weighted residual estimate is needed to control the unbounded moment in U_M. Put $L_{-1}^{(1)} = 0$ and

$$U_j^* = L_j(x) + L_{j+1}(x), \qquad V_j^* = L_{j-1}^{(1)}(x) + L_{j+1}^{(1)}(x).$$

Direct substitution of the quadratic taper gives

$$g_j = \sqrt{j+1}\left[\frac{2(D-j)}{D^2}U_j^* - \frac{1}{D^2}V_j^*\right] \ (0 \le j \le N), \qquad g_{N+1} = -\frac{\sqrt{N+2}}{D^2}L_N^{(1)}(x).$$

All later entries vanish. The identities $L_{j+1}^{(1)} - L_j^{(1)} = L_{j+1}$ and $L_{j+1} - L_j = -xL_j^{(1)}/(j+1)$, together with (101), imply

$$U_j^* = O((j+1)^{-1/4}), \quad \Delta U_j^* = O((j+1)^{-3/4}), \qquad V_j^* = O((j+1)^{1/4}), \quad \Delta V_j^* = O((j+1)^{-1/4}).$$

Hence $g_j = O((j+1)^{1/4}/D)$ in the bulk and $g_N, g_{N+1} = O(D^{-5/4})$. Summing squares gives $\|g\|^2 = O(N^{-1/2})$. Moreover the Jacobi energy identity is

$$\langle g, Jg\rangle = g_0^2 + \sum_{j\ge 0}(\sqrt{j+1}\,g_j - \sqrt{j+2}\,g_{j+1})^2.$$

In the bulk the displayed difference is $O((j+1)^{1/4}/D)$: apply the bounds on $\Delta U_j^*, \Delta V_j^*$ and the elementary first differences of $(j+1)(D-j)$ and $j+1$. The two endpoint differences are $O(D^{-3/4})$. Their squared sum therefore proves

$$\langle g, (J+1)g\rangle = O(N^{-1/2}). \tag{104}$$

The exact moment identities now give

$$B_M = (x^2-x)S + (2x-1)\delta + \|g\|^2, \qquad T_M = (x^2+x)S + (2x+1)\delta + \|g\|^2.$$

Using $\sqrt{N} = M + O(M^{-1})$, $x = 1 - k/M$, and $\kappa(1) = \kappa$, equations (102)– (104) give the asserted B_M, T_M asymptotics. Finally, since $(J-1)b = g + (x-1)b$ and $J+1$ is positive,

$$U_M = \langle (J-1)b, (J+1)(J-1)b\rangle \le 2\langle g, (J+1)g\rangle + 2(x-1)^2\langle b, (J+1)b\rangle = O_k(M^{-1}).$$

The last inner product is $(x+1)S + \delta = O(M)$. The lower sandwich in (100) is asymptotic to $k^2\kappa/M$, and the upper sandwich is $O_k(M^{-1})$. This proves $I_M = \Theta_k(M^{-1})$. Since $|B_M| \to k\kappa$, the remaining assertions follow, including the exact primitive denominator threshold. □

21 Odd primes at a zero of the rational node

The rational forms A, B, their Laguerre Gram matrices, the exact polynomial-schedule identities, and the real localization threshold are derived in Section 20.

The earlier two-adic theorem concerns the particular node $1 - 1/M$. Changing it to $1 - 3/M$ can cancel that leading two-adic residue. The following result tracks primes in the numerator of the new node and proves that such a cancellation can leave a much larger factor in the actual reduced denominator.

For an integer $D \ge 2$ and a rational x, put

$$z_j(x) = (D-j)^2 L_j^{(1)}(x), \qquad Q_{D,x} = \sum_{j=0}^{D-1} z_j(x)(L_j - L_{j+1}).$$

The common factor D^{-2} in the normalized localization family is omitted: it has no effect on $A(Q)/B(Q)$ or its primitive error. The rational forms satisfy

$$A(Q) - \gamma B(Q) = \int_0^\infty Q(t)^2(t-1)\log t\, e^{-t}\,dt > 0.$$

Define three rational polynomials

$$P(D) = \frac{D(D+1)(D+2)(2D^2-11D-31)}{15},$$

$$R(D) = -\frac{D(334D^4-1915D^3-9660D^2-9425D-934)}{900},$$

$$V(D) = -\frac{D(D+1)(D+2)(2D^2-11D-1)}{15}.$$

The finite-sum identities proved in Section 20 give

$$A(Q_{D,0}) = P(D)H_D + R(D), \qquad B(Q_{D,0}) = V(D).$$

Indeed the ordinary Laguerre coefficients at $x = 0$ are $c_j = 3j^2-(4D+1)j+D^2$, and substitution into the exact Gram matrices gives these identities by finite sums.

Theorem 21.1 (A node-zero prime survives the complete gcd)**.** *Let $p \geq 7$ be prime, assume $D \geq p$, let $v_p(x) = s \geq 1$, and put*

$$a = \lfloor \log_p D \rfloor, \qquad h = \lfloor D/p^a \rfloor.$$

Suppose $P(D)$, $V(D)$, and H_h are p-adic units. Then $B(Q_{D,x}) \neq 0$, and

$$v_p(A(Q_{D,x})) = -a, \qquad v_p(B(Q_{D,x})) = 0.$$

Consequently the actual reduced denominator satisfies

$$\boxed{v_p(\mathrm{den}(A(Q_{D,x})/B(Q_{D,x}))) = a.}$$

This is a valuation after complete reduction, not a common-denominator estimate.

Proof. For every integer $r \geq 1$, $v_p(x^r/r!) \geq s$, since $v_p(r!) \leq r-1$ and $s \geq 1$. The finite Laguerre formula therefore gives, uniformly in j,

$$L_j^{(1)}(x) \equiv j+1 \pmod{p^s\mathbb{Z}_p}.$$

Thus every ordinary Laguerre coefficient of $Q_{D,x} - Q_{D,0}$ belongs to $p^s\mathbb{Z}_p$. The half-lcm clearing theorem 11.1 says that p^aA has p-integral quadratic coefficients; B has integral quadratic coefficients. It follows that

$$v_p(A(Q_{D,x}) - A(Q_{D,0})) \geq s-a, \qquad v_p(B(Q_{D,x}) - B(Q_{D,0})) \geq s.$$

In H_D, the terms with denominator divisible by p^a sum to $p^{-a}H_h$. All other terms have valuation at least $1-a$. The unit hypothesis gives $v_p(H_D) = -a$. Since $a \geq 1$, $P(D)$ is a unit and $R(D)$ is p-integral, so $v_p(A(Q_{D,0})) = -a$; its perturbation has strictly larger valuation. Likewise $V(D)$ is a unit and its perturbation is divisible by p. These facts prove the assertion. □

Corollary 21.2 (A necessary harmonic exception for a shifted node)**.** *Fix a positive integer k and an integer $M > k$, and use $D = M^2$, $x = 1-k/M$. Let $p \geq 7$ divide $M-k$, with $p \nmid k$, and set $a = \lfloor \log_p(M^2) \rfloor$, $h = \lfloor M^2/p^a \rfloor$. If $P(k^2)$, $V(k^2)$, and H_h are p-adic units, then*

$$v_p(q_M) = a, \qquad q_M \geq p^a > \frac{M^2}{p} > M.$$

Thus any sequence satisfying $q_M = o(M)$ must eventually arrange, at every such numerator prime except the fixed coefficient exceptions, the congruence $H_h \equiv 0 \pmod{p}$.

Proof. Here $v_p(x) = v_p(M-k) \geq 1$, since $M \equiv k \not\equiv 0 \pmod{p}$. The polynomial unit hypotheses reduce to their values at k^2. Also $p \leq M - k < M$, so $D > p^2$ and the theorem applies with $a \geq 2$. Finally $M^2 < p^{a+1}$ proves the displayed lower bounds. □

Corollary 21.3 (An all-exponent obstruction for bounded cofactors)**.** *Fix positive integers k, c. Apart from an explicit finite set of primes, every prime*

$$p > \max\{c^2, 2kc + k^2, 5\}$$

has the following property. For every $e \geq 1$, let $M = k + cp^e$ and $x = 1 - k/M$. Then

$$\boxed{v_p(q_M) = 2e, \qquad q_M \geq p^{2e} = ((M-k)/c)^2.}$$

The excluded primes are precisely those additionally dividing the numerators of H_{c^2}, $P(k^2)$, or $V(k^2)$.

Proof. The size condition gives $c^2 < M^2/p^{2e} < c^2 + 1 \leq p$. Therefore $a = 2e$ and $h = c^2$, while $v_p(x) = e$. The preceding theorem proves the result. The two fixed polynomial values are nonzero, since neither $2D^2 - 11D - 31$ nor $2D^2 - 11D - 1$ has an integer root. Hence the exceptional prime set is finite. □

For example, for $k = 3, c = 4$,

$$P(9) = 2112 = 2^6 \cdot 3 \cdot 11, \qquad V(9) = -4092 = -2^2 \cdot 3 \cdot 11 \cdot 31,$$

$$H_{16} = \frac{2436559}{720720}, \qquad 2436559 = 17^2 \cdot 8431.$$

Thus every prime $p \geq 37$, $p \neq 8431$, and every $e \geq 1$ gives $v_p(q_{3+4p^e}) = 2e$. In particular primes $p \equiv 3 \pmod{4}$ at $e = 1$ give infinitely many obstructed parameters $M \equiv 15 \pmod{16}$, a class in which the leading two-adic residue had canceled. Infinitude of primes 3 modulo 4 follows from the usual Euclid argument and is unaffected by deleting a finite set.

Corollary 21.4 (The full dyadic $k = 3$ schedule does not have decaying error)**.** *Take $k = 3$ and $M_s = 2^s - 1$. Along infinitely many s,*

$$q_{M_s} > M_s^2/7.$$

The primitive errors of this full schedule therefore have infinite limsup; in particular they do not tend to zero. This statement does not exclude every further subsequence of the schedule.

Proof. For $s \equiv 2 \pmod{3}$, $7 \mid M_s - 3$. The numbers $P(9)$ and $V(9)$ are 7-adic units, and

$$(H_1, H_2, H_3, H_4, H_5, H_6) \equiv (1, 5, 3, 5, 1, 0) \pmod{7}.$$

Consequently, if the leading base-7 digit of M_s^2 is at most 5, the theorem gives $v_7(q_{M_s}) = \lfloor \log_7(M_s^2) \rfloor$ and $q_{M_s} > M_s^2/7$.

There are infinitely many such $s \equiv 2 \pmod{3}$. Indeed, for $s = 3n + 2$,

$$\log_7(M_s^2) = 6n \log_7 2 + 4 \log_7 2 + o(1).$$

The number $6 \log_7 2$ is irrational by unique factorization. Density of an irrational rotation supplies infinitely many n for which the fractional part of the main term belongs to $(1/4, 1/2)$. The vanishing perturbation preserves membership in a slightly larger interval strictly contained in $(0, \log_7 6)$; this is exactly the required leading-digit condition. Density can be proved by the

elementary pigeonhole argument already used in the Farey-selection theorem, and every tail of the orbit is dense.

For any fixed $k > 0$, the normalized localization proof in Section 20 applies to $D^{-2}Q_{D,x}$ at $x = 1 - k/M$ and gives

$$B_M = -k\kappa + O(\log M/M), \quad T_M = 2\kappa M + O(\log M), \quad U_M = O(M^{-1}), \qquad \kappa = 128e/(315\pi).$$

The exact rational sandwich then gives $\gamma - A_M/B_M = \Theta(M^{-1})$ and primitive error $\mathcal{E}_M = \Theta(q_M/M)$. The lower bound $q_{M_s} > M_s^2/7$ on the stated infinite subsequence therefore makes $\mathcal{E}_{M_s}$ tend to infinity on that subsequence. □

Relation to the existing target-realization results. Positive rational mixing of two nearby certificates can cancel all their old denominator factors. That observation is already contained in the earlier exact target-realization theorem, its determinant/gcd identity, and the Farey optimization theorems; it is not a new solution to condition 3. The new contribution here is the local survival criterion at numerator primes, including its all-exponent and dyadic consequences. It concerns this single localized-square family and does not exclude arbitrary positive mixtures or other constructions.

A finite full-denominator audit of the apparent two-adic escape. For $k = 3$, $M = 11$, $D = 121$, $x = 8/11$, the earlier modular computation gave $v_2(q) = 0$. Complete rational evaluation of A/B gives an odd reduced denominator with 633 decimal digits:

$$10^{632} \le q < 10^{633}.$$

The exact moment calculation also gives $10|B| > T$, where $T = \mathsf{H}_2 + \mathsf{H}_1$ is the rational moment quantity from the localization sandwich. Consequently

$$\mathcal{E} \ge \frac{2q|B|}{T} > \frac{q}{5} > 10^{631}.$$

The accompanying exact-arithmetic script and its full integer/rational certificate record the complete denominator and both error bounds. No decimal value of γ enters this calculation. This finite audit is not an infinite obstruction theorem; it shows concretely why eliminating the factor 2 at one parameter did not satisfy condition 3.

Part III

Proper kernels, prime support, and exact cancellation

22 Kernel identities and analytic inputs

22.1 Proper Binet kernels and rational error bounds

We reproduce the kernel calculation to fix the normalization of these certificates. Put

$$d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}.$$

Binet's identity [4] gives, for every integer $j \geq 1$,

$$\int_0^\infty \frac{d\mu(u)}{u + j^2} = \gamma + \log j - H_j + \frac{1}{2j}.$$

For $n \geq 1$, define

$$R_n(u) = \frac{\prod_{k=0}^{2n-1}(4u + k^2)}{\prod_{j=n}^{3n}(u + j^2)}, \qquad I_n = \int_0^\infty R_n(u)\, d\mu(u) > 0.$$

The residues, constant coefficient, and prime-log coefficients are

$$A_{n,j} = (-1)^{j-n} \frac{4j^2(2j + 2n - 1)!(n + j - 1)!}{(2j - 2n)!(j - n)!(3n - j)!(3n + j)!},$$
$$p_n = \sum_{j=n}^{3n} A_{n,j} \left(H_j - \frac{1}{2j}\right), \qquad E_{n,\ell} = \sum_{j=n}^{3n} A_{n,j} v_\ell(j).$$

There is no polynomial part in the partial fraction decomposition, and $\sum_j A_{n,j} = 16^n$. Thus Binet's formula gives exactly

$$I_n = 16^n \gamma - p_n + \sum_{\ell \leq 3n} E_{n,\ell} \log \ell. \tag{105}$$

Only primes occur in the last sum.

Lemma 22.1 (Uniform height and raw integral bounds)**.** *For every $n \geq 1$ and $n \leq j \leq 3n$,*

$$2A_{n,j} \in \mathbb{Z}, \qquad 2|A_{n,j}| \leq 6\, 2^{12n}.$$

Moreover

$$0 < I_n < \mathcal{U}_n := \frac{(4n)!((n-1)!)^2}{8\, 16^n((3n)!)^2}, \qquad \mathcal{U}_{n+1} < \mathcal{U}_n, \tag{106}$$

and, with $\kappa_0 = \log(729/16)$,

$$\log \mathcal{U}_n = -\kappa_0 n + O(\log(n+1)).$$

Proof. Put $a = j + n$ and $b = j - n$. With

$$B = \binom{2a}{b}, \quad C = \binom{a}{2b}, \quad B' = \binom{2a-1}{b-1}, \quad C' = \binom{a-1}{2b-1},$$

where $B' = C' = 0$ at $b = 0$, direct factorial simplification gives

$$2|A_{n,j}| = BC + 4B'C + B'C'.$$

This proves integrality. Every product in this expression is at most $2^{3a} \le 2^{12n}$, proving the stated height bound.

For the integral estimate, put $d\nu(u) = du/(2\sinh(2\pi\sqrt{u}))$. Then $1 < d\mu/d\nu = 1+e^{-2\pi\sqrt{u}} < 2$. Let $T_j(u) = P_{2j}^{(1)}(2\sqrt{u}; \pi/2)$ be the even Meixner–Pollaczek polynomials in their usual normalization. Their generating function and orthogonality [9, 10] imply

$$S_r(u) := \sum_{j=0}^{r} \binom{r}{j} T_j(u) = \frac{4^r}{(2r)!} \prod_{k=0}^{r-1} (4u + k^2), \qquad \int_0^\infty S_r \, d\nu = \frac{1}{16}.$$

For clarity, the generating function of the binomial transform is $\cosh(4\sqrt{u}\arcsin\sqrt{z})$, whose coefficient gives the displayed product. In the integral, all T_j with $j > 0$ vanish by orthogonality; the constant norm is $\int d\nu = 1/16$. The product is positive for $u > 0$, so $\int S_r d\mu < 1/8$. Since

$$R_n(u) = \frac{(4n)!}{16^n} \frac{S_{2n}(u)}{\prod_{j=n}^{3n}(u + j^2)},$$

replacing each denominator factor by j^2 gives (106). A direct quotient estimate gives

$$\frac{\mathcal{U}_{n+1}}{\mathcal{U}_n} = \frac{(4n+1)(4n+2)(4n+3)(4n+4)n^2}{16(3n+1)^2(3n+2)^2(3n+3)^2} \le \frac{16}{729}\left(1 + \frac{1}{n}\right)^4 \le \frac{256}{729} < 1.$$

Finally Stirling's formula [21] gives the logarithmic estimate. □

We use $L_j = \sum_\ell E_{j,\ell} \log \ell$. The latter denotes a logarithmic value, not a least common multiple. The factor-two denominator result proved in Section 54 gives $d_j p_j, d_j E_{j,\ell} \in \mathbb{Z}$ for $d_j = 2\,\mathrm{lcm}(1, \ldots, 3j)$. The asymptotic formula, including its prefactor, is

$$r_n := \frac{\mathcal{U}_n}{16^n} \sim \frac{\sqrt{2\pi}}{12} \, n^{-3/2} 729^{-n}. \tag{107}$$

This follows directly from Stirling's formula [21]. The quotient bound above also gives $r_{n+1} \le (16/729) r_n$.

22.2 Positive kernels from weighted prefixes

Write $f_j = 16^{-j} R_j$, and retain $L_j = \sum_\ell E_{j,\ell} \log \ell$ and $\rho_j = 16^{-j} L_j$.

Lemma 22.2 (Pointwise ordering and logarithmic ordering)**.** *For every $j \ge 1$ and every $u > 0$,*

$$0 < \frac{f_{j+1}(u)}{f_j(u)} = \frac{(u + j^2)^2 (u + (j + \frac{1}{2})^2)}{(u + (3j+1)^2)(u + (3j+2)^2)(u + (3j+3)^2)} < 1.$$

Moreover $\rho_{j+1} > \rho_j$.

Proof. Cancel the common numerator and denominator factors in R_{j+1}/R_j. Each of the three resulting numerator factors is smaller than its paired denominator factor. The strict inequality follows. The partial fractions and $\sum_k A_{j,k} = 16^j$ give

$$2\rho_j = \int_0^\infty \left(\frac{1}{u+1} - f_j(u) \right) du,$$

since $\int_0^\infty ((u+1)^{-1} - (u+k^2)^{-1})\,du = 2\log k$. Subtracting consecutive identities gives

$$2(\rho_j - \rho_{j+1}) = -\int_0^\infty (f_j(u) - f_{j+1}(u))\,du < 0.$$

This integral converges: each f_j has leading term $1/u$ at infinity, and each is regular at zero. Its integrand is strictly positive. □

Theorem 22.3 (The cumulative positivity cone)**.** *Let $c = (c_n, \dots, c_M)$ be a finite real vector and set*

$$a_j = 16^j c_j, \qquad S_k = \sum_{j=n}^{k} a_j, \qquad K_c(u) = \sum_{j=n}^{M} c_j R_j(u).$$

Then

$$K_c(u) = \sum_{k=n}^{M-1} S_k(f_k(u) - f_{k+1}(u)) + S_M f_M(u).$$

If $c \neq 0$ and every $S_k \geq 0$, then $K_c(u) > 0$ for every $u > 0$. If in addition c cancels all prime-log coordinates, then necessarily $S_M > 0$, and

$$0 < \int_0^\infty K_c\,d\mu = S_M\gamma - \sum_{j=n}^{M} c_j p_j < \sum_{j=n}^{M} |c_j|\mathcal{U}_j.$$

No arithmetic assumption about γ is used.

Proof. The displayed decomposition is finite summation by parts. Every function appearing there is strictly positive on $(0,\infty)$. Not all S_k vanish unless $c = 0$, proving positivity of K_c. Log cancellation and a second summation by parts give

$$0 = \sum_{j=n}^{M} a_j\rho_j = \sum_{k=n}^{M-1} S_k(\rho_k - \rho_{k+1}) + S_M\rho_M.$$

If $S_M = 0$, one of the earlier S_k is positive and the strict increase of ρ_j makes the right-hand side negative. This proves $S_M > 0$. Binet's identity, positivity, and the strict individual upper bounds $0 < I_j < \mathcal{U}_j$ finish the proof. □

22.3 Quantitative logarithmic increments

Retain $f_k = R_k/16^k$, $\rho_k = L_k/16^k$, and put

$$\Delta_k = \rho_{k+1} - \rho_k, \qquad T_k = 2 + \frac{3}{2}H_{k-1}, \qquad H_0 = 0.$$

Lemma 22.4 (Explicit rational bounds for the logarithmic increments)**.** *For every integer $k \geq 1$,*

$$\boxed{\Delta_k > \frac{1}{66k}, \qquad 0 < \rho_k < T_k.}$$

Proof. Set

$$B_k = \sum_{j=k}^{3k} j^2 = \frac{26k^3 + 15k^2 + k}{3} \leq 14k^3.$$

Factoring powers of u out of numerator and denominator gives

$$f_k(u) = \frac{1}{u}\frac{\prod_{h=0}^{2k-1}(1 + (h/2)^2/u)}{\prod_{j=k}^{3k}(1 + j^2/u)}.$$

For $u \geq 14k^3$, the numerator factors are at least one and $\prod_j (1 + j^2/u) \leq \exp(B_k/u) \leq e < 3$. Hence

$$f_k(u) > \frac{1}{3u}.$$

The exact ratio from the preceding positivity lemma satisfies

$$\frac{f_{k+1}(u)}{f_k(u)} \leq \frac{u + k^2}{u + (3k+1)^2}.$$

On $14k^3 \leq u \leq 28k^3$, it follows that

$$1 - \frac{f_{k+1}(u)}{f_k(u)} \geq \frac{8k^2 + 6k + 1}{u + (3k+1)^2} \geq \frac{8k^2}{44k^3} = \frac{2}{11k}.$$

Frullani's identity therefore yields

$$\begin{aligned} \Delta_k &= \frac{1}{2} \int_0^\infty (f_k - f_{k+1})\, du \\ &> \frac{1}{2}\,(14k^3) \frac{1}{84k^3} \frac{2}{11k} = \frac{1}{66k}. \end{aligned}$$

For the upper bound, the elementary product inequality $\prod_i (1 - t_i) \geq 1 - \sum_i t_i$ for $0 \leq t_i \leq 1$ gives

$$\prod_{j=k}^{3k} \frac{1}{1 + j^2/u} \geq 1 - \sum_{j=k}^{3k} \frac{j^2}{u + j^2} \geq 1 - \frac{B_k}{u}.$$

Consequently

$$f_k(u) \geq \frac{1}{u} - \frac{B_k}{u^2} \qquad (u > 0).$$

Here, if the right-hand side is negative, the inequality follows already from $f_k(u) > 0$; if it is nonnegative, multiplying the product bound by the numerator factors preserves it. Split the Frullani integral at B_k. On its first part drop the nonnegative f_k; on its second part use the last estimate. This gives

$$\begin{aligned} 2\rho_k &= \int_0^\infty \left(\frac{1}{u+1} - f_k(u) \right) du \\ &< \log(B_k + 1) + \int_{B_k}^\infty \frac{B_k}{u^2}\, du = \log(B_k + 1) + 1 \\ &\leq \log(15k^3) + 1 < 4 + 3H_{k-1} = 2T_k. \end{aligned}$$

The last strict inequality uses $\log 15 < 3$ and $\log k \leq H_{k-1}$. The elementary estimate $e > 5/2$ proves $\log 15 < 3$. Positivity of ρ_k was proved by the earlier Frullani separation bound, and also follows directly from $f_k(u) < 1/(u + 9k^2) \leq 1/(u+1)$. □

For later compactness, exact cancellation gives

$$\sum_{k=n}^{M-1} S_k \Delta_k = q\rho_M.$$

If $S_k \geq 0$ and $q = S_M > 0$, each term on the left is nonnegative, so $S_k < 66kT_M q$ for $k < M$. Thus all normalized prefixes $q = 1$ are bounded in every fixed window.

22.4 A sharper uniform residue height

Lemma 22.5 (A sharper uniform residue height)**.** *Put*

$$G = \frac{16399}{1296}, \qquad \beta = G^2.$$

For all $m \le j \le 3m$,

$$2|A_{m,j}| \le 6\beta^m, \qquad \beta^3 < 2^{22}.$$

Proof. Set $a = j + m$, $b = j - m$, so $a - b = 2m$ and $0 \le 2b \le a$. The exact binomial identity used to prove residue integrality is

$$2|A_{m,j}| = BC + 4B'C + B'C',$$

where

$$B = \binom{2a}{b}, \quad C = \binom{a}{2b}, \quad B' = \binom{2a-1}{b-1}, \quad C' = \binom{a-1}{2b-1};$$

put $B' = C' = 0$ when $b = 0$. Since $B' \le B$ and $C' \le C$, the right side is at most $6BC$.

For any positive t, s, the binomial theorem gives

$$BC \le \frac{(1+t)^{2a}(1+s)^a}{t^b s^{2b}}.$$

Choose $t = 5/18$ and $s = 27/4$. Then

$$(1+t)^2(1+s) = G, \qquad \frac{G}{ts^2} = \frac{32798}{32805} < 1.$$

Consequently

$$BC \le G^{a-b}\left(\frac{G}{ts^2}\right)^b \le G^{2m} = \beta^m.$$

The remaining strict inequality is an exact integer comparison:

$$2^{22}1296^6 - 16399^6 = 424901949697141879058783 > 0.$$

□

22.5 The matrices and the two arithmetic coordinates

For each N, set

$$V_N = (2E_{n,\ell})_{\substack{\ell \le 3N \text{ prime}, \\ 1 \le n \le N}}, \qquad \Psi_N(c) = \left(\sum_{n=1}^{N} 16^n c_n,\ \sum_{n=1}^{N} p_n c_n\right).$$

Rows of V_N are ordered by increasing prime. All entries of V_N are integers; those of Ψ_N are rational. The phrase *log-null* means $V_N c = 0$, coordinate by coordinate. It is stronger than a numerical cancellation test on a real sum of logarithms. For such a vector,

$$\sum_{n=1}^{N} c_n I_n = L_\gamma(\Psi_N(c)), \qquad L_\gamma(q,p) = q\gamma - p.$$

For rational x, $\mathrm{den}(x)$ denotes its positive reduced denominator, with $\mathrm{den}(0) = 1$.

23 Exact rank and a recursive basis for logarithm cancellation

This section gives an infinite construction of all finite logarithm cancellations. The rank certificates at its two starting indices use only arithmetic in finite fields. No numerical value of γ is used.

Lemma 23.1 (One new prime at a time)**.** *For $N \geq 2$, the interval $(3N-3, 3N]$ contains at most one prime. If it contains a prime ℓ_N, then*

$$V_{\ell_N,n} = 0 \quad (n < N), \qquad V_{\ell_N,N} = 2A_{N,\ell_N} \neq 0.$$

Proof. The integer $3N$ is composite, and one of the consecutive integers $3N-2, 3N-1$ is even and greater than two. Thus at most one is prime. If ℓ_N occurs, then $\ell_N > 3(N-1) \geq N$ and $2\ell_N > 3N$. Earlier pole intervals contain no multiple of ℓ_N, whereas the current pole interval contains exactly the single multiple $j = \ell_N$. The residue formula shows that its coefficient is nonzero. □

Theorem 23.2 (Exact logarithm rank)**.** *For every $N \geq 9$,*

$$\boxed{\operatorname{rank}_{\mathbb{Q}} V_N = \pi(3N), \qquad \dim_{\mathbb{Q}} \ker V_N = N - \pi(3N).}$$

The same rank holds over $\mathbb{R}$. Here $\pi(x)$ denotes the number of primes not exceeding x.

Proof. The 9 by 9 matrix V_9 is nonsingular: the exact finite-field certificate in Appendix A gives $\det V_9 \equiv 97 \pmod{101}$. Suppose the claimed row rank holds for $N-1$. If no new prime occurs, the old matrix is a full-row-rank submatrix of V_N. If a new prime occurs, its new row is zero in all old columns and nonzero in the new column by Lemma 23.1, so the rank increases by one. This proves the assertion by induction. Rational and real rank agree for a rational matrix. □

The starting index is minimal: for $1 \leq N \leq 8$ one has $\pi(3N) = N+1$, so full row rank is impossible. The kernel dimension increases by exactly one at every index with no new prime, and stays unchanged at every index with a new prime.

Theorem 23.3 (Exact rank of the arithmetic output)**.** *Put*

$$W_N = \begin{pmatrix} V_N \\ (16^n)_{n=1}^{N} \\ (p_n)_{n=1}^{N} \end{pmatrix}.$$

For every $N \geq 17$,

$$\operatorname{rank}_{\mathbb{Q}} W_N = \pi(3N) + 2, \qquad \Psi_N(\ker_{\mathbb{Q}} V_N) = \mathbb{Q}^2.$$

Proof. The matrix W_{17} is square. Its entries have denominators prime to 211, and the exact certificate in Appendix A gives $\det W_{17} \equiv 65 \pmod{211}$. Thus the output rank is two at $N = 17$. This rank cannot decrease when zero coordinates are appended to old kernel vectors. Theorem 23.2, together with the elementary rank identity

$$\operatorname{rank} \begin{pmatrix} V_N \\ \Psi_N \end{pmatrix} = \operatorname{rank} V_N + \operatorname{rank}(\Psi_N|_{\ker V_N}),$$

proves both assertions. □

23.1 A canonical recursion with growing support

Let $\mathcal{P}_9 = \{1, \dots, 9\}$. For $N > 9$, define the pivot set by

$$\mathcal{P}_N = \begin{cases} \mathcal{P}_{N-1} \cup \{N\}, & \pi(3N) > \pi(3N-3), \\ \mathcal{P}_{N-1}, & \pi(3N) = \pi(3N-3). \end{cases}$$

Rows are ordered by increasing prime and columns by increasing index. The square matrix $B_N = V_N[:, \mathcal{P}_N]$ is invertible. More precisely,

$$\det B_N = \det V_9 \prod_{\substack{10 \le k \le N \\ \pi(3k) > \pi(3k-3)}} 2A_{k,\ell_k}. \tag{108}$$

This follows from the block upper triangular form at every new-prime step.

At an index $m \ge 10$ with no new prime, write v_m for the new column, with the same row set as V_{m-1}. Define a rational vector $h^{(m)}$ by

$$h^{(m)}_m = 1, \qquad h^{(m)}_{\mathcal{P}_{m-1}} = -B^{-1}_{m-1} v_m, \qquad h^{(m)}_j = 0 \quad (j \notin \mathcal{P}_{m-1} \cup \{m\}). \tag{109}$$

Append zeros when this vector is regarded as a vector in a larger block.

Theorem 23.4 (All finite cancellations from a triangular recursion)**.** *For every $N \ge 9$, the vectors*

$$\left\{ h^{(m)} : 10 \le m \le N,\ \pi(3m) = \pi(3m-3) \right\}$$

form a rational basis of $\ker V_N$*. Their number is exactly $N - \pi(3N)$.*

Proof. At its creation, (109) cancels every existing prime row. A later prime row is identically zero on its support, so the vector remains in every larger kernel. Its coordinate at the nonpivot index m is one, while all other displayed basis vectors have zero in that coordinate. The vectors are therefore independent. Their number is $N - |\mathcal{P}_N| = N - \pi(3N)$, the full kernel dimension from Theorem 23.2. □

This is a constructive infinite recursion for exact logarithm cancellation. All matrices and residues are rational and explicitly given. Multiplying any $h^{(m)}$ by the least common multiple of its entry denominators gives an integer vector. A second finite clearing factor makes both arithmetic output coordinates integral. These operations prove exact integrality; they do not bound the resulting primitive error. In particular, neither the growth of the kernel dimension nor the determinant product (108) implies that the second primitive minimum tends to zero.

24 What kind of recurrence the integrals cannot satisfy

Theorem 24.1 (The exact rational span of the integrals)**.** *For every $N \ge 17$,*

$$\operatorname{span}_{\mathbb{Q}}\{I_1, \dots, I_N\} = \operatorname{span}_{\mathbb{Q}}\big(\{1, \gamma\} \cup \{\log \ell : \ell \le 3N,\ \ell \text{ prime}\}\big). \tag{110}$$

In particular the dimension of the left side is at least $\pi(3N)$ and tends to infinity.

Proof. The Binet identity gives the inclusion from left to right. For the other inclusion, Theorem 23.3 says that the formal coefficient columns, whose coordinates correspond to $\{1, \gamma, \log \ell\}$, span the full rational coordinate space. The factors 2 in V_N and the sign of the p_n row are invertible rational row scalings. Thus a rational combination of the I_n recovers each displayed generator separately.

The logarithms of distinct primes are rationally independent. Indeed, clearing denominators in $\sum_\ell a_\ell \log \ell = 0$ and exponentiating gives $\prod_\ell \ell^{a_\ell} = 1$ with integral exponents; unique factorization gives $a_\ell = 0$ for every ℓ. No assertion about the independence of γ, 1, and the logarithms is required. □

Theorem 24.2 (No eventually nonsingular rational recurrence)**.** *There do not exist a fixed $r \geq 0$, an index n_0, and rational numbers $a_{k,n}$, for $0 \leq k \leq r$ and $n \geq n_0$, satisfying*

$$a_{r,n} \neq 0, \qquad \sum_{k=0}^{r} a_{k,n} I_{n+k} = 0 \quad \text{for every } n \geq n_0.$$

Consequently (I_n) is not P-recursive over $\mathbb{Q}$, and its ordinary generating series is not D-finite over $\mathbb{Q}(z)$.

Proof. Such a recurrence would express each subsequent term as a rational linear combination of the preceding r terms. Including the finite initial segment, all terms would lie in a fixed finite-dimensional rational vector space, contrary to Theorem 24.1. The case $r = 0$ would give $I_n = 0$, also impossible.

In a polynomial-coefficient recurrence over $\mathbb{Q}$, the highest nonzero coefficient polynomial has only finitely many integer zeros. It would therefore give the excluded recurrence after a finite starting index. Finally, a differential equation with coefficients in $\mathbb{Q}[z]$ for the generating series gives a polynomial-coefficient recurrence by equating coefficients. This is the usual P-recursive/D-finite correspondence; see Stanley [22, Theorem 1.5]. □

Corollary 24.3 (No closed finite rational state)**.** *A representation of (I_n) by a fixed finite-dimensional state, updated by rational matrices and read by rational linear output maps, is impossible if the initial state has finitely many real coordinates. The same conclusion holds if fixed finitely many real constants appear as inhomogeneous inputs.*

Proof. Every state and output lies in the rational span of the initial coordinates and the finitely many fixed inputs. This contradicts Theorem 24.1. □

The restriction to rational coefficients is essential. These results concern this particular family of proper Binet integrals. Other families of approximants to γ can satisfy polynomial recurrences; see, for example, [24]. The recursive kernel construction in Section 23 is not excluded, because its dimension and support increase.

25 Total positivity and the necessary cancellation bandwidth

The polynomial degree bound in the prime-band argument can be replaced by the dimension bound. The reason is that the cleared residue polynomials belong to a strictly totally positive generalized Bernstein basis. We include an elementary proof of the required total positivity; in particular, no numerical sign test is used.

Lemma 25.1 (Positive solid minors imply strict total positivity)**.** *Let X be a finite real matrix. If every square minor with consecutive rows and consecutive columns is strictly positive, then every square minor of X is strictly positive.*

Proof. We first prove the following completion step for an $N \times m$ matrix Y, with $N \geq m$. Suppose all its $(m-1)$-minors on its first $m-1$ columns are positive, and all its m-minors with consecutive rows are positive. Then all its m-minors are positive. For $m = 1$ there is nothing to prove. For $m \geq 2$, induct on the dispersion $i_m - i_1 - m + 1$ of a row set $I = \{i_1 < \cdots < i_m\}$. If

this dispersion is nonzero, insert a missing interior index to obtain $J = \{j_1 < \cdots < j_{m+1}\}$ with $I = J \setminus \{j_t\}$, $1 < t < m+1$. Set

$$\Delta_s = \det Y[J \setminus \{j_s\}, \{1, \dots, m\}], \qquad \delta_{uv} = \det Y[J \setminus \{j_u, j_v\}, \{1, \dots, m-1\}].$$

The determinantal identity

$$\delta_{1,m+1}\Delta_t = \delta_{t,m+1}\Delta_1 + \delta_{1,t}\Delta_{m+1}$$

is the elementary three-term Plücker identity (and also follows by expanding each determinant in its last column). All δ's are positive by hypothesis; Δ_1 and Δ_{m+1} have smaller dispersion, so they are positive by induction. This proves the completion step.

Now induct on the order m of minors of X. All smaller minors may be assumed positive. Apply the completion step first to every submatrix formed by m consecutive columns: its full-size minors are positive. Next fix any m rows and apply the completion step to the transpose of the submatrix on those rows. Its consecutive-row full-size minors are precisely the minors just proved positive. Hence all m-minors of X are positive, completing the induction. □

Lemma 25.2 (Separated-knot Bernstein basis)**.** *Let $N \geq 0$ and let $a_1, \dots, a_N, b_1, \dots, b_N$ be real numbers such that the interval*

$$\mathcal{I} = (\max_i a_i, \min_i b_i)$$

is nonempty; when $N = 0$, take any nonempty open interval instead. Repeated knots on the same side are allowed. Define

$$B_j(x) = \prod_{i=1}^{j}(x - a_i) \prod_{i=j+1}^{N}(b_i - x), \qquad 0 \leq j \leq N.$$

For every finite increasing list of distinct nodes in $\mathcal{I}$, the collocation matrix with rows indexed by nodes and columns indexed by j is strictly totally positive. Consequently every subfamily of $s+1$ of the B_j is a Chebyshev system: a nonzero real linear combination has at most s distinct zeros in $\mathcal{I}$.

Proof. For a block of consecutive columns $j = v, v+1, \dots, v+k$, factor the positive common function

$$W(x) = \prod_{i \leq v}(x - a_i) \prod_{i > v+k}(b_i - x)$$

out of each row. The remaining $k+1$ polynomials are

$$g_j(x) = \prod_{i=v+1}^{v+j}(x - a_i) \prod_{i=v+j+1}^{v+k}(b_i - x), \qquad 0 \leq j \leq k.$$

Their coefficient determinant, in increasing monomial order, is

$$\det\big([x^i]g_j(x)\big)_{0 \leq i,j \leq k} = \prod_{v < i \leq j \leq v+k}(b_j - a_i) > 0. \tag{CB}$$

For completeness, evaluate the polynomials at a_{v+1} and express the remainder in the basis $1, (x - a_{v+1}), \dots, (x - a_{v+1})^k$. The monomial change of basis has determinant one. Only the first polynomial has a nonzero constant coefficient, namely $\prod_{j=v+1}^{v+k}(b_j - a_{v+1})$. Deleting its row and column and dividing the remaining polynomials by $x - a_{v+1}$ leaves the same problem with $k-1$ pairs of knots. Induction proves (CB), including repeated a_i.

At any $k+1$ increasing nodes $x_0 < \cdots < x_k$ in $\mathcal{I}$, the minor on these consecutive columns is therefore

$$\left(\prod_{h=0}^{k} W(x_h)\right)\left(\prod_{0\le h<l\le k}(x_l - x_h)\right)\left(\prod_{v<i\le j\le v+k}(b_j - a_i)\right) > 0.$$

In particular all solid minors are positive, so Lemma 25.1 proves strict total positivity. If a combination of $s+1$ selected columns vanished at $s+1$ distinct nodes, their strictly positive square collocation determinant would make every coefficient zero. This gives the zero bound. □

Theorem 25.3 (Sharp dimension bound for the prime band)**.** *Let $n \ge 1$, $r \ge 0$, and suppose a nonzero vector $(c_0, \dots, c_r) \in \mathbb{R}^{r+1}$ cancels every prime-log coordinate of the proper kernels $R_n, \dots, R_{n+r}$. Then*

$$\boxed{r \ge \pi(3n) - \pi\left(\frac{3(n+r)}{2}\right).} \tag{PB}$$

In particular, along any such sequence of cancellations with $n \to \infty$ and $r = o(n)$,

$$r \ge \left(\frac{3}{2} + o(1)\right)\frac{n}{\log n}.$$

Proof. For primes $3(n+r)/2 < \ell \le 3n$, the only multiple of ℓ in each pole interval $[n+k, 3(n+k)]$ is ℓ itself. The corresponding log coefficient equals $A_{n+k,\ell}$. Write $x = \ell^2$. The exact residue ratio is

$$\frac{A_{m+1,\ell}}{A_{m,\ell}} = \frac{-16(x-m^2)^2(x-(m+\frac{1}{2})^2)}{((3m+1)^2-x)((3m+2)^2-x)((3m+3)^2-x)}.$$

Use $N = 3r$ pairs of knots, indexed in blocks $m = n, \dots, n+r-1$:

$$(a_{3(m-n)+1}, a_{3(m-n)+2}, a_{3(m-n)+3}) = (m^2, m^2, (m+\tfrac{1}{2})^2),$$

$$(b_{3(m-n)+1}, b_{3(m-n)+2}, b_{3(m-n)+3}) = ((3m+1)^2, (3m+2)^2, (3m+3)^2).$$

Every prime-band node $x = \ell^2$ lies strictly between every a_i and every b_i. If this band contains no primes, (PB) is immediate. Otherwise this separation interval is nonempty, and Lemma 25.2 applies. Clearing the common positive denominator from the residue ratios converts the cancellation equation at each prime into

$$\sum_{k=0}^{r} c_k(-16)^k B_{3k}(\ell^2) = 0.$$

The subfamily $B_0, B_3, \dots, B_{3r}$ is a Chebyshev system, so a nonzero combination has at most r distinct roots there. This proves (PB). The case $r = 0$ is the same assertion for one positive constant polynomial and may also be checked directly. For $r = o(n)$ the prime number theorem [7] gives

$$\pi(3n) - \pi\left(\frac{3(n+r)}{2}\right) = \left(\frac{3}{2} + o(1)\right)\frac{n}{\log n},$$

as claimed. The term "sharp dimension bound" refers to the number r of possible zeros of a system of dimension $r+1$; it makes no claim that a logarithm-free vector exists at this bandwidth. □

The proof of Lemma 25.1 is the classical Fekete solid-minor criterion, included to make the application self-contained; see also [23, Section 4.6]. Total positivity itself is not a new principle; its consequence here is the stronger necessary bandwidth for exact prime-log cancellation.

26 The exact asymptotic size of the positive integral

Retain the proper Binet kernels and their rational upper weights:

$$R_n(u)=\frac{\prod_{k=0}^{2n-1}(4u+k^2)}{\prod_{j=n}^{3n}(u+j^2)},\qquad I_n=\int_0^\infty \frac{R_n(u)}{e^{2\pi\sqrt{u}}-1}\,du,\qquad \mathcal{U}_n=\frac{(4n)!((n-1)!)^2}{8\,16^n((3n)!)^2}.$$

Both the integrals and the upper weights are positive. The previously proved inequality is $I_n<\mathcal{U}_n$. The next theorem determines their precise asymptotic ratio. It makes no assumption about the arithmetic nature of Euler's constant.

Theorem 26.1 (Exact limiting ratio and the first corrections)**.** *As $n\to\infty$,*

$$\boxed{\frac{I_n}{\mathcal{U}_n}=\frac{3\sqrt{3}}{16}-\frac{13\sqrt{3}}{256n}+\frac{\sqrt{2}}{\pi^{7/2}n^{3/2}}+O(n^{-2}).}$$

In particular the rational bound $\mathcal{U}_n$ and the actual integral I_n have the same exponential decay rate, and $I_n/\mathcal{U}_n\to 3\sqrt{3}/16$.

Proof. Euler's product for the hyperbolic sine [20] gives

$$\prod_{k=1}^{\infty}\left(1+\frac{4u}{k^2}\right)=\frac{\sinh(2\pi\sqrt{u})}{2\pi\sqrt{u}}.$$

Define

$$a_n=\frac{8\,16^n}{n^2\binom{4n}{2n}},\qquad E_n(u)=\prod_{k=2n}^{\infty}\left(1+\frac{4u}{k^2}\right)^{-1}\prod_{j=n}^{3n}\left(1+\frac{u}{j^2}\right)^{-1}.$$

The infinite product converges for each $u\ge 0$. Direct cancellation of the factorials in $R_n/\mathcal{U}_n$, followed by the displayed Euler product, gives the exact identity

$$\frac{I_n}{\mathcal{U}_n}=\frac{a_n}{4\pi}\int_0^\infty \sqrt{u}\left(1+e^{-2\pi\sqrt{u}}\right)E_n(u)\,du. \tag{111}$$

For example, before using Euler's product the prefactor is $32u\,16^n((2n-1)!)^2/(4n)!=a_nu$; this records the normalization.

Put $e_n(x)=E_n(nx)$. For every fixed $x\ge 0$,

$$\log e_n(x)\longrightarrow -\frac{8}{3}x.$$

Indeed, the linear terms in the two logarithmic products are $4nx\sum_{k\ge 2n}k^{-2}\to 2x$ and $nx\sum_{j=n}^{3n}j^{-2}\to 2x/3$, and their total quadratic remainder is $O(x^2/n)$. A uniform integrable majorant is available without an asymptotic assumption on x. The $2n$ factors with $2n\le k<4n$ give

$$0<e_n(x)\le\left(1+\frac{x}{4n}\right)^{-2n}\le\left(1+\frac{x}{8}\right)^{-4}\qquad(n\ge 2).$$

The last inequality follows because $(1+y/t)^t$ increases with $t>0$. Stirling's formula [21] yields

$$\frac{a_nn^{3/2}}{4\pi}=2\sqrt{\frac{2}{\pi}}\left(1+\frac{1}{16n}+O(n^{-2})\right). \tag{112}$$

Thus dominated convergence already proves

$$\lim_{n\to\infty}\frac{I_n}{\mathcal{U}_n}=2\sqrt{\frac{2}{\pi}}\int_0^\infty\sqrt{x}\,e^{-8x/3}\,dx=\frac{3\sqrt{3}}{16}.$$

For the corrections, let $L_n(x) = -\log e_n(x)$. Elementary tail-sum expansions, obtainable from Euler–Maclaurin with its remainder, give

$$4n\sum_{k=2n}^{\infty}\frac{1}{k^2}+n\sum_{j=n}^{3n}\frac{1}{j^2}=\frac{8}{3}+\frac{19}{18n}+O(n^{-2}),$$
$$16n^2\sum_{k=2n}^{\infty}\frac{1}{k^4}+n^2\sum_{j=n}^{3n}\frac{1}{j^4}=\frac{80}{81n}+O(n^{-2}),$$
$$64n^3\sum_{k=2n}^{\infty}\frac{1}{k^6}+n^3\sum_{j=n}^{3n}\frac{1}{j^6}=O(n^{-2}).$$

The first two identities can be checked from

$$\sum_{k=2n}^{\infty}k^{-2}=\frac{1}{2n}+\frac{1}{8n^2}+O(n^{-3}),\qquad\sum_{j=n}^{3n}j^{-2}=\frac{2}{3n}+\frac{5}{9n^2}+O(n^{-3}),$$

and the corresponding leading fourth-power integrals. Since every argument of the logarithms is at most x/n, Taylor's formula is uniform for $0\le x\le n^{1/4}$ and gives

$$L_n(x)=\frac{8}{3}x+\frac{1}{n}\left(\frac{19}{18}x-\frac{40}{81}x^2\right)+O\left(\frac{x+x^2+x^3}{n^2}\right).$$

Consequently on the same range

$$e_n(x)=e^{-8x/3}\left[1-\frac{1}{n}\left(\frac{19}{18}x-\frac{40}{81}x^2\right)+O\left(\frac{(1+x)^4}{n^2}\right)\right].$$

These expansions may be integrated with error $O(n^{-2})$. To justify this assertion explicitly, for $n^{1/4}\le x\le n$ the earlier product bound is at most $e^{-2x/5}$. For $x\ge n$,

$$\int_n^{\infty}\sqrt{x}\left(1+\frac{x}{4n}\right)^{-2n}dx\le Cn^{3/2}(5/4)^{-n}\qquad(n\ge 4),$$

by putting $x=ny$ and retaining four powers for the integrable factor. Both tails are smaller than every negative power of n.

Under the density proportional to $\sqrt{x}\,e^{-8x/3}$, the first two moments are $9/16$ and $135/256$. Combining the integrated expansion with (112), the relative coefficient of $1/n$ is

$$\frac{1}{16}-\frac{19}{18}\frac{9}{16}+\frac{40}{81}\frac{135}{256}=-\frac{13}{48}.$$

This yields the first correction $-13\sqrt{3}/(256n)$.

The part of (111) containing the extra exponential can be handled before the substitution $u=nx$. Positivity of the product factors implies

$$0\le 1-E_n(u)\le u\left(4\sum_{k=2n}^{\infty}k^{-2}+\sum_{j=n}^{3n}j^{-2}\right)\le\frac{Cu}{n}.$$

Therefore

$$\int_0^{\infty}\sqrt{u}\,e^{-2\pi\sqrt{u}}E_n(u)\,du=\int_0^{\infty}\sqrt{u}\,e^{-2\pi\sqrt{u}}\,du+O(n^{-1})=\frac{1}{2\pi^3}+O(n^{-1}).$$

Multiplication by $a_n/(4\pi)$ gives $\sqrt{2}/(\pi^{7/2}n^{3/2})+O(n^{-5/2})$. This proves the complete stated expansion. □

Lemma 26.2 (An explicit uniform rational improvement)**.** *For every $n \geq 1$,*

$$0 < \frac{I_n}{\mathcal{U}_n} < \frac{488}{945}.$$

Proof. Use the polynomials and measure from the proper-kernel bound:

$$S_r(u) = \frac{4^r}{(2r)!} \prod_{k=0}^{r-1} (4u + k^2), \qquad d\nu(u) = \frac{du}{2\sinh(2\pi\sqrt{u})}, \qquad \int_0^\infty S_r \, d\nu = \frac{1}{16}.$$

The last normalization is the identity already proved in establishing $I_n < \mathcal{U}_n$. Thus $dP_r = 16 S_r \, d\nu$ is a probability measure. The likelihood ratio

$$g_r(u) = \frac{S_{r+1}(u)}{S_r(u)} = \frac{4(4u + r^2)}{(2r+2)(2r+1)}$$

is strictly increasing in u, and its P_r-mean is one. For the strictly decreasing function $f(u) = e^{-2\pi\sqrt{u}}$,

$$\mathbb{E}_{P_{r+1}} f - \mathbb{E}_{P_r} f = \mathrm{Cov}_{P_r}(g_r, f) < 0.$$

Indeed, twice this covariance is the expectation of $(g_r(u) - g_r(v))(f(u) - f(v))$ for two independent P_r samples; the product is negative off the diagonal. Hence $J_r = \int S_r e^{-2\pi\sqrt{u}} \, d\nu$ decreases with r. Since $e^{-2\pi\sqrt{u}} d\nu = du/(e^{4\pi\sqrt{u}} - 1)$ and $S_2(u) = 32u^2/3 + 8u/3$, the elementary Bose moments give

$$J_2 = \frac{32}{3} \frac{1}{16128} + \frac{8}{3} \frac{1}{1920} = \frac{1}{1512} + \frac{1}{720} = \frac{31}{15120}.$$

For completeness, the moments here follow from

$$\int_0^\infty \frac{u^k \, du}{e^{4\pi\sqrt{u}} - 1} = \frac{2\Gamma(2k+2)\zeta(2k+2)}{(4\pi)^{2k+2}}$$

at $k = 1, 2$, with $\zeta(4) = \pi^4/90$ and $\zeta(6) = \pi^6/945$. Writing $D_n(u) = \prod_{j=n}^{3n} (u + j^2)$, exact normalization yields

$$\frac{I_n}{\mathcal{U}_n} = 8 \int_0^\infty S_{2n}(u) \frac{D_n(0)}{D_n(u)} (1 + e^{-2\pi\sqrt{u}}) \, d\nu(u) < \frac{1}{2} + 8J_{2n} \leq \frac{1}{2} + 8J_2 = \frac{488}{945}.$$

The first inequality is strict because $D_n(0) < D_n(u)$ for $u > 0$. □

Corollary 26.3 (A constant improvement and its precise scope)**.** *Put $\rho_n = I_n/\mathcal{U}_n$ and $\rho_* = \sup_{n \geq 1} \rho_n$. Then*

$$\frac{3\sqrt{3}}{16} \leq \rho_* < \frac{488}{945} < 1.$$

For every finite real coefficient vector c,

$$\left| \sum_n c_n I_n \right| \leq \rho_* \sum_n \mathcal{U}_n |c_n|.$$

For vectors supported on indices $n \geq M$, the best universal constant in this inequality is $\sup_{n \geq M} \rho_n$, and it converges to $3\sqrt{3}/16$ as $M \to \infty$.

Proof. Lemma 26.2 and the limit strictly below 488/945 give $\rho_* < 488/945$ by separating a finite prefix from a tail. The triangle inequality proves the estimate. Single-coordinate vectors show that the stated constants are optimal for the indicated unrestricted coefficient spaces. This last optimality assertion does not impose prime-log cancellation and hence is not an optimality assertion for the quotient norms. □

27 Thin-block cancellation and the formal-zero obstruction

The coefficients $E_{n,\ell}$ and p_n are those of the common kernel identity. The sufficient clearing factor $4\,\mathrm{lcm}(1,\dots,3M)$ used in this construction can be replaced by the factor two proved below; the exponential rates are unchanged.

Theorem 27.1 (Thin-block cancellation with an exponential bound, possibly zero)**.** *For integers $N \geq 2$, define using integer operations*

$$h_N = \max\left(1, \left\lfloor \sqrt{\lfloor \log_2 N \rfloor} \right\rfloor\right), \quad M = N + \left\lceil \frac{N}{h_N} \right\rceil, \quad d = M - N + 1,$$

$$D_N = 4\,\mathrm{lcm}(1,\dots,3M), \qquad \delta_* = \log(729/16) - 3 > 0.$$

For every sufficiently large N, there is a nonzero integer vector $c = (c_N, \dots, c_M)$ satisfying

$$\sum_{n=N}^{M} c_n E_{n,\ell} = 0 \quad \textit{for every prime } \ell, \qquad \log \max(1, \|c\|_\infty) = O\left(\frac{N}{\sqrt{\log N}}\right).$$

Its exactly log-free form is

$$F_N = D_N \sum_{n=N}^{M} c_n I_n = Q_N \gamma - P_N, \qquad Q_N, P_N \in \mathbb{Z}, \tag{113}$$

and satisfies

$$|F_N| \leq \exp(-\delta_* N + o(N)). \tag{114}$$

This is an upper bound; F_N is allowed to be exactly zero.

Indeed, for every sufficiently large N, there also exists a nonzero integer vector of the same height order for which all prime-log coefficients, Q_N, and P_N are zero simultaneously. Its rational kernel $\sum c_n R_n$ is nonzero and changes sign on $(0,\infty)$.

Proof. Use the integer log matrix

$$V_{\ell,n} = 2E_{n,\ell}, \qquad \ell \leq 3M \text{ prime}, \quad N \leq n \leq M.$$

It has $r = \pi(3M)$ rows and d columns, and its entries have absolute value at most the integer

$$H_N = 6(2M+1)\lceil \log_2(3M) \rceil 2^{12M}.$$

The prime number theorem [7] and the chosen block give

$$M/N \to 1, \quad d \asymp \frac{N}{\sqrt{\log N}}, \quad r \asymp \frac{N}{\log N}, \quad \log H_N = O(N).$$

Thus $d > r$ for all sufficiently large N.

Here is the full counting argument. For any r-by-d integer matrix with entries bounded by H, let K be a positive integer such that

$$K^{d-r} > (3dH)^r.$$

The $(K+1)^d$ vectors in $\{0,\dots,K\}^d$ have images with at most $(2dKH+1)^r \leq (3dKH)^r < K^d$ possible values. Two images coincide. Subtracting the input vectors gives a nonzero integer null vector of supremum norm at most K. Taking the smallest admissible integer K yields

$$\log K \leq \frac{r}{d-r} \log(3dH) + O(1) = O\left(\frac{N}{\sqrt{\log N}}\right).$$

This proves exact log cancellation and the height estimate. It also gives a finite, though not efficient, construction by enumeration.

Since $2A_{n,j}$ is integral, every denominator in p_n divides $4\,\mathrm{lcm}(1,\dots,3M) = D_N$. Thus the coefficients

$$Q_N = D_N \sum c_n 16^n, \qquad P_N = D_N \sum c_n p_n$$

in (113) are integers. Positivity and monotonicity of the raw bounds give

$$|F_N| \le D_N dK \mathcal{U}_N.$$

The same prime number theorem gives $\log D_N = 3M + o(M) = 3N + o(N)$, while $\log(dK) = o(N)$. Lemma 22.1 therefore proves (114). The margin is positive since $e < 3$ gives $e^3 < 27 < 729/16$.

To see exactly why nonvanishing is missing, append to V the two integer rows $(16^n)_{n=N}^{M}$ and $(D_N p_n)_{n=N}^{M}$. Their entries still have logarithmic height $O(N)$: the residue bound, $H_j \le j$, and $\log D_N = O(N)$ suffice. Replacing r by $r+2$ does not change the asymptotic height estimate. The same counting argument therefore gives a nonzero vector annihilating all three types of coefficients. Its integral is exactly zero, as a formal identity, independently of the arithmetic nature of γ.

Let n_* be the largest index with nonzero coefficient. The pole at $u = -(3n_*)^2$ occurs in R_{n_*} with nonzero residue and in none of the smaller-index kernels. Hence $\sum c_n R_n$ is not the zero rational function. It is continuous on $(0,\infty)$ and has zero integral against a strictly positive measure with full support. It must take both signs there. This proves the last assertion. □

27.1 A fixed-degree positive construction from two independent pairs

Let $\mathcal{K}_N = \ker_{\mathbb{Z}} V$ and define the integer coefficient map

$$\Phi_N(c) = \left(D_N \sum c_n 16^n,\ D_N \sum c_n p_n \right).$$

For $c \in \mathcal{K}_N$, the corresponding form is $F_c(\gamma) = Q_c\gamma - P_c$. Define $\sigma_2(N)$ as the least value of $\max(\|c^{(1)}\|_\infty, \|c^{(2)}\|_\infty)$ among pairs in $\mathcal{K}_N$ with

$$\det \begin{pmatrix} Q_1 & P_1 \\ Q_2 & P_2 \end{pmatrix} \neq 0.$$

Set $\sigma_2(N) = \infty$ if no such pair exists. This definition is a property of explicitly given integer and rational matrices. It does not compare any computed value to γ.

Proposition 27.2 (A positive quadratic certificate and its exact missing bound)**.** *For every pair with nonzero determinant, the integer polynomial*

$$W_N(X) = (Q_1 X - P_1)^2 + (Q_2 X - P_2)^2$$

is strictly positive for every real X. If, along an infinite sequence of indices tending to infinity,

$$\limsup \frac{\log \sigma_2(N)}{N} < \delta_*, \tag{115}$$

then, choosing a pair attaining $\sigma_2(N)$ at each of those indices, the resulting polynomials satisfy $W_N(\gamma) \to 0$, and γ is irrational. No estimate (115) is proved here.

Proof. Two independent coefficient pairs cannot vanish at the same real argument. Equivalently the leading coefficient is positive and the discriminant of W_N is

$$-4(Q_1P_2 - Q_2P_1)^2 < 0.$$

For a pair attaining $\sigma_2(N)$, the raw estimates give

$$|F_{c^{(i)}}(\gamma)| \le D_N d\sigma_2(N)\mathcal{U}_N.$$

The logarithm of this upper bound is $-\delta_* N + \log \sigma_2(N) + o(N)$. Under (115) both forms tend to zero, and so does their sum of squares. If $\gamma = a/b$ were rational, then

$$b^2 W_N(\gamma) = (aQ_1 - bP_1)^2 + (aQ_2 - bP_2)^2 \in \mathbb{Z}_{>0},$$

contradicting convergence to zero. The degree is the fixed number 2, so this integerization introduces the fixed factor b^2 only. □

The preceding construction separates positivity from the sign of an integral kernel. It replaces that sign problem by an exact determinant test. It does not supply cheap independent pairs: the counting theorem controls a short null vector, which can lie in $\ker \Phi_N$.

Proposition 27.3 (Why dimension and entry height alone do not bound the pair)**.** *For integers $H \ge 2$, $s \ge 1$, and any number of free coordinates $z_1, \dots, z_k$ with $k \ge 1$, impose*

$$x_i - Hx_{i+1} = 0, \quad y_i - Hy_{i+1} = 0 \quad (0 \le i < s), \qquad \Phi(x, y, z) = (x_s, y_s).$$

The constraint matrix has rank $2s$, entries of absolute value at most H, and arbitrarily many independent null vectors of height 1 with $\Phi = 0$. Appending the two rows of Φ raises its rank by 2. Nevertheless the least height of two null vectors with independent Φ-images is exactly H^s.

Proof. Every null vector has $x_i = H^{s-i}x_s$ and $y_i = H^{s-i}y_s$. If its height is less than H^s, both x_s and y_s are zero. The free coordinate vectors have height 1, and the two choices $(x_s, y_s) = (1, 0), (0, 1)$ give independent images at height exactly H^s. The rank assertions follow directly from the independent chain constraints. □

This is not a failure theorem for the specific proper-kernel matrix. It proves that its additional arithmetic structure must be used: dimension, entry size, and formal rank alone cannot upgrade the counting theorem to the required independent-pair estimate.

28 An actual-matrix height barrier for independent signed outputs

The counting theorem produces short vectors in the integer kernel of the prime-log matrix, but it does not provide two independent coefficient pairs. The following additional synthesis result shows that this distinction is unavoidable for the actual matrices considered here: subexponential-height log-null vectors can have formal outputs of rank at most one. The proof combines the common gamma-coefficient factor with the integral estimate already established above. The underlying integer determinant argument and Smith normal form are standard tools; no historical-priority claim is made for them.

Lemma 28.1 (An integer-output determinant bound)**.** *Let $\Lambda \subseteq \mathbb{Z}^d$ be a lattice, let $a, b : \Lambda \to \mathbb{Z}$ be homomorphisms, and let θ be real. Suppose that, for constants $A, E > 0$ and every $c \in \Lambda$,*

$$|a(c)| \le A\|c\|_\infty, \qquad |\theta a(c) - b(c)| \le E\|c\|_\infty.$$

Put $\mathcal{L} = \{(a(c), b(c)) : c \in \Lambda\} \subseteq \mathbb{Z}^2$. If $\mathcal{L}$ has rank two, let $j = [\mathbb{Z}^2 : \mathcal{L}]$. Then every pair with independent outputs satisfies

$$\|c_1\|_\infty \|c_2\|_\infty \ge \frac{j}{2AE}. \tag{116}$$

In particular, if $2AEK^2 < 1$, all outputs with $\|c\|_\infty \le K$ span a space of dimension at most one, whether or not the full image has rank two.

When the full image has rank two, j is the greatest common divisor of the determinants of all pairs of image vectors. Thus it concerns the actual image lattice, which need not be saturated in $\mathbb{Z}^2$.

Proof. Set $e_i = \theta a(c_i) - b(c_i)$. Cancellation of θ gives

$$a(c_1)b(c_2) - a(c_2)b(c_1) = a(c_2)e_1 - a(c_1)e_2.$$

Its absolute value is at most $2AE\|c_1\|_\infty\|c_2\|_\infty$. A nonzero determinant of two vectors of $\mathcal{L}$ is a nonzero integer multiple of the covolume j, proving (116).

For completeness, a rank-two sublattice of $\mathbb{Z}^2$ has a basis $(a_0, b_0), (0, b_1)$ with $a_0, b_1 > 0$: choose a_0 as the positive generator of its first-coordinate image, choose a vector above it, and choose b_1 as the positive generator of its vertical kernel. Subtracting a multiple of the first vector proves generation by these two vectors. The index and the absolute basis determinant are both a_0b_1. Every pair determinant is a multiple of this number, and the basis pair attains it, proving the gcd assertion.

Finally, if $2AEK^2 < 1$, every determinant of two outputs in the specified height ball is an integer of absolute value less than one, and therefore is zero. If one such output is nonzero, every other one is collinear with it; if none is nonzero, the assertion is immediate. □

Lemma 28.2 (The exact image index and the fibres of the output map)**.** *Let $C \in \mathbb{Z}^{r\times d}$ have row rank r, let $B \in \mathbb{Z}^{2\times d}$, and suppose* $\operatorname{rank}\binom{C}{B} = r + 2$. *For a full-row-rank integer matrix T with s rows, write $\Delta_s(T)$ for the positive gcd of its s-by-s minors; use $\Delta_0(C) = 1$ when $r = 0$. Then*

$$\left[\mathbb{Z}^2 : B(\ker_{\mathbb{Z}} C)\right] = \frac{\Delta_{r+2}\left(\binom{C}{B}\right)}{\Delta_r(C)}. \tag{117}$$

For any attained output $y = Bc_0$ with $Cc_0 = 0$, its entire fibre is

$$\{c \in \mathbb{Z}^d : Cc = 0,\ Bc = y\} = c_0 + \ker_{\mathbb{Z}} \begin{pmatrix} C \\ B \end{pmatrix}.$$

Both integer kernels in this statement are saturated in $\mathbb{Z}^d$. This does not assert that the image $B(\ker_{\mathbb{Z}} C)$ is saturated in $\mathbb{Z}^2$.

Proof. Put $\mathcal{H} = \binom{C}{B}\mathbb{Z}^d$, $\mathcal{C} = C\mathbb{Z}^d$, and $\mathcal{L} = B(\ker_{\mathbb{Z}} C)$. Projection to the first r coordinates gives the exact sequence of finite abelian groups

$$0 \longrightarrow \mathbb{Z}^2/\mathcal{L} \longrightarrow \mathbb{Z}^{r+2}/\mathcal{H} \longrightarrow \mathbb{Z}^r/\mathcal{C} \longrightarrow 0.$$

Indeed, a class whose first coordinate lies in $\mathcal{C}$ can be represented by $(0, y)$ after subtracting a vector of $\mathcal{H}$, and $(0, y)$ lies in $\mathcal{H}$ exactly when $y \in \mathcal{L}$. Taking group orders gives the ratio of the two image indices.

For a full-row-rank integer matrix, its image index is the gcd of its maximal minors. To see this, apply integer unimodular row and column operations to obtain Smith normal form. Such operations preserve the image index and the gcd of the maximal minors; on the diagonal form both are the product of the nonzero diagonal entries. This proves (117). The fibre identity follows by subtracting c_0. An integer kernel is saturated because $kTc = 0$, for a nonzero integer k, implies $Tc = 0$ in the torsion-free group $\mathbb{Z}^s$. □

Theorem 28.3 (Independent signed outputs have an exponential height cost)**.** *Use the proper kernels, their rational bounds $\mathcal{U}_n$ in (106), and the thin blocks of Theorem 27.1. Thus*

$$\mathcal{K}_N = \ker_{\mathbb{Z}} V, \quad D_N = 4\,\mathrm{lcm}(1,\dots,3M), \quad \delta_* = \log(729/16) - 3.$$

Set the explicit positive rational quantities

$$\mathcal{S}_N = \sum_{n=N}^{M} 16^{n-N} = \frac{16^{M-N+1}-1}{15}, \qquad \mathcal{T}_N = \sum_{n=N}^{M} \mathcal{U}_n, \qquad G_N = D_N 16^N.$$

For $c \in \mathcal{K}_N$, write the integer form as

$$F_c(\gamma) = Q_c\gamma - P_c, \qquad a_N(c) = \frac{Q_c}{G_N} = \sum_{n=N}^{M} c_n 16^{n-N} \in \mathbb{Z}.$$

Let $\sigma_2(N)$ be the least possible maximum input height of two vectors whose original coefficient pairs (Q_c, P_c) are independent, and set it to ∞ if no such pair exists. Then, for every N for which the block is defined,

$$\sigma_2(N) \geq \left(2D_N\mathcal{S}_N\mathcal{T}_N\right)^{-1/2}. \tag{118}$$

If the normalized image

$$\mathcal{L}_N = \{(a_N(c), P_c) : c \in \mathcal{K}_N\}$$

has rank two and index j_N in $\mathbb{Z}^2$, the sharper bound is

$$\sigma_2(N) \geq \sqrt{\frac{j_N}{2D_N\mathcal{S}_N\mathcal{T}_N}}. \tag{119}$$

In particular, with the extended-real convention for $\log\infty$,

$$\liminf_{N\to\infty} \frac{\log\sigma_2(N)}{N} \geq \frac{\delta_*}{2} = \frac{\log(729/16)-3}{2} > 0. \tag{120}$$

More strongly, if $K_N \geq 1$ and

$$\limsup_{N\to\infty} \frac{\log K_N}{N} < \frac{\delta_*}{2},$$

then, for all sufficiently large N, the formal outputs of all *$c \in \mathcal{K}_N$ with $\|c\|_\infty \leq K_N$ span a space of dimension at most one. In particular this holds for every subexponential height schedule $K_N = \exp(o(N))$, including the heights produced by the counting argument.*

Proof. The exact formulas preceding (114) give

$$P_c = D_N \sum_{n=N}^{M} c_n p_n \in \mathbb{Z}, \qquad F_c(\gamma) = D_N \sum_{n=N}^{M} c_n I_n.$$

Consequently

$$|a_N(c)| \leq \mathcal{S}_N\|c\|_\infty, \qquad |G_N\gamma a_N(c) - P_c| \leq D_N\mathcal{T}_N\|c\|_\infty.$$

The two original pairs are independent exactly when the normalized pairs are independent, since their first coordinates differ by the same nonzero factor G_N. Apply Lemma 28.1 with

$$\theta = G_N\gamma, \quad A = \mathcal{S}_N, \quad E = D_N\mathcal{T}_N.$$

Every nonzero normalized determinant is an integer, and in the rank-two case it is a nonzero multiple of j_N. This proves both finite bounds; if there is no independent pair, (118) holds by the convention $\sigma_2(N) = \infty$.

The thin-block choice gives $M - N = o(N)$, so

$$\log \mathcal{S}_N = o(N).$$

Since $\mathcal{U}_n$ decreases, $\mathcal{U}_N \le \mathcal{T}_N \le (M - N + 1)\mathcal{U}_N$. The proved asymptotic for $\mathcal{U}_N$ therefore gives

$$\log \mathcal{T}_N = -\log(729/16)N + o(N).$$

The already used least-common-multiple asymptotic gives $\log D_N = 3N + o(N)$. Hence

$$2D_N\mathcal{S}_N\mathcal{T}_N = \exp\{-\delta_* N + o(N)\}. \tag{121}$$

Taking logarithms of (118) proves (120). Under the stated hypothesis on K_N, the product of (121) with K_N^2 tends to zero. The final conclusion follows from the last assertion of Lemma 28.1. □

Exact arithmetic and scope. The index j_N, whenever the normalized image has rank two, can be computed without any evaluation of γ. Choose a maximal independent set of rows of the integer log matrix as C, and let the two rows of B be $(16^{n-N})_{n=N}^{M}$ and $(D_N p_n)_{n=N}^{M}$. Then Lemma 28.2 gives its exact value. All vectors with a fixed formal output differ by the saturated integer kernel of the stacked matrix; changing a vector inside that fibre changes neither its formal linear form nor its primitive arithmetic error.

The new conclusion is a lower height bound for the actual proper-kernel matrices, not just for an unrelated model matrix. It does not prove that every short output is zero, and it does not exclude independent outputs at larger heights. In particular, the previously stated sufficient target $\limsup \log \sigma_2(N)/N < \delta_*$ is not contradicted: an exponential-rate window between $\delta_*/2$ and δ_* remains. Neither existence of suitable pairs in that window nor irrationality of γ is established here.

29 A separation obstruction across different cancellation blocks

Put $\kappa_0 = \log(729/16)$ and $\delta_* = \kappa_0 - 3$. Retain the proper-kernel identity

$$I_n = 16^n\gamma - p_n + \sum_{\ell} E_{n,\ell} \log \ell, \qquad 0 < I_n < \mathcal{U}_n, \qquad \log \mathcal{U}_n = -\kappa_0 n + O(\log(n+1)).$$

The constants and kernels in this section are exactly those of the proper-kernel construction. In particular $4\,\mathrm{lcm}(1, \dots, 3M)$ clears all p_n for $n \le M$.

Lemma 29.1 (An exact comparison of two blocks)**.** *Let $N_1 \le N_2$ and let $c^{(i)}$ be integer vectors supported on $[N_i, M_i]$ whose prime-log coefficients vanish. Put*

$$H_i = \max(1, \|c^{(i)}\|_\infty), \quad S_i = \sum_{n=N_i}^{M_i} 16^{n-N_i}, \quad T_i = \sum_{n=N_i}^{M_i} \mathcal{U}_n,$$

$$a_i = \sum_{n=N_i}^{M_i} c_n^{(i)} 16^{n-N_i}, \qquad b_i = \sum_{n=N_i}^{M_i} c_n^{(i)} p_n, \qquad D = 4\,\mathrm{lcm}(1, \dots, 3\max(M_1, M_2)).$$

If

$$DH_1H_2\left(16^{N_2-N_1}S_2T_1 + S_1T_2\right) < 1, \tag{122}$$

then the two coefficient pairs $(16^{N_i}a_i, b_i)$ are proportional. If both a_i are nonzero, this means exactly

$$\frac{b_1}{16^{N_1}a_1} = \frac{b_2}{16^{N_2}a_2}.$$

Proof. Set $J_i = \sum_n c_n^{(i)} I_n$. Log cancellation gives $J_i = 16^{N_i} a_i \gamma - b_i$, while $|J_i| \le H_i T_i$. The integer

$$\Delta = a_1(Db_2) - 16^{N_2-N_1} a_2(Db_1) = D\left(16^{N_2-N_1} a_2 J_1 - a_1 J_2\right)$$

has absolute value at most the left side of (122), since $|a_i| \le H_i S_i$. It must therefore vanish. Scaling its first coordinates back by the nonzero common factor 16^{N_1} gives the asserted proportionality. □

Theorem 29.2 (Persistent short outputs force a rational limiting slope)**.** *Fix $0 \le \alpha < \delta_*/2$ and set*

$$\rho_\alpha = \frac{\log 729 - \alpha}{3 + \log 16 + \alpha} > 1.$$

Let $N_k \to \infty$ be strictly increasing integers. Suppose that integer prime-log-null vectors $c^{(k)}$ are supported on $[N_k, M_k]$, where

$$M_k = N_k + o(N_k), \qquad \log \max(1, \|c^{(k)}\|_\infty) \le \alpha N_k + o(N_k),$$

and that every formal output

$$\left(\sum_n c_n^{(k)} 16^n,\ \sum_n c_n^{(k)} p_n\right)$$

is nonzero. If

$$\limsup_{k\to\infty} \frac{N_{k+1}}{N_k} < \rho_\alpha, \tag{123}$$

then γ is rational. More precisely, the rational slopes of the formal outputs are eventually equal to γ, so all their evaluated linear forms are eventually exactly zero.

Proof. Let $D_k = 4\,\mathrm{lcm}(1, \dots, 3M_k)$ and $J_k = \sum_n c_n^{(k)} I_n$. The least-common-multiple asymptotic and the integral bound give

$$|D_k J_k| \le \exp\{-(\delta_* - \alpha)N_k + o(N_k)\} \longrightarrow 0.$$

If the first formal coefficient were zero, $D_k J_k$ would be an integer. It would vanish for sufficiently large k, contradicting nonzero formal output. Hence $a_k = \sum c_n^{(k)} 16^{n-N_k} \ne 0$ for all sufficiently large k.

For two consecutive blocks, write $N = N_k$ and $N' = N_{k+1}$. Condition (123) bounds N'/N, and $\max(M_k, M_{k+1}) = N' + o(N)$. In Lemma 29.1,

$$\log D = 3N' + o(N), \quad \log S_i = o(N), \quad \log T_1 = -\kappa_0 N + o(N), \quad \log T_2 = -\kappa_0 N' + o(N).$$

The larger of the two exponents on the left of (122) is

$$(3 + \log 16 + \alpha)N' - (\log 729 - \alpha)N + o(N).$$

Indeed its difference from the other exponent is $\log 729\,(N' - N) \ge 0$. By (123) this larger exponent tends to minus infinity linearly in N. Thus consecutive rational slopes are equal for every sufficiently large k; call the eventual rational value r.

Since a_k is a nonzero integer,

$$|\gamma - r| = \frac{|J_k|}{16^{N_k}|a_k|} \le \frac{H_k T_k}{16^{N_k}} \le \exp\{-(\log 729 - \alpha)N_k + o(N_k)\} \longrightarrow 0.$$

Therefore $r = \gamma$. The claimed eventual zero evaluation follows from the exact identities. □

Corollary 29.3 (A necessary separation for successful short forms)**.** *Suppose the vectors in Theorem 29.2 instead satisfy $J_k \neq 0$ at every index. Then necessarily*

$$\limsup_{k\to\infty} \frac{N_{k+1}}{N_k} \geq \rho_\alpha.$$

In particular, at subexponential height ($\alpha = 0$), the threshold is

$$\rho_0 = \frac{\log 729}{3 + \log 16} = 1.141892147\ldots.$$

Thus a subexponential-height nonzero-valued form at every sufficiently large block is impossible for this construction. This conclusion does not rule out sufficiently separated infinite subsequences or larger coefficient heights.

Proof. The opposite strict inequality would make the evaluated forms eventually zero by the theorem, a contradiction. □

Scope. The displayed separation threshold is sufficient for this determinant argument; no optimality is claimed. The theorem does not show that γ is rational, because persistent nonzero formal outputs have not been constructed. It does not show that γ is irrational, because nonzero evaluated forms on an infinite admissible subsequence have not been constructed. It strengthens the single-block rank obstruction by comparing different blocks.

30 Primitive lifts and the remaining quantitative target

The following formulation specializes the primitive quotient optimization of corpus manuscript [019]. It is included to state the remaining target after division by the actual output gcd; no new general optimization theorem is claimed.

Fix a finite set of indices and an exact identity

$$e_j = a_j\gamma - b_j + \sum_\ell V_{\ell j} \log \ell, \qquad a_j, b_j, V_{\ell j} \in \mathbb{Z}, \qquad |e_j| \leq \varepsilon_j \in \mathbb{Q}_{>0}.$$

Put

$$\Lambda = \ker V \cap \mathbb{Z}^s, \qquad \Phi(w) = (a^T w, b^T w), \qquad C(w) = \sum_{j=1}^{s} \varepsilon_j |w_j|.$$

Assume that $\Phi : \ker_{\mathbb{Q}} V \longrightarrow \mathbb{Q}^2$ has rank two. A nonzero integer pair $v = (q, p)$ is called primitive when $\gcd(|q|, |p|) = 1$.

Lemma 30.1 (Exact primitive lifting)**.** *For a primitive pair v, define the rational quotient cost*

$$F(v) = \min_{Vw=0,\ \Phi(w)=v} C(w), \qquad w \in \mathbb{R}^s.$$

The minimum has a rational optimizer, and

$$F(v) = \min_{\substack{c\in\Lambda,\ \Phi(c)=gv \\ g\in\mathbb{Z}_{>0}}} \frac{C(c)}{g}, \qquad g = \gcd(|a^T c|, |b^T c|). \tag{124}$$

In particular $|q\gamma - p| \leq F(v)$.

There is also an explicit lift from any independent output pair. If $u_1, u_2 \in \Lambda$, $\Phi(u_i) = (Q_i, P_i)$, and $\Delta = Q_1P_2 - Q_2P_1 \neq 0$, then

$$c = \operatorname{sgn}(\Delta)\big((P_2q - Q_2p)u_1 + (Q_1p - P_1q)u_2\big)$$

satisfies $\Phi(c) = |\Delta|v$. *Its actual output gcd is exactly* $|\Delta|$, *and therefore*

$$F(v) \le \frac{|P_2q - Q_2p|C(u_1) + |Q_1p - P_1q|C(u_2)}{|\Delta|}. \tag{125}$$

Proof. The feasible fiber is nonempty by the rank assumption. Positive weights make its intersections with sublevel sets of C compact. The minimum exists and can be attained rationally: after writing $w = w^+ - w^-$, this is a feasible rational linear program with an attained optimum. Every integral lift in (124) gives the rational lift c/g. Conversely, multiply a rational optimizer by a positive common denominator g of its coordinates. Its output is gv; primitivity of v makes the output gcd exactly g. Homogeneity of C proves the equality, including attainment. Log cancellation gives $e^T w = q\gamma - p$, proving the error bound. Direct multiplication of the two output columns proves the adjugate lift formula and its bound. □

The same proof with the supremum norm gives the exact normalized coefficient cost

$$\tau(v) := \min_{Vw=0,\ \Phi(w)=v} \|w\|_\infty = \min_{\substack{c\in\Lambda,\ \Phi(c)=gv \\ g\in\mathbb{Z}_{>0}}} \frac{\|c\|_\infty}{g}, \qquad F(v) \le \left(\sum_j \varepsilon_j\right) \tau(v). \tag{126}$$

Thus τ charges the coefficient size *after* the exact output gcd has been removed. It is not the raw coefficient height.

Proposition 30.2 (A sufficient test in the primitive quotient cost)**.** *For an unbounded sequence of blocks, let* F_N *be the cost above. If there are primitive pairs* $v_N^{(1)}, v_N^{(2)}$ *such that*

$$\det(v_N^{(1)}, v_N^{(2)}) \neq 0, \qquad \max_{i=1,2} F_N(v_N^{(i)}) \longrightarrow 0,$$

then γ *is irrational. Alternatively, one primitive pair per block suffices if its evaluated form is proved nonzero and its cost tends to zero.*

Proof. Two independent pairs cannot both lie in the kernel of the nonzero linear functional $(q,p) \mapsto q\gamma - p$. Hence at each block at least one evaluated integer form is nonzero and has absolute value at most the displayed maximum. If $\gamma = A/B$ with integers A, B, $B > 0$, every nonzero integer form has absolute value at least $1/B$, which is a contradiction. The one-pair version is identical. □

For the proper kernels on $[N, M]$, take

$$D_M = 4\,\mathrm{lcm}(1,\dots,3M), \qquad e_n = D_M I_n, \qquad \varepsilon_n = D_M \mathcal{U}_n,$$
$$\Phi_N(c) = \left(D_M \sum c_n 16^n,\ D_M \sum c_n p_n\right).$$

The log matrix is $(D_M E_{n,\ell})_{\ell,n}$; it has exactly the same rational and integer kernels as $(2E_{n,\ell})_{\ell,n}$. For $M = N + o(N)$ the established bounds imply

$$\sum_{n=N}^{M} \varepsilon_n = \exp\{-(\log(729/16) - 3)N + o(N)\} = \exp\{-\delta_* N + o(N)\}.$$

Consequently a sufficient concrete target is an infinite sequence of such blocks with independent primitive pairs and, for one fixed $\eta > 0$,

$$\max_{i=1,2} \tau_N(v_N^{(i)}) \le \exp\{(\delta_* - \eta)N\}. \tag{127}$$

Equations (126) and (127) would give primitive integer errors tending to zero, while independence would supply the required nonvanishing.

Scope. Rank two makes every primitive rational slope accessible through a multiple, as the explicit adjugate lift shows. It proves neither a small quotient cost nor (127). The latter is an unproved sufficient target, not an irrationality result. A lower bound for the raw height of two independent integer lifts cannot be transferred to F_N or τ_N without controlling their actual output gcds. Conversely, merely allowing large gcds gives no quantitative upper bound for these costs.

31 Reverse design from a denominator-cleared decay budget

Put

$$\kappa = \log(729/16), \qquad \delta = \kappa - 3 > 0, \qquad \varepsilon_2 = \delta - \log 2 = \log(729/32) - 3 > 0.$$

All logarithms in asymptotic expressions are natural. The last positivity follows, for example, from $e < 11/4$ and $(11/4)^3 = 1331/64 < 729/32$. For $m \geq 1$ define

$$D_m = 4\,\mathrm{lcm}(1, \dots, 3m).$$

The proper kernels, integrals, and rational bounds are

$$R_m(u) = \frac{\prod_{k=0}^{2m-1}(4u + k^2)}{\prod_{j=m}^{3m}(u + j^2)}, \qquad I_m = \int_0^\infty \frac{R_m(u)}{e^{2\pi\sqrt{u}} - 1}\, du, \qquad \mathcal{U}_m = \frac{(4m)!((m-1)!)^2}{8\,16^m((3m)!)^2}.$$

Retain the exact identity

$$I_m = 16^m\gamma - p_m + \sum_\ell E_{m,\ell} \log \ell, \qquad p_m = \sum_{j=m}^{3m} A_{m,j}\left(H_j - \frac{1}{2j}\right), \quad E_{m,\ell} = \sum_{j=m}^{3m} A_{m,j} v_\ell(j),$$

where

$$A_{m,j} = (-1)^{j-m} \frac{4j^2(2j + 2m - 1)!(m + j - 1)!}{(2j - 2m)!(j - m)!(3m - j)!(3m + j)!}.$$

The established bounds used below are

$$2A_{m,j} \in \mathbb{Z}, \qquad 0 < I_m < \frac{488}{945}\mathcal{U}_m, \qquad \frac{\mathcal{U}_{m+1}}{\mathcal{U}_m} \leq \frac{256}{729}.$$

The weaker inequality $I_m < \mathcal{U}_m$ would give all the same exponential conclusions; the improved constant only sharpens the finite budget.

Theorem 31.1 (An unconditional positive seed after denominator clearing)**.** *For every $n \geq 1$, all coefficients in*

$$D_n I_n = (D_n 16^n)\gamma - D_n p_n + \sum_\ell D_n E_{n,\ell} \log \ell$$

are integers. Moreover

$$0 < D_n I_n = \exp\{-\delta n + o(n)\} \longrightarrow 0.$$

Thus denominator-cleared decay and nonvanishing are already available for this raw family. Its prime-logarithm terms have not been removed.

Proof. Each $A_{n,j}$ is a half-integer. Harmonic denominators divide $\mathrm{lcm}(1, \dots, 3n)$, while the additional $1/(2j)$ requires at most one further factor two. Hence $D_n p_n$ is integral, as are $D_n E_{n,\ell}$ and $D_n 16^n$.

Writing $\psi(x) = \sum_{\ell^a \leq x,\ \ell \text{ prime}} \log \ell$, one has $\log D_n = \log 4 + \psi(3n) = 3n + o(n)$ by the prime number theorem. Stirling's formula gives

$$\mathcal{U}_n = \frac{\sqrt{2\pi}}{12}\, n^{-3/2} \left(\frac{16}{729}\right)^n (1 + O(n^{-1})).$$

The proved limit $I_n/\mathcal{U}_n \to 3\sqrt{3}/16 > 0$ therefore gives the claimed exponential rate. Positivity follows directly from the integral. □

Theorem 31.2 (A uniform budget fixed before cancellation)**.** *Let $n \geq 1$, $r \geq 0$, $M = n + r$, and $H \geq 1$. For every integer vector $c = (c_0, \ldots, c_r)$ with $\|c\|_\infty \leq H$, define*

$$F_c = D_M \sum_{k=0}^{r} c_k I_{n+k}.$$

All coefficients of γ, 1, and the prime logarithms in F_c are integers, and

$$\boxed{|F_c| < \frac{13176}{16555}\, D_M H \mathcal{U}_n.} \tag{RB}$$

If $\sum_{k=0}^{r} c_k E_{n+k,\ell} = 0$ for every prime, then

$$F_c = Q_c\gamma - P_c, \qquad Q_c = D_M \sum_{k=0}^{r} c_k 16^{n+k}, \quad P_c = D_M \sum_{k=0}^{r} c_k p_{n+k} \in \mathbb{Z},$$

with precisely the same bound.

Suppose along an infinite sequence $n \to \infty$ that $r/n \leq \rho + o(1)$ and $\log H/n \leq \eta + o(1)$, where ρ, η are finite nonnegative constants. If

$$\boxed{\eta + 3(1 + \rho) < \kappa,}$$

then the bound in (RB) tends to zero exponentially, uniformly over every coefficient vector in the specified integer box.

Proof. The denominator argument in Theorem 31.1 applies simultaneously to every index at most M. The triangle inequality and the geometric upper bound give

$$|F_c| \leq D_M H \sum_{k=0}^{r} I_{n+k} < \frac{488}{945} D_M H \mathcal{U}_n \sum_{k=0}^{\infty} (256/729)^k = \frac{13176}{16555} D_M H \mathcal{U}_n.$$

No cancellation hypothesis was used in this estimate. Imposing exact prime-log cancellation therefore changes the form of the output without spending any extra budget, provided the coefficients remain in the preassigned box. Finally,

$$\log(D_M H \mathcal{U}_n) \leq \{\eta + 3(1 + \rho) - \kappa + o(1)\}n.$$

This proves the assertion. In particular for $r = o(n)$ and $H = 2^n$ the uniform bound is $\exp\{-\varepsilon_2 n + o(n)\}$. □

32 An exact residue bound and a constructive integer box

The sharper residue-height input is Lemma 22.5, with $\beta = (16399/1296)^2$.

Lemma 32.1 (Counting with separate row heights)**.** *Let A be an integer matrix with s rows and d columns, and suppose that the entries of row i have absolute value at most $B_i \geq 1$. If an integer $K \geq 1$ satisfies*

$$K^{d-s} > \prod_{i=1}^{s} (3dB_i),$$

there exists a nonzero integer vector c with $Ac = 0$ and $\|c\|_\infty \leq K$.

Proof. The $(K+1)^d$ vectors in $\{0,\dots,K\}^d$ have at most

$$\prod_{i=1}^{s}(2dKB_i+1) \le K^s \prod_{i=1}^{s}(3dB_i) < K^d$$

different images. Two images coincide; their difference is the required vector. Enumeration of this finite input set gives a deterministic construction whenever the displayed inequality holds. □

Theorem 32.2 (Exact cancellation inside a preassigned decay budget)**.** *For $n \ge 2$ put*

$$r_n = \left\lceil \frac{25n}{\log n} \right\rceil, \qquad M_n = n + r_n, \qquad K_n = 2^n.$$

For every sufficiently large n, there exists a nonzero integer vector $c = (c_0,\dots,c_{r_n})$ with

$$\|c\|_\infty \le 2^n, \qquad \sum_{k=0}^{r_n} c_k E_{n+k,\ell} = 0 \quad \text{for every prime } \ell.$$

It produces an exactly log-free integer form

$$F_n = D_{M_n} \sum_{k=0}^{r_n} c_k I_{n+k} = Q_n\gamma - P_n, \qquad Q_n, P_n \in \mathbb{Z},$$

satisfying the unconditional bound

$$\boxed{|F_n| \le \exp\{-\varepsilon_2 n + o(n)\}.}$$

The construction uses no digits of γ. It does not assert $F_n \ne 0$ or $(Q_n,P_n) \ne (0,0)$.

In fact, the same block and the same height bound also contain, for every sufficiently large n, a nonzero integer vector for which all prime-log coefficients and both formal output coordinates vanish. The corresponding rational kernel is nonzero and changes sign on $(0,\infty)$.

Proof. Write $M = M_n$, $d = r_n + 1$, and $s = \pi(3M)$. The integer log matrix $V = (2E_{n+k,\ell})_{\ell,k}$ has its entries bounded by

$$B_M = \left\lceil 6(2M+1)\lceil \log_2(3M)\rceil\, \beta^M \right\rceil.$$

This follows from Lemma 22.5, summing the $2m+1$ possible residues and using $v_\ell(j) \le \log_2(3M)$. The prime number theorem and the definition of the block give

$$M/n \to 1, \qquad d = (25+o(1))\frac{n}{\log n}, \qquad s = (3+o(1))\frac{n}{\log n}, \qquad \log B_M = M\log\beta + O(\log M).$$

For $K = 2^n$, the two sides of the logarithmic counting inequality have respective asymptotics

$$(d-s)\log K = (22\log 2 + o(1))\frac{n^2}{\log n},$$
$$s\log(3dB_M) = (3\log\beta + o(1))\frac{n^2}{\log n}.$$

Lemma 22.5 gives the strict margin $3\log\beta < 22\log 2$. Thus the counting inequality holds for every sufficiently large n, proving the required log-null vector. Since $r_n = o(n)$, Theorem 31.2 proves the claimed error bound uniformly for all such vectors.

For the formal-zero assertion, append the two integer rows

$$(16^k)_{k=0}^{r_n}, \qquad (D_M p_{n+k})_{k=0}^{r_n}.$$

The first has row height $B_Q = 16^{r_n}$. The second may be bounded by

$$B_P = \max\left(1, \left\lceil 3D_M(2M+1)(3M+1)\beta^M \right\rceil\right).$$

Indeed $|A_{m,j}| \le 3\beta^M$ and $H_j + 1/(2j) \le 3M+1$. Therefore $\log B_Q = O(r_n)$ and $\log B_P = O(n)$. Use separate row heights in Lemma 32.1. Its logarithmic inequality now reads

$$(d-s-2)\log K > s\log(3dB_M) + \log(3dB_Q) + \log(3dB_P).$$

Compared with the previous inequality, the changes on both sides are $O(n)$, while its positive margin is of order $n^2/\log n$. Hence it too holds for every sufficiently large n. The first appended row kills the original gamma coefficient, since $\sum c_k 16^{n+k} = 16^n \sum c_k 16^k$; the second kills the constant coefficient. The resulting integral is exactly zero.

Let m be the greatest index with nonzero coefficient in this vector. The pole at $u = -(3m)^2$ occurs with nonzero residue only in R_m among the selected kernels. Thus the combined rational function is not zero. It is continuous on $(0,\infty)$, has zero integral, and the integrating measure is positive with full support. It must consequently take both signs. □

An integer-operation version. The width may instead be defined by

$$\widetilde{r}_n = \left\lceil \frac{37n}{\lfloor \log_2 n \rfloor} \right\rceil \qquad (n \ge 2).$$

Its asymptotic constant is $37\log 2 > 25$, so the same proof applies. Here $\lfloor \log_2 n \rfloor$ is the binary length of n minus one, and all indices and coefficient-height bounds are computed by integer operations. One may first check the exact counting inequality and then enumerate inputs in lexicographic order until a repeated log image appears. This yields a completely deterministic finite selection procedure for every sufficiently large index, with no evaluation of γ or of the integrals.

33 Which extension preserves the budget and guarantees nonvanishing?

Proposition 33.1 (The exact height window for independent outputs)**.** *For a block $[n, n+r]$, put*

$$S_r = \sum_{k=0}^{r} 16^k, \qquad T_{n,r} = \sum_{k=0}^{r} \mathcal{U}_{n+k}, \qquad M = n+r.$$

If two integer log-null vectors $c^{(1)}, c^{(2)}$ have independent formal output pairs, then

$$\|c^{(1)}\|_\infty \|c^{(2)}\|_\infty \ge \frac{1}{2D_M S_r T_{n,r}}.$$

For $r = o(n)$, a common height bound H_n for such a pair therefore necessarily satisfies

$$\liminf \frac{\log H_n}{n} \ge \frac{\delta}{2}.$$

On the other hand, a common height exponent strictly below δ is sufficient to preserve denominator-cleared decay. In particular, $H_n = 2^n$ lies strictly between these two exponential thresholds.

Proof. Define integral normalized outputs

$$a(c) = \sum_{k=0}^{r} c_k 16^k, \qquad P(c) = D_M \sum_{k=0}^{r} c_k p_{n+k}.$$

The original first coordinate is $G_n a(c)$, with $G_n = D_M 16^n$, so independence is unchanged. For $\theta_n = G_n\gamma$ the error is $e(c) = \theta_n a(c) - P(c) = F_c$, and

$$|a(c)| \le S_r \|c\|_\infty, \qquad |e(c)| \le D_M T_{n,r} \|c\|_\infty.$$

Cancellation of θ_n in the two-by-two determinant gives

$$1 \le |a(c^{(1)})P(c^{(2)}) - a(c^{(2)})P(c^{(1)})| \le 2D_M S_r T_{n,r} \|c^{(1)}\|_\infty \|c^{(2)}\|_\infty.$$

When $r = o(n)$, one has $\log S_r = o(n)$, $\log T_{n,r} = -\kappa n + o(n)$, and $\log D_M = 3n + o(n)$. This proves the lower threshold. The upper threshold follows from Theorem 31.2. Finally $\delta/2 < \log 2 < \delta$; the right inequality was checked above, and the left follows, for example, from $e > 5/2$. □

A concrete remaining target. For the explicit width in Theorem 32.2, prove that the integer box $\|c\|_\infty \le 2^n$ contains two log-null vectors with independent formal output pairs for infinitely many n. This is a purely exact-arithmetic statement about specified matrices and integer boxes. It has not been established here. If it holds, choose such pairs and form

$$W_n(X) = (Q_{1,n}X - P_{1,n})^2 + (Q_{2,n}X - P_{2,n})^2.$$

Their nonzero determinant makes $W_n(X) > 0$ for every real X, while the already secured budget gives $W_n(\gamma) \le \exp\{-2\varepsilon_2 n + o(n)\} \to 0$. Under $\gamma = a/b$, the number $b^2 W_n(\gamma)$ is then a positive integer tending to zero, a contradiction. This final paragraph is an implication from the stated independent-pair target, not a claim that the target or irrationality has been proved.

Status of the five requirements in this construction. The concrete counting theorem supplies exact integer coefficients with no remaining logarithms, a denominator-cleared bound tending to zero, and a uniform existence proof for every sufficiently large index, without an unproved arithmetic assumption. Its unresolved requirement is a nonzero final value. The existence of formal-zero vectors in the very same box proves why that requirement cannot be inferred merely from nonzero input coefficients. The independent-pair target supplies a specific route to nonvanishing while preserving the budget already secured.

Part IV

Primitive quotient norms and exact transfer geometry

34 Primitive quotient costs and exact finite certificates

For $N \geq 17$ and $v \in \mathbb{R}^2$, define

$$C_N(c) = \sum_{n=1}^{N} \mathcal{U}_n |c_n|, \qquad F_N(v) = \min_{\substack{V_N c = 0 \\ \Psi_N(c) = v}} C_N(c), \qquad K_N = \{v : F_N(v) \leq 1\}. \tag{128}$$

Theorem 23.3 guarantees feasibility for every real target. A bounded-cost set of coefficients is compact, so the minimum is attained. The norm C_N and surjectivity of Ψ_N on the kernel show that F_N is a norm. Its unit ball is a centrally symmetric rational polygon: it is the image of a rational polytope obtained by intersecting a weighted cross-polytope with $\ker V_N$. For a rational target, an optimal rational lift exists by linear programming over rational data.

Lemma 34.1 (The quotient cost includes the whole output gcd)**.** *Let $(q, p) \in \mathbb{Z}^2$ be primitive, and let $c \in \mathbb{Q}^N$ be a log-null lift with $\Psi_N(c) = (q, p)$. If a positive integer g clears every denominator of c, then*

$$\Psi_N(gc) = (gq, gp), \qquad \gcd(|gq|, |gp|) = g.$$

Consequently the upper cost after this entire gcd has been removed is exactly $C_N(c)$, and

$$|q\gamma - p| \leq F_N(q, p).$$

Proof. Linearity gives the output, primitiveness gives its gcd, and $C_N(gc) = gC_N(c)$. The integral identity and the triangle inequality give the bound for every lift; minimize over lifts. □

Define the first and second lattice minima by

$$\mu_1(N) = \min_{0 \neq v \in \mathbb{Z}^2} F_N(v), \qquad \mu_2(N) = \min_{\substack{v, w \in \mathbb{Z}^2 \\ \det(v,w) \neq 0}} \max\{F_N(v), F_N(w)\}.$$

Minimizers can be taken primitive. Both minima are nonincreasing in N, because an old lift can be extended by zeros. The independence condition in μ_2 is what supplies nonvanishing in Corollary 1.2.

35 Certified finite minima and the area criterion

The notation F_N, its unit ball K_N, and the lattice successive minima is fixed in the preceding section. The following finite certificates use that same primitive quotient norm.

Theorem 35.1 (Finite global optima for three proper-kernel blocks)**.** *The first and second successive minima are attained, respectively, at the following pairs; a change of sign has no effect:*

N	$v_1 = (q_1, p_1)$	$v_2 = (q_2, p_2)$
20	26685 , 15403	85313 , 49244
22	559595 , 323007	2739347 , 1581194
26	15700603 , 9062634	7040223 , 4063727

The determinant of the two displayed targets has absolute value one in every row. The following are strict rational inequalities:

N	*bounds for* $\mu_1(N)$	*bounds for* $\mu_2(N)$
20	$607002/10^{10} < \mu_1 < 607003/10^{10}$	$883918/10^{10} < \mu_2 < 883919/10^{10}$
22	$189360/10^{11} < \mu_1 < 189361/10^{11}$	$333400/10^{11} < \mu_2 < 333401/10^{11}$
26	$345529/10^{13} < \mu_1 < 345530/10^{13}$	$210024/10^{12} < \mu_2 < 210025/10^{12}$

In particular the two independent primitive forms at $N = 26$ both have certified absolute value less than $1/4700000$. At least one is nonzero, solely by their independence. These are finite certificates; no infinite decay conclusion is asserted.

Exact finite verification and its completeness. All the following operations are over the rational numbers. Row reduce V_N to obtain a basis of its rational kernel, of dimension d. Identify this kernel with $\mathbb{R}^d$. Its input unit ball is the section defined by $C_N(c) \le 1$.

Every vertex of this section lies on at least $d-1$ independent coordinate hyperplanes $c_i = 0$. Indeed, if the vanishing coordinate functionals had rank at most $d-2$, their common nullspace would contain a nonzero direction also annihilating the active linear cost functional. A sufficiently short segment in that direction would preserve all other coordinate signs and the cost, contradicting extremality. Thus all vertices are included by enumerating the one-dimensional intersections of $d-1$ independent coordinate hyperplanes with the kernel, and normalizing each of the two directions to have cost one. Cofactor determinants generate these directions.

Project those normalized directions by Ψ_N and take their exact convex hull. The preceding argument proves that this gives the entire polygon K_N, rather than an inner approximation. For $N = 20, 22, 26$, the kernel dimensions are $3, 4, 5$, and the hulls have $34, 36, 40$ vertices, respectively.

If x, y are consecutive counterclockwise vertices, the corresponding facet functional is

$$f_{x,y}(v) = \frac{\det(v, y-x)}{\det(x, y)}, \qquad F_N(v) = \max_{(x,y)} f_{x,y}(v).$$

Use the following unimodular bases $B_N = (b_1, b_2)$:

N	b_1	b_2	coordinate bounds
20	$-26685,\ -15403$	$58628,\ 33841$	$(\lvert m\rvert, \lvert n\rvert) \le (1, 1)$
22	$-559595, -323007$	$2179752,\ 1258187$	$(\lvert m\rvert, \lvert n\rvert) \le (2, 1)$
26	$15700603,\ 9062634$	$-85543238, -49376897$	$(\lvert m\rvert, \lvert n\rvert) \le (7, 1)$

Put $R_N = \max(F_N(b_1), F_N(b_2))$. For every lattice point $v = mb_1 + nb_2$ in $R_N K_N$,

$$|m| \le R_N \max_{x \in K_N} |\det(x, b_2)|, \qquad |n| \le R_N \max_{x \in K_N} |\det(b_1, x)|.$$

The maxima occur at vertices. Exact evaluation and taking integer parts give the displayed coordinate bounds. Enumerating the primitive pairs in these finite rectangles, up to simultaneous change of sign, proves the asserted global minima: $\mu_2 \le R_N$ already follows from the independent basis, so a point outside the rectangles cannot improve either minimum. The first shortest vector and the first vector not collinear with it give μ_1 and μ_2 in dimension two. The rational interval comparisons in the theorem are then direct exact comparisons of the facet values.

Attainment is verified on the input side as well. If a target v attains its gauge on the edge (x, y), then

$$v = \alpha x + \beta y, \quad \alpha = \frac{\det(v, y)}{\det(x, y)} \ge 0, \quad \beta = \frac{\det(x, v)}{\det(x, y)} \ge 0, \quad \alpha + \beta = F_N(v).$$

The same combination of the normalized rational input lifts of x, y is a rational optimizer for v. Clearing its denominators verifies the exact output gcd identities stated above.

The complete verifier embedded in Appendix C implements precisely these finite rational operations using only the Python standard library. It constructs all residues, log matrices, kernel bases, polygons, lattice rectangles, and optimizing lifts from their displayed formulas; it reads no saved basis or external certificate. Decimal output is informational only. No floating-point optimization or numerical value of γ enters any assertion. □

The old $N = 20$ certificate already used $(26685, 15403)$, with upper bound approximately $6.42358 \cdot 10^{-5}$. The exact quotient minimum is approximately $6.07002131 \cdot 10^{-5}$; the new result here is its global optimality for this norm, the independent second direction, and the improvements at larger blocks. The $N = 26$ second minimum is about 421 times smaller than the $N = 20$ second minimum. Finite improvement does not establish a rate for an infinite sequence.

Proposition 35.2 (A quantitative geometric condition for primitive decay)**.** *Let* $\mathcal{A}_N = \operatorname{area}(K_N)$. *Then*

$$2 \le \mathcal{A}_N \mu_1(N) \mu_2(N) \le 4, \qquad \frac{2}{\mathcal{A}_N \mu_1(N)} \le \mu_2(N) \le \frac{4}{\mathcal{A}_N \mu_1(N)}. \tag{129}$$

Consequently $\mu_2(N) \to 0$ *is equivalent to* $\mathcal{A}_N \mu_1(N) \to \infty$. *For these cumulative blocks, even unboundedness of* $\mathcal{A}_N \mu_1(N)$ *suffices. For example, bounds*

$$\mathcal{A}_N \ge e^{\alpha N}, \qquad \mu_1(N) \ge e^{-\beta N}, \qquad \alpha > \beta,$$

on an unbounded sequence of indices would prove primitive-error decay for independent pairs, and hence irrationality of γ.

Proof. The first inequality is the two-dimensional case of Minkowski's second theorem for the lattice $\mathbb{Z}^2$; see, for example, [16, Eq. (1.4)]. Rearranging gives the other inequalities and the first equivalence. Zero extension of an input lift shows $K_N \subseteq K_{N+1}$, so $\mu_2(N)$ is nonincreasing. Unboundedness of $\mathcal{A}_N \mu_1(N)$ gives a subsequence with $\mu_2 \to 0$; monotonicity gives the full limit. The final statement follows from $|q\gamma - p| \le F_N(q, p)$ and the independent-pair criterion. □

The condition controls two quantities, not area alone. For example, $K_T = [-T, T] \times [-1, 1]$ has area $4T$, first minimum $1/T$, and second minimum 1. Moreover, the actual quotient polygons satisfy $K_N \subseteq \{(q, p) : |q\gamma - p| \le 1\}$, so their ordinary Euclidean inradii cannot tend to infinity. The relevant remaining task concerns lattice directions and the product in (129). The computed values of $\mathcal{A}_N \mu_1(N)$ are approximately 39167.12, 952101.84, and 16875495.18 at $N = 20, 22, 26$; these three values are not a proof of unboundedness.

Lemma 35.3 (Exact plateaus when a new prime occurs)**.** *If a prime* ℓ *satisfies* $3N < \ell \le 3(N+1)$, *then* $F_{N+1} = F_N$ *and* $K_{N+1} = K_N$.

Proof. The ℓ-log coefficient vanishes in every column indexed by $n \le N$. In column $N + 1$ it is $A_{N+1,\ell} \ne 0$: the interval ends below 2ℓ, so its only multiple of ℓ is ℓ itself. Log cancellation therefore forces $c_{N+1} = 0$. The feasible quotient problem is exactly the preceding one. □

Thus the cumulative costs are monotone but need not improve at each step; infinitely many exact plateaus occur because there are infinitely many primes. The finite optimality theorem prevents arbitrary gcd rescaling or further linear recombination within one of the fixed blocks from beating its stated minima. It does not preclude better analytic weights, correlated bounds, or improvement on unbounded blocks. An infinite quantitative estimate sufficient for condition 3 remains unproved.

36 Exact facet transfer and adaptive polygon certificates

Let V have full row rank, let $\Psi(c) = (a^T c, b^T c)$ have rank two on $\ker V$, and let $w_n > 0$ be rational error weights. Put

$$K = \{\Psi(c) : Vc = 0,\ \sum_n w_n |c_n| \le 1\}.$$

This is the exact primitive quotient unit polygon: arbitrary rational lifts, and hence arbitrary actual output gcds, are included.

Lemma 36.1 (A finite primal–dual polygon certificate)**.** *Suppose a centrally symmetric polygon P contains the origin in its interior. For every vertex v of P, suppose a rational vector c_v satisfies $Vc_v = 0$, $\Psi(c_v) = v$, and $\sum w_n|(c_v)_n| \le 1$. For every edge inequality $\ell \cdot v \le 1$ of P, suppose a rational vector y satisfies*

$$|\ell_1 a_n + \ell_2 b_n - (V^T y)_n| \le w_n \qquad \text{for every } n.$$

Then $P = K$.

Proof. The vertex lifts and convexity give $P \subseteq K$. Conversely, for $Vc = 0$ the dual inequality gives

$$\ell \cdot \Psi(c) = \sum_n (\ell_1 a_n + \ell_2 b_n - (V^T y)_n) c_n \le \sum_n w_n |c_n|.$$

Every point of K lies in every edge half-plane of P, so $K \subseteq P$. □

This lemma permits verification of the whole polygon without enumerating all intersections of coordinate hyperplanes. A support oracle can locate new vertices, while only the final rational lifts and edge inequalities are needed for the proof. The computations here use an exact rational simplex method even for the search; no floating-point optimization is required.

Theorem 36.2 (An exact one-column facet-survival rule)**.** *Assume that a new input column introduces no new prime-log row. Write v for its log column, (a, b) for its output column, and $w > 0$ for its error weight. Let $\ell \cdot x \le 1$ support an edge of the old unit polygon. Define the nonempty rational polytope*

$$Y(\ell) = \{y : |\ell_1 a_n + \ell_2 b_n - (V^T y)_n| \le w_n \text{ for all old columns } n\},$$

and put

$$L = \min_{y \in Y(\ell)} v^T y, \qquad H = \max_{y \in Y(\ell)} v^T y.$$

The old supporting line remains valid for the enlarged unit polygon if and only if

$$\boxed{\ell_1 a + \ell_2 b \in [L - w, H + w].}$$

If $Y(\ell)$ consists of a single vector y, this reduces to the single exact residual test $|\ell_1 a + \ell_2 b - v^T y| \le w$.

Proof. Finite-dimensional linear-programming duality gives the nonemptiness of $Y(\ell)$, since the old support value equals one. Full row rank of V makes $Y(\ell)$ bounded: $V^T y$ has bounded coordinates and V^T is injective. Thus $Y(\ell)$ is a compact rational polytope, and the image of the linear functional $v^T y$ is exactly $[L, H]$. The new dual inequality is feasible precisely when this interval meets $[\ell_1 a + \ell_2 b - w, \ell_1 a + \ell_2 b + w]$. Strong duality gives equivalence with the new support value being at most one. The old polygon is contained in the new one, so this value is already at least one. This proves the assertion. □

At a prime-entry index, the new log row is zero on all old columns and nonzero on the new column. Its free dual multiplier can absorb the entire new residual. Every old edge survives, giving another proof of the exact plateau. At the other indices, the theorem provides a finite rational test for each old edge. It does not bound how often edges change or the infinite rate of area growth.

36.1 Further exact global minima

Use the proper-kernel weights $w_n = \mathcal{U}_n$ on $1 \le n \le N$. The adaptive primal–dual certificate gives complete unit polygons with $46, 50, 52$ vertices at $N = 29, 31, 32$, respectively. The first and second primitive successive minima are attained at

N	$v_1 = (q_1, p_1)$	$v_2 = (q_2, p_2)$
29	$(652385103, 376566901)$	$(15700603, 9062634)$
31	$(24091147002, 13905787435)$	$(16278226369, 9396047257)$
32	$(200542096649, 115756039658)$	$(2181871916137, 1259410648803)$.

Every row has determinant of absolute value one. Exact rational bounds are

N	bounds for μ_1	bounds for μ_2
29	$861897/10^{15} < \mu_1 < 861898/10^{15}$	$495784/10^{14} < \mu_2 < 495785/10^{14}$
31	$898911/10^{16} < \mu_1 < 898912/10^{16}$	$143837/10^{15} < \mu_2 < 143838/10^{15}$
32	$188658/10^{17} < \mu_1 < 188659/10^{17}$	$640768/10^{17} < \mu_2 < 640769/10^{17}$.

In particular,

$$\boxed{\mu_2(32) < 1/156000000000.}$$

The two independent primitive integer forms therefore both have certified absolute value below this bound, and at least one is nonzero. The optimizing rational lifts at $N = 32$ clear to integer relations whose unscaled output gcds each have 770 decimal digits. This normalization-dependent size is reported only to make explicit that large gcds were included; the invariant quantity is the quotient cost.

Completeness of the finite proof. The support search is a linear program in $x^+, x^- \ge 0$ and a nonnegative slack variable, with equalities

$$V(x^+ - x^-) = 0, \qquad \sum_n \mathcal{U}_n(x_n^+ + x_n^-) + s = 1.$$

Every vertex certificate specifies a feasible basis of this rational system. Every edge certificate specifies a feasible basis with support value one and nonpositive reduced costs. These are exact primal and dual certificates for Lemma 36.1. An independent verifier reconstructs all matrices and checks the certificates by rational basis exchange; it does not rerun the support search or trust its numerical summaries.

The integer-minimum check then uses exact facet gauges, as in the preceding paper. Suitable unimodular bases are

N	b_1	b_2	$(\lvert m\rvert, \lvert n\rvert)$ bound
29	$(-652385103, -376566901)$	$(-1972855912, -1138763337)$	$(8, 1)$
31	$(-24091147002, -13905787435)$	$(-40369373371, -23301834692)$	$(2, 1)$
32	$(200542096649, 115756039658)$	$(1981329819488, 1143654609145)$	$(4, 1)$.

Writing $R = \max(F_N(b_1), F_N(b_2))$, the bounds enclose every integer point of RK_N, by the exact determinant coordinate bounds. Since $\mu_2 \le R$, exhaustive enumeration in these rectangles

proves global optimality for the specified weighted quotient norm. The verifier checks the target determinants, all six rational intervals, all optimal lifts, exact log cancellation, and the final output gcds.

The product area$(K_N)\mu_1(N)$ is approximately $5.5255322\cdot10^8$, $2.5505733\cdot10^{10}$, and $4.6707794\cdot10^{11}$ on these three blocks. Its unboundedness has not been proved. The finite second minimum at 32 is roughly 32777 times smaller than at 26, but no infinite rate is inferred from these values. The infinite rank law and recursive basis construction are proved for all indices; primitive-error decay for two independent integer outputs remains a separate, unproved quantitative assertion.

Corollary 36.3 (Explicit bounds for the evaluated forms)**.** *For the two primitive outputs in the $N = 32$ row,*

$$|q_1\gamma - p_1| < \frac{1}{1026000000000}, \qquad |q_2\gamma - p_2| < \frac{1}{302000000000}.$$

At least one of these values is nonzero.

Proof. Lemma 26.2 gives $|q\gamma - p| < (488/945)F_N(q,p)$ for every nonzero target. Use the exact bounds $\mu_1(32) < 1/530000000000$ and $\mu_2(32) < 1/156000000000$ from the finite certificate. The rational comparisons are

$$\frac{488}{945\cdot 530000000000} < \frac{1}{1026000000000}, \qquad \frac{488}{945\cdot 156000000000} < \frac{1}{302000000000}.$$

Independence supplies the nonzero member, as in Corollary 1.2. □

37 The limiting quotient seminorm and its exact infinite dual

All coefficient vectors in this section have finite support unless otherwise specified. Write

$$(Ec)_\ell = \sum_n E_{n,\ell}c_n, \qquad \Psi(c) = \left(\sum_n 16^n c_n, \sum_n p_n c_n\right), \qquad C(c) = \sum_n \mathcal{U}_n|c_n|.$$

Each column of E has only finitely many prime coordinates. The identities defining the proper kernels give

$$Ec = 0 \quad\Longrightarrow\quad \sum_n c_n I_n = \gamma\,\Psi(c)_1 - \Psi(c)_2.$$

The condition $Ec = 0$ is the formal cancellation of every prime-log coefficient; it is not replaced by a numerical smallness test on a sum of logarithms.

For $N \geq 17$, let F_N be the quotient norm previously defined using coefficients supported on $1,\dots,N$. Output rank two at $N = 17$ is provided by Theorem 23.3. Put

$$L_\gamma(q,p) = \gamma q - p, \qquad F_\infty(v) = \inf_{N\geq 17} F_N(v).$$

Theorem 37.1 (A continuous limit and an exact infinite multiplier test)**.** *The functions F_N decrease to a finite continuous seminorm F_∞ and converge uniformly on every compact subset of $\mathbb{R}^2$. Moreover,*

$$F_\infty(v) \geq \frac{|L_\gamma(v)|}{\rho_*},$$

where $\rho_* < 1$ is defined in Corollary 26.3. In particular F_∞ is not identically zero.

For a real pair (α,β), the following are equivalent:

1. *$\alpha q + \beta p \leq F_\infty(q,p)$ for every real target (q,p);*

2. there are real numbers y_ℓ, one for each prime, such that

$$\left|\alpha 16^n + \beta p_n + \sum_\ell y_\ell E_{n,\ell}\right| \le \mathcal{U}_n \qquad \text{for every } n \ge 1. \tag{130}$$

Every sum in (130) is finite; no growth restriction or convergence convention for the sequence y_ℓ is being suppressed.

Proof. Zero extension proves $F_{N+1} \le F_N$. Homogeneity and the triangle inequality pass to the decreasing limit. Since $F_\infty \le F_{17}$,

$$|F_\infty(v) - F_\infty(w)| \le F_{17}(v - w),$$

so the limit is continuous. In fact the same modulus of continuity works for every F_N. Compactness and a finite net, or Dini's monotone convergence theorem, then give uniform convergence on compact sets. Applying the estimate of Corollary 26.3 to every feasible lift proves the displayed lower bound.

Assume the first condition. On the subspace $M = \ker E$ of the normed space $c_{00}(\mathbb{R})$ with norm C, define

$$f(c) = \alpha\Psi(c)_1 + \beta\Psi(c)_2.$$

For $c \in M$, a finite block containing its support gives $|f(c)| \le F_\infty(\Psi(c)) \le C(c)$; symmetry was used for the absolute value. Hahn–Banach extends f to a linear functional r on all of c_{00} satisfying $|r(c)| \le C(c)$. If e_n is the nth coordinate vector, then $r_n = r(e_n)$ satisfies $|r_n| \le \mathcal{U}_n$.

The linear functional $c \mapsto \alpha\Psi(c)_1 + \beta\Psi(c)_2 - r(c)$ vanishes on $\ker E$. It therefore factors through the algebraic image of E. Extend that factor linearly to the direct sum of all prime-coordinate spaces, and write its value on the ℓth coordinate vector as $-y_\ell$. Evaluation on e_n gives precisely

$$r_n = \alpha 16^n + \beta p_n + \sum_\ell y_\ell E_{n,\ell},$$

and proves the second condition. No continuity of E or boundedness of the multiplier sequence was used or needed.

Conversely, if (130) holds and $Ec = 0$, then $|\alpha\Psi(c)_1 + \beta\Psi(c)_2| \le C(c)$. Take the infimum over every finite-block lift and then over N. This gives the first condition. □

Corollary 37.2 (Exact termwise integral weights have the same decay question)**.** *Let $\widehat{F}_N$ be the quotient norm obtained by replacing the cost $\sum_n \mathcal{U}_n|c_n|$ by $\sum_n I_n|c_n|$, with precisely the same logarithm-cancellation constraints and output map. There is a constant $\rho_- > 0$, independent of N, such that*

$$\rho_- F_N(v) \le \widehat{F}_N(v) \le \rho_* F_N(v) \qquad (N \ge 17,\ v \in \mathbb{R}^2).$$

Consequently the second successive minima for these two weight systems tend to zero simultaneously. The same conclusion holds for any intermediate positive weights $I_n \le W_n \le \mathcal{U}_n$.

Proof. Set $\rho_- = \inf_{n\ge1} I_n/\mathcal{U}_n$. Each ratio is positive, and Theorem 26.1 gives a positive limit, so $\rho_- > 0$ by separating a finite prefix from a tail. The two cost inequalities hold for every feasible coefficient vector. Taking infima gives the norm inequalities, which pass to both successive minima by their definitions. Intermediate weights are sandwiched between the two costs. □

This corollary concerns termwise positive bounds and the weighted triangle inequality. It says that replacing the current weights by the exact positive values of all the integrals cannot by itself change the answer to the uniform primitive-decay question. It does not exclude estimates that use cancellation between different integrals before taking absolute values.

Theorem 37.3 (Constructive infinite dual from bounded pivot residuals)**.** *Use the pivot set $\mathcal{P}_\infty$ and the rational log-null basis $h^{(n)}$ of Theorem 23.4. Fix real α, β and $B > 0$. A prime-multiplier sequence satisfying*

$$\left|\alpha 16^n + \beta p_n + \sum_\ell y_\ell E_{n,\ell}\right| \le B\mathcal{U}_n \qquad (n \ge 1)$$

exists if and only if one can choose numbers r_b, for $b \in \mathcal{P}_\infty$, such that

$$|r_b| \le B\mathcal{U}_b \qquad (b \in \mathcal{P}_\infty), \tag{131}$$

$$\left|\alpha\Psi(h^{(n)})_1 + \beta\Psi(h^{(n)})_2 - \sum_{b\in\mathcal{P}_{n-1}} h_b^{(n)} r_b\right| \le B\mathcal{U}_n \qquad (n \ge 10,\ n \notin \mathcal{P}_\infty). \tag{132}$$

Each constraint involves only finitely many variables. The infinite system (131)–(132) is feasible if and only if every finite prefix is feasible.

Given its pivot residuals, the multiplier sequence is unique and is constructed by rational linear operations applied to the residuals, α, and β. In particular, no compactness assumption on an unbounded sequence of prime multipliers is required.

Proof. Write $d_n = \alpha 16^n + \beta p_n$. Start by solving the invertible nine-by-nine system

$$\sum_{\ell\le 27} E_{b,\ell} y_\ell = r_b - d_b \qquad (1 \le b \le 9).$$

At a later prime-entry index b, its unique new prime ℓ_b has $E_{b,\ell_b} = A_{b,\ell_b} \ne 0$. Define

$$y_{\ell_b} = \frac{r_b - d_b - \sum_{\ell<\ell_b} E_{b,\ell} y_\ell}{E_{b,\ell_b}}.$$

Every prime eventually occurs, so this specifies all multipliers uniquely. Each step is finite, and the entries of the coefficient matrices are rational.

At a nonpivot index n define the resulting residual $r_n = d_n + \sum_\ell E_{n,\ell} y_\ell$. Since $Eh^{(n)} = 0$ and $h_n^{(n)} = 1$,

$$r_n + \sum_{b\in\mathcal{P}_{n-1}} h_b^{(n)} r_b = d_n + \sum_{b\in\mathcal{P}_{n-1}} h_b^{(n)} d_b = \alpha\Psi(h^{(n)})_1 + \beta\Psi(h^{(n)})_2.$$

Thus its required bound is exactly (132). Conversely any feasible multipliers supply pivot residuals satisfying these equations, proving the first assertion.

The pivot residuals lie in the countable product of compact intervals $\prod_{b\in\mathcal{P}_\infty}[-B\mathcal{U}_b, B\mathcal{U}_b]$. Each slab is a closed cylinder set. Feasibility of every prefix gives the finite-intersection property, hence a common point. Equivalently, one may choose feasible solutions on increasing prefixes, extend their unused coordinates by zero, and successively extract convergent subsequences in each pivot coordinate; the diagonal limit satisfies every fixed slab. The converse is restriction to a prefix. □

Corollary 37.4 (A rational finite test for each proposed extra-dual budget)**.** *For a fixed positive rational B, the extra-dual condition (133) with this budget is equivalent to feasibility of every finite prefix of (131)–(132) with $(\alpha, \beta) = (1, 0)$. All coefficients of these finite linear inequality systems are rational. If no infinite solution with this budget exists, a finite prefix already witnesses infeasibility.*

Proof. Apply Theorem 37.3. The numbers $\mathcal{U}_b$ and the coordinates of $h^{(n)}$ and $\Psi(h^{(n)})$ are rational. □

Theorem 37.5 (Complete geometric alternatives)**.** *Exactly one of the following occurs.*

1. *F_∞ is a norm. Both successive minima of F_N converge to the positive successive minima of F_∞ on $\mathbb{Z}^2$. In particular $\mu_2(N)$ does not tend to zero.*

2. *For a finite constant $\kappa \geq 1/\rho_*$,*

$$F_\infty(q,p) = \kappa|\gamma q - p|.$$

The closure of $\bigcup_{N\geq 17} K_N$ is exactly the strip $|\gamma q - p| \leq 1/\kappa$. Every compact subset of its strict interior is contained in K_N for all sufficiently large N.

In the second case, if $\gamma = a/b$ is rational in lowest terms with $b > 0$, then

$$\mu_1(N) \longrightarrow 0, \qquad \mu_2(N) \longrightarrow \kappa/b.$$

If γ is irrational, then both successive minima tend to zero. These last two statements are alternatives, not assumptions about which arithmetic case holds for Euler's constant.

Proof. The lower bound in Theorem 37.1 puts the kernel of F_∞ inside the one-dimensional kernel of L_γ. If that kernel is trivial, compactness of the Euclidean unit circle gives $F_\infty(v) \geq a\|v\|$ for some $a > 0$. Since $F_N \geq F_\infty$, all lattice vectors capable of improving the fixed F_{17} minima lie in one bounded Euclidean set. Only finitely many integer vectors occur there. Pointwise convergence on that finite set proves convergence of both successive minima to those of the limiting norm.

If the kernel is nontrivial, it equals $\ker L_\gamma$. Factoring the seminorm through the one-dimensional quotient gives $F_\infty = \kappa|L_\gamma|$, with $\kappa = F_\infty(0,-1) \geq 1/\rho_*$. If $F_\infty(v) < 1$, then $F_N(v) < 1$ eventually. If $F_\infty(v) = 1$, apply this observation to $(1-\varepsilon)v$ and let $\varepsilon \downarrow 0$. This proves the closure statement, and uniform convergence on compact sets proves the interior statement.

In the rational case the primitive vector $v = (b,a)$ satisfies $F_N(v) \to 0$. Every independent pair of integer vectors contains a vector w outside its line, for which $|L_\gamma(w)| \geq 1/b$. Thus $\mu_2(N) \geq \kappa/b$. Choose an integer w with $|\det(v,w)| = 1$ by Bezout's identity. Then $|L_\gamma(w)| = 1/b$ and $F_N(w) \to \kappa/b$, proving the matching upper limit.

In the irrational case elementary rational approximation gives infinitely many primitive integer vectors with $|L_\gamma(v)| \to 0$. Distinct such vectors, up to sign, are not collinear. For each $\varepsilon > 0$, choose two independent ones whose limiting costs are less than ε. Pointwise convergence on these two fixed targets gives $\mu_2(N) < 2\varepsilon$ eventually. The first minimum is at most the second. □

Proposition 37.6 (An additional dual functional is exactly the remaining geometry obstruction)**.** *The limiting seminorm is a norm if and only if there exist a finite constant $B > 0$ and real prime multipliers y_ℓ such that*

$$\left|16^n + \sum_\ell y_\ell E_{n,\ell}\right| \leq B\mathcal{U}_n \qquad (n \geq 1). \tag{133}$$

Equivalently, the limiting polar contains a functional linearly independent of $(q,p) \mapsto \gamma q - p$ if and only if (133) has such a solution.

Proof. If F_∞ is a norm, its unit ball is bounded, so $|q| \leq BF_\infty(q,p)$ for some $B > 0$. Apply Theorem 37.1 to $(1/B, 0)$ and rescale the multipliers. Conversely (133) gives $F_\infty(q,p) \geq |q|/B$. Together with its lower bound by $|L_\gamma|/\rho_*$, this makes its kernel trivial. The limiting polar always contains the nontrivial segment of evaluation functionals $\tau(\gamma,-1)$ for $|\tau| \leq 1/\rho_*$. An independent functional makes its support seminorm a norm, while the strip case has only that one dual direction. □

What these theorems leave undecided. The infinite dual can contain more than the evaluation direction: excluding that possibility for these specific kernels requires ruling out (133). The finite polygons and the asymptotic ratio theorem do not do so. Conversely, proving strip exhaustion would not establish $\mu_2(N) \to 0$ without also excluding the rational alternative in Theorem 37.5. Thus the theorem separates a geometric limit question from the arithmetic question; it does not count either unresolved assertion as a proved irrationality result.

Connection to the five arithmetic requirements. The triangular basis and the rational quotient optimization enforce exact cancellation of logarithms. For an integer output (q, p) the remaining value is exactly $q\gamma - p$, so under $\gamma = a/b$ the scaled value $b(q\gamma - p)$ is an integer. Two independent integer outputs ensure that at least one such value is nonzero, even under that rationality assumption. A proof that $\mu_2(N) \to 0$ would supply, for every positive integer k, a finite block and two independent integer outputs with rational log-null lifts of cost less than $1/k$. Rational lifts exist because the finite optimization data are rational. Their remainders are bounded by ρ_*/k, and the scale b is fixed under the rationality assumption, so the nonzero integer would eventually have absolute value less than one.

Conversely, that uniform supply of two small-cost integer outputs is exactly the assertion $\mu_2(N) \to 0$, by monotonicity of the finite quotient norms. The pivot recursion proves exact feasibility and makes every finite budget test explicit; it does not supply the required uniform decay. No asymptotic arithmetic condition has been replaced by finite numerical evidence in these statements.

38 An exact dilation recursion and its primitive arithmetic cost

This section tests a natural way to extend a finite logarithm-free certificate to infinitely many larger supports. The recursion cancels its new logarithms identically and makes the raw integrals geometrically small. Nevertheless its actual primitive errors diverge. The result applies to every fixed nonzero starting kernel, including a finite prime-log-null combination of the proper kernels.

Let

$$d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1}, \qquad h_m = H_m - \frac{1}{2m}, \qquad F(u) = \sum_{m \in S} \frac{b_m}{u + m^2},$$

where S is a finite nonempty set of positive integers, every $b_m \in \mathbb{Q} \setminus \{0\}$, and

$$q_0 = \sum_{m \in S} b_m \neq 0, \qquad \sum_{m \in S} b_m v_\ell(m) = 0 \quad \text{for every prime } \ell.$$

Thus Binet's formula gives $\int F\, d\mu = q_0\gamma - P_0$, where $P_0 = \sum_m b_m h_m$ is rational. Fix an integer $a \geq 2$, and define

$$(\mathscr{D}_a F)(u) = a^{-2} F(u/a^2), \qquad \mathscr{T}_a = 2\mathscr{D}_a - \mathscr{D}_{a^2}, \qquad F_n = \mathscr{T}_a^n F, \quad J_n = \int F_n\, d\mu.$$

These identities define rational kernels without referring to γ.

Theorem 38.1 (Exact preservation of the gamma and prime-log coordinates)**.** *For every $n \geq 0$,*

$$F_n(u) = \sum_{r=0}^{n} (-1)^r \binom{n}{r} 2^{n-r} \sum_{m \in S} \frac{b_m}{u + (ma^{n+r})^2},$$

and

$$J_n = q_0\gamma - P_n, \qquad P_n = \sum_{r=0}^{n} (-1)^r \binom{n}{r} 2^{n-r} \sum_{m \in S} b_m h_{ma^{n+r}} \in \mathbb{Q}.$$

In particular every auxiliary logarithm vanishes exactly at every step. The gamma coordinate q_0 is unchanged; the rational coordinate P_n is generally changed.

Proof. The dilation operators commute and $\mathscr{D}_a^j = \mathscr{D}_{a^j}$, giving the binomial expansion. Put $C(z) = 2z - z^2$. Its coefficient vector at the n-th stage is that of $C(z)^n$. Since $C(1) = 1$ and $C'(1) = 0$, the weights satisfy

$$\sum_{r=0}^{n}(-1)^r\binom{n}{r}2^{n-r} = 1, \qquad \sum_{r=0}^{n}(n+r)(-1)^r\binom{n}{r}2^{n-r} = 0.$$

Applying Binet's identity to each pole, the old prime-log coordinates are multiplied by the first sum, while the new $\log a$ coefficient is q_0 times the second. Hence the displayed form is exact. In fact the operator preserves every old prime-log coordinate, even before the initial log-null hypothesis is imposed. □

Theorem 38.2 (Raw contraction for every nonzero finite starting kernel)**.** *Write the Taylor expansion at zero as*

$$F(u) = \sum_{j\ge 0} f_j u^j, \qquad f_j = (-1)^j \sum_{m\in S}\frac{b_m}{m^{2j+2}},$$

and let $\nu = \operatorname{ord}_{u=0} F < \infty$. *Define*

$$\mu_j = \int_0^\infty u^j\, d\mu(u) = \frac{|B_{2j+2}|}{2j+2} > 0, \qquad \beta_j = 2a^{-2j-2} - a^{-4j-4}.$$

There is a constant $0 \le \eta < \beta_\nu < 1$ *such that*

$$\boxed{J_n = f_\nu \mu_\nu \beta_\nu^n + O_F(\eta^n).}$$

Thus J_n *has a fixed nonzero sign for every sufficiently large* n. *The statement does not require pointwise positivity of* F_n. *Here* B_j *are Bernoulli numbers. This raw asymptotic applies to every nonzero finite rational simple-pole kernel, even without a nonzero gamma coordinate or a prime-log-null hypothesis.*

Proof. For every integer $R \ge 0$ and every $u \ge 0$, the finite geometric identity for each simple pole gives

$$\left|F(u) - \sum_{j=0}^{R} f_j u^j\right| \le C_R u^{R+1}, \qquad C_R = \sum_m \frac{|b_m|}{m^{2R+4}}.$$

After dilation and the binomial expansion, integration yields the entirely explicit bound

$$\left|J_n - \sum_{j=0}^{R} f_j \mu_j \beta_j^n\right| \le C_R \mu_{R+1}\left(2a^{-2R-4} + a^{-4R-8}\right)^n.$$

The moment formula follows by $v = 2\pi\sqrt{u}$, expansion of $(e^v - 1)^{-1}$, and the classical even-zeta evaluation. Since $2t - t^2$ is strictly increasing on $(0, 1)$, $\beta_0 > \beta_1 > \cdots > 0$ and $\beta_j \to 0$. Choose $R \ge \nu$ so large that $2a^{-2R-4} + a^{-4R-8} < \beta_\nu$. All terms below ν vanish, and all other bases are strictly smaller than β_ν. This proves the asymptotic and its nonvanishing assertion. □

Theorem 38.3 (The full primitive cost of the recursion diverges)**.** *Let* $Q_n = \operatorname{den}(P_n/q_0)$ *be the actual reduced denominator, and put* $M_n = m_* a^{2n}$, *where* $m_* = \max S$. *If* S *has more than one element let* m_{**} *be its second largest; otherwise put* $m_{**} = 0$. *Set*

$$\theta = \max\left\{\frac{1}{2}, \frac{1}{a}, \frac{m_{**}}{m_*}\right\} < 1.$$

For every sufficiently large n,

$$\boxed{\prod_{\substack{\theta M_n < \ell \le M_n \\ \ell \text{ prime}}} \ell \;\mid\; Q_n.}$$

Consequently the prime number theorem gives

$$\log Q_n \geq (1-\theta+o(1))M_n, \qquad Q_n\left|\gamma-\frac{P_n}{q_0}\right| \longrightarrow +\infty.$$

This conclusion already incorporates every common factor removed from the integrated output pair.

Proof. The largest pole parameter is M_n, arising only from $m = m_*$ and $r = n$. Its residue is $(-1)^n b_{m_*}$. Every other pole parameter is at most θM_n. For sufficiently large n, primes in $(\theta M_n, M_n]$ divide none of the fixed numerator or denominator factors of the b_m or q_0. Such a prime ℓ occurs exactly once in H_{M_n} and does not occur in any other harmonic number in P_n. Also $\ell \nmid 2M_n$: for $n \geq 1$ the number M_n is composite, whereas a prime divisor exceeding $M_n/2$ would have to equal M_n. Therefore

$$v_\ell(P_n) = -1, \qquad v_\ell(P_n/q_0) = -1.$$

This proves the product divisibility for the reduced denominator. The prime number theorem [7] gives $\sum_{\theta M_n<\ell\leq M_n} \log \ell = (1-\theta+o(1))M_n$. Meanwhile the preceding theorem gives $\log|J_n| = n\log\beta_\nu + O_F(1)$. Since the primitive error equals $Q_n|J_n|/|q_0|$, its logarithm tends to positive infinity. □

Corollary 38.4 (A finite independent pair cannot be bootstrapped this way)**.** *Start from two finite prime-log-null kernels with nonzero gamma coordinates and distinct rational targets. Apply the same fixed operator $\mathscr{T}_a$ repeatedly to both. Their targets remain distinct for every sufficiently large stage, but both primitive errors tend to positive infinity.*

Proof. Apply the raw asymptotic to the nonzero difference of the two kernels after dividing each by its gamma coordinate. Its evaluated integral is the difference of the two rational targets, so that difference is eventually nonzero. The primitive-cost theorem applies to each kernel. □

An elementary one-pole version requiring only Bertrand's theorem

Take $F(u) = 1/(u+1)$. Then $q_0 = 1$ and

$$\beta = \frac{2a^2-1}{a^4}, \qquad \tau = \frac{2a^4+1}{a^8} < \beta.$$

The explicit Taylor remainder at $R = 0$ gives, for every $n \geq 0$,

$$\frac{\beta^n}{12} - \frac{\tau^n}{120} \leq J_n \leq \frac{\beta^n}{12} + \frac{\tau^n}{120}.$$

Thus $J_n \geq (3/40)\beta^n > 0$. For $n \geq 1$, Bertrand's theorem [25] gives a prime $a^{2n}/2 < \ell < a^{2n}$. It sees only the last harmonic pole and has valuation -1 in the rational target. Hence the actual reduced denominator exceeds $a^{2n}/2$, and

$$\boxed{\mathcal{E}_n = Q_n J_n > \frac{3}{80}(2-a^{-2})^n \longrightarrow +\infty.}$$

This exact log-free recursion satisfies integrality after reduction, nonvanishing, and an infinite unconditional construction, but fails primitive-error decay even under a lower bound using just one prime.

Pointwise positivity is not silently assumed. For example, at $a = 2$,

$$(\mathscr{T}_2^4(1/(u+1)))\big|_{u=4^6} = -\frac{7}{348160} < 0,$$

while its integral is strictly positive by the displayed bound.

Scope and comparison with earlier constructions. The earlier corpus constructs positive multiscale kernels and proves prime-survival bounds for particular schedules. The present result concerns a fixed autonomous dilation operator applied to *any* finite starting log-null kernel; the raw asymptotic is determined by its exact order of vanishing at zero. It therefore tests the proposed bootstrap from a finite optimized proper-kernel certificate. It does not exclude stage-dependent operators, growing cancellation width, variable dilations, or other output-changing circuits.

Part V
Positive and signed prefixes, output intervals, and repairs

39 Why coefficientwise positivity cannot supply nonvanishing

An integer log-null vector whose cleared weighted error tends to zero still needs a nonzero final value. One attractive sufficient condition would be nonnegative combination coefficients, because every proper kernel is strictly positive. The following exact separation theorem excludes that particular completion of the construction.

Theorem 39.1 (A separating functional for all proper-kernel log columns)**.** *For $n \geq 1$, put*

$$R_n(u) = \frac{\prod_{k=0}^{2n-1}(4u+k^2)}{\prod_{j=n}^{3n}(u+j^2)}, \qquad q_n = 16^n, \qquad L_n = \sum_{\ell \leq 3n} E_{n,\ell} \log \ell = \sum_{j=n}^{3n} A_{n,j} \log j.$$

Then

$$\boxed{L_n > 16^n \log(3n) > 0.}$$

Consequently, on any finite nonempty set of indices, a nonzero real vector cancelling every prime-log coordinate must have both a positive and a negative coefficient. In particular no nonzero coefficientwise nonnegative combination of these proper kernels is log-null.

Proof. The partial fractions and their leading coefficient give

$$R_n(u) = \sum_{j=n}^{3n} \frac{A_{n,j}}{u+j^2}, \qquad \sum_{j=n}^{3n} A_{n,j} = q_n.$$

Integrating each finite partial-fraction difference yields the elementary identity

$$\int_0^\infty \left(\frac{q_n}{u+1} - R_n(u) \right) du = \sum_{j=n}^{3n} A_{n,j} \int_0^\infty \left(\frac{1}{u+1} - \frac{1}{u+j^2} \right) du = 2L_n.$$

Each difference is integrable; the cancellation of the $1/u$ terms also makes the summed integrand $O(u^{-2})$ at infinity. On the other hand, for every $u \geq 0$,

$$\frac{R_n(u)}{q_n} = \frac{1}{u+9n^2} \prod_{k=0}^{2n-1} \frac{u+(k/2)^2}{u+(n+k)^2} < \frac{1}{u+9n^2},$$

since $0 \leq k/2 < n+k$. The inequality is strict throughout, including at zero. Thus

$$\begin{aligned} 2L_n &> q_n \int_0^\infty \left(\frac{1}{u+1} - \frac{1}{u+9n^2} \right) du \\ &= q_n \log(9n^2) = 2q_n \log(3n). \end{aligned}$$

The strict comparison is legitimate because the difference between the two integrands is positive and integrable.

If $c_n \geq 0$ and at least one coefficient is positive, then $\sum_n c_n L_n > 0$. Exact cancellation of every prime-log coordinate would instead give

$$\sum_n c_n L_n = \sum_\ell \left(\sum_n c_n E_{n,\ell} \right) \log \ell = 0,$$

a contradiction. Replacing a nonpositive vector by its negative proves the full assertion. □

This theorem concerns the signs of the combination coefficients. It does not exclude a signed combination whose resulting rational kernel is nonnegative, and it does not exclude other positive integral constructions. Its proof uses no arithmetic assumption about γ.

40 Positive certificates and finite feasibility

The cumulative-positivity theorem proved with the common kernels allows mixed coefficient signs; it controls the resulting kernel.

40.1 An exact finite nonvanishing certificate at small primitive cost

The first exact lift stored in `primitive32_lift.json` has primitive output

$$(q, p) = (200542096649, 115756039658).$$

The rational coefficient vector and its weighted prefix sums have been extracted to `cumulative_positive_kernel32.json`. Regenerating the residues independently from their factorial formula and using exact rational arithmetic verifies

$$V_{32}c = 0, \qquad \sum_{j=1}^{32} 16^j c_j = q, \qquad \sum_{j=1}^{32} p_j c_j = p, \qquad \gcd(q, p) = 1,$$

$$S_1 = \cdots = S_6 = 0, \qquad S_7, \ldots, S_{32} > 0, \qquad \sum_{j=1}^{32} |c_j| \mathcal{U}_j < \frac{1}{530000000000}.$$

Theorem 22.3 therefore gives the strict certified inequality

$$\boxed{0 < 200542096649\gamma - 115756039658 < \frac{1}{530000000000}.}$$

The sign certificate concerns the entire rational kernel and does not use digits of γ. The standard-library verifier also clears every coefficient denominator and checks that the complete gcd of the two resulting integer output coordinates is exactly that clearing factor. Thus the quoted cost is already the primitive cost.

This is one finite target. Repeating or scaling it does not establish decay along an infinite sequence.

40.2 The first canonical relation: positivity without small cost

At $N = 12$, the canonical log-null relation is unique up to scaling. It is defined exactly by

$$h_{12} = 1, \qquad (h_1, \ldots, h_{11})^T = -V_{11}^{-1} V_{12}[:, 12].$$

Clear denominators and remove the coefficient gcd to obtain the integer vector in `cumulative_positive_kernel12.json`. If Q denotes its positive gamma coefficient, all twelve prefix sums satisfy

$$\frac{99}{100} Q < S_k < 3Q, \qquad S_1 > Q.$$

This is an exact mixed-coefficient positive-kernel certificate. Its primitive target, however, has denominator $q > 10^{105}$ and primitive error greater than 10^{98}. Thus positivity by itself does not solve the arithmetic-cost problem.

For the last lower bound, after normalization to the primitive target the first Abel coefficient is $(q/Q)S_1 > q$. On $1 \le u \le 2$,

$$f_1(u) = \frac{u(u+1/4)}{(u+1)(u+4)(u+9)} \ge \frac{5}{792} > \frac{1}{160}, \qquad \frac{f_2(u)}{f_1(u)} \le \frac{153}{4 \cdot 17 \cdot 26 \cdot 37} < \frac{1}{2}.$$

Also $2\pi\sqrt{2} < 9$ and $e < 3$ give $d\mu/du > 1/20000$ there. Consequently

$$\int_1^2 (f_1 - f_2)\, d\mu > \frac{1}{320 \cdot 20000} > 10^{-7}.$$

All other Abel summands are nonnegative, so the primitive error exceeds $q10^{-7} > 10^{98}$. This check uses no numerical enclosure of γ.

40.3 An exact feasibility alternative

For a fixed window let $e_j = (E_{j,\ell}/16^j)_\ell$ be the rational normalized log column. By Theorem 22.3, a nonzero log-null vector in the cumulative cone can be scaled to $S_M = 1$. It exists if and only if

$$-e_M \in \operatorname{cone}\{e_n - e_{n+1}, \dots, e_{M-1} - e_M\}.$$

Equivalently, by the finite-dimensional Farkas alternative, it fails to exist if and only if a real row vector y satisfies

$$y \cdot e_n \ge y \cdot e_{n+1} \ge \cdots \ge y \cdot e_M > 0.$$

All data in the feasibility problem are rational, so a feasible point can be chosen rational and cleared to an integer certificate. The functional $y_\ell = \log \ell$ that excluded nonnegative combination coefficients does not satisfy this dual obstruction: it makes $y \cdot e_j = \rho_j$ strictly increasing.

For a prescribed integer target (q, p), the whole finite intersection is a rational linear program in the prefix variables. Put $S_{n-1} = 0$, $d_j = p_j/16^j$, and impose

$$S_k \ge 0, \qquad S_M = q,$$

$$\sum_{k=n}^{M-1} S_k(e_k - e_{k+1}) + qe_M = 0, \qquad \sum_{k=n}^{M-1} S_k(d_k - d_{k+1}) + qd_M = p.$$

The weighted primitive cost to minimize, for a primitive target, is

$$\sum_{j=n}^{M} \frac{\mathcal{U}_j}{16^j} |S_j - S_{j-1}|.$$

Absolute values are linearized in the usual way. These constraints simultaneously enforce exact log cancellation, the prescribed integer target, strict final nonvanishing when feasible and nonzero, and the actual weighted cost. The remaining infinite problem is to produce primitive targets with such feasible lifts and costs tending to zero.

40.4 Why ordinary pigeonhole collisions do not ensure comparability

Define $x \preceq y$ when every weighted prefix sum of $y - x$ is nonnegative. On any fixed log fiber, a nontrivial comparable pair must have strictly different gamma coordinates, by Theorem 22.3. Consequently every joint fiber of the log map and the gamma-coordinate map is an antichain. The same is true, a fortiori, for fibers of the full formal coefficient map.

This gives a precise obstruction to simply adding an order condition to the elementary collision count. For the reverse-budget window $M - n = \lceil 25n/\log n \rceil$ and grid height $H = 2^n$, let

$\beta = (16399/1296)^2$. The row-specific bounds used in the reverse-budget construction show that some full formal-coordinate fiber of $\{0, \dots, H\}^{M-n+1}$ contains at least

$$\exp\left((\eta + o(1))\frac{n^2}{\log n}\right) \quad \text{points}, \qquad \eta = 22\log 2 - 3\log\beta > 0.$$

Indeed the grid logarithmic cardinality is $(25\log 2 + o(1))n^2/\log n$; the prime-log image count contributes at most $(3\log\beta + 3\log 2 + o(1))n^2/\log n$; and the gamma and denominator-cleared rational rows each contribute only $O(n)$. All points in that large fiber form an antichain. Thus the already proved small-input collision count does not supply the missing comparable pair. A stronger order-sensitive estimate or a direct construction would still be needed.

41 A cost bound with no linear-horizon restriction

Retain the proper kernels, their rational bounds $\mathcal{U}_j$, and the normalized logarithmic values ρ_j defined above. Write

$$r_j = \mathcal{U}_j/16^j, \qquad \alpha = 16/729, \qquad T_j = 2 + \tfrac{3}{2}H_{j-1}, \qquad \Delta_j = \rho_{j+1} - \rho_j.$$

The established estimates are $r_{j+1} \le \alpha r_j$, $\Delta_j > 1/(66j)$, and $0 < \rho_j < T_j$.

Theorem 41.1 (Weighted variation removes the horizon factor)**.** *Let a nonzero real vector c on $[n, M]$ be exactly log-null and have nonnegative weighted prefixes $S_k = \sum_{j=n}^{k} 16^j c_j$. Put $q = S_M > 0$ and $C(c) = \sum_{j=n}^{M} \mathcal{U}_j|c_j|$. Then*

$$\boxed{\frac{C(c)}{q} < r_M + \frac{66\cdot 745}{729}\, nT_M r_n.}$$

In particular, uniformly for all integers $M \ge n \ge 2$ for which such a vector exists,

$$C(c) = O\left(q\, 729^{-n}\frac{\log(M+1)}{\sqrt{n}}\right),$$

with an absolute implied constant.

Proof. Put $S_{n-1} = 0$. The weighted total variation satisfies

$$\begin{aligned} C(c) &= \sum_{j=n}^{M} r_j|S_j - S_{j-1}| \\ &\le qr_M + \sum_{k=n}^{M-1} S_k(r_k + r_{k+1}). \end{aligned}$$

Moreover kr_k is decreasing, since $((k+1)/k)\alpha \le 2\alpha < 1$. Thus for $k \ge n$,

$$\frac{r_k + r_{k+1}}{\Delta_k} < 66k(1+\alpha)r_k \le 66n(1+\alpha)r_n.$$

The exact summation-by-parts identity from logarithm cancellation is $\sum_{k=n}^{M-1} S_k\Delta_k = q\rho_M$. Its right side is positive. Multiplying the last strict bound by $S_k\Delta_k$ and summing therefore gives

$$C(c) < qr_M + 66n(1+\alpha)r_n q\rho_M < qr_M + 66n(1+\alpha)r_n qT_M.$$

Finally $r_M \le r_n$, $T_M = O(\log(M+1))$, and $r_n \sim (\sqrt{2\pi}/12)n^{-3/2}729^{-n}$. □

Corollary 41.2 (A larger admissible support horizon)**.** *Suppose for unbounded n there are rational vectors as above with primitive integer outputs (q_n, p_n) and*

$$q_n \log(M_n + 1) = o(729^n \sqrt{n}).$$

Then $0 < q_n\gamma - p_n < C(c^{(n)}) \to 0$, and hence γ is irrational. In particular, the conditions $q_n \le 729^n$ and $\log(M_n + 1) = o(\sqrt{n})$ suffice; the earlier restriction $M_n \le An$ is not needed.

Proof. Apply Theorem 41.1 and cumulative positivity. If $\gamma = a/b$, the positive integers $aq_n - bp_n$ would tend to zero. The complete denominator clearing and primitive gcd reduction are exactly those described in Section 1. □

This is a stronger conditional completion criterion. It does not assert an infinite supply of the required primitive outputs. For example, a horizon as large as $M_n \le \exp(\sqrt{n}/\log n)$ is permitted with $q_n \le 729^n$.

Proposition 41.3 (The logarithmic horizon factor in a real relaxation)**.** *Fix $n \ge 1$ and relax separate prime-log cancellation to the single real equation $\sum_j c_j L_j = 0$, allowing real coefficients. For every $M \ge n + 2$ there is a cumulative-positive vector on $[n, M]$ with $q = 1$ and exact cost*

$$C = r_M + \frac{\rho_M}{\Delta_n}(r_n + r_{n+1}).$$

For fixed n this cost is $\Theta_n(\log M)$ as $M \to \infty$. Thus no bound independent of M can hold on this relaxation.

Proof. Set $S_n = \rho_M/\Delta_n$, set all intervening prefixes to zero, and set $S_M = 1$. The only nonzero coefficients are

$$c_n = \frac{\rho_M}{16^n \Delta_n}, \qquad c_{n+1} = -\frac{\rho_M}{16^{n+1}\Delta_n}, \qquad c_M = 16^{-M}.$$

Their aggregate log sum equals $\rho_M(\rho_n - \rho_{n+1})/\Delta_n + \rho_M = 0$. Their weighted total variation is the displayed cost, and the kernel is $S_n(f_n - f_{n+1}) + f_M > 0$.

Pairing all numerator factors of f_M with the first $2M$ denominator factors gives $f_M(u) < 1/(u + 9M^2)$. Hence Frullani's identity gives

$$2\rho_M > \int_0^\infty \left(\frac{1}{u+1} - \frac{1}{u+9M^2}\right) du = \log(9M^2).$$

Together with $\rho_M < T_M = O(\log M)$, this proves the two-sided order statement for fixed n. □

The three displayed coefficients are real; they are not asserted to be rational or to cancel the individual prime-log coordinates. This proposition establishes sharpness only for the stated relaxed problem. It does not exclude a stronger bound that uses the separate prime rows or the arithmetic of rational coefficients.

41.1 A signed prefix budget with no linear restriction on the window

Write

$$r_j = \frac{\mathcal{U}_j}{16^j}, \quad \alpha = \frac{16}{729}, \quad \rho_j = \frac{\sum_\ell E_{j,\ell} \log \ell}{16^j}, \quad \Delta_j = \rho_{j+1} - \rho_j, \quad \mathcal{H}_j = 2 + \frac{3}{2}H_{j-1}.$$

The established exact estimates are

$$r_{j+1} \le \alpha r_j, \qquad \Delta_j > \frac{1}{66j}, \qquad 0 < \rho_j < \mathcal{H}_j.$$

Theorem 41.4 (Uniform cost for bounded negative prefixes)**.** *Let rational coefficients $c_n, \dots, c_M$ satisfy exact prime-log cancellation. Assume explicitly that*

$$q = \sum_{j=n}^{M} 16^j c_j > 0.$$

Let $\lambda \geq 0$ and suppose

$$S_k = \sum_{j=n}^{k} 16^j c_j \geq -\lambda q \qquad (n \leq k \leq M).$$

Then, with $C(c) = \sum_{j=n}^{M} \mathcal{U}_j |c_j|$,

$$\boxed{\frac{C(c)}{q} < \lambda r_n + (1+\lambda) r_M + 66n(1+\alpha)(1+\lambda)\mathcal{H}_M r_n.}$$

Uniformly over $M \geq n$, $\lambda \geq 0$, and all such lifts,

$$\boxed{C(c) = O\!\left((1+\lambda) q\, 729^{-n} \frac{\log(M+1)}{\sqrt{n}}\right).}$$

The implied constant is absolute. Strict positivity of the integral is not asserted in this signed-prefix theorem.

Proof. Put $S_{n-1} = 0$ and $T_k = S_k + \lambda q$ for $n-1 \leq k \leq M$. Then all $T_k \geq 0$, $T_{n-1} = \lambda q$, and $T_M = (1+\lambda)q$. Exact log cancellation and summation by parts give

$$\sum_{k=n}^{M-1} S_k \Delta_k = q \rho_M.$$

Consequently

$$\begin{aligned} \sum_{k=n}^{M-1} T_k \Delta_k &= q\rho_M + \lambda q(\rho_M - \rho_n) \\ &= q((1+\lambda)\rho_M - \lambda \rho_n) < (1+\lambda) q \mathcal{H}_M. \end{aligned}$$

The additive offset does not change successive differences. Hence

$$\begin{aligned} C(c) &= \sum_{j=n}^{M} r_j |T_j - T_{j-1}| \\ &\leq r_n T_{n-1} + r_M T_M + \sum_{k=n}^{M-1} (r_k + r_{k+1}) T_k. \end{aligned}$$

For $k \geq n$,

$$\frac{r_k + r_{k+1}}{\Delta_k} < 66k(1+\alpha) r_k \leq 66n(1+\alpha) r_n,$$

because $(k+1) r_{k+1}/(k r_k) \leq (1+1/k)\alpha \leq 2\alpha < 1$. Multiplying the preceding log-budget identity by this uniform bound proves the asserted cost inequality. If an empty sum occurs, the same strict bound follows directly from the positive final added term.

Finally $r_M \leq r_n$, $\mathcal{H}_M = O(\log(M+1))$, and

$$r_n \sim \frac{\sqrt{2\pi}}{12} n^{-3/2} 729^{-n}.$$

Since $\lambda \leq 1 + \lambda$, these estimates prove the uniform asymptotic bound. No comparison $M = O(n)$ was used. □

Corollary 41.5 (Joining the signed budget to determinant nonvanishing)**.** *Suppose that for unbounded integers n there are two rational log-null lifts $c^{(1,n)}, c^{(2,n)}$ on a common window $[n, M_n]$. Suppose their primitive integer outputs $(q_{i,n}, p_{i,n})$ satisfy*

$$q_{i,n} > 0, \qquad \det \begin{pmatrix} q_{1,n} & q_{2,n} \\ p_{1,n} & p_{2,n} \end{pmatrix} \neq 0,$$

and their prefixes satisfy $S_k(c^{(i,n)}) \geq -\lambda_{i,n} q_{i,n}$ with $\lambda_{i,n} \geq 0$. If

$$\max_{i=1,2} \frac{(1+\lambda_{i,n}) q_{i,n} \log(M_n+1)}{729^n \sqrt{n}} \longrightarrow 0,$$

then γ is irrational. In particular it is sufficient that

$$(1+\lambda_{i,n}) q_{i,n} \leq 729^n \quad (i=1,2), \qquad \log(M_n+1) = o(\sqrt{n}).$$

Proof. The theorem gives costs $C_{i,n} \to 0$, and hence errors $|q_{i,n}\gamma - p_{i,n}| < C_{i,n}$. The nonzero determinant implies that the two linear forms have no common real zero, so

$$0 < W_n(\gamma) := (q_{1,n}\gamma - p_{1,n})^2 + (q_{2,n}\gamma - p_{2,n})^2 < C_{1,n}^2 + C_{2,n}^2 \longrightarrow 0.$$

If $\gamma = a/b$ with integers a, b and $b > 0$, then $b^2 W_n(\gamma)$ is a positive integer tending to zero, a contradiction. Clearing a common denominator of each rational lift and then removing the complete output gcd returns exactly its primitive target and cost. No separate assumption $q_{i,n} \to \infty$ is needed for this paired argument. □

For example, the sufficient window condition allows every fixed polynomial horizon $M_n \leq n^A$ and also $M_n \leq \exp(n^{1/3})$ for all sufficiently large n. The corollary is a conditional completion statement. The simultaneous infinite supply of such integer outputs and bounded negative prefixes is not established here. Its improvement is that strict kernel positivity and linearly bounded windows are no longer requirements for this cost-and-nonvanishing route.

42 A sharp budget for negative prefixes

The nonnegative-prefix condition can be relaxed by an exact rational budget. The budget is sharp for the class specified only by lower bounds on its prefixes. Sharpness here is not a statement about the smaller class satisfying prime-log cancellation. Its useful consequence for that smaller class is unconditional positivity of every admitted kernel, including the boundary of the budget.

For integers $j \geq 1$ and $u > 0$, set

$$R_j(u) = \frac{\prod_{h=0}^{2j-1}(4u+h^2)}{\prod_{h=j}^{3j}(u+h^2)}, \qquad f_j(u) = 16^{-j} R_j(u).$$

Fix integers $1 \leq n < M$. Given real coefficients $c_n, \ldots, c_M$, write

$$S_{n-1} = 0, \qquad S_k = \sum_{j=n}^{k} 16^j c_j, \qquad q = S_M, \qquad K_c(u) = \sum_{j=n}^{M} c_j R_j(u).$$

Summation by parts gives the exact identity

$$K_c(u) = \sum_{k=n}^{M-1} S_k(f_k(u) - f_{k+1}(u)) + q f_M(u). \tag{134}$$

Conversely, arbitrary values of the prefixes and of $q = S_M$ are realized uniquely by $c_j = (S_j - S_{j-1})/16^j$.

Lemma 42.1 (Initial slopes and monotone gap ratios)**.** *The positive rational numbers*

$$a_j := \lim_{u\downarrow 0} \frac{f_j(u)}{u} = \frac{1}{16^j j^4 \binom{3j}{j}^2} \tag{135}$$

satisfy $0 < a_{j+1} < a_j/729$*. The functions*

$$\vartheta_j(u) := \frac{f_{j+1}(u)}{f_j(u)} = \frac{(u+j^2)^2(u+(j+\frac{1}{2})^2)}{(u+(3j+1)^2)(u+(3j+2)^2)(u+(3j+3)^2)}$$

are strictly increasing on $(0,\infty)$*, have values in* $(0,1)$*, and tend to* 1 *at infinity. For* $n \le k < M$*, the functions*

$$g_{k,M}(u) := \frac{f_k(u) - f_{k+1}(u)}{f_M(u)} = (\vartheta_k(u)^{-1} - 1) \prod_{\ell=k+1}^{M-1} \vartheta_\ell(u)^{-1} \tag{136}$$

are strictly decreasing, with limits

$$g_{k,M}(0+) = \frac{a_k - a_{k+1}}{a_M}, \qquad g_{k,M}(+\infty) = 0.$$

Proof. The single zero factor $4u$ in the numerator of R_j gives

$$a_j = \frac{4((2j-1)!)^2((j-1)!)^2}{16^j((3j)!)^2} = \frac{((2j)!)^2(j!)^2}{16^j j^4((3j)!)^2},$$

which is (135). Cancellation of the common factors of R_{j+1}/R_j gives the displayed quotient ϑ_j. Pair its numerator factors with its denominator factors in the displayed order. Each ratio has the form $(u+A)/(u+B)$ with $0 < A < B$, so it is positive, less than one, and strictly increasing. Their product has the same properties and tends to one. At zero,

$$\frac{a_{j+1}}{a_j} = \vartheta_j(0) = \frac{1}{729}\left(\frac{j}{j+\frac{1}{3}}\right)^2\left(\frac{j}{j+\frac{2}{3}}\right)^2\left(\frac{j+\frac{1}{2}}{j+1}\right)^2 < \frac{1}{729}.$$

In (136), the first factor is positive and strictly decreasing, and all remaining factors are positive and decreasing. The product is therefore strictly decreasing. Its limits follow from (135) and $\vartheta_j(u) \to 1$. □

Theorem 42.2 (Sharp nonuniform prefix budget)**.** *Fix real numbers* $\lambda_k \ge 0$ *for* $n \le k < M$*. The following statements are equivalent:*

1. *Every real coefficient vector with* $q > 0$ *and* $S_k \ge -q\lambda_k$ *for* $n \le k < M$ *satisfies* $K_c(u) > 0$ *for every* $u > 0$*.*
2. *The rationally weighted budget satisfies*

$$\boxed{\sum_{k=n}^{M-1} \lambda_k(a_k - a_{k+1}) \le a_M.} \tag{137}$$

The equivalence remains true if "every real coefficient vector" is replaced by "every rational coefficient vector".

If the budget is exceeded, the extremal real prefix vector $S_k = -q\lambda_k$ *has a kernel that is negative near zero and positive for sufficiently large* u*, with exactly one zero on* $(0,\infty)$*. No prime-log cancellation is imposed in this sharpness assertion.*

Proof. Put $T_k = S_k + q\lambda_k \geq 0$. Equation (134) becomes

$$K_c(u) = \sum_{k=n}^{M-1} T_k(f_k - f_{k+1}) + qf_M(u)\left(1 - \sum_{k=n}^{M-1} \lambda_k g_{k,M}(u)\right). \tag{138}$$

Every term in the first sum is nonnegative. If some λ_k is positive, Lemma 42.1 gives, for every $u > 0$,

$$\sum_{k=n}^{M-1} \lambda_k g_{k,M}(u) < \frac{\sum_{k=n}^{M-1} \lambda_k(a_k - a_{k+1})}{a_M} \leq 1.$$

If all λ_k vanish, the same sum is zero. Thus the last term in (138) is strictly positive even when (137) is an equality. This proves sufficiency.

If the budget is exceeded, take $T_k = 0$ for every $k < M$. Then the parenthesized sum in (138) is $1 - G(u)$, where G is continuous and strictly decreasing from a value greater than one to zero. There is exactly one point with $G(u) = 1$, and the claimed signs follow. These prefixes give a real coefficient vector by the inverse formula preceding the lemma. When the λ_k are rational, choose $q = 1$ to get a rational counterexample directly. For arbitrary real λ_k, choose rational numbers $0 \leq \lambda'_k \leq \lambda_k$ sufficiently close from below that their weighted budget still exceeds a_M. The extremal prefixes $S_k = -\lambda'_k$, $q = 1$, then give a rational counterexample in the original prefix class. This proves necessity also for that class. □

Corollary 42.3 (Exact uniform tolerance)**.** *Put*

$$\kappa_{n,M} = \frac{a_M}{a_n} = 16^{n-M}\left(\frac{n}{M}\right)^4 \left(\frac{\binom{3n}{n}}{\binom{3M}{M}}\right)^2 \in (0,1).$$

For $\lambda \geq 0$, every coefficient vector with $q > 0$ and $S_k \geq -\lambda q$ has a strictly positive kernel on $(0,\infty)$ if and only if

$$\boxed{0 \leq \lambda \leq \frac{\kappa_{n,M}}{1 - \kappa_{n,M}}.}$$

At equality positivity remains strict. The same statement holds for real or rational coefficient vectors, without imposing prime-log cancellation.

Proof. Set $\lambda_k = \lambda$ in Theorem 42.2 and use $\sum_{k=n}^{M-1}(a_k - a_{k+1}) = a_n - a_M$. Equivalently, the extremal kernel is $q((1+\lambda)f_M - \lambda f_n)$ and $f_M(u)/f_n(u) > a_M/a_n$ for $u > 0$. □

Corollary 42.4 (A certificate for an individual coefficient vector)**.** *Let $(x)_+ = \max(x, 0)$. If $q > 0$ and*

$$\boxed{\sum_{k=n}^{M-1}(a_k - a_{k+1})(-S_k)_+ \leq qa_M,} \tag{139}$$

then $K_c(u) > 0$ for all $u > 0$. Every such vector also satisfies

$$S_k > -q/728 \qquad (n \leq k < M). \tag{140}$$

Condition (139) is sufficient for an individual kernel; it is not asserted to be necessary for its positivity.

Proof. Apply Theorem 42.2 with $\lambda_k = (-S_k)_+/q$. Moreover, each summand in the budget is nonnegative, so

$$\frac{(-S_k)_+}{q} \leq \frac{a_M}{a_k - a_{k+1}} \leq \frac{a_{k+1}}{a_k - a_{k+1}} < \frac{1/729}{1 - 1/729} = \frac{1}{728}.$$

This proves (140). □

42.1 The complete budget is a compact rational interval problem

We now impose exact prime-log cancellation. Retain the established linear forms

$$I_j = \int_0^\infty R_j(u)\, d\mu(u) = 16^j\gamma - p_j + \sum_\ell E_{j,\ell} \log \ell, \qquad d\mu(u) = \frac{du}{e^{2\pi\sqrt{u}} - 1},$$

where $p_j, E_{j,\ell}$ are rational and only primes $\ell \le 3M$ need be retained on this window. Define

$$e_j = (E_{j,\ell}/16^j)_\ell, \quad \bar{p}_j = p_j/16^j, \quad \rho_j = \sum_\ell e_{j,\ell} \log \ell, \quad \Delta_j = \rho_{j+1} - \rho_j > 0.$$

The strict positivity of Δ_j is the established logarithmic ordering property of these kernels. For a normalized prefix vector $S_M = 1$, put

$$t(S) = \bar{p}_M + \sum_{k=n}^{M-1} S_k(\bar{p}_k - \bar{p}_{k+1}).$$

Theorem 42.5 (A rational interval for all admissible negative budgets)**.** *The set*

$$\mathcal{B}_{n,M} = \Big\{ (S,\lambda) : \lambda_k \ge 0, \quad S_k + \lambda_k \ge 0 \quad (n \le k < M), \\ \sum_{k=n}^{M-1} (a_k - a_{k+1})\lambda_k \le a_M, \\ \sum_{k=n}^{M-1} S_k(e_k - e_{k+1}) = -e_M \Big\}$$

is either empty or a compact rational polytope. Its projection onto the S coordinates is exactly the set of normalized log-null prefix vectors satisfying (139) with $q = 1$. If it is nonempty, its output image under $t(S)$ is a closed rational interval $[A_{n,M}, B_{n,M}]$, possibly a singleton, and

$$B_{n,M} < \gamma.$$

Every rational point of this interval has a rational feasible lift. Consequently, for relatively prime integers $q > 0, p$, a rational log-null lift with output (q,p) satisfying (139) exists if and only if $p/q \in [A_{n,M}, B_{n,M}]$.

Proof. All defining equations and inequalities have rational coefficients. The existence of nonnegative λ_k with $\lambda_k \ge (-S_k)_+$ and the displayed budget is equivalent to the negative-part budget itself; for the reverse direction choose $\lambda_k = (-S_k)_+$. This proves the projection statement.

Because $a_k - a_{k+1} > 0$, the budget bounds each λ_k. Taking the scalar product of the prime-row equation with $(\log \ell)_\ell$ gives

$$\sum_{k=n}^{M-1} S_k \Delta_k = \rho_M.$$

For $T_k = S_k + \lambda_k \ge 0$ this becomes

$$\sum_{k=n}^{M-1} T_k \Delta_k = \rho_M + \sum_{k=n}^{M-1} \lambda_k \Delta_k.$$

The right side is bounded on the already bounded set of possible λ, and every Δ_k is strictly positive. Thus each T_k, and hence each S_k, is bounded. Closedness proves compactness.

A nonempty bounded rational polyhedron has rational vertices. Linear projection onto one coordinate is a compact interval with rational endpoints, attained at rational feasible points. For every feasible point, Corollary 42.4 gives

$$\gamma - t(S) = \int_0^\infty K_c(u)\, d\mu(u) > 0.$$

The upper endpoint is attained, so its inequality is strict as well. For any rational point of a nonsingleton image interval, rational convex interpolation of rational endpoint lifts gives a rational lift. A singleton already has a rational attaining lift.

Finally, multiply a normalized rational lift of p/q by q to obtain the integer output (q, p). Conversely, divide a lift of (q, p) by $q > 0$ to obtain a normalized feasible point. Both the prime-row equations and the negative-part budget are homogeneous under this positive scaling. □

This polytope contains the nonnegative-prefix polytope, by taking $\lambda = 0$. It also contains every uniform-offset polytope whose offset is at most the sharp value in Corollary 42.3. The usual exact minimum-denominator calculation for a closed rational interval therefore applies to this entire budget at once. When the old nonnegative-prefix output interval is nondegenerate, its enlargement can be made explicit without another linear optimization.

Corollary 42.6 (Constructive strict enlargement of an output interval)**.** *Suppose that the normalized nonnegative-prefix polytope on $[n, M]$ has a nondegenerate output interval $[A, B]$. Choose rational nonnegative endpoint lifts S^-, S^+, with outputs A, B, and set*

$$D_k = S_k^+ - S_k^-, \qquad Z_+ = \sum_{k=n}^{M-1} (a_k - a_{k+1})(D_k)_+, \qquad Z_- = \sum_{k=n}^{M-1} (a_k - a_{k+1})(-D_k)_+.$$

Then $Z_+, Z_- > 0$, and the new weighted-budget output interval contains the strictly larger rational interval

$$\boxed{\left[A - \frac{a_M}{Z_+}(B - A), \quad B + \frac{a_M}{Z_-}(B - A)\right].} \tag{141}$$

Both displayed endpoints have explicit rational log-null lifts and are strictly below γ. The least primitive denominator in the new interval cannot exceed its value in the old interval.

Proof. Subtracting the prime-row equations for S^+ and S^- gives a homogeneous prime-row equation for D. In particular,

$$\sum_{k=n}^{M-1} D_k \Delta_k = 0.$$

The difference of the outputs is $B - A > 0$, so $D \neq 0$. As every $\Delta_k > 0$, the vector D has both positive and negative coordinates. Hence $Z_+, Z_- > 0$.

Set $\varepsilon_+ = a_M/Z_+$ and $\varepsilon_- = a_M/Z_-$. The extrapolated prefixes

$$\widetilde{S}^- = S^- - \varepsilon_+ D, \qquad \widetilde{S}^+ = S^+ + \varepsilon_- D$$

still satisfy the normalized prime-row equation. Nonnegativity of S^- and S^+ gives the componentwise bounds

$$(-\widetilde{S}_k^-)_+ \le \varepsilon_+ (D_k)_+, \qquad (-\widetilde{S}_k^+)_+ \le \varepsilon_- (-D_k)_+.$$

Their weighted negative masses are at most a_M, so both are feasible for the new polytope. Linearity of the output map gives the endpoints in (141); positivity and the strict inequalities below γ follow from Theorem 42.5. Convexity supplies every point between them. The denominator assertion follows from containment; strict enlargement alone does not imply a strict decrease in the minimum denominator. □

42.2 A conditional completion using one output at each index

Write $\mathcal{U}_j$ for the established rational bounds $0 < I_j < \mathcal{U}_j$ and $C(c) = \sum_{j=n}^{M} \mathcal{U}_j |c_j|$. The signed-prefix estimate gives

$$C(c) = O\left((1+\lambda) q \, 729^{-n} \frac{\log(M+1)}{\sqrt{n}}\right) \tag{142}$$

for every rational log-null vector with $q > 0$ and $S_k \geq -\lambda q$, uniformly in $M \geq n$, $\lambda \geq 0$, and the chosen lift. This is the earlier signed cost estimate, with an absolute implied constant.

Corollary 42.7 (Single-output completion under the sharp budget)**.** *Suppose that for unbounded integers n there are integers $M_n > n$ and rational log-null lifts $c^{(n)}$ on $[n, M_n]$ with integer outputs*

$$q_n = \sum_{j=n}^{M_n} 16^j c_j^{(n)} > 0, \qquad p_n = \sum_{j=n}^{M_n} p_j c_j^{(n)} \in \mathbb{Z}.$$

Suppose their prefixes satisfy

$$\sum_{k=n}^{M_n - 1} (a_k - a_{k+1})(-S_k^{(n)})_+ \leq q_n a_{M_n},$$

and

$$\frac{q_n \log(M_n + 1)}{729^n \sqrt{n}} \longrightarrow 0. \tag{143}$$

Then γ is irrational. It is sufficient, for example, to have $q_n \leq 729^n$ and $\log(M_n + 1) = o(\sqrt{n})$. No second output, determinant condition, or separate assumption $q_n \to \infty$ is needed.

Proof. Corollary 42.4 gives a strictly positive kernel and $S_k^{(n)} > -q_n/728$. Apply (142) with the fixed value $\lambda = 1/728$. Exact prime-log cancellation and the strict integral bound give

$$0 < q_n \gamma - p_n < C(c^{(n)}) = O\left(q_n 729^{-n} \frac{\log(M_n + 1)}{\sqrt{n}}\right) \longrightarrow 0.$$

If $\gamma = a/b$ with integers a, b and $b > 0$, then $b(q_n \gamma - p_n) = a q_n - b p_n$ would be a positive integer tending to zero, which is impossible. □

For primitive targets the rational-input formulation already accounts for clearing denominators and removing the complete output gcd: if D clears a rational lift of a coprime pair (q, p), its output is (Dq, Dp) and the complete gcd is D. Dividing the form and its cost by D restores exactly the target and cost used above.

Scope and provenance. The ratio calculations, the exact negative-prefix budget, its sharp endpoint, and its use with the signed cost bound are proved here. Summation by parts, monotonicity of products of positive functions, and rational-polytope projection are classical ingredients. No claim of historical priority for a general principle is made. Theorem 42.2 settles positivity for its specified prefix class; it does not produce log-null vectors in that class. Corollary 42.7 leaves explicit the unproved existence of integer outputs obeying its conditions at unbounded indices. It is not a proof of the irrationality of γ.

Example 42.8 (The first nontrivial window and its sharp boundary)**.** For $n = 1$, $M = 2$, the initial slopes are

$$a_1 = \frac{1}{144}, \qquad a_2 = \frac{1}{921600}, \qquad \kappa_{1,2} = \frac{1}{6400}.$$

Thus the exact uniform tolerance is $\lambda_* = 1/6399$. At its boundary, take $q = 1$, $S_1 = -1/6399$, $S_2 = 1$. The unique coefficients are

$$c_1 = -\frac{1}{102384}, \qquad c_2 = \frac{25}{6399}.$$

Direct simplification gives

$$K_c(u) = \frac{6400 f_2(u) - f_1(u)}{6399} = \frac{u^2(4u+1)(6399u^2 + 27123u + 33324)}{25596 \prod_{r=1}^{6}(u + r^2)} > 0 \qquad (u > 0).$$

The leading order at zero has changed from u to u^2, with

$$\lim_{u \downarrow 0} \frac{K_c(u)}{u^2} = \frac{2777}{1105747200} > 0.$$

For a general uniform offset λ, the extremal kernel is $(1+\lambda)f_2 - \lambda f_1$ and

$$\lim_{u \downarrow 0} \frac{(1+\lambda)f_2(u) - \lambda f_1(u)}{u} = \frac{1 - 6399\lambda}{921600}.$$

It is therefore negative near zero whenever $\lambda > 1/6399$. This example checks strict positivity at the budget boundary and the sharp failure immediately beyond it. It does not satisfy prime-log cancellation: $E_{1,5} = 0$, $E_{2,5} = -2600$, so its coefficient of $\log 5$ is $-65000/6399 \neq 0$.

43 An exact affine correction joins primitive arithmetic and nonvanishing

For a finite rational coefficient vector c define

$$Ec = \left(\sum_j c_j E_{j,\ell} \right)_{\ell}, \qquad \Psi(c) = \left(\sum_j 16^j c_j, \sum_j p_j c_j \right), \qquad C(c) = \sum_j \mathcal{U}_j |c_j|.$$

If $Ec = 0$ and $\Psi(c) = (q, p)$, then

$$\sum_j c_j I_j = q\gamma - p, \qquad |q\gamma - p| \le C(c).$$

All vectors below may be padded with zeros, so their sums and common supports are unambiguous.

Lemma 43.1 (A fixed exact correction of cost less than four)**.** *There is a rational vector d_0 supported on $1, \dots, 17$ with*

$$Ed_0 = 0, \qquad \Psi(d_0) = (0, 1), \qquad B_0 := C(d_0) < 4.$$

The vector is uniquely specified by

$$d_0 = \begin{pmatrix} V_{17} \\ (16^j)_{j=1}^{17} \\ (p_j)_{j=1}^{17} \end{pmatrix}^{-1} \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

Proof. The displayed matrix is the nonsingular rational 17-by-17 augmented matrix whose nonsingularity is certified by the exact elimination below. Thus the vector is defined and has the required formal output. The standard-library verifier `verify_affine_completion.py`

regenerates every matrix entry from the residue and harmonic-number formulas, solves the displayed system by exact rational elimination, checks all seventeen equations, and checks the strict rational bounds

$$\frac{3775898}{10^6} < C(d_0) < \frac{3775899}{10^6} < 4.$$

No numerical value of γ or of a logarithm is used. The qualitative correction theorem below only needs the finite constant B_0; the additional inequality $B_0 < 4$ is this explicit finite arithmetic certificate. □

Theorem 43.2 (Primitive completion with exact cost and error bounds)**.** *Suppose c is rational, $Ec = 0$, and*

$$\Psi(c) = (q, p) \in \mathbb{Z}^2, \qquad q > 0, \qquad \gcd(q, p) = 1.$$

Put $\varepsilon = C(c)$ and $e = q\gamma - p$. Choose integers r, s satisfying

$$qs - pr = 1, \qquad |r| \le q/2.$$

Such a choice exists by Bezout's identity and reduction modulo q. The explicit rational vector

$$\boxed{c' = \frac{r}{q}\, c + \frac{1}{q}\, d_0}$$

has

$$Ec' = 0, \qquad \Psi(c') = (r, s), \qquad \det \begin{pmatrix} q & r \\ p & s \end{pmatrix} = 1.$$

Its weighted cost and its evaluated error satisfy the distinct bounds

$$\boxed{C(c') \le \frac{\varepsilon}{2} + \frac{B_0}{q} < \frac{\varepsilon}{2} + \frac{4}{q},}$$

$$\boxed{e' := r\gamma - s = \frac{r}{q}\, e - \frac{1}{q}, \qquad |e'| \le \frac{|e|}{2} + \frac{1}{q}.}$$

The correction has support contained in the support of c together with $1, \dots, 17$. Both output pairs are primitive.

Proof. For $q > 1$, take the centered representative $r \equiv -p^{-1} \pmod q$, and set $s = (1 + pr)/q$. For $q = 1$, take $(r, s) = (0, 1)$. Linearity gives

$$\Psi(c') = \frac{r}{q}(q, p) + \frac{1}{q}(0, 1) = \left(r, \frac{rp + 1}{q}\right) = (r, s),$$

and $Ec' = 0$. The cost bound is the weighted triangle inequality. Evaluation at γ gives the displayed identity for e', because the evaluated value of d_0 is -1. The determinant is one, which also proves primitiveness of (r, s). □

Lemma 43.3 (Clearing coefficients introduces no hidden primitive cost)**.** *If a rational log-null lift h has primitive integer output (a, b), and a positive integer g clears its coefficient denominators, then $u = gh$ is integral,*

$$\Psi(u) = g(a, b), \qquad \gcd(|\Psi(u)_1|, |\Psi(u)_2|) = g, \qquad C(u)/g = C(h).$$

Thus Theorem 43.2 bounds the actual cost after the complete output gcd has been removed. It does not assume that the rational coefficient denominators may be ignored.

Proof. Linearity gives the output identity, and $\gcd(a,b) = 1$ gives the exact gcd. Positive homogeneity gives the cost identity. If a further common normalization makes the entire output map integral, it multiplies both the cost and the exact output gcd by the same factor and cancels in the same way. □

Proposition 43.4 (A general transverse-anchor correction)**.** *In place of d_0, let d be any rational log-null vector with output w and cost B. Let c have primitive integer output u, and assume $\Delta = \det(u,w) \neq 0$. There is an explicit rational log-null lift h of an integer vector v with*

$$\det(u,v) = 1, \qquad C(h) \leq \frac{C(c)}{2} + \frac{B}{|\Delta|}.$$

Proof. Choose any integer v_0 with $\det(u,v_0) = 1$. Since $\det(u, w/\Delta) = 1$, there is a rational t with $w/\Delta = v_0 + tu$. Choose an integer k with $|k - t| \leq 1/2$. Then

$$h = d/\Delta + (k-t)c, \qquad v = v_0 + ku$$

have the claimed output, determinant, and cost. □

44 One compatible primitive family is enough

Theorem 44.1 (All five conditions from one primitive family)**.** *Suppose a sequence of rational log-null lifts c_m has primitive integer outputs (q_m, p_m) with*

$$q_m > 0, \qquad q_m \longrightarrow \infty, \qquad \varepsilon_m := C(c_m) \longrightarrow 0.$$

Apply Theorem 43.2 separately at every index and write the complementary output as (r_m, s_m). The resulting integer polynomials

$$W_m(X) = (q_m X - p_m)^2 + (r_m X - s_m)^2$$

have discriminant exactly -4 and satisfy

$$W_m(X) > 0 \quad \textit{for every real } X,$$

$$0 < W_m(\gamma) \leq \varepsilon_m^2 + \left(\frac{\varepsilon_m}{2} + \frac{1}{q_m}\right)^2 \longrightarrow 0.$$

All logarithms cancel exactly, both linear outputs have primitive integer coefficients, and the polynomial degree is the fixed number two. If $\gamma = a/b$ were rational, the numbers $b^2 W_m(\gamma)$ would be positive integers tending to zero. Thus the stated hypotheses imply irrationality.

Proof. The determinant of the two output columns is one. They therefore cannot vanish at the same real argument, proving the strict positivity of the sum of squares. Equivalently, its discriminant is $-4(q_m s_m - p_m r_m)^2 = -4$, and its minimum on the real line is $1/(q_m^2 + r_m^2) > 0$. The error estimate follows from the separate evaluation bounds in Theorem 43.2. Under a rationality assumption the common integerizing factor is the fixed number b^2. The preceding lemma verifies that coefficient clearing and removal of the complete output gcd cause no unrecorded loss in the cost. □

Corollary 44.2 (A positive primitive family makes denominator growth automatic)**.** *Suppose the c_m in Theorem 44.1 instead satisfy*

$$q_m > 0, \qquad C(c_m) \to 0, \qquad \sum_j c_{m,j} I_j > 0,$$

where the last inequality is proved by an exact kernel certificate. Then $q_m \to \infty$, so the same conclusion follows.

Proof. If $q_m \le H$ on infinitely many indices, then $0 < q_m\gamma - p_m < 1$ eventually on those indices. Only finitely many integer pairs (q_m, p_m) are possible. Their strictly positive errors form a finite set and hence have a positive minimum, contradicting convergence of those errors to zero. This holds for every fixed H, proving $q_m \to \infty$. □

Corollary 44.3 (A divisibility escape is an alternative joining condition)**.** *Suppose rational log-null lifts h_m have integer outputs (Q_m, P_m), satisfy $C(h_m) \to 0$, and obey*

$$2^{t_m} \mid Q_m, \qquad P_m \neq 0, \qquad v_2(P_m) \le t_m - a_m, \qquad a_m \to \infty.$$

Here $0 \le a_m \le t_m$ are integers and the valuation inequality is an exact arithmetic condition, not a sign or size test on γ. For all sufficiently large m, $Q_m \neq 0$, and the primitive output denominator satisfies

$$q_m = \frac{|Q_m|}{\gcd(|Q_m|, |P_m|)} \ge 2^{a_m}.$$

After primitive normalization and orientation, the hypotheses of Theorem 44.1 hold.

Proof. If $Q_m = 0$, then the evaluated error is the nonzero integer $-P_m$, whose absolute value is at least one. This contradicts $|Q_m\gamma - P_m| \le C(h_m) \to 0$ for large m. For nonzero Q_m, the two-adic valuation of the primitive denominator is

$$v_2(q_m) = \max\{v_2(Q_m) - v_2(P_m), 0\} \ge a_m.$$

Dividing the input by the complete output gcd and changing its sign if necessary cannot increase its cost. The preceding theorem applies. □

For a concrete reverse-design box, put

$$M_n = n + \left\lceil \frac{25n}{\log n} \right\rceil, \qquad |c_k| \le 2^n \quad (0 \le k \le M_n - n), \qquad c_k \in \mathbb{Z}.$$

For every vector in this box the fully cleared cost satisfies

$$D_{M_n} \sum_k \mathcal{U}_{n+k}|c_k| \le \frac{729}{473} D_{M_n} 2^n \mathcal{U}_n = \exp\{-\varepsilon n + o(n)\} \longrightarrow 0, \qquad \varepsilon = \log(729/32) - 3 > 0.$$

Indeed $M_n = n + o(n)$ and the prime number theorem [7] gives $\log D_{M_n} = 3n + o(n)$, while the factorial expression for $\mathcal{U}_n$ gives $\log \mathcal{U}_n = -\log(729/16)n + O(\log n)$. If the input is log-null, its integer gamma coordinate is $Q_n = D_{M_n} 16^n \sum_k 16^k c_k$. Thus one may take $t_n = 4n + v_2(D_{M_n})$ in the preceding corollary. Finding an exactly log-null vector in this box with

$$P_n \not\equiv 0 \pmod{2^{\,t_n - a_n + 1}}, \qquad a_n \to \infty,$$

would supply the divisibility escape without changing the secured denominator-cleared decay. No such infinite surviving family is established here. This box estimate does not identify which inputs have exact log cancellation, nonzero outputs, or positive kernels.

The cumulative-prefix criterion

$$\sum_{j \le k} 16^j c_{m,j} \ge 0 \quad \text{for every prefix } k, \qquad c_m \neq 0,$$

together with exact log cancellation, supplies the strict positivity and $q_m > 0$ in Corollary 44.2 by the cumulative cone theorem. The latter strict inequality uses more than prefix nonnegativity: the normalized log values $\rho_j = L_j/16^j$ increase strictly, and summation by parts gives $0 = \sum_{k<M} S_k(\rho_k - \rho_{k+1}) + S_M\rho_M$. If $S_M = q_m = 0$ and the input is nonzero, this sum would be negative. Thus the criterion can be checked using rational arithmetic, with its implication justified by that structural theorem. What remains to be proved is an infinite supply of primitive integer targets with such lifts and costs tending to zero. The existence of one finite certificate does not supply that infinite family.

Compatibility and the scope of the correction. The matrices $B_m = \left(\begin{smallmatrix} q_m & r_m \\ p_m & s_m \end{smallmatrix}\right)$ lie in $\mathrm{SL}_2(\mathbb{Z})$. Hence consecutive stages are automatically related by the exact nonautonomous integral transfer $B_{m+1} = B_m(B_m^{-1}B_{m+1})$, with $B_m^{-1}B_{m+1} \in \mathrm{SL}_2(\mathbb{Z})$. This is compatibility of already constructed stages, not a uniform bound on those transfer matrices or an algorithm for finding the next cheap positive lift.

The correction uses rational coefficients and the fixed old block $1, \dots, 17$, followed by exact primitive normalization. It therefore does not assert the existence of two small-height integer input vectors supported in the same thin moving block. The previously proved raw-height obstruction for such pairs is unaffected.

45 Finite brackets and the fixed-output trap

Proposition 45.1 (An exact adjacent rational bracket)**.** *Suppose $q > 1$, $\gcd(q,p) = 1$, and an exact certificate gives $0 < e = q\gamma - p < 1/q$. Choose the uncentered solution $0 < r < q$ of $pr \equiv -1 \pmod q$, and put $s = (1+pr)/q$. Then*

$$\frac{p}{q} < \gamma < \frac{s}{r}, \qquad \frac{s}{r} - \frac{p}{q} = \frac{1}{qr}.$$

The corresponding proper-kernel lift is $c' = (r/q)c + d_0/q$ and has cost at most $C(c) + B_0/q$.

Proof. The lower inequality is the assumed positive error. The exact complementary error is $(re-1)/q < 0$, since $r < q$ and $qe < 1$. Dividing by positive r gives the upper inequality. The determinant identity gives the interval width, and the triangle inequality gives the cost. □

For the certified positive primitive target at $N = 32$,

$$(q,p) = (200542096649, 115756039658), \qquad C(c) < \frac{1}{530000000000},$$

one has $qC(c) < 1$. The uncentered complement is exactly

$$(r,s) = (176450949647, 101850252223),$$

while the centered complement is $(-24091147002, -13905787435)$. The standalone verifier checks both the determinant arithmetic and the complete output-gcd normalization of the centered proper-kernel lift. These are finite certificates; no infinite assertion follows from their repetition or scaling.

Proposition 45.2 (Why contraction in a fixed output fiber is insufficient)**.** *If rational log-null lifts h_m all have the same integer output $v = (q,p)$ and $C(h_m) \to 0$, then $q\gamma - p = 0$. Two independent fixed integer outputs cannot both have such lifts.*

Proof. The same evaluated value satisfies $|q\gamma - p| \le C(h_m)$ for every m and hence is zero. The kernel of the nonzero linear functional $(q,p) \mapsto q\gamma - p$ is one-dimensional, so it cannot contain two independent outputs. □

Thus a nested-fiber or compactness construction cannot obtain the desired certificate merely by preserving one fixed formal output while shrinking its cost. The primitive outputs must move, or strict nonvanishing must be supplied by another condition. The affine correction theorem makes that movement compatible with exact integrality and determinant-one independence; it does not supply the unresolved infinite cost decay in the positive cone.

46 The entire finite output problem is a rational interval problem

For fixed integers $1 \le n < M$ put $e_j = (E_{j,\ell}/16^j)_\ell$ and $\bar{p}_j = p_j/16^j$. Let $\mathcal{P}_{n,M}$ be the normalized prefix polytope

$$\left\{(S_n, \dots, S_{M-1}) : \ S_k \ge 0, \quad \sum_{k=n}^{M-1} S_k(e_k - e_{k+1}) = -e_M \right\}, \qquad S_M = 1.$$

Define its output coordinate

$$t(S) = \bar{p}_M + \sum_{k=n}^{M-1} S_k(\bar{p}_k - \bar{p}_{k+1}).$$

Theorem 46.1 (Exact interval reduction, including boundary points)**.** *Either $\mathcal{P}_{n,M}$ is empty or it is a compact rational polytope. In the latter case its output image is a closed rational interval $[a_{n,M}, b_{n,M}]$, possibly a singleton, and $b_{n,M} < \gamma$. For relatively prime integers $q > 0, p$, the following assertions are equivalent:*

1. *There is a rational log-null lift on $[n, M]$ with all prefixes nonnegative and output (q, p).*
2. *$p/q \in [a_{n,M}, b_{n,M}]$.*

Consequently the least possible primitive denominator is exactly the least denominator of a rational in this closed interval.

Proof. The prefix bound already proved gives $0 \le S_k < 66MT_M$ throughout this rational polyhedron; it is therefore bounded and closed, hence compact. Linear projection to one real coordinate gives a compact interval. A bounded nonempty rational polyhedron has rational vertices, so both extrema are rational and have rational attaining prefix vectors S^-, S^+.

Every normalized feasible vector has nonzero cumulative-positive kernel and hence evaluated error $\gamma - t(S) > 0$. The maximum is attained, so $b_{n,M} < \gamma$, including the endpoint.

Normalize a lift of (q, p) by q to prove the first implication. Conversely, for a rational $t = p/q$ in a nonsingleton interval, put

$$\lambda = \frac{t - a_{n,M}}{b_{n,M} - a_{n,M}}, \qquad S = (1 - \lambda)S^- + \lambda S^+.$$

This is a rational feasible prefix vector. Multiplying its associated coefficients $(S_j - S_{j-1})/16^j$ by q gives the required lift. For a singleton, use either rational attaining vector. No strict interior assumption is used. □

Proposition 46.2 (A short certificate of denominator minimality)**.** *Suppose $[a, b]$ is contained strictly between Farey neighbors $L = u/v < R = w/z$, where $v, z > 0$ and $wv - uz = 1$. If their mediant*

$$t_* = (u + w)/(v + z)$$

belongs to $[a, b]$, then $v + z$ is the least denominator of a rational in $[a, b]$. The mediant is the unique such rational of that denominator. Integer translations preserve this assertion.

Proof. For a reduced rational p/q strictly between L and R, the integers $r = wq - zp$ and $s = vp - uq$ are both positive. The determinant identity gives $q = rv + sz \ge v + z$. Equality forces $r = s = 1$, giving the mediant. The general integer and singleton cases are handled separately. □

Subtract $\lfloor a \rfloor$ first. If the interval contains an integer, its minimum denominator is one. Otherwise the translated interval lies strictly between zero and one. The accelerated Stern–Brocot algorithm starts from $0/1, 1/0$ and moves the left or right endpoint by the largest integer multiple of the other that keeps the bracket strictly outside the closed target interval. It stops at the first mediant in the interval. Equivalently the jump on the left is

$$k = \left\lceil \frac{av - u}{w - az} \right\rceil - 1, \qquad (u, v) \leftarrow (u + kw, v + kz),$$

and the right jump is the analogous formula with b. The strict-bracket condition treats rational endpoints correctly. A rational singleton is returned directly; an interval containing an integer has least denominator one. Standard Stern–Brocot termination also follows by retaining any fixed rational in the interval: every skipped node is strictly outside, and a finite path reaches that rational unless an earlier mediant already lies inside. The final neighboring pair is a four-integer exact minimality witness; no numerical gamma approximation is needed.

For a one-dimensional affine prefix fiber, two endpoint linear programs are unnecessary. Exact elimination writes every solution as $S = x + tz$. Nonnegativity gives the closed parameter interval

$$\max_{z_i > 0}(-x_i/z_i) \le t \le \min_{z_i < 0}(-x_i/z_i),$$

provided every fixed coordinate with $z_i = 0$ is nonnegative. Projection of its two endpoints yields $[a, b]$ exactly. This is the case in the finite moving-window certificates accompanying this note.

Scope and provenance. Compact-polytope projection and Farey denominator minimality are classical. The point here is their exact use for this gamma-kernel cone: a fixed-window mixed-integer search becomes two rational linear optimizations and one rational-interval calculation. Neither the reduction nor a finite list of successful intervals proves that the required height inequality holds at unbounded indices.

Proposition 46.3 (The signed-prefix interval is also compact)**.** *Fix rational $\lambda \ge 0$ and replace $S_k \ge 0$ in $\mathcal{P}_{n,M}$ by $S_k \ge -\lambda$. The resulting rational polyhedron is either empty or compact, and its output is a closed rational interval with rational lifts of every rational point. The inequality $b_{n,M} < \gamma$ is not asserted for this offset case.*

Proof. Put $T_k = S_k + \lambda$. The exact identity used in Theorem 41.4 gives $\sum_{k=n}^{M-1} T_k \Delta_k = (1 + \lambda)\rho_M - \lambda\rho_n$. Its summands are nonnegative and $\Delta_k > 0$, so every T_k, and hence every S_k, is bounded. Closedness and rationality give compactness and rational endpoint lifts. Convex interpolation proves the assertion for all intermediate rational outputs. □

46.1 A determinant-one pair costs no extra denominator height in an interval

Theorem 46.4 (The two smallest denominator levels in a rational interval)**.** *Let $I = [a, b]$ be a nondegenerate closed rational interval containing no integer. Let $F = P/Q$ be its least-denominator rational, and let $L = u/v < R = w/z$ be its final enclosing Farey neighbors, translated back if necessary, so that*

$$L < a \le F \le b < R, \qquad wv - uz = 1, \qquad (P, Q) = (u + w, v + z), \qquad v, z > 0.$$

If $a < F$, define

$$k_- = \left\lceil \frac{av - u}{P - aQ} \right\rceil, \qquad F_- = \frac{u + k_- P}{v + k_- Q}.$$

If $F < b$, define

$$k_+ = \left\lceil \frac{w - bz}{bQ - P} \right\rceil, \qquad F_+ = \frac{w + k_+ P}{z + k_+ Q}.$$

At least one candidate is defined. Each defined candidate belongs to I, is distinct from F, and has determinant of absolute value one with F. The candidate of smaller denominator has the least possible denominator among all rationals in $I \setminus \{F\}$. Consequently the pair formed by F and that candidate minimizes the maximum denominator among all pairs of distinct rationals in I.

Proof. The defining thresholds are strictly positive, so every $k_{\pm} \geq 1$. The left candidate lies strictly below F and at least a by the ceiling inequality; the right one lies strictly above F and at most b. Closed endpoints are therefore included correctly.

Every reduced rational p/q in $[a, F)$ lies strictly between the Farey neighbors L and F. It has the unique representation

$$(p, q) = r(u, v) + s(P, Q), \qquad r, s \in \mathbb{Z}_{>0}.$$

The inequality $p/q \geq a$ implies $s/r \geq (av-u)/(P-aQ)$. Hence $s \geq k_-$ and $q = rv+sQ \geq v+k_-Q$, with equality attained by $r = 1, s = k_-$. The corresponding right-side argument uses the Farey neighbors F, R and gives $q \geq z + k_+Q$. The determinants with F are ± 1 because adding an integer multiple of F to either neighbor preserves that determinant. This also proves that the candidate fractions are already reduced.

Any pair of distinct fractions in I must contain a fraction other than the unique minimum-denominator fraction F. Thus its largest denominator is at least the smaller candidate denominator, and the displayed pair attains this bound. □

Integer and singleton cases. A singleton interval contains no pair of distinct fractions. If an interval contains two integers, it contains two consecutive integers; these give a determinant-one pair of maximum denominator one. If it contains exactly one integer m, the candidates distinct from m of least denominator on its two sides are

$$m - \frac{1}{\lceil 1/(m-a) \rceil} \quad (a < m), \qquad m + \frac{1}{\lceil 1/(b-m) \rceil} \quad (m < b).$$

For example, every reduced $p/q < m$ in I has $1/q \leq m-p/q \leq m-a$, proving the left denominator bound; the other case is identical. Each candidate is adjacent to m. Thus every nondegenerate rational interval admits an explicitly computable determinant-one pair that minimizes the largest denominator among all distinct rational pairs in that interval.

Corollary 46.5 (No extra height cost for determinant nonvanishing)**.** *Suppose a normalized rational prefix polytope has output interval I. For an integer $B \geq 1$, it has two primitive integer outputs of positive denominators at most B and nonzero determinant if and only if it has such a pair with determinant of absolute value one. The minimum possible value of the largest denominator is given by the preceding interval algorithm.*

Proof. Distinct rational output slopes are equivalent to nonzero output determinant. Every rational point of the interval has a rational lift by convex interpolation of rational endpoint lifts. Apply the theorem and the explicitly handled integer cases, then scale each normalized lift to its reduced integer target. Interchanging the pair gives determinant positive one if desired. □

The corollary applies to the nonnegative-prefix polytope. It also applies to its rational offset variant $S_k \geq -\lambda$ for any rational $\lambda \geq 0$, by Proposition 46.3. It is a finite exact height statement and does not itself bound the resulting height as the window moves.

47 Extension schedules, failed shifts, and exact finite repairs

The prime-entry rank theorem in Part III supplies the rank input for this schedule.

47.1 A guaranteed schedule of genuine extension indices

If $(3N-3, 3N]$ contains a new prime, its row forces $c_N = 0$ in every log-null vector supported on $1, \dots, N$. Thus both the full quotient norm and the cumulative-positive feasible cone are exactly unchanged at that step.

There are nevertheless unconditional free extension indices with bounded gaps. For $N \geq 12$, every $N \equiv 2$ or $9 \pmod{10}$ has no new prime: the sole possible odd prime candidate is respectively $3N-1$ or $3N-2$, and it is a multiple of five greater than five. The other candidate is even, and $3N$ is composite. The gaps between these guaranteed indices alternate between seven and three. The exact kernel dimension therefore increases on an infinite explicit schedule with gaps at most seven.

In particular the six indices $33, \dots, 38$ introduce respectively $97, 101, 103, 107, 109, 113$ and are exact plateaus. Index 39 is the first genuine new direction after 32. The next proposition tests that direction using a particular canonical repair. It does not exclude a different, cost-controlled repair or an optimized primitive lift at the same index. Kernel dimension growth by itself gives no bound on primitive cost.

48 A canonical shift preserves positivity but destroys the arithmetic budget

Let V_N be the integer prime-log matrix, and let $\mathcal{P}_N$ be its canonical pivot set: the first nine indices, followed by every new-prime index. Write $B_N = V_N[:, \mathcal{P}_N]$. The exact rank theorem makes B_N invertible. If ι_N inserts pivot coordinates into the full coefficient vector, define the rational projection

$$\mathcal{R}_N(z) = z - \iota_N B_N^{-1} V_N z.$$

It cancels every prime-log coefficient, fixes every vector already in $\ker V_N$, and leaves all nonpivot coordinates of z unchanged. This gives one fully specified log-repair rule.

Proposition 48.1 (Exact failure of the first available shifted repair)**.** *Let c be the certified first primitive lift at $N = 32$, with output* $(200542096649, 115756039658)$ *and cost below* $1/530000000000$. *Shift its coefficients by seven indices: $z_{j+7} = c_j$, with its first seven entries zero. Form $\widetilde{c} = \mathcal{R}_{39}(z)$ and normalize its rational output to a primitive integer output with positive first coordinate. Call the resulting rational lift c^*. Exact arithmetic gives*

$$C(z) < 10^{-23}, \qquad C(\widetilde{c}) > 10^{73}, \qquad C(c^*) > 10^{1088}.$$

Every weighted prefix $S_k^* = \sum_{j=1}^{k} 16^j c_j^*$ is strictly positive, and

$$S_1^* > 10^{1091}.$$

In particular the repaired lift has a positive final kernel, but its actual primitive evaluated error satisfies

$$\boxed{\sum_j c_j^* I_j > 10^{1084}.}$$

The failure is therefore present in the evaluated arithmetic form, not only in the chosen weighted upper bound.

Proof. The script `shift_repair.py` regenerates V_{39}, its canonical pivots, and all rational weights. It solves the repair equations by reverse substitution on new-prime pivots and exact elimination on the initial nine-by-nine block. It checks $V_{39}\widetilde{c} = 0$, computes the rational output and its complete

primitive normalization, and verifies all displayed rational inequalities and all thirty-nine prefix signs. The exact data are in `shift32_to39_certificate.json`. Neither the computation nor the following lower bound evaluates γ or a prime logarithm.

Put $f_j = 16^{-j} R_j$. Finite summation by parts gives

$$\sum_j c_j^* R_j = \sum_{j=1}^{38} S_j^* (f_j - f_{j+1}) + S_{39}^* f_{39}.$$

Every term is positive. On $1 \le u \le 2$ one has

$$f_1(u) \ge \frac{5}{792} > \frac{1}{160}, \qquad \frac{f_2(u)}{f_1(u)} \le \frac{153}{4 \cdot 17 \cdot 26 \cdot 37} < \frac{1}{2}.$$

Also $2\pi\sqrt{2} < 9$ and $e < 3$ imply $d\mu/du > 1/20000$ there. Hence

$$\int_1^2 (f_1 - f_2)\, d\mu > \frac{1}{320 \cdot 20000} > 10^{-7}.$$

The first Abel coefficient alone therefore gives a strict lower bound $S_1^* 10^{-7} > 10^{1084}$. □

49 A certified arithmetic extension at dimension 39

The failure of the canonical repair is specific to that repair rule. At the same dimension, rational quotient optimization produces a much smaller compatible pair. The following finite theorem needs only the displayed witnesses and their exact checks, not a claim of global optimality of the search procedure.

Proposition 49.1 (An exact one-direction update)**.** *Let $N \ge 9$, let $M > N$ introduce no new prime, and suppose every intervening index $N+1, \dots, M-1$ introduces one. Let $h^{(M)}$ be the canonical log-null vector whose last coordinate is one and whose other nonpivot coordinates are zero. Then*

$$\ker_{\mathbb{Q}} V_M = (\ker_{\mathbb{Q}} V_N) \oplus \mathbb{Q} h^{(M)},$$

where old vectors are padded with zeros. Every new lift has the unique form $c = u + t h^{(M)}$ and exact cost

$$C_M(c) = \sum_{j=1}^{N} \mathcal{U}_j |u_j + t h_j^{(M)}| + |t| \sum_{j=N+1}^{M} \mathcal{U}_j |h_j^{(M)}|.$$

The output coordinates and cumulative prefixes are affine functions of (u, t).

Proof. The invertible pivot submatrix defines $h^{(M)}$. Subtract $c_M h^{(M)}$ from any $c \in \ker V_M$. The result has last coordinate zero; the intervening fresh-prime rows successively force its last remaining coordinates to zero until its support is contained in $1, \dots, N$. Uniqueness follows from $h_M^{(M)} = 1$. The cost formula is its definition applied to that decomposition. □

Theorem 49.2 (An exact finite contraction)**.** *There are rational log-null lifts $c^{(1)}, c^{(2)}$ supported on $1, \dots, 39$ with primitive integer outputs*

$$(q_1, p_1) = (18585954467984, 10728104066067),$$

$$(q_2, p_2) = (3361033349029, 1940041099316)$$

such that

$$q_1 p_2 - q_2 p_1 = 1, \qquad C(c^{(1)}) < \frac{1}{7000000000000}, \qquad C(c^{(2)}) < \frac{1}{5600000000000}.$$

Writing $e_i = q_i\gamma - p_i$, the polynomial

$$W_{39}(X) = (q_1X - p_1)^2 + (q_2X - p_2)^2$$

has integer coefficients and discriminant exactly -4. In particular

$$0 < W_{39}(\gamma) < \frac{1}{70000000000002} + \frac{1}{56000000000002}.$$

Both costs are less than one tenth of the cost of the positive $N = 32$ lift, and their maximum is less than one thirty-sixth of the cost of the second primitive certificate previously constructed at $N = 32$.

Proof. The supplement supplies the exact rational vectors in `pair39_witness.json` and a standalone checker `verify_pair39.py`. The checker regenerates each residue from its factorial formula, verifies every prime-log coordinate, both outputs, primitiveness, the full output gcd after coefficient clearing, and the displayed strict rational cost inequalities. The rational comparisons with the prior $N = 32$ data are also checked. No value of γ or of a prime logarithm is evaluated.

For any determinant-one pair the coefficients of W are $A = q_1^2 + q_2^2$, $-2B = -2(q_1p_1 + q_2p_2)$ and $C = p_1^2 + p_2^2$. The identity

$$AC - B^2 = (q_1p_2 - q_2p_1)^2 = 1$$

gives discriminant -4 and the positive real minimum $1/A$. The individual integral bounds give the estimate at γ. □

The exact costs are approximately $1.4211968134 \cdot 10^{-13}$ and $1.7713337217 \cdot 10^{-13}$; these decimals are explanatory only. The two particular optimized lifts have mixed prefix signs and do not supply cumulative-prefix positivity certificates. Their nonvanishing certificate is the determinant-one sum of squares.

This is one successful arithmetic extension after six exact plateaus. Neither the decrease factor nor the existence of another such decrease at every future extension index is inferred from it.

50 An exact obstruction for the two new targets at index 39

The two unconstrained primitive quotient targets at $N = 39$ are

$$u_1 = (q_1, p_1) = (18585954467984, 10728104066067),$$
$$u_2 = (q_2, p_2) = (3361033349029, 1940041099316).$$

They satisfy $q_1p_2 - p_1q_2 = 1$. The first target has an exact log-null lift with weighted cost less than $1/(7 \cdot 10^{12})$. Neither target has a cumulative-positive lift on $[1, 39]$. The proof does not infer infeasibility merely from the signs of the previously optimized lifts.

Proposition 50.1 (A rational dual certificate for the first target)**.** *There is no rational or real vector $c \in \mathbb{R}^{39}$ satisfying*

$$V_{39}c = 0, \qquad \Psi_{39}c = u_1, \qquad S_k(c) = \sum_{j=1}^{k} 16^j c_j \geq 0 \quad (1 \leq k \leq 39).$$

Proof. The accompanying file `cone39_exclusion_certificate.json` specifies primitive integer vectors $y \in \mathbb{Z}^{30}$ and $w \in \mathbb{Z}^{39}$ and integers $\alpha > 0$, $\beta < 0$ such that

$$w_k \geq 0, \qquad \{k : w_k > 0\} = \{1, 2, 3, 4, 5, 6, 7, 10\},$$

and the following is an identity of rational-linear forms in every $c \in \mathbb{R}^{39}$:

$$\boxed{\sum_{k=1}^{39} w_k S_k(c) = y^T V_{39} c + \alpha Q(c) + \beta P(c).}$$

It also certifies the strict integer inequality $\alpha q_1 + \beta p_1 < 0$. The standalone standard-library verifier regenerates all residues from their factorial formula and checks the displayed identity in all 39 coordinate columns, the nonnegative integer weights, the primitive gcd, and the negative right side. Thus this is an exact rational Farkas certificate. On a putative feasible lift the left side would be nonnegative and the right side strictly negative, a contradiction.

For a direct reproducible definition of the certificate, put $W = [V_{39}; (16^j); (p_j)]$ and $B = W[:, 8{:}39]$. Exact elimination shows that this 32 by 32 matrix is invertible. Let $h_j = 16^j$ for $8 \le j \le 10$ and $h_j = 0$ for $11 \le j \le 39$; solve $B^T z = h$. If $r = z^T W$ and $r_{40} = 0$, set $\widetilde{w}_j = r_j/16^j - r_{j+1}/16^{j+1}$. Clearing the common denominators of $(z, \widetilde{w})$ and removing their total gcd gives exactly (y, α, β, w) in the certificate. The verifier checks the resulting identity directly and does not need to trust the elimination that discovered it. □

Proposition 50.2 (The second target has negative value)**.** *The exact value $q_2\gamma - p_2$ is strictly negative. Consequently it has no cumulative-positive lift at any index.*

Proof. Write $e_i = q_i\gamma - p_i$. The determinant identity gives

$$q_2 e_1 - q_1 e_2 = 1.$$

The independently verified first lift proves $|e_1| < 1/(7 \cdot 10^{12})$. If $e_2 \ge 0$, then

$$1 = q_2 e_1 - q_1 e_2 \le q_2 |e_1| < \frac{3361033349029}{7000000000000} < 1,$$

a contradiction. Every nonzero cumulative-positive log-null lift has strictly positive evaluated value, so none can represent u_2. □

The first obstruction concerns the specified cumulative-prefix cone through index 39; it does not exclude a different positivity certificate for a resulting kernel, or a cumulative-positive lift of u_1 after adding later indices. The second obstruction is an exact sign statement for its oriented target. Neither result is inferred from a numerical approximation to γ.

51 Exact moving-window certificates and a one-step positive repair

Theorem 51.1 (Eighteen exact moving-window tests)**.** *For each row of the following table, the normalized cumulative-prefix polytope is nonempty and has affine dimension one. Its complete output interval has the indicated least possible primitive denominator q. There is a rational attaining lift with output (q, p), nonnegative prefixes, and cost strictly below the indicated bound. Each row satisfies $q \le 729^n$.*

n	M	*Least primitive denominator* q	*Cost upper bound*
1	*17*	*149*	10^{-1}
2	*19*	*5653*	10^{-3}
3	*22*	*58628*	10^{-5}
4	*26*	*2179752*	10^{-5}
5	*29*	*15700603*	10^{-8}
6	*31*	*652385103*	10^{-9}
7	*32*	*24091147002*	10^{-10}
8	*39*	*2181871916137*	10^{-11}
9	*40*	*40532942284997*	10^{-12}
10	*41*	*1072041369684608*	10^{-14}
11	*42*	*5491888775325118*	10^{-17}
12	*45*	*366862559589281285*	10^{-17}
13	*48*	*1316860892378685043 4*	10^{-19}
14	*49*	*1598521279946132865 15*	10^{-22}
15	*52*	*2079281126942205252047 13*	10^{-20}
16	*54*	*3711482189101324715412673*	10^{-21}
17	*57*	*8510054686470304629242998*	10^{-25}
18	*59*	*1547806150903404730398417 49*	10^{-26}

In particular the last row gives the strictly positive integer-coefficient form

$$0 < 154780615090340473039841749\,\gamma - 893417956532391068205964 73 < 10^{-26}.$$

Proof. For each window, the supplementary certificate gives an exact affine particular solution x, a nonzero homogeneous solution z, and the complete allowed interval for the parameter in $x + tz$. The independent verifier regenerates residues from their factorial formula (the search uses their binomial formula). It proves the full affine rank using an odd-prime modular minor, and checks every entry of the affine equations. It then computes the nonnegativity interval exactly, projects its two endpoints, and checks the final Farey-neighbor bracket and its mediant. The interval theorem and denominator-minimality proposition certify the claimed global minimum denominator within that particular cone.

The verifier checks the chosen integer outputs, every prime-log row, all prefix signs, every displayed strict rational cost bound, and the complete output gcd after coefficient clearing. It also checks the new horizon-free upper bound on each finite lift. No numerical value of γ or of any logarithm is used. □

The endpoint intervals are closed: a zero prefix at an endpoint does not invalidate strict positivity, because $S_M = 1$. The costs in this table need not be monotone; for example the cost of the supplied $n = 15$ lift is greater than that of the supplied $n = 14$ lift. Neither the finite list nor denominator minimality on each finite window implies the existence of the infinite family in the completion theorem.

Proposition 51.2 (An excluded target becomes positive at the next index)**.** *The primitive target*

$$(q, p) = (18585954467984, 10728104066067)$$

has a rational exactly log-null lift on $[8, 40]$ *with all its* 33 *weighted prefixes strictly positive and*

$$C(c) < \frac{1}{7000000000000}.$$

Together with the previously certified exclusion through $N = 39$*, this shows that index* 40 *is the first possible terminal index for a cumulative-positive lift of this target.*

Proof. The fixed-target prefix equations on $[8, 40]$ have a one-dimensional affine solution space. Its nonnegative parameter interval has nonempty interior; the rational midpoint defines the coefficient vector in `positive_repair40.json`. The independent factorial verifier checks all prime-log equations, both target coordinates, the 33 strict prefix inequalities, the primitive gcd, and the displayed cost bound. The earlier integer Farkas certificate excludes every lift on $[1, 39]$, hence every smaller terminal index. The cumulative positivity theorem gives the required strict evaluated sign at index 40. □

This repair theorem concerns one fixed target. It is not a uniform repair bound for every subsequent target or window.

Proposition 51.3 (Exact pairs in the eighteen finite windows)**.** *For each of the eighteen windows in the preceding table, the interval algorithm supplies two rational cumulative-positive lifts with determinant of absolute value one and both primitive denominators at most* 729^n*. Their largest denominator is the minimum achievable by any independent integer output pair in that same normalized cone. For* $[n, M] = [18, 59]$*, the second denominator is*

$$q_2 = 513098698738978141923058164,$$

and the two supplied costs satisfy $C_1 < 10^{-26}$ *and* $C_2 < 10^{-27}$*.*

Proof. The rational data in `minimum_height_pairs.json` record the chosen side, its exact ceiling multiple, both integer targets, and the second interpolated lift. The independent checker validates the full first-interval certificate, both candidate formulas and their denominator comparison, every logarithm and output coordinate, all prefix signs, costs, and the complete output gcd. The theorem proves the asserted finite minimax property. □

No uniform denominator bound on an infinite list of intervals is inferred from these eighteen exact calculations.

For clarity the complete ordered pair in the last window is

$$(q_1, p_1) = (154780615090340473039841749,\ 89341795653239106820596473),$$
$$(q_2, p_2) = (513098698738978141923058164,\ 29616860655273056851923417 7).$$

It satisfies $q_1p_2 - q_2p_1 = 1$, both kernels are positive, and

$$0 < q_1\gamma - p_1 < 10^{-26}, \qquad 0 < q_2\gamma - p_2 < 10^{-27}.$$

In particular the two positive forms, their exact cancellations, their primitive normalization, and their costs belong to the same finite certificates.

52 Exact target transfer and rational repair

52.1 Separating integral target transfer from formal-zero repair

Retain the proper-kernel maps E, Ψ and weighted cost $C(c) = \sum_j \mathcal{U}_j|c_j|$. All spaces and lifts below are over $\mathbb{Q}$. Put

$$K_N = \ker E_N, \qquad Z_N = K_N \cap \ker \Psi_N.$$

Here K_N denotes a coefficient space, not the image of a cost ball. Old coefficient vectors are padded with zeros when the index increases.

Theorem 52.1 (An exact normal form for a one-direction extension)**.** *Suppose $K_M = K_N \oplus \mathbb{Q}h$ and rational lifts $c_1, c_2 \in K_N$ have output matrix*

$$B = [\Psi(c_1)\ \Psi(c_2)] \in \mathrm{SL}_2(\mathbb{Z}).$$

Set $\eta = B^{-1}\Psi(h)$ and

$$g = h - \eta_1 c_1 - \eta_2 c_2.$$

Then $Z_M = Z_N \oplus \mathbb{Q}g$. A pair $d_1, d_2 \in K_M$ has integer outputs of determinant one if and only if it has the form

$$\boxed{d_i = \sum_{j=1}^{2} c_j T_{ji} + z_i + t_i g \quad (i = 1, 2), \quad T \in \mathrm{SL}_2(\mathbb{Z}), \quad z_i \in Z_N, \quad t_i \in \mathbb{Q}.}$$

For a given pair the matrix T, the two scalars t_i, and the two vectors z_i are unique.

Proof. The vector g has zero formal output and the same nonzero class as h modulo K_N. For a vector $z \in Z_M$, write $z = x + th$ with $x \in K_N$; then $z - tg \in K_N$ and has zero output, proving the direct-sum identity. For a determinant-one integer output matrix D, the matrix $T = B^{-1}D$ lies in $\mathrm{SL}_2(\mathbb{Z})$. Subtracting the corresponding combinations of c_1, c_2 leaves two vectors in Z_M, which have unique decompositions in its displayed direct sum. The converse follows by applying Ψ. □

The theorem puts integrality entirely into T. The repair parameters t_i are rational and need not be integral because the repairs have zero formal output. In particular, requiring every intermediate output-changing correction to be integral is an additional restriction; it is not a consequence of integrality of the final pair.

Lemma 52.2 (Exact one-variable repair by a weighted median)**.** *Let $h \neq 0$ be a rational vector and let all $\mathcal{U}_j > 0$ be rational. For fixed rational c, the minimum of $C(c + th)$ over real t is attained at a rational weighted median of*

$$r_j = -c_j/h_j, \qquad w_j = \mathcal{U}_j|h_j| \quad (h_j \neq 0).$$

Define

$$A = \sum_{c_j \neq 0} \mathcal{U}_j h_j \operatorname{sgn}(c_j), \qquad Z = \sum_{c_j = 0} \mathcal{U}_j |h_j|.$$

There is a strict cost decrease from $t = 0$ if and only if $|A| > Z$. When this holds, put $\sigma = -\operatorname{sgn}(A)$, $\Delta = |A| - Z$, and

$$\tau = \min_{c_j \sigma h_j < 0} |c_j/h_j|.$$

Then τ exists, is positive and rational, and

$$\boxed{C(c + \sigma\tau h) = C(c) - \Delta\tau.}$$

If $Eh = 0$ and $\Psi(h) = 0$, every such repair preserves all log cancellations and the exact primitive target of c.

Proof. The coordinates with $h_j = 0$ contribute a constant, while the others give $\sum_j w_j|t - r_j|$. Its minimizers are exactly its weighted medians. A median can be chosen among the rational knots. The left and right derivatives at zero are $A - Z$ and $A + Z$. Convexity therefore gives strict improvement exactly when $|A| > Z$. In the indicated direction the derivative is $-\Delta$ until the first nonzero coordinate crosses zero. Such a crossing must exist: otherwise every nonzero coordinate would have nonnegative derivative in that direction, contradicting the strict negative total derivative. Integrating the constant derivative through the first knot proves the exact gain formula. The last assertion follows from linearity. □

This lemma supplies an exact finite descent step. For a fixed target whose evaluated value $q\gamma - p$ is nonzero, the inequality $C(c) \geq |q\gamma - p|$ prevents its cost from tending to zero. For an arbitrary fixed target, such decay would force $q\gamma - p = 0$ and would not furnish a nonzero irrationality certificate. Integral target transfer and the choice of repair directions must be controlled together in an infinite argument.

52.2 An arithmetic obstruction to a restricted shared-direction update

Theorem 52.3 (Classification and cost of integral shared-direction updates)**.** *Let c_1, c_2 have an integer determinant-one output matrix B and support in $1, \dots, N$. Let h be rational and log-null, with primitive integer output $u = \Psi(h)$. Write*

$$(a, b)^T = B^{-1}u \in \mathbb{Z}^2, \qquad \gcd(a, b) = 1.$$

The two lifts $c_i + t_i h$ have integer outputs of determinant one if and only if, for an integer k,

$$(t_1, t_2) = k(b, -a).$$

The resulting output transfer matrix is

$$\begin{pmatrix} 1 + kab & -ka^2 \\ kb^2 & 1 - kab \end{pmatrix}.$$

If

$$T_{\text{new}}(h) = \sum_{j>N} \mathcal{U}_j |h_j|,$$

then every nonzero such update satisfies

$$\boxed{\max_i C(c_i + t_i h) \geq |k| \max(|a|, |b|) T_{\text{new}}(h).}$$

Proof. If $t_i u$ is integral and u is primitive, Bezout's identity forces t_i to be an integer. Expansion of the output determinant gives $1 + at_1 + bt_2$. Since (a, b) is primitive, the integer solutions of $at_1 + bt_2 = 0$ are precisely $(t_1, t_2) = k(b, -a)$. Multiplication by B gives the displayed transfer matrix, whose determinant is one. On coordinates beyond N, the old lifts vanish, so their new cost contributions are exactly $|t_i| T_{\text{new}}(h)$. Dropping the remaining nonnegative terms proves the lower bound. □

Proposition 52.4 (A certified obstruction at the next extension)**.** *Use the determinant-one pair at $N = 39$ and the canonical log-null direction at $N = 40$: its last coordinate is one, its earlier nonpivot coordinates vanish, and its pivot coordinates are determined by the prime-log equations. Normalize its output to a primitive integer pair. Then every nonzero integral shared-direction update from Theorem 52.3 has*

$$\max_i C(c_i + t_i h) > 10^{595}.$$

Proof. The file `rank_one_shear40_certificate.json` specifies the direction and all rational quantities in the lower bound. The standard-library script `verify_rank_one_shear.py` regenerates the residues and canonical pivot solution, performs the exact primitive normalization, and checks

$$\max(|a|, |b|) \mathcal{U}_{40} |h_{40}| > 10^{595}.$$

The preceding theorem applies to every nonzero integer k. □

This is a lower bound on the specified weighted cost, not a lower bound on the actual absolute error. It excludes this restricted primitive-direction update rule. It does not exclude rational formal-zero repairs or the full normal form of Theorem 52.1.

52.3 A new determinant-one pair at index 40

Write the preceding pair as

$$u_1 = (18585954467984, 10728104066067), \qquad u_2 = (3361033349029, 1940041099316),$$

with $\det(u_1, u_2) = 1$.

Theorem 52.5 (An exact finite transfer with strict contraction)**.** *There are rational log-null lifts at index* 40 *of the primitive integer vectors*

$$w = (1072041369684608, 618799072004451), \qquad v = (21946987817013, 12668145165383)$$

whose costs satisfy

$$C_w < \frac{1}{289000000000000}, \qquad C_v < \frac{1}{332000000000000}.$$

Their exact transfer is

$$\boxed{[w\ v] = [u_1\ u_2]\begin{pmatrix} 49 & 1 \\ 48 & 1 \end{pmatrix}, \qquad \det(w, v) = 1.}$$

The maximum new cost is less than one fifty-first of the maximum cost of the preceding supplied pair. In particular, if $w = (q_w, p_w)$ and $v = (q_v, p_v)$, then

$$0 < (q_w\gamma - p_w)^2 + (q_v\gamma - p_v)^2 < \frac{1}{289000000000000^2} + \frac{1}{332000000000000^2}.$$

Proof. The exact vectors in `pair40_witness.json` are checked by `verify_pair40.py`. The checker regenerates all residues and weights, verifies every prime-log equation, both output coordinates, and all strict rational cost bounds. It clears every input denominator and verifies that the complete output gcd returns the displayed primitive pair and the same cost. The previous reference lift is also rechecked for the strict factor-51 comparison. The transfer matrix has determinant one, so the sum-of-squares polynomial has discriminant -4 and is strictly positive on $\mathbb{R}$. Its upper bound follows from the integral estimates. No value of γ or of a logarithm is evaluated. □

The costs are approximately $3.4509735587 \cdot 10^{-15}$ and $3.0083861504 \cdot 10^{-15}$; these decimals are not proof inputs. The finite theorem uses the supplied witnesses and does not require any global optimality assertion for the unrestricted cost problem.

Corollary 52.6 (An exact sign transfer and an adjacent rational bracket)**.** *The new second form is negative, and the previous first form is positive:*

$$q_v\gamma - p_v < 0, \qquad 18585954467984\gamma - 10728104066067 > 0.$$

Consequently

$$\boxed{\frac{10728104066067}{18585954467984} < \gamma < \frac{12668145165383}{21946987817013}.}$$

The exact bracket width is

$$\frac{1}{407905716276403181959011792}.$$

Proof. Put $e_w = q_w\gamma - p_w$, $e_v = q_v\gamma - p_v$, and $e_1 = q_1\gamma - p_1$ for the first preceding output. The determinant identity gives $q_v e_w - q_w e_v = 1$. If $e_v \geq 0$, it would imply $1 \leq q_v|e_w| < q_v/(289000000000000) < 1$. Thus $e_v < 0$. Since $v = u_1 + u_2$ and $\det(u_1, v) = 1$,

$$q_v e_1 - q_1 e_v = 1, \qquad e_1 > \frac{1 - q_1/(332000000000000)}{q_v} > 0.$$

The signs prove the bracket. Its cross determinant is one, so its width is $1/(q_1 q_v)$, equal to the displayed fraction. □

The previous first target therefore has positive actual value despite its proved exclusion from the cumulative-prefix cone through index 39. This distinguishes scalar sign from that particular sufficient kernel certificate. The transfer matrix and its cost decrease are proved for this step only. No iteration with uniformly shrinking cost is inferred, and the infinite-family requirement remains open.

Proposition 52.7 (The new finite pair lies in this enlarged region)**.** *The supplied index-*40 *pair* (w, v) *has a common starting index* 8*. The lift of* w *has its first eight prefixes zero and every later prefix positive. The lift of* v *satisfies*

$$S_k \geq -\frac{q_v}{800} \qquad (1 \leq k \leq 40).$$

Both targets satisfy

$$q_w \leq 729^8, \qquad \frac{801}{800} q_v \leq 729^8.$$

In particular $q_w\gamma - p_w > 0$*, and the pair has one positive and one negative evaluated form.*

Proof. The exact weighted prefixes and inequalities are checked from the rational witnesses by `verify_pair40.py`. The first sign follows from the cumulative-prefix positivity theorem, and the second sign was established by determinant transfer. This verifies one finite instance of the combined region; it does not establish unbounded starting indices. □

Part VI

Columnwise budgets, binary arithmetic, and output classes

53 Clearing each column separately removes the terminal-index penalty

The common denominator of an entire window need not be imposed on every column before searching for integer coefficients. Put

$$d_j = 2\,\mathrm{lcm}(1,\dots,3j), \qquad J_j = d_j I_j, \qquad v_j = d_j \mathcal{U}_j, \qquad \delta = \log(729/16) - 3 > 0.$$

The factor-two denominator theorem (Section 54) gives the exact integer expansion

$$J_j = a_j\gamma - b_j + \sum_{\ell} F_{j,\ell}\log\ell, \qquad a_j = d_j 16^j, \quad b_j = d_j p_j, \quad F_{j,\ell} = d_j E_{j,\ell} \in \mathbb{Z}.$$

For an integer vector c of finite support define

$$\mathcal{B}(c) = \sum_j v_j |c_j|.$$

This is the cost after these individual denominator clearings; it is not an unnormalized integral estimate.

Theorem 53.1 (A terminal-index-independent integer budget)**.** *Fix $0 \le \eta < \delta$. Uniformly over all finite terminal indices $M \ge n$ and all integer vectors satisfying*

$$\mathrm{supp}(c) \subseteq [n, M], \qquad |c_j| \le e^{\eta j},$$

one has

$$\boxed{\mathcal{B}(c) \le \exp\{-(\delta-\eta)n + o(n)\} \longrightarrow 0.}$$

If $Fc = 0$, the resulting form is exactly $Q\gamma - P$ with $Q, P \in \mathbb{Z}$, and the same bound holds. No claim that this form is nonzero is part of the theorem.

Proof. The prime number theorem [7] gives $\log d_j = 3j + o(j)$, and Stirling's formula [21] gives $\log \mathcal{U}_j = -\log(729/16)j + O(\log j)$. Thus $\log v_j = -\delta j + o(j)$. Define

$$\epsilon_n = \sup_{j \ge n} \frac{|\log v_j + \delta j|}{j},$$

so $\epsilon_n \to 0$. For large n, the ratio $\exp(-\delta + \eta + \epsilon_n)$ is less than one. Hence

$$\mathcal{B}(c) \le \sum_{j=n}^{\infty} e^{-(\delta-\eta-\epsilon_n)j} = \frac{e^{-(\delta-\eta-\epsilon_n)n}}{1 - e^{-(\delta-\eta-\epsilon_n)}} = e^{-(\delta-\eta)n + o(n)}.$$

Every coefficient in the displayed expansion of J_j is integral. Exact prime-row cancellation therefore leaves $Q\gamma - P$, and the triangle inequality gives its absolute-value bound. □

Theorem 53.2 (Short exact cancellations on a doubling window)**.** *For $n \geq 2$ put*

$$\mathfrak{h}_n = \left\lceil \exp\left(\frac{128n}{\log n}\right) \right\rceil.$$

For every sufficiently large n the individually cleared log matrix on $[n, 2n-1]$ has a nonzero integer kernel vector with $\|c\|_\infty \leq \mathfrak{h}_n$. Every such vector satisfies

$$\left| \sum_{j=n}^{2n-1} c_j J_j \right| \leq \mathcal{B}(c) \leq e^{-\delta n + o(n)}.$$

The same box also contains a nonzero vector for which both output coordinates Q and P vanish exactly.

Proof. Use Lemma 22.5

$$2|A_{j,k}| \leq 6\beta^j, \qquad \beta = (16399/1296)^2, \qquad \beta^3 < 2^{22}.$$

The matrix has n columns and $s_n = \pi(6n-3) = (6+o(1))n/\log n$ rows. Its entries have a common integer bound B_n satisfying

$$\log B_n \leq 2(3 + \log\beta)n + o(n).$$

Indeed the log coefficient is a sum of at most $2j+1$ residues times valuations bounded by $\log_2(3j)$, followed by multiplication by d_j. All polynomial factors contribute only $O(\log n)$.

For completeness, an integer matrix with d columns, s rows, and row bounds $B_i \geq 1$ has a nonzero integer kernel vector of height at most H whenever

$$H^{d-s} > \prod_{i=1}^{s} (3dB_i).$$

The $(H+1)^d$ vectors in $\{0, \ldots, H\}^d$ have fewer than H^d possible images under this inequality; subtract two with the same image.

For the present matrix the logarithms of the two sides have leading coefficients 128 and $12(3+\log\beta)$, respectively, on the scale $n^2/\log n$. The second is strictly smaller, since

$$12(3 + \log\beta) < 36 + 88\log 2 < 124 < 128.$$

The counting condition therefore holds eventually. Furthermore $\log \mathfrak{h}_n = o(n)$, so

$$\mathcal{B}(c) \leq \mathfrak{h}_n \sum_{j=n}^{2n-1} v_j \leq e^{-\delta n + o(n)}$$

by the same tail argument as in Theorem 53.1.

Append the two integer rows (a_j) and (b_j). Their entry logarithms are $O(n)$ on this window. The two extra row factors and the change from $d-s$ to $d-s-2$ alter the logarithmic counting inequality by only $O(n)$, whereas its positive margin has order $n^2/\log n$. Thus the same box contains a nonzero formal-zero vector. □

An integer-operation alternative is

$$\widehat{\mathfrak{h}}_n = 2^{\lceil 288n/\lfloor \log_2 n \rfloor \rceil}.$$

Its logarithm is asymptotic to $288(\log 2)^2 n/\log n$, whose coefficient exceeds 128 because $\log 2 > 2/3$. The same argument applies.

Proposition 53.3 (What the change of normalization does and does not do)**.** *Let $D = d_M$, and suppose $z_n, \dots, z_M$ are integers. Then*

$$c_j = (D/d_j)z_j \in \mathbb{Z}$$

represents the same form in the individually cleared system:

$$\sum_j c_j J_j = D \sum_j z_j I_j, \qquad \mathcal{B}(c) = D \sum_j |z_j| \mathcal{U}_j.$$

Thus changing the notation for a fixed form does not lower its cost. It changes the admissible search space: individually cleared columns permit integer coefficients without paying D on every column. In particular, an attainable odd constant output in the common-D system remains attainable in the columnwise system, but its new input height may be large.

Proof. For $j \le M$ the least common multiple defining d_j divides the one defining D. Substitute the displayed integer c_j and take absolute values to obtain both identities. □

Corollary 53.4 (A concrete parity-coset completion target)**.** *Suppose that for unbounded integers n there are exactly log-null integer inputs on $[n, 2n-1]$ with height at most $\mathfrak{h}_n$ as above and*

$$P_n = \sum_j d_j p_j c_j \quad \textit{odd.}$$

Then γ is irrational. More generally it suffices that $P_n \neq 0$ and $4n - v_2(P_n) \to \infty$.

Proof. The cost tends to zero, and $2^{4n} \mid Q_n$. If $Q_n = 0$, the error is the nonzero integer $-P_n$, which is impossible for large n. Write $g_n = \gcd(|Q_n|, |P_n|)$. The primitive positive denominator satisfies

$$v_2(|Q_n|/g_n) = \max\{v_2(Q_n) - v_2(P_n), 0\} \longrightarrow \infty.$$

If $\gamma = a/b$ were rational, the integers $aQ_n - bP_n$ would tend to zero and hence would vanish eventually. The reduced denominator of $P_n/Q_n = \gamma$ would then be the fixed denominator b, contradicting the proved unbounded binary order. This argument also covers the odd-output special case. □

The short log-null vectors and the attainability of a particular output congruence class are separate statements. The missing step is a short representative of that class at unbounded starting indices. The formal-zero vectors in the same height box make this compatibility issue explicit.

54 Binary arithmetic foundations and supercongruences

This section consolidates the binary arithmetic results proved in the preceding research note [30]. They are included with their proofs so that the present dyadic reflection and output obstructions have no unproved internal arithmetic dependencies. They are background results, not additional new claims of this paper.

Write $v = v_2$, with $v(2) = 1$ and $v(0) = +\infty$. For a rational number, membership in $\mathbb{Z}_2$ means that its reduced denominator is odd. Polynomial congruences are coefficientwise. Retain

$$R_n(u) = \frac{\prod_{h=0}^{2n-1}(4u + h^2)}{\prod_{j=n}^{3n}(u + j^2)},$$

$$p_n = \sum_{j=n}^{3n} A_{n,j} \left(H_j - \frac{1}{2j} \right), \quad E_{n,\ell} = \sum_{j=n}^{3n} A_{n,j} v_\ell(j).$$

Here $H_j = \sum_{k=1}^{j} 1/k$ and $v_\ell(j)$ is the exponent of ℓ in j.

Lemma 54.1 (Residue formula, half-integrality, and total mass)**.** *The proper partial fractions of R_n are*

$$R_n(u) = \sum_{j=n}^{3n} \frac{A_{n,j}}{u + j^2},$$

$$A_{n,j} = (-1)^{j-n} \frac{4j^2(2j+2n-1)!(n+j-1)!}{(2j-2n)!(j-n)!(3n-j)!(3n+j)!}.$$

Moreover $B_{n,j} := 2A_{n,j}$ is integral and $\sum_{j=n}^{3n} A_{n,j} = 16^n$.

Proof. The poles are distinct and the numerator degree is one less than the denominator degree. Taking residues gives

$$A_{n,j} = \frac{\prod_{h=0}^{2n-1}(h^2 - 4j^2)}{\prod_{n \le k \le 3n,\ k \ne j}(k^2 - j^2)},$$

and factoring each difference of squares gives the displayed factorial formula. Put $a = j + n$, $b = j - n$, and

$$U = \binom{2a}{b}, \quad V = \binom{a}{2b}, \quad U' = \binom{2a-1}{b-1}, \quad V' = \binom{a-1}{2b-1},$$

with $U' = V' = 0$ when $b = 0$. Direct cancellation gives

$$B_{n,j} = (-1)^b(UV + 4U'V + U'V') = (-1)^b UV \left(\frac{2j}{a}\right)^2.$$

The first expression proves integrality. Finally comparison of the u^{-1} coefficients at infinity, where $R_n(u) = 16^n u^{-1} + O(u^{-2})$, gives the total mass. □

We use the elementary factorial valuation identity $v(m!) = \sum_{a \ge 1} \lfloor m/2^a \rfloor = m - s_2(m)$, where $s_2(m)$ is the number of ones in the binary expansion of m. The first equality counts powers of two in each factorial factor; the second follows by writing m in binary. In particular the valuation of a binomial coefficient is unchanged when both of its arguments are multiplied by the same power of two.

54.1 The exact common denominator and dyadic constant coefficient

Put $L_m = \operatorname{lcm}(1, \dots, 3m)$.

Lemma 54.2 (Parity of the residue coefficients)**.** *For $n \le j \le 3n$, set $a = j + n$ and $b = j - n$. Then*

$$B_{n,j} = (-1)^b \binom{2a}{b} \binom{a}{2b} \left(\frac{2j}{a}\right)^2.$$

If $v_2(j) \ne v_2(n)$, then $4 \mid B_{n,j}$. More precisely, if $v_2(j) > v_2(n)$, then

$$v_2(B_{n,j}) \ge 2 + 2\big(v_2(j) - v_2(n)\big).$$

Proof. The displayed identity follows by cancellation of factorials in the residue formula. The two binomial coefficients are integers. For unequal valuations one has $v_2(j+n) = \min(v_2(j), v_2(n))$. Hence the valuation of the squared rational factor is $2 + 2v_2(j) - 2\min(v_2(j), v_2(n))$, which gives both claims. □

Theorem 54.3 (A factor-two improvement in the common denominator)**.** *For every $n \geq 1$,*

$$\boxed{2L_n p_n \in \mathbb{Z}.}$$

Consequently $\widetilde{D}_M = 2L_M$ simultaneously clears all coefficients of 1, γ, and the prime logarithms in I_n for $1 \leq n \leq M$.

Proof. The defining expression gives

$$2L_n p_n = \sum_{j=n}^{3n} L_n B_{n,j} H_j - \sum_{j=n}^{3n} \frac{L_n B_{n,j}}{2j}.$$

Every term in the first sum is integral. Put $a = \lfloor \log_2(3n) \rfloor = v_2(L_n)$. The integer L_n/j is even unless $v_2(j) = a$, in which case $j = 2^a$ is the unique possibility. Since $v_2(n) \leq a - 1$, Lemma 54.2 makes $B_{n,2^a}$ even. Thus each term in the second sum is also integral. For $n \leq M$, L_n divides L_M. Finally $2E_{n,\ell}$ is integral, so $\widetilde{D}_M E_{n,\ell}$ is integral as well. □

Theorem 54.4 (Exact constant-coefficient valuation at dyadic indices)**.** *For every integer $s \geq 0$,*

$$\boxed{v_2(p_{2^s}) = -s - 2.}$$

In particular $\widetilde{D}_{2^s} p_{2^s}$ is odd. More generally, if $n = 2^s$ and $n \leq M < 4n/3$, then $\widetilde{D}_M p_n$ is odd, whereas $v_2(\widetilde{D}_M 16^n) = s + 2 + 4n$.

Proof. Set $n = 2^s$. The only $j \in [n, 3n]$ with $v_2(j) = s$ are $j = n, 3n$. All other residue coefficients are even by Lemma 54.2. Moreover

$$B_{n,n} = 1, \qquad B_{n,3n} = \frac{9}{4}\binom{8n}{2n}$$

are both odd. Indeed the elementary factorial identity $v_2(k!) = k - s_2(k)$ gives $v_2\binom{8n}{2n} = 1+2-1 = 2$.

For $n \leq j \leq 3n$, the 2-adic integer $2^{s+1} H_j$ is even when $j < 2n$ and odd when $j \geq 2n$: only the summand $1/(2n)$ can have denominator with 2-adic valuation $s + 1$. Therefore

$$2^{s+1} \sum_{j=n}^{3n} B_{n,j} H_j \equiv 1 \pmod 2.$$

On the other hand,

$$2^s \sum_{j=n}^{3n} \frac{B_{n,j}}{j} \equiv 0 \pmod 2.$$

To verify this, the endpoint terms are $B_{n,n}$ and $B_{n,3n}/3$, both odd, so their sum is even. If $v_2(j) < s$, the remaining term has valuation at least $s - v_2(j) + 2 > 0$. The only j with valuation greater than s is $j = 2n$; there $16 \mid B_{n,2n}$ and the term has valuation at least three. Combining these two congruences in

$$2^{s+2} p_n = 2^{s+1} \sum_j B_{n,j} H_j - 2^s \sum_j \frac{B_{n,j}}{j}$$

proves that the left side is odd. Finally $v_2(\widetilde{D}_M) = s + 2$ when $n \leq M < 4n/3$, proving the remaining statements. □

54.2 Residue valuations and binary dilation

Lemma 54.5 (A lower-order residue bound)**.** *Put $t = v(n)$ and $u = v(j)$. If $u < t$, then*

$$v(A_{n,j}) \geq t - u + 2.$$

If $u > t$, then $v(A_{n,j}) \geq 1 + 2(u - t)$.

Proof. Use

$$2A_{n,j} = (-1)^{j-n}\binom{2a}{b}\binom{a}{2b}\left(\frac{2j}{a}\right)^2, \qquad a = j + n, \quad b = j - n.$$

The second assertion follows at once from $v(a) = t$ and integrality of the binomial product. For the first assertion, divide n, j, a, b by 2^u. The valuations of both binomial coefficients are unchanged, as follows from the factorial valuation formula. Thus assume that j is odd and $v(n) = h = t - u \geq 1$. Then a, b are odd and $a \equiv b \pmod{2^{h+1}}$.

For $q = 2^k$, $1 \leq k \leq h + 1$, write $a = Aq + r$, $b = Bq + r$, with $0 < r < q$. Direct substitution gives

$$\left\lfloor \frac{2a}{q} \right\rfloor + \left\lfloor \frac{a}{q} \right\rfloor - \left\lfloor \frac{b}{q} \right\rfloor - \left\lfloor \frac{2a - b}{q} \right\rfloor - \left\lfloor \frac{2b}{q} \right\rfloor - \left\lfloor \frac{a - 2b}{q} \right\rfloor = 1.$$

This is the contribution of level k to the valuation of the binomial product. Contributions at all other levels are nonnegative, being the sum of two binomial carry counts. Hence that product has valuation at least $h + 1$. The squared factor has valuation two, giving $v(2A_{n,j}) \geq h + 3$, as required. □

Lemma 54.6 (The exact signed doubling identity)**.** *For $0 \leq k \leq m$ put*

$$T(m, k) = \frac{\binom{2m}{2k}}{\binom{m}{k}} = (-1)^k \prod_{r=1}^{k} \left(1 - \frac{2m}{2r - 1}\right).$$

For $n \leq j \leq 3n$, $a = j + n$, and $b = j - n$, one has

$$\frac{A_{2n,2j}}{A_{n,j}} = (-1)^b T(2a, b)T(a, 2b).$$

Proof. Separating the odd and even factors in the factorial expressions for the binomial ratio gives the displayed product for T. In the binomial formula of Lemma 54.2, simultaneous doubling of n, j leaves $(2j/a)^2$ unchanged and changes the sign by $(-1)^b$. The two remaining binomial ratios are precisely $T(2a, b)$ and $T(a, 2b)$. □

Lemma 54.7 (Residues at a higher binary order)**.** *If $v_2(j) > v_2(n)$, then*

$$v_2(A_{n,j}) \geq 2 + 2\big(v_2(j) - v_2(n)\big).$$

Proof. In the exact binomial formula for $2A_{n,j}$, divide n, j by $2^{v_2(n)}$. The binomial valuations are unchanged under this simultaneous scaling. Now n is odd and j is even, so $a = j + n$ and $b = j - n$ are odd. The binomial coefficient $\binom{2a}{b}$ is even: this follows from $\binom{2a}{b} = (2a/b)\binom{2a-1}{b-1}$. The squared rational factor has valuation $2 + 2(v_2(j) - v_2(n))$. Subtracting one for A instead of $2A$ and adding the extra even binomial factor proves the claim. □

54.3 A coefficientwise product congruence

Lemma 54.8 (A balanced odd-product polynomial)**.** *For an integer $b \geq 1$ define*

$$\mathcal{R}_b(X) = \prod_{r=1}^{b} \left(1 - \frac{4X}{2r-1}\right) \prod_{r=1}^{2b} \left(1 - \frac{2X}{2r-1}\right).$$

Then

$$\boxed{\mathcal{R}_b(X) - 1 \in 2^{v_2(b)+2} X(X-b)\mathbb{Z}_2[X].}$$

Proof. Pair the consecutive factors in the second product. With $t = 2r-1$, the resulting factor is

$$F_r(X) = \left(1 - \frac{4X}{t}\right)\left(1 + \frac{4X(X-2t)}{4t^2-1}\right), \qquad \mathcal{R}_b = \prod_{r=1}^{b} F_r.$$

Every denominator is odd, and $F_r \in 1 + 4\mathbb{Z}_2[X]$. More generally, if t, t' are odd and $t \equiv t' \pmod{2^h}$, the coefficients after dividing $F(X,t) - F(X,t')$ by four are congruent to zero modulo 2^h. Indeed those coefficients are rational functions of t with integral coefficients and denominators that are units at odd t; subtracting two such values preserves the indicated congruence. Thus for $h \geq 1$,

$$F_{r+2^{h-1}} \equiv F_r \pmod{2^{h+2}}.$$

Put $Q_h = \prod_{r=1}^{2^h} F_r$. The last congruence gives

$$Q_h \equiv Q_{h-1}^2 \pmod{2^{h+2}} \qquad (h \geq 1).$$

Starting from $Q_0 = F_1 \equiv 1 \pmod 4$, induction yields $Q_h \equiv 1 \pmod{2^{h+2}}$. If $b = 2^h m$ with m odd, split the product into m consecutive blocks of length 2^h. The same coefficientwise periodicity makes every block congruent to Q_h modulo 2^{h+2}. Hence

$$\mathcal{R}_b(X) - 1 \in 2^{h+2}\mathbb{Z}_2[X].$$

Finally $\mathcal{R}_b(0) = 1$ and $\mathcal{R}_b(b) = 1$ exactly. For the second identity put $O_m = \prod_{r=1}^{m}(2r-1)$. At $X = b$ the two products have values $(-1)^b O_{2b}/O_b^2$ and $(-1)^b O_b^2/O_{2b}$, respectively. The polynomial $(\mathcal{R}_b - 1)/2^{h+2}$ therefore vanishes at both 0 and b. Division by the monic factors X and $X - b$ preserves $\mathbb{Z}_2[X]$, proving the assertion. In the second division the first quotient vanishes at b because $b \neq 0$ in $\mathbb{Q}_2$; no assumption that b is a unit is needed. □

Theorem 54.9 (A stronger residue supercongruence)**.** *For every $n \geq 1$ and $n \leq j \leq 3n$,*

$$\boxed{A_{2n,2j} - A_{n,j} \in 2^{2v_2(n)+5}\mathbb{Z}_2.}$$

Proof. At $j = n$ the difference is zero. Otherwise put $a = j + n$, $b = j - n > 0$. The exact signed doubling identity gives

$$\frac{A_{2n,2j}}{A_{n,j}} = \mathcal{R}_b(a).$$

The polynomial lemma therefore gives

$$v_2(A_{2n,2j} - A_{n,j}) \geq v_2(A_{n,j}) + v_2(a) + v_2(b) + v_2(n) + 3. \tag{S}$$

Write $t = v_2(n)$ and $u = v_2(j)$. If $u < t$, then $v_2(a) = v_2(b) = u$ and Lemma 54.5 gives $v_2(A_{n,j}) \geq t - u + 2$. The right side of (S) is therefore at least $2t + u + 5 \geq 2t + 5$. If $u > t$, it is at least $1 + 2(u-t) + 2t + t + 3 = 3t + 4 + 2(u-t) \geq 2t + 5$. If $u = t$, the sum and difference of the two odd integers $j/2^t, n/2^t$ have valuations at least one and two in some order. Thus $v_2(a) + v_2(b) \geq 2t + 3$. Since $v_2(A_{n,j}) \geq -1$, (S) is at least $3t + 5 \geq 2t + 5$. □

Corollary 54.10 (Sharper log-coordinate congruences)**.** *For every odd prime ℓ and every $n \geq 1$,*

$$E_{2n,\ell} - E_{n,\ell} \in 2^{v_2(n)+3}\mathbb{Z}_2.$$

For $n \geq 2$,

$$E_{2n,2} - E_{n,2} \in 2^{2v_2(n)+5}\mathbb{Z}_2.$$

In particular,

$$\boxed{E_{2^s,2} \equiv -32 \pmod{128} \qquad (s \geq 1).}$$

Proof. For odd ℓ, the even new indices are controlled by the residue theorem, while each odd new index has residue valuation at least $v_2(n) + 3$ by Lemma 54.5. For $\ell = 2$, use the exact identity

$$E_{2n,2} - E_{n,2} = 16^n + \sum_{j=n}^{3n} (A_{2n,2j} - A_{n,j})(v_2(j) + 1).$$

When $n \geq 2$, one has $4n \geq 2v_2(n) + 5$, so every term has the required valuation. Finally direct computation gives $E_{2,2} = 3808 \equiv -32 \pmod{128}$, and the proved difference is divisible by 128 at every subsequent dyadic step. □

Scope. The congruences concern individual residue and prime-log columns. They neither provide exact log cancellation nor an integer input in a prescribed small box with a surviving constant coefficient. The full parity distinction between powers of two and nonpowers is proved separately in Theorem 54.13. Its necessary integer-kernel congruence also does not supply a short surviving affine representative.

54.4 Exact moments and the complete log-two parity law

Lemma 54.11 (Exact even-moment denominators)**.** *For every integer $r \geq 0$,*

$$M_r(n) := \sum_{j=n}^{3n} A_{n,j} j^{2r} \in 16^n 4^{-r}\mathbb{Z}.$$

Proof. At infinity the proper partial-fraction identity gives

$$R_n(u) = \sum_{r \geq 0} (-1)^r M_r(n) u^{-r-1}.$$

On the other hand,

$$R_n(u) = \frac{16^n}{u} \frac{\prod_{k=0}^{2n-1}(1 + k^2/(4u))}{\prod_{j=n}^{3n}(1 + j^2/u)}.$$

The coefficient of u^{-r} in the last quotient belongs to $4^{-r}\mathbb{Z}$: numerator coefficients have those denominators, and denominator reciprocals have integral coefficients. Comparison proves the result. □

Lemma 54.12 (The two base parity classes)**.** *If $n \geq 3$ is odd, or if $n \geq 6$ and $v_2(n) = 1$, then*

$$E_{n,2} \in 64\mathbb{Z}_2.$$

Proof. First let $n \geq 3$ be odd and set $f(X) = (X-1)(X-9)$. For odd j, one has $v_2(f(j^2)) \geq 7$: both factors are divisible by eight, and their difference is eight, so at least one is divisible by sixteen. Since $v_2(A_{n,j}) \geq -1$, the product vanishes modulo 64. For $v_2(j) = 1$, one has $j^2 \equiv 4 \pmod{16}$ and $f(j^2) \equiv 1 \pmod{16}$, while Lemma 54.7 gives $v_2(A_{n,j}) \geq 4$. For $v_2(j) \geq 2$, that lemma gives $v_2(A_{n,j}) \geq 6$. These cases prove, term by term,

$$E_{n,2} \equiv \sum_{j=n}^{3n} A_{n,j} f(j^2) \pmod{64}.$$

The polynomial has degree two with integer coefficients, so Lemma 54.11 makes the right side divisible by 2^{4n-4}, which is divisible by 64 for $n \geq 3$.

Now assume $v_2(n) = 1$ and $n \geq 6$. Put

$$g(X) = (X-4)(X-36), \qquad F(X) = -g(X) + \frac{(X-1)(X-9)g(X)}{16}.$$

We verify

$$\sum_j A_{n,j}(v_2(j) - 1) \equiv \sum_j A_{n,j} F(j^2) \pmod{64} \tag{P}$$

term by term. If j is odd, then $g(j^2) \equiv 1 \pmod 8$ and $v_2((j^2-1)(j^2-9)) \geq 7$, hence $F(j^2) \equiv -1 \pmod 8$. Lemma 54.5 gives $v_2(A_{n,j}) \geq 3$, as required. If $v_2(j) = 1$, write $j = 2k$ with k odd. Then

$$g(j^2) = 16(k^2-1)(k^2-9), \qquad v_2(g(j^2)) \geq 11.$$

It follows that $v_2(F(j^2)) \geq 7$, which is sufficient even when $v_2(A_{n,j}) = -1$. If $v_2(j) = 2$, write $j = 4k$. Then

$$\frac{g(j^2)}{16} = (4k^2-1)(4k^2-9) \equiv 1 \pmod 4, \qquad (j^2-1)(j^2-9) \equiv 1 \pmod 4.$$

Thus $F(j^2) \equiv 1 \pmod 4$, and $v_2(A_{n,j}) \geq 4$ proves the needed congruence. Finally, if $v_2(j) \geq 3$, then $v_2(A_{n,j}) \geq 6$ and $F(j^2) \in \mathbb{Z}_2$, since $16 \mid g(j^2)$ for even j. This proves (P).

The difference between $E_{n,2}$ and the left side of (P) is $\sum_j A_{n,j} = 16^n$. The polynomial F has degree four and its coefficients belong to $\frac{1}{16}\mathbb{Z}$, so its moment sum has valuation at least $4n - 12 \geq 12$. Hence $E_{n,2}$ is divisible by 64, completing both cases. □

Theorem 54.13 (The complete log-two parity pattern)**.** *The exact congruences are*

$$E_{1,2} = -16, \qquad \boxed{E_{n,2} \equiv \begin{cases} 32 \pmod{64}, & n = 2^s,\ s \geq 1, \\ 0 \pmod{64}, & n \geq 2 \text{ is not a power of two.} \end{cases}}$$

In particular $E_{n,2}/32$ is an integer for every $n \geq 2$, and it is odd exactly at the powers of two.

Proof. The factorial residue formula gives $A_{1,2} = -16$, hence $E_{1,2} = -16$. The dyadic assertion was proved in Corollary 54.10. For a nonpower, write $n = 2^t m$ with $m \geq 3$ odd. If $t = 0$, use the first base class. If $t \geq 1$, the second base class gives $E_{2m,2} \equiv 0 \pmod{64}$. At each further doubling the previous index has valuation at least one, so the proved difference has valuation at least $2v_2(\text{previous index}) + 5 \geq 7$. Induction proves the result. Each $E_{n,2}$ is already a half-integer by its residue formula; the asserted 2-adic divisibility therefore gives ordinary integral divisibility by 32. □

Corollary 54.14 (A necessary congruence in the actual integer log kernel)**.** *Let (c_n) be a finitely supported integer log-null vector with support in $n \geq 2$. Then*

$$\boxed{\sum_{s\geq 1} c_{2^s} \equiv 0 \pmod 2.}$$

Consequently, if its support is contained in $[2^s, M]$ with $s \geq 1$ and $M < 2^{s+1}$, then c_{2^s} is even. In this moving block the normalized gamma coordinate $\sum_{j=2^s}^{M} 16^{j-2^s} c_j$ is also even.

Proof. The exact log-two equation is $\sum_n c_n E_{n,2} = 0$. Divide its integer row by 32 and reduce modulo two using the theorem. This is a necessary condition for every actual integer kernel vector; it does not assume that every modular kernel vector lifts to an integer one. The indicated interval contains exactly one power of two. All later terms in its normalized gamma coordinate are divisible by sixteen. □

Use in the present paper. The factor-two denominator theorem, the dyadic valuation of p_n, and the complete log-two parity theorem above are the reproduced inputs for Section 56. That section adds the thin-block reflection and constant-row parity results. Neither these background statements nor their modular consequences supply a short representative of a prescribed coset in the actual integer log kernel.

55 A lacunary support cannot cancel the new prime rows

Proposition 55.1 (A lacunary obstruction to exact cancellation)**.** *Let $n_1 < \cdots < n_r$ be positive integers with $n_{i+1} \geq 2n_i$. The corresponding prime-log columns are linearly independent over $\mathbb{Q}$. In particular no nonzero finite exactly log-null relation is supported entirely on powers of two.*

Proof. Let $M = n_r$. For $M \geq 2$, Bertrand's postulate [25] applied to $\lfloor 3M/2 \rfloor$ supplies a prime $3M/2 < \ell < 3M$. Every previous pole index is at most $3n_{r-1} \leq 3M/2$, so the earlier ℓ-rows vanish. In the last column the only pole index divisible by ℓ is $j = \ell$, and its coefficient is $A_{M,\ell} \neq 0$. The last coefficient of an exactly log-null relation must therefore vanish. Descending induction proves the claim, with the possible singleton $M = 1$ checked using $E_{1,2} = -16$. □

The distinction between congruence and exact cancellation can thus be seen on a two-column example. Put $n = 2^s$ with $s \geq 1$. By the sharper log-coordinate congruences,

$$V_{2n}(e_{2n} - e_n) \equiv 0 \pmod{2^{s+4}},$$

where e_j is the jth standard coordinate vector. Its input height is one, and the modulus tends to infinity. Nevertheless the exact prime-log kernel on these two columns is zero. Increasing congruence precision alone cannot produce the exact log-null inputs required in the irrationality construction.

56 A dyadic reflection law and an obstruction to odd constant output

Retain the proper-kernel residues $A_{n,j}$, put $B_{n,j} = 2A_{n,j}$, and write

$$p_n = \sum_{j=n}^{3n} A_{n,j} \left(H_j - \frac{1}{2j} \right), \qquad E_{n,2} = \sum_{j=n}^{3n} A_{n,j} v_2(j).$$

All congruences involving rational numbers below are in $\mathbb{Z}_2$. The already proved residue identity is

$$B_{n,j} = (-1)^b \left\{ \binom{2a}{b}\binom{a}{2b} + 4\binom{2a-1}{b-1}\binom{a}{2b} + \binom{2a-1}{b-1}\binom{a-1}{2b-1} \right\}.$$

Here $a = j + n$ and $b = j - n$. Binomial coefficients with negative lower argument are zero.

Lemma 56.1 (Two copies of a smaller residue row modulo two)**.** *Let $N = 2^s$ and $1 \le k < N/3$. For $N + k \le j \le 3(N + k)$,*

$$B_{N+k,j} \equiv \begin{cases} B_{k,r} \pmod 2, & j = N + r, \quad k \le r \le 3k, \\ B_{k,r} \pmod 2, & j = 3N - r, \quad k \le r \le 3k, \\ 0 \pmod 2, & \textit{otherwise.} \end{cases}$$

The two indicated ranges are disjoint and lie on opposite sides of $2N$.

Proof. We use two consequences of Lucas's theorem. For a power of two L,

$$\binom{qL + a}{b} \equiv \binom{a}{b} \pmod 2 \quad (0 \le a, b < L),$$

and

$$\binom{L - a}{L - b} \equiv \binom{b - 1}{a - 1} \pmod 2 \quad (1 \le a, b \le L).$$

For the second identity, both sides are zero if $a > b$; otherwise replace the lower argument by $b - a$ and compare coefficients below degree L in $(1 + X)^{L-a} \equiv (1 + X)^{-a} \pmod{2, X^L}$. The middle term with coefficient four in the residue identity can be discarded throughout this proof.

First write $j = N + r < 2N$, so $k \le r < N$. Put $d = r + k$ and $h = r - k$. Then $a = 2N + d$, $b = h$. Removing the high binary digits in the four binomial factors gives, in order,

$$\binom{2d}{h}, \qquad \binom{d}{2h}, \qquad \binom{2d-1}{h-1}, \qquad \binom{d-1}{2h-1}.$$

These are precisely the four factors for $B_{k,r}$. If $r > 3k$, both factors with lower argument $2h$ or $2h - 1$ vanish, so the residue is even.

Next write $j = 3N - r$ with $1 \le r \le N$. If $r > k$, put again $d = r + k$, $h = r - k$; now $a = 4N - h$, $b = 2N - d$. Digit removal followed by the complement identity transforms the same four factors into

$$\binom{d-1}{2h-1}, \qquad \binom{2d-1}{h-1}, \qquad \binom{d}{2h}, \qquad \binom{2d}{h}.$$

Their two products give the same parity as $B_{k,r}$. Again it is zero for $r > 3k$. If $r = k$, the first two original factors are even and the last two are odd, giving $B_{k,k} = 1$. If $r < k$, both factors with upper arguments $2a$ and $2a - 1$ are even: after removal of the $8N$ digit their upper arguments are $2(k - r)$ and $2(k - r) - 1$, smaller than the respective lower arguments $2N - r - k$ and $2N - r - k - 1$.

Finally write $j = 3N + r$ with $0 \le r \le 3k$. Here $a = 4N + k + r$ and $b = 2N + r - k$. After removing the $8N$ digit from $2a$ and $2a - 1$, both resulting upper arguments are smaller than the respective lower arguments, since $2N - 3k - r \ge 2N - 6k > 0$. Both products are even. These cases cover the full pole interval. The separation follows from $N + 3k < 2N < 3N - 3k$. □

Theorem 56.2 (The constant row has the same thin-block parity as the log-two row)**.** *Let $N = 2^s$, and let $N \le n < 4N/3$. Then*

$$\boxed{2^{s+2}p_n \equiv \begin{cases} 1 \pmod 2, & n = N, \\ 0 \pmod 2, & N < n < 4N/3. \end{cases}}$$

For $s \ge 1$, this is equivalently

$$2^{s+2}p_n \equiv E_{n,2}/32 \pmod 2.$$

Proof. The half-denominator theorem ensures $2^{s+2}p_n \in \mathbb{Z}_2$. For $j < 4N$, the value $2^{s+1}H_j$ is congruent to zero when $j < 2N$ and to one when $j \ge 2N$: among the harmonic denominators only $2N$ has binary order $s+1$. Moreover

$$2^s \sum_{j=n}^{3n} \frac{B_{n,j}}{j} \equiv 0 \pmod 2.$$

Indeed terms with $v_2(j) < s$ are even. At $j = 2N$ the residue is divisible by at least four, so division by two still leaves an even term. The only other possible denominators of order at least s are $j = N, 3N$. If $n > N$, only $3N$ lies in the pole interval and its residue is even because $v_2(n) < s = v_2(3N)$. If $n = N$, both endpoint residues are odd and their two contributions cancel modulo two. Thus

$$2^{s+2}p_n = 2^{s+1}\sum_j B_{n,j}H_j - 2^s \sum_j B_{n,j}/j \equiv \sum_{j \ge 2N} B_{n,j} \pmod 2.$$

For $n = N$, the only odd residue in this tail is $B_{N,3N}$. For $n = N + k$, $1 \le k < N/3$, Lemma 56.1 gives

$$\sum_{j \ge 2N} B_{N+k,j} \equiv \sum_{r=k}^{3k} B_{k,r} = 2 \cdot 16^k \equiv 0 \pmod 2.$$

This proves the first assertion. The second follows from the complete log-two parity theorem, since N is the only power of two in this block. □

Corollary 56.3 (Odd cleared constant output is impossible in the thin dyadic block)**.** *Let $N = 2^s$ with $s \ge 1$, let $N \le M < 4N/3$, and let integers $c_N, \dots, c_M$ cancel every prime-log coordinate exactly. Put*

$$D = 2\,\mathrm{lcm}(1, \dots, 3M), \qquad Q = D\sum_{n=N}^{M} 16^n c_n, \qquad P = D\sum_{n=N}^{M} p_n c_n.$$

Then

$$\boxed{2 \mid c_N, \qquad 2 \mid P, \qquad 2^{s+3+4N} \mid Q.}$$

Consequently the two raw integer outputs have a common factor two.

Proof. The exact log-two relation and its complete parity law give c_N even. Since $v_2(D) = s + 2$, the quotient $D/2^{s+2}$ is odd. Theorem 56.2 therefore gives $P \equiv c_N \equiv 0 \pmod 2$. Finally

$$Q = D16^N\left(c_N + \sum_{n>N} 16^{n-N}c_n\right)$$

has the claimed divisibility. □

Corollary 56.4 (The obstruction persists under individual column clearing)**.** *In the same thin block, put* $d_n = 2\,\mathrm{lcm}(1,\dots,3n)$*. Suppose integers* h_n *cancel the individually cleared log rows,* $\sum_n h_n d_n E_{n,\ell} = 0$ *for every prime* ℓ*. For* $Q = \sum_n h_n d_n 16^n$ *and* $P = \sum_n h_n d_n p_n$*, one still has*

$$2 \mid h_N, \qquad 2 \mid P, \qquad 2^{s+3+4N} \mid Q.$$

Proof. Every d_n in this block has binary order $s+2$; write $d_n = 2^{s+2}\eta_n$ with η_n odd. Divide the exact log-two equation by 2^{s+7}. Its coefficient $\eta_n E_{n,2}/32$ is odd exactly at $n = N$, so h_N is even. By the constant-row parity theorem, $P \equiv \eta_N h_N \equiv 0 \pmod 2$. Finally $Q/(2^{s+2}16^N) = \eta_N h_N + \sum_{n>N} 16^{n-N}\eta_n h_n$ is even. □

Scope. This is an all-index restriction on the actual integer log kernel; it does not replace exact cancellation by congruence cancellation. It excludes the previously tempting choice of an odd cleared constant output in these blocks. It does not bound $v_2(P)$ from above, does not exclude outputs with $v_2(P) = 1$, and does not determine the complete output gcd. In particular it does not provide a bounded-height surviving family or an irrationality proof.

57 An exact image invariant of an actual integer log kernel

Lemma 57.1 (Determinantal divisors measure the output image)**.** *Let* $A \in \mathbb{Z}^{r\times d}$ *have row rank* r*, let* $T \in \mathbb{Z}^{k\times d}$*, and suppose* $[A;T]$ *has row rank* $r+k$*. Write* $\delta_j(B)$ *for the positive gcd of all* j *by* j *minors of* B*. Then the full-rank output lattice*

$$\Lambda = T(\ker_{\mathbb{Z}} A) \subseteq \mathbb{Z}^k$$

has index

$$\boxed{[\mathbb{Z}^k : \Lambda] = \frac{\delta_{r+k}([A;T])}{\delta_r(A)}.}$$

For $k = 1$ *this index is the positive generator of the image ideal.*

Proof. Put A in integer Smith normal form using unimodular row and column operations. Its nonzero diagonal entries are $a_1,\dots,a_r$. Apply the same column operations to T, and let T_0 denote its last $d-r$ columns. Those columns generate the output lattice. A nonzero $(r+k)$-minor of the augmented matrix must contain all first r columns, and equals $\pm(a_1\cdots a_r)$ times a k-minor of T_0. Thus $\delta_{r+k}([A;T]) = \delta_r(A)\delta_k(T_0)$. The last determinantal divisor is the index of its column lattice, again by Smith normal form. □

Proposition 57.2 (Finite simultaneous binary survival on a wider dyadic block)**.** *Let the pole-index block be* $256 \le n \le 511$*, put*

$$D = 2\,\mathrm{lcm}(1,\dots,1533), \qquad \mathcal{L} = \left\{ c \in \mathbb{Z}^{256} : \sum_{n=256}^{511} c_n E_{n,\ell} = 0 \text{ for every prime } \ell \right\},$$

and define integer output maps

$$P(c) = D\sum_{n=256}^{511} p_n c_n, \qquad G(c) = \sum_{n=256}^{511} 16^{n-256} c_n.$$

The lattice $\mathcal{L}$ *has rank* 14*. Its two-coordinate image has rank two and satisfies*

$$\min_{P(c)\neq 0} v_2(P(c)) = 0, \qquad \min_{G(c)\neq 0} v_2(G(c)) = 4,$$

and

$$v_2([\mathbb{Z}^2 : (P,G)(\mathcal{L})]) = 4.$$

In particular an actual integer log-null vector exists with P odd and $G \equiv 16 \pmod{32}$ simultaneously.

Proof. Take $A = (2E_{n,\ell})$ with rows the 242 primes at most 1533. The attached exact verifier regenerates the matrix from the factorial residue formula and performs invertible Smith operations modulo 2^{64}. The following table records the ranks and the sums of the binary orders of the nonzero Smith entries:

rows	rank	$\sum v_2$(Smith entry)
A	242	288
$[A;P]$	243	288
$[A;G]$	243	292
$[A;P;G]$	244	292.

Every nonzero Smith entry in all four calculations has binary order at most 11, strictly below the modulus precision 64; hence the displayed ranks and valuations are exact, not lower bounds from a truncated computation. To justify this precision claim explicitly, lift the performed invertible modular operations to $\mathbb{Z}_2$. Their resulting matrix is $[\mathrm{diag}(2^{e_i}) \mid 0]$ modulo 2^{64}. Factor 2^{e_i} from row i. The remaining first square block is the identity modulo two, since $e_i < 64$, and is therefore invertible over $\mathbb{Z}_2$; all remaining columns are integral. This gives precisely the same local Smith invariants as the displayed diagonal matrix. In regenerating the P row, the verifier uses $2^{11}p_n$ instead of Dp_n. Since $D/2^{11}$ is odd, this is an invertible row scaling over $\mathbb{Z}_2$ and does not alter any binary determinantal-divisor valuation.

The lemma gives the three asserted image invariants by subtraction. Let $\Lambda = (P,G)(\mathcal{L})$. The second invariant implies $\Lambda \subseteq \mathbb{Z} \times 16\mathbb{Z}$, and the third implies that its index in this latter lattice is odd. Therefore its reduction modulo $2(\mathbb{Z} \times 16\mathbb{Z})$ is surjective. In particular it contains a vector with P odd and $G/16$ odd. This concerns the image of the actual integer kernel, not a kernel computed merely modulo a power of two. □

For the vector in the proposition, the unnormalized gamma coordinate is $Q = D16^{256}G$. Thus $v_2(Q) = 11 + 4 \cdot 256 + 4 = 1039$, while P is odd; primitive reduction leaves the binary order of the gamma denominator equal to 1039. No bound on the height of a representative c is obtained.

Per-column normalization. Let $d_n = 2\,\mathrm{lcm}(1,\dots,3n)$. Since $d_n \mid D$, the integer vector $c'_n = (D/d_n)c_n$ gives the same exactly log-free form through $\sum c'_n d_n I_n$. Consequently the proven finite odd-constant attainability also holds in the basis with individually cleared columns. The preceding proposition does not control the height of c' or its weighted error.

Scope. The block $[256, 511]$ is outside the thin-block reflection theorem, so odd constant output does not contradict that obstruction. It is also outside the former common-denominator decay budget; individual column clearing removes that particular analytic restriction. The surviving difficulty is a small representative in the correct integer-kernel congruence class, uniformly along unbounded blocks. The classical determinantal-divisor lemma and this finite image certificate do not prove that height bound or an irrationality result.

58 Two precise infinite completion targets

The new reductions expose two different compatibility assertions. They must not be replaced by a count of successful finite examples.

58.1 Heights in moving output intervals

For a nonempty normalized offset-prefix interval on $[n, M]$, let $h_2(n, M, \lambda)$ be the minimum possible largest primitive denominator of two distinct rational points. Set it to infinity when the interval is empty or a singleton. The interval theorem computes this quantity exactly in every finite instance. The sufficient infinite target is

$$\frac{(1+\lambda_n)h_2(n, M_n, \lambda_n)\log(M_n+1)}{729^n\sqrt{n}} \longrightarrow 0 \tag{144}$$

along unbounded n, with rational $\lambda_n \geq 0$. The minimum-height pair theorem and the signed-prefix estimate would then give a positive integer-under-rationality sum of squares that tends to zero. A particularly concrete sufficient special case is

$$(1+\lambda_n)h_2(n, M_n, \lambda_n) \leq 729^n, \qquad \log(M_n+1) = o(\sqrt{n}).$$

The eighteen exact positive windows satisfy the displayed height inequality at their own indices. They do not establish it along an unbounded sequence. Extending an old lift by zeros preserves its starting index and error, so it does not solve this target.

58.2 Short representatives of an attainable output class

For the individually cleared columns on $[n, 2n-1]$, the target is

$$Fc = 0, \qquad c \in \mathbb{Z}^n, \qquad \|c\|_\infty \leq \left\lceil e^{128n/\log n} \right\rceil, \qquad \sum_{j=n}^{2n-1} d_j p_j c_j \text{ odd}, \tag{145}$$

again for unbounded n. The counting theorem supplies a nonzero vector satisfying the first three conditions eventually. The image calculation proves that the odd-output class is attainable at one larger block, without a height bound. These are different existential statements; they do not give a vector satisfying all four conditions.

The weaker survival requirement $P_n \neq 0$ and $4n - v_2(P_n) \to \infty$ would also suffice. The thin-block obstruction rules out odd P on those thin blocks, but it does not rule out every weaker valuation escape.

58.3 The logical status of the five requirements

For the certified positive finite windows, exact arithmetic, nonvanishing, and the stated small fully normalized errors are proved without extra assumptions. No theorem here supplies infinitely many such windows with vanishing costs. Conversely the columnwise counting theorem is an unconditional infinite statement about small log-null inputs, but it does not exclude formal-zero outputs. Therefore the five requirements have not yet been met by one infinite family.

The reductions do remove some previous restrictions. Signed prefixes can be used with a determinant pair; the terminal index need not be linear in the starting index; determinant one does not impose extra maximum denominator height in a finite output interval; and an individual denominator budget permits doubling windows. None of these facts supplies the missing infinite height or coset representative bound by itself.

59 The consolidated outcome and the exact remaining junction

The earlier stages now fit into one arithmetic picture. Positive target realization and sharp degree bounds settle existence and analytic complexity inside specified classes. Exact prime valuations show that several natural realizations pay too much in primitive denominators. Output quotient norms remove irrelevant coefficient denominators, while cumulative-prefix geometry makes positivity and cost simultaneously tractable. The new nonuniform theorem enlarges the certified positive class by a precisely weighted allowance for negative prefixes. Columnwise integerization secures smallness on long windows before a useful output class is selected.

59.1 Which directions are closed, and which are conditional?

Direction	Current conclusion
Canonical dyadic targets and interpolation	Exact kernels, brackets, valuations, and full-class obstructions are proved. Failure for dyadic interpolation does not exclude arbitrary rational interpolation.
Repaired rational localization	The exact threshold is $q_M = o(M)$, and the two-adic argument excludes it for every parameter of the repaired $k = 1$ family. The threshold is a failed route for that family, not an outstanding existence problem within it.
Bessel mirror and acceleration	Large harmonic-core costs and the displayed mirror/accelerated obstructions are proved. The accelerated core-cancelled family X_j has exact two-adic cancellation and odd-prime survival criteria; the latter are not a universal denominator estimate for every possible subsequence.
Autonomous dilation of a fixed admissible kernel	Raw error decays, but the actual primitive error diverges under the stated fixed-kernel and nonzero-output hypotheses.
Finite quotient optimization	Exact finite minima and transfer certificates are proved. The infinite seminorm and its dual have precise alternatives; the favorable alternative is not established.
Nonuniform negative-prefix positivity	The universal prefix threshold, including its boundary, is settled. The individual-vector sufficient test and compact rational output interval are unconditional. Infinite cheap output existence is not supplied by those geometric facts.
Columnwise small vectors	Short exact log-null integer inputs exist eventually, and their fully cleared cost tends to zero. Equally short formal-zero vectors also exist. A useful bounded-height output class is still needed.
Wide-block binary output class	The desired simultaneous congruence class exists in the actual integer kernel on the stated finite window. No height bound or unbounded sequence follows from that image computation.

59.2 Two explicit ways to complete the same-family argument

The positive route can now be stated without a determinant partner. For unbounded n, find a rational log-null lift on $[n, M_n]$ with integer output (q_n, p_n), $q_n > 0$, such that

$$\sum_{k=n}^{M_n-1} (a_k - a_{k+1})(-S_k)_+ \le q_n a_{M_n}, \qquad \frac{q_n \log(M_n + 1)}{729^n \sqrt{n}} \longrightarrow 0.$$

The new positivity theorem and the signed cost estimate would then give $0 < q_n\gamma - p_n \to 0$. This is a concrete sufficient theorem, not a claim that the required lifts have been found infinitely often.

The integer-column route starts with $d_j = 2\,\mathrm{lcm}(1,\dots,3j)$ and $J_j = d_j I_j$. The uniform budget and counting arguments already provide short log-null inputs. It remains to choose such inputs with output (Q_n, P_n) satisfying, for example,

$$P_n \neq 0, \qquad 4n - v_2(P_n) \longrightarrow +\infty.$$

Small cost and $P_n \neq 0$ force $Q_n \neq 0$ eventually. The gamma coordinate is divisible by 2^{4n}; the displayed valuation gap forces the primitive denominator to diverge. The affine-completion theorem then produces a determinant-one companion and a positive quadratic form tending to zero. A finite surviving class and an unrelated short input do not establish their intersection.

59.3 Five-condition audit

	Requirement	What has and has not been established
1	Integer coefficients, no residual logs	Established exactly for the stated finite lifts and the infinite log-cancellation constructions.
2	Nonzero evaluated value	Established for the positive-prefix and new budget-certified kernels, and for independent-output sums of squares. It is not established for the short vectors supplied by counting alone.
3	Decay after arithmetic normalization	Uniform sufficient bounds and short-vector budgets are proved; many concrete finite costs are certified. Their compatibility with the required output/sign class at unbounded indices is unresolved.
4	Infinitely many useful indices	Infinite log-null inputs and free extension indices are proved. Infinite useful outputs obeying all the same cost and nonvanishing conditions are not.
5	No unproved premise in the final contradiction	The general theorems and exact finite certificates keep their hypotheses explicit. The proposed irrationality conclusions remain conditional on the missing simultaneous existence assertion.

Thus it is not accurate to say that only requirement 3 remains independently of the others. The unresolved statement joins a cost bound to a useful output class for infinitely many indices. The new sharp theorem improves that joining problem by providing an exact, larger positive class. It does not by itself resolve the infinite arithmetic selection problem.

A Exact starting certificates for the infinite rank theorems

The two finite calculations below close the base cases of Theorems 23.2 and 23.3. They take place in finite fields and do not require numerical integration. The residue entries $2A_{n,j}$ are computed by the integral binomial identity from Section 2, and independently checked against their factorial expression. The row p_n is computed using modular inverses of the harmonic denominators. The moduli exceed $6N$, so all factorial and harmonic denominators appearing in these checks are units.

The following complete Python program uses only the standard library. Its determinant outputs are

$$\det V_9 = 97 \pmod{101}, \qquad \det W_{17} = 65 \pmod{211}.$$

They are nonzero, which is all the infinite induction needs.

```python
#!/usr/bin/env python3
"""Standard-library modular certificates for the infinite rank induction."""
from math import comb,factorial,isqrt
from pathlib import Path
import json

def primes(n):return [p for p in range(2,n+1) if all(p%d for d in range(2,isqrt(p)+1))]
def valuation(n,p):
    v=0
    while n%p==0:n//=p;v+=1
    return v

def matrix(N,mod,augmented):
    assert mod>6*N and mod in primes(mod)
    rows=[];constants=[]
    for n in range(1,N+1):
        residues={};h=0;pn=0
        for j in range(1,3*n+1):
            h=(h+pow(j,-1,mod))%mod
            if j<n:continue
            a=j+n;b=j-n
            B=comb(2*a,b);C=comb(a,2*b)
            Bp=comb(2*a-1,b-1) if b else 0
            Cp=comb(a-1,2*b-1) if b else 0
            w=(-1)**b*(B*C+4*Bp*C+Bp*Cp)%mod
            numerator=(-1)**b*8*j*j*factorial(2*j+2*n-1)*factorial(n+j-1)
            denominator=factorial(2*j-2*n)*factorial(j-n)*factorial(3*n-j)*factorial(3*n+j)
            assert w==numerator*pow(denominator,-1,mod)%mod
            residues[j]=w
            pn=(pn+w*pow(2,-1,mod)*(h-pow(2*j,-1,mod)))%mod
        assert sum(residues.values())%mod==2*pow(16,n,mod)%mod
        rows.append(residues);constants.append(pn)
    ps=primes(3*N)
    V=[[sum(w*valuation(j,p) for j,w in row.items())%mod for row in rows]for p in ps]
    if augmented:V+=[[pow(16,n,mod) for n in range(1,N+1)],constants]
    return V,ps

def determinant(A,mod):
    A=[row[:] for row in A];n=len(A);assert all(len(row)==n for row in A)
    product=1;pivots=[];swaps=[]
    for j in range(n):
        k=next((k for k in range(j,n) if A[k][j]),None)
        assert k is not None
        if k!=j:A[j],A[k]=A[k],A[j];product=-product;swaps.append((j,k))
        p=A[j][j];pivots.append(p);product=product*p%mod;inv=pow(p,-1,mod)
        for k in range(j+1,n):
            f=A[k][j]*inv%mod
            for l in range(j,n):A[k][l]=(A[k][l]-f*A[j][l])%mod
    return product%mod,pivots,swaps

def verify():
    out=[]
    for N,mod,augmented,expected in [(9,101,False,97),(17,211,True,65)]:
        A,ps=matrix(N,mod,augmented);det,piv,swaps=determinant(A,mod)
        assert det==expected
        out.append({'N':N,'modulus':mod,'augmented':augmented,'primes':ps,'matrix':A,'determinant':det,'
 pivots':piv,'swaps':swaps})
        print('N',N,'modulus',mod,'determinant',det,'pivots',piv,'swaps',swaps)
    assert all(len(primes(3*n))>n for n in range(1,9))
    assert all(n-len(primes(3*n))<2 for n in range(9,17))

if __name__=='__main__':verify()
```

B Source coverage, corrections, and proof dependencies

This is a mathematical consolidation of the recovered research record, not a transcript of every conversational sentence. The scope starts at the corpus gamma-TOP10 discussion and includes the six subsequent source manuscripts below and the additional junction theory stored with their

exact verification material. The early lemniscate exchange and the separate theta-sum project are contextual background, not additional claims proved by this paper.

ID	Recovered source	Material retained in the consolidation
S1	Positive Certificates and Primitive Arithmetic for Euler's Constant, 31 October 2026	Positive realization, sharp degree law, exact interval optimization, Laguerre arithmetic, support compression, localization, thin cancellation, and the raw independent-output height barrier.
S2	Primitive Denominators in Approximations to Euler's Constant, 1 November 2026	All-parameter repaired-localization obstruction, opposite Bessel limits, harmonic-core escape, cross-block separation, and primitive lift normalization.
S3	Arithmetic Costs after Acceleration in Approximations to Euler's Constant, 1 November 2026	Uniform two-adic cancellation, odd-prime survival, exact finite quotient minima, fixed-shift localization, and node-zero arithmetic.
S4	Prime-Support Recursions and Primitive Errors in Approximations to Euler's Constant, 1 November 2026	Exact rank and basis, non-P-recursion, total positivity and bandwidth, sharp integral asymptotics, finite facets, infinite dual geometry, and autonomous-dilation obstruction.
S5	Intrinsic Cost Bounds and Certified Transfers for Euler's Constant, 1 November 2026	Positive finite certificate, affine and primitive completion, fixed-output trap, extension schedule, failed canonical shift, finite repairs, and the stronger all-prime binary congruences.
S6	Rational Intervals, Signed Prefixes, and Columnwise Arithmetic Budgets for Euler's Constant, 2 November 2026	The strongest shared kernel foundations, horizon-free cost bound, minimax Farey pairs, eighteen finite windows, rational repair, column clearing, thin-block reflection, and actual integer-image invariants.
J	Five-condition junction theory and certificates, 1 November 2026	Explicit reverse-budget boxes, the formal-zero comparison, coefficient-sign obstruction, the first positive but expensive relation, Farkas feasibility, and fiber antichains. Affine and binary lemmas duplicated in S5/S6 are proved only once.
N	New synthesis, 2 November 2026	Sharp nonuniform negative-prefix budget, its boundary and universal sharpness, individual rational positivity certificate, compact output polytope, and a single-output conditional completion criterion.

The historical ten-result-group list in S1 comprised: least-denominator selection; exact primitive-error optimization; positive target realization; the polynomial-square image; Laguerre integerization; primitive-cost obstructions; support compression; localization; adverse-prime removal; and integerized signed cancellation. Every group is included above. The original rank numbers are not reused as a ranking of the substantially enlarged manuscript.

Editorial and mathematical corrections. An imprecise sentence in S1 asserted rationality of all displayed kernel masses, although one display contained $\gamma - 1/2$. The corrected assertion

concerns only the rational masses of K_a. No argument depends on rationality of $\gamma - 1/2$. Meyer's rational-isotropy theorem is explicitly attributed to its classical literature [36]. The exact factorial moment is

$$\int_0^\infty t^s(t-1)\log t\, e^{-t}\, dt = s!(1 + sH_s - s\gamma).$$

The earlier redundant display with an extra s is not retained. Historical "remaining target" statements are superseded where later proofs settle the corresponding family, particularly the repaired localization obstruction. The proper-kernel majorant is renamed $\mathcal{U}_j$ to separate it from the core-cancelled approximant U_j.

Proof status. The numbered statements are proved deductions, explicitly conditional completion results, or exact finite certificates as identified in their statements. Standard external inputs are cited where used. Computer-assisted finite statements are defined by exact rational equations or inequalities and accompanied by reproducible verifiers. The compilation and checks do not constitute independent peer review, a formal proof-assistant development, or a historical novelty search.

B.1 An exact finite test of the new interval enlargement

The strict-enlargement formula can be applied directly to an existing certificate, without a new linear-programming optimization.

Proposition B.1 (The constructed extension from 18 to 59)**.** *Take the certified normalized endpoint lifts of the nonnegative-prefix interval on* $[18, 59]$*, as specified in* `cone_interval_18_59.json` *inside the embedded interval archive. Apply the two explicit extrapolations in Corollary 42.6. Both new endpoint lifts cancel every prime-log row exactly and satisfy the weighted positivity budget. The resulting closed* extrapolated interval *strictly contains the old interval and has the same least primitive denominator*

$$q_{\min} = 154780615090340473039841749,$$

attained at numerator 893417956532391068205964 73*. This assertion does not compute the least denominator in the entire new sharp-budget polytope.*

Proof. The standard-library verifier `verify_window18_extension.py` reads the specified rational endpoint vectors and regenerates every residue from the factorial formula. It checks the forty prime rows, the normalization $q = 1$, the output slopes, and the negative-part budgets at both extrapolated endpoints. The exact positive extension lengths follow from the formula of Corollary 42.6. It then checks that both new endpoints remain strictly inside the original certified Farey-parent bracket. The determinant-one parent property gives the lower bound $q_{\min}$ for every rational in that bracket, while the previously certified fraction still belongs to the enlarged interval. This proves equality of the least denominator. All comparisons use exact fractions; no digits of γ are used. □

For scale only, the old width is approximately $1.767715027228 \cdot 10^{-53}$. The displayed extrapolation extends it to the left by approximately $3.529919840930 \cdot 10^{-171}$ and to the right by approximately $3.120903973075 \cdot 10^{-174}$. The exact comparisons, not those rounded decimals, certify the strict enlargement. Thus this particular constructed interval gives a positive feasible-region gain without a denominator improvement. Optimizing the full new polytope is a different finite problem and has not been performed here.

C One-source compilation and embedded exact verification

Compile the single TeX file with a standard TeX installation:

```
pdflatex gamma_consolidated_positive_kernels_and_arithmetic_2026_11_02.tex
pdflatex gamma_consolidated_positive_kernels_and_arithmetic_2026_11_02.tex
```

The source has no external `\input`, `\include`, figure, or BibTeX dependencies. Its post-document comments contain one Base64-encoded ZIP. That collection preserves the five earlier verification archives byte-for-byte and supplies a combined runner, the additional consolidation checks, and a manifest.

The following extractor uses only the Python standard library. Run it beside the TeX file; the name of the archive is not an external dependency of compilation.

```
from pathlib import Path
import base64, hashlib, re

p = Path("gamma_consolidated_positive_kernels_and_arithmetic_2026_11_02.tex")
s = p.read_text(encoding="utf-8")
meta, rest = s.split("% BEGIN EMBEDDED VERIFICATION\n", 1)[1].split("\n", 1)
expected = meta.removeprefix("% SHA256 ")
payload = rest.split("% END EMBEDDED VERIFICATION", 1)[0]
data = base64.b64decode("".join(
    line[2:] for line in payload.splitlines() if line.startswith("% ")
), validate=True)
if hashlib.sha256(data).hexdigest() != expected:
    raise SystemExit("Embedded archive hash mismatch")
Path("gamma_consolidated_verification.zip").write_bytes(data)
print("Verified embedded archive:", expected)
```

Extract the ZIP into a new directory and follow its `README.txt`. The top-level runner checks the inner archive hashes, rejects unsafe paths and optimized Python, and uses fresh temporary extraction directories. Any older checkers that generate local reports do so inside those temporary directories. The source archives are not modified. Some full lattice checks are substantially slower than the elementary sanity checks.

What the computational checks certify. The collection checks the finite canonical brackets, exact quotient minima and primal–dual certificates, affine corrections, prefix signs, transfer identities, binary arithmetic, the finite integer kernel-image invariants, and the new boundary example. It does not turn a finite collection of windows into an all-index theorem. The all-index identities and inequalities are established by the written mathematical proofs.

References

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[2] Keio–Shaitetsu research notes, *Eighteen Auxiliary Theorems for Gamma Approximation*, 31 October 2026, internal source manuscript.

[3] Keio–Shaitetsu research notes, *Primitive Costs for Euler's Constant*, 31 October 2026, internal source manuscript.

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[5] NIST Digital Library of Mathematical Functions, §5.5, equation 5.5.8 (digamma duplication). https://dlmf.nist.gov/5.5.E8.

[6] NIST Digital Library of Mathematical Functions, §18.12, equation 18.12.13 (Laguerre generating function), and §18.3 (orthogonality and normalization). https://dlmf.nist.gov/18.12.E13; https://dlmf.nist.gov/18.3.

[7] NIST Digital Library of Mathematical Functions, §27.2, equation 27.2.3 (the prime number theorem). The corresponding $\vartheta(x) \sim x$ follows by partial summation. https://dlmf.nist.gov/27.2.E3.

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